

Gravitation, torsion, and cosmological dynamics

Arithmetic memory, field equations,
and transition from contraction to expansion

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Abstract

Torsional amplitude is determined as an explicit functional of the material state through the spinorial action, elimination of contorsion, and metric variation. A material sector is constructed in which this amplitude is positive and conserved. Its energetic contribution produces a^{-6} dilution and, in the flat homogeneous realization with $\Lambda_{\text{eff}} = 0$, positive densities, and $w < 1$, a unique transverse bounce threshold. For dust, an exact solution is obtained for all time, with finite curvature and Levi–Civita causal geodesics complete in both directions.

The construction starts from a formal kernel called Triadic Modular Holography, henceforth HMT. *All Possible Paths*, henceforth APP, comprises arithmetic paths on $(\mathbb{Z}/9\mathbb{Z})^2$, with adjacency given by the additive Cayley graph with cardinal generators and a toroidal cellular realization, distinguishing sum and product operations and evaluations; TRIT is the unit of oriented signature $\{-1, 0, +1\}$, determining local regimes; the *Topological Phase Kernel*, henceforth TPK, composes the dynamics of transport and memory. The enriched state preserves sheets, orientation, remainder, carry, and history during phase return. Dimensional Mechanics expresses its internal results through physical quantities and observables, preserving the structural provenance of the couplings.

Gravitational normalization is first developed through the longitudinal composition $L = (5759/23040)\alpha^{16}L_*$ and its positive radial metric. In the realization defined by these operations and by the identification of the reduced gravitational radius, reciprocity with the action section fixes $G = c^3 L^2 / \hbar_{\text{tet}}$ in every admissible mode. The proof preserves the full archive and establishes the conditions for static and dynamic reduction of memory. The dimensional choice $L_* = 1 \text{ m}$, the longitudinal and metric rules, and the radial identification are stated in the theorem itself.

Route transport admits a Cartan realization compatible with composition, inversion, and refinement. Two material actions are distinguished on that realization: the minimizing extension of the memory screen and the spinorial realization. Each retains its own variation with respect to the connection. In the spinorial sector, affine in contorsion, inversion of the Holst and torsion operators determines the effective correction; for the minimal extension, the nonlinear dependence of the current is preserved and a local criterion is proved through the Hessian. The current, quartic observable, and its normalization explain the value of the amplitude as well as its role in cosmological evolution.

Persistence of the bounce under perturbations is proved through distinct local bounds, whereas causal completeness belongs to the explicit global dust solution. Residual balances, horizon information, and the observer's operations preserve the conditions on sources, couplings, and charts that they use. The relational response of the joint state is connected to advanced and retarded propagation through Green operators for finitely supported sources, associated with the covariant difference constructed from route transport and a boundary compatibility condition. The result relates generated memory, its geometric realization, and the dynamical and informational consequences of that realization.

The composition of action, the Barbero–Immirzi area increment, and the horizon determines the conversion $T_{\text{cos}}(L_\alpha)/\Theta_{\text{clk}} = \log 3/(2\pi)$. The angular response Hamiltonian yields a finite free-energy balance through relative entropy and a first law at fixed radius. A relative thermometric section also characterizes the positive transducer and the criterion for identifying another reader on that same section.

Keywords: gravitation; torsion; arithmetic memory; field equations; Planck length; transition from contraction to expansion.

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1 Introduction

The gravitational question organizing this work is how a structure preserving the history of its operations can determine a material current and, through it, a contribution to spacetime evolution. The answer is constructed from route transport through to variation of the action: the central result is an explicit torsional amplitude, accompanied by a sector of conserved positivity and a global cosmological solution in the dust realization. The thesis connects arithmetic memory to a geometric response determined by operations; phase return preserves an observable periodicity while the state accumulates orientation, carry, and depth.

A formal mathematical kernel called Triadic Modular Holography, henceforth HMT, is introduced for this purpose. *All Possible Paths*, henceforth APP, is an arithmetic path structure on $(\mathbb{Z}/9\mathbb{Z})^2$. Its adjacency is described by the additive Cayley graph with cardinal generators $\pm(1, 0)$ and $\pm(0, 1)$, and its periodic cellular realization is toroidal. Additive accumulation, multiplicative composition, and their evaluations retain distinct roles within that structure. TRIT is the unit of oriented signature $\{-1, 0, +1\}$, typing an operation's local regime. The *Topological Phase Kernel*, henceforth TPK, constitutes its dynamics of selection, transport, memory, and prolongation. The enriched state and the joint discrete structure of the continuum preserve sheets and residual data through those operations. Dimensional Mechanics realizes their outputs in action sections, geometric quantities, and physical observables; internal values transmit to each stage the relations determining them.

The central question is how transported memory comes to enter gravitational evolution. Dilution, the bounce, and its persistence describe the cosmological consequences of an amplitude that must first be constructed from the material action. The first relationship is expressed through two readers of one and the same route,

$$\frac{a_n}{a_{\text{ref}}} = \varphi^{n/3}, \quad M_n = \varphi^{-2n}, \quad M_n = \left(\frac{a_n}{a_{\text{ref}}}\right)^{-6}. \quad (1.1)$$

Here $a_{\text{ref}} > 0$ fixes the normalization of the scale factor. The power expresses the relationship between the growth of the spatial realization and the quadratic character of the memory reader. The gravitational contribution also requires determining the current and amplitude accompanying that scale dependence. Both constructions occupy a section of their own before being introduced into the cosmological equations.

The first movement of the article brings together construction and geometry. The decomposition of transport distinguishes symmetric memory from oriented circulation. The representation of the route groupoid in Cartan geometry is developed with composition, inversion, covariance, and refinement. The action, the Barbero–Immirzi parameter, the gravitational constant, and the areal unit preserve their structural antecedents; their reuse transmits the conditions individuating each section.

In particular, Section 15 first constructs the length L , composes its positive reader with the same character that determines energy, and obtains the gravitational coefficient. The circular clock then receives $t_0 = \pi_{\text{HMT}} L / (54c)$; its radius R_{108} coincides with L in this realization. The proof distinguishes longitudinal composition, operator response, and evaluation in physical units, in that order. The longitudinal and metric rules are compositions made explicit in this development; the theorem proves rigidity for the realizations preserving them, with the stated radial identification.

The second movement develops amplitude, evolution, and persistence. The screen current is carried to a differential with respect to the connection through transported incidence and the minimizing extension. The spinorial realization is developed, in turn, from the internal Clifford matrices and its material action. The minimizing-extension action and the spinorial action are distinct functionals: the former generally depends nonlinearly on the connection, whereas the latter is affine in contorsion with cotetrad and spinorial field fixed. Their variations act on each functional's own arguments; any composition of the two must preserve that dependence. Tracial normalization fixes the corresponding density reading and remains distinguished from an extensive sum. Variation of the connection determines the torsional response of the minimal action. As the cotetrad and material fields evolve, the torsion algebraically associated with their spin current also changes.

The amplitude of the spinorial sector is determined by

$$s_0[\varrho, V_c] = \frac{3\kappa c^2 \hbar^2}{16} \frac{\gamma^2}{1 + \gamma^2} \frac{\text{Tr}(\varrho \hat{Q}_{\text{sp}})}{V_c^2}, \quad \kappa = \frac{8\pi G}{c^4}. \quad (1.2)$$

Here ϱ is the normalized material state, V_c the comoving volume, and $\hat{Q}_{\text{sp}}$ the quartic observable constructed from the current. Theorem 28.4 derives the formula, and Corollary 28.5 exhibits states in which the amplitude is positive and remains constant. Dependence on the second moment preserves information about the state even when the mean current vanishes. The material datum and its volume fix a realization; the coupling constants come from the structural sections constructed previously.

Elimination of torsion is also treated for the minimal-extension action, whose dependence on the connection is generally nonlinear. Its differential determines the stationary equation; the Hessian controls local continuation, and an integral identity determines the effective action. This formulation preserves the full dependence of the minimum when variations are performed.

The bounce formulation preserves the positive densities, the sign of the torsional contribution, and the exponent inequality of the homogeneous realization. Its local persistence is proved through two separate controls: a uniform perturbation bound preserves the sign change at the endpoints of an interval; a uniform bound on its derivative preserves monotonicity and transversality. Realizations of the cosmological selectors are stated with their own domains, together with the invariants and balances they produce.

The flat dust solution with $\Lambda_{\text{eff}} = 0$ also provides a global development:

$$a(t) = a_b \left(1 + \frac{(t - t_b)^2}{\tau^2} \right)^{1/3}, \quad a_b^3 = \frac{s_0}{\rho_0}, \quad \tau^2 = \frac{s_0}{6\pi G \rho_0^2}, \quad c = 1.$$

The proof establishes curvature regularity and completeness of Levi-Civita causal geodesics in that homogeneous realization. Its integration and local persistence of the threshold describe two complementary properties, with their respective explicit domains.

Conservation of the residual tensor is formulated in the selected metric chart, with constant couplings on that chart and a covariantly conserved material source. Its decomposition into trace and traceless parts identifies exchange between the two sectors; constant trace characterizes the case of separate conservation.

The third movement studies horizons and relational response. The reduced state, accessible records, and purification describe what information a reading represents and what information remains in the correlations of the joint state. The operational definition of the observer comprises orientation and trivialization of transport, restriction to observables, and projection onto the subsystem. The self-inertial response is connected to those operations through factorization through the global trace and the geometric projection defect.

Horizon balances likewise preserve the conventions of area, energy, temperature, and state that they use. Sections 32.7–32.9 compose the area quantum with the calendar's decision energy and obtain the conversion between chronological and cosmological temperatures. The reduced state preserves the occupations of both angular sectors. At fixed radius, its energetic response has Hamiltonian $H_R = \tau_R \Delta_0 D_\partial$; its variations give the first law, while finite differences retain the relative-entropy term. This development constructs a relative thermometric section and establishes on it a test of equality between linear conversion maps. Sectorial provenance remains accessible in both area and energetic cost. Advanced and retarded propagation for finitely supported sources is composed on unitary route transport. Equality of their boundary readings is expressed through a linear constraint on sources; its projection determines the variational response in that domain.

The argument's unity lies in the relationship between source, current, amplitude, and geometry. Memory preserves information not exhausted by the phase reading; its material realization produces a second moment capable of contributing to the dynamics even with zero mean current. Cosmological evolution expresses that contribution globally in the resolved sector, and the observer's operations specify which part of the information remains accessible. Structure, equation, and solution are thereby connected within realizations with explicit domains.

I. Construction and geometry

2 Formal kernel of Triadic Modular Holography

Triadic Modular Holography, abbreviated HMT, begins with a finite structure of positions and operations. The order of construction matters: positions receive arithmetic evaluations, transitions acquire a ternary signature, and the dynamics preserves the information needed to prolong them. Numerical realizations are formed only afterwards. A constant recognized at the end of this process does not replace the rule that produced its region or the history that determines its evaluation.

The name *All Possible Paths*, APP, expresses the role of paths on the support. It is adopted as the name of the construction, with a conceptual reference to the path formulation of quantum mechanics; it does not imply that its arithmetic law is Feynman's path integral. The *Topological Phase Kernel*, TPK,¹ denotes the composition of selection, transport, memory update, and prolongation. The acronyms are retained only after the objects they represent have been defined.

2.1 APP: positional support and arithmetic evaluations

Consider the nine interior marks on a line graduated from zero to ten,

$$D_9 = \{1, 2, 3, 4, 5, 6, 7, 8, 9\}.$$

Their horizontal and vertical arrangement produces the eighty-one positions of D_9^2 . The interior marks must be distinguished from the ten unit intervals of the reference line. This arrangement fixes an alphabet and two directions; it does not yet use a real expansion to determine a trajectory.

Cyclic identification of the positions yields the support

$$X_9 = (\mathbb{Z}/9\mathbb{Z})^2.$$

The chart of positive representatives retains the symbol nine for the zero class. For every positive integer, define

$$\rho_9(n) = 1 + ((n - 1) \bmod 9), \quad q_9(n) = \frac{n - \rho_9(n)}{9}, \quad n = \rho_9(n) + 9q_9(n). \quad (2.1)$$

The positive digital root is therefore a choice of representative of the residue class. By itself, it does not preserve the integer before reduction; the coordinate q_9 supplies precisely the missing information.

At each position (i, j) , accumulating i , repeated j times, produces ij . The two integer evaluations and their projections are

$$S(i, j) = i + j, \quad P(i, j) = ij, \quad \Sigma(i, j) = \rho_9(i + j), \quad \Pi(i, j) = \rho_9(ij).$$

The symmetry $(i, j) \mapsto (j, i)$ interchanges the directions of accumulation and preserves both evaluations. This is a property of a single support, not a subsequent identification between two independent matrices.

Definition 2.1 (Lifted arithmetic evaluation). The APP evaluation that preserves both sheets is

$$\mathcal{A}(i, j) = ((R_{ij}^+, q_{ij}^+), (R_{ij}^\times, q_{ij}^\times)) = ((\rho_9(i + j), q_9(i + j)), (\rho_9(ij), q_9(ij))). \quad (2.2)$$

Here the term *sheet* denotes an evaluation on the same support: additive or multiplicative. The two quotient coordinates belong to the arithmetic state and are not discarded when the residues are reduced.

Proposition 2.2 (Reconstruction of the evaluations). *The two pairs in (2.2) determine $i + j$ and ij exactly. The residues alone do not determine these integer evaluations or their quotients.*

¹The authors also dedicate the initials TPK to Tan Pengkiang. This dedication is independent of the mathematical definition.

	1	2	3	4	5	6	7	8	9
1	1	2	3	4	5	6	7	8	9
2	2	4	6	8	1	3	5	7	9
3	3	6	9	3	6	9	3	6	9
4	4	8	3	7	2	6	1	5	9
5	5	1	6	2	7	3	8	4	9
6	6	3	9	6	3	9	6	3	9
7	7	5	3	1	8	6	4	2	9
8	8	7	6	5	4	3	2	1	9
9	9	9	9	9	9	9	9	9	9

One support: 81 positions.

Two evaluations:
 $S(i, j) = i + j$, $P(i, j) = ij$.

Each row is generated by accumulation:
 $P(i, j + 1) - P(i, j) = i$.

The chart represents 0 (mod 9) by 9.

The highlighted block is
 $B_c = \begin{pmatrix} 7 & 2 \\ 2 & 7 \end{pmatrix}$,
 with eigenvalues 9 and 5.

Figure 1: APP multiplication table in positive representatives. Non-unit rows and columns are distinguished by gray shading; the frame identifies the adjacent diagonal block with equal diagonals. The displayed table is an observable of the arithmetic torus; its representation edges do not constitute an intrinsic topological boundary of the torus.

Proof. The first statement is (2.1) applied to each sheet. For the second, $10 = 1 + 9$ and $19 = 1 + 2 \cdot 9$ have the same residue and different quotients. On the two-dimensional support, positions (5, 5) and (8, 2) give the same sum and the same product residue, but products $25 = 7 + 18$ and $16 = 7 + 9$. The residue projection identifies data that the lifted evaluation distinguishes. $\square$

2.1.1 Cayley graph, toroidal topology, and winding number

The cardinal translations are given, in row and column coordinates, by

$$\nu_N = (-1, 0), \quad \nu_E = (0, 1), \quad \nu_S = (1, 0), \quad \nu_O = (0, -1), \quad T_d(x) = x + \nu_d.$$

The first coordinate increases southwards and the second eastwards. The graph

$$\mathcal{G}_9 = \text{Cay}(X_9, \{\nu_N, \nu_E, \nu_S, \nu_O\})$$

is the two-dimensional cyclic lattice: it has eighty-one vertices and four outgoing oriented edges per vertex. It is a Cayley graph for the *additive* structure of the support. Multiplication of residues has zero divisors and does not turn all APP rows into a multiplicative group.

A cardinal word $w = d_1 \cdots d_r$ and an origin x_0 determine

$$x_k = x_0 + \sum_{j=1}^k \nu_{d_j} \pmod{9}.$$

There are $81 \cdot 4^r$ oriented paths of length r , including returns and backtracking. When the path closes, its integer displacement before reduction defines

$$\text{wind}(w) = \frac{1}{9}(\#S(w) - \#N(w), \#E(w) - \#O(w)) \in \mathbb{Z}^2. \quad (2.3)$$

Path reversal changes the sign of this vector. Returning to the same residue position therefore does not require the winding to vanish. This is the first example of a distinction that will reappear in phase return: one coordinate may close while another records a nontrivial evolution.

2.1.2 Transport quotient and cocycle identity

Let $s_9 : \mathbb{Z}/9\mathbb{Z} \rightarrow D_9$ be the section of positive representatives. The transport quotient associated with additive composition, also called the *carry* in positional arithmetic, is

$$c_9(a, b) = \frac{s_9(a) + s_9(b) - s_9(a + b)}{9}.$$

The compositional character of this memory is expressed by an exact identity.

Lemma 2.3 (Cocycle identity). *For $a, b, c \in \mathbb{Z}/9\mathbb{Z}$,*

$$c_9(a, b) + c_9(a + b, c) = c_9(b, c) + c_9(a, b + c).$$

Proof. Both sums equal

$$\frac{s_9(a) + s_9(b) + s_9(c) - s_9(a + b + c)}{9}.$$

Associativity of the sum of lifted representatives requires precisely this additional term when composing in the quotient. $\square$

With positive representatives, this cocycle is not normalized at the identity: $c_9(0, a) = 1$. If s_0 takes representatives in $\{0, \dots, 8\}$, its normalized carry c_0 satisfies

$$c_9(a, b) = c_0(a, b) + \mathbf{1}_{a=0} + \mathbf{1}_{b=0} - \mathbf{1}_{a+b=0}.$$

The difference is the change of section; it does not alter the cocycle identity. The subsequent decomposition of the calendar into phase and window will use representatives from zero to eight.

The totals of both evaluations distinguish the visible contribution from that retained in the quotient:

$$\begin{aligned} \sum S &= 810, & \sum P &= 2025, & \sum \Sigma &= 405, & \sum \Pi &= 459, \\ \sum q^+ &= 45, & \sum q^\times &= 174. \end{aligned}$$

It follows that

$$2025 - 810 = (459 - 405) + 9(174 - 45) = 54 + 9 \cdot 129. \quad (2.4)$$

The sum-product difference is not exhausted by the visible defect 54. The identity also preserves the quotient defect 129. Removing it would change the arithmetic being transported, even if the residue table remained identical.

2.1.3 Ternary reduction and multiplicative radical

The ring $R = \mathbb{Z}/9\mathbb{Z}$ has unit group $\{1, 2, 4, 5, 7, 8\}$ and maximal ideal

$$\mathfrak{m} = (3) = \{0, 3, 6\}, \quad \mathfrak{m}^2 = 0.$$

Every additive row is a permutation of the nine representatives. A multiplicative row is such a permutation exactly when its index is a unit. Rows three and six contain three repeated values; row nine contains only the representative of the zero class. Thus multiplicative folding already distinguishes the units from the nilpotent sector before any continuous geometry is introduced.

The map $x \mapsto 3x$ has kernel and image both equal to $\mathfrak{m}$. It induces an isomorphism of vector spaces over $\mathbb{F}_3$,

$$R/\mathfrak{m} \longrightarrow \mathfrak{m}, \quad [x] \longmapsto 3x,$$

but not a ring isomorphism. In particular, the visible sequence 3, 6, 9 admits two different readings: its direct reduction modulo three is 0, 0, 0, whereas extracting the factor three and then reducing its internal coordinate gives 1, 2, 0. This second operation preserves a coordinate of the ideal; it is not the same projection as directly reducing the original number.

2.1.4 Latin realization of the additive evaluation

The difference between the two evaluations can also be expressed through operators. Let $(e_r)_{r \in \mathbb{Z}/9\mathbb{Z}}$ be the orthonormal basis of $\mathbb{C}^9$. The assignment $v_{ab} = e_{a+b}$ produces an orthonormal basis in every row and column. The projectors $P_{ab} = v_{ab}v_{ab}^*$ satisfy

$$\sum_b P_{ab} = I_9 = \sum_a P_{ab}, \quad P_{ab}^2 = P_{ab}.$$

Indeed, each translation $b \mapsto a + b$ permutes the nine residues. This yields a realization of a Latin square by rank-one projectors, in the terminology of quantum Latin squares [3, 5]. This construction starts from the additive sheet already defined and makes explicit the map to its operator representation.

The multiplicative evaluation preserves different content: for $a = 3$, the vectors e_{3b} repeat the residues 0, 3, 6, and

$$\sum_{b \in \mathbb{Z}/9\mathbb{Z}} |e_{3b}\rangle\langle e_{3b}| = 3(|e_0\rangle\langle e_0| + |e_3\rangle\langle e_3| + |e_6\rangle\langle e_6|).$$

The multiplicity is recorded in the operator and characterizes the multiplicative radical. Thus the toroidal support, the Latin property, and the operator resolution of the identity are properties requiring different definitions. The quantum magic squares studied by De las Cuevas, Drescher, and Netzer require positive entries and row and column sums equal to the identity; their theory also distinguishes dilations with commuting entries[4]. This reference provides a precise language for comparing realizations of APP.

2.2 TRIT: orientation and quadratic regimes

Definition 2.4 (Ternary signature and local lift). The TRIT signature uses the balanced identification

$$\mathbb{F}_3 \longrightarrow \{-1, 0, +1\}, \quad 0 \mapsto 0, \quad 1 \mapsto +1, \quad 2 \mapsto -1.$$

For $n \in \mathbb{Z}$, its lift preserves $n = r + 3c$, where $r \in \{-1, 0, +1\}$ and $c \in \mathbb{Z}$. In a transition of bounded local amplitude, the three states $(0, -1)$, $(0, 0)$, $(0, +1)$ may exhibit the same zero residue and different carries.

The signature does not replace the trajectory. A zero residue does not mean an absence of orientation, phase, or quotient information. Nor does it introduce a physical quantity by itself. Its function is to distinguish algebraically the transport classes that will be realized next.

2.2.1 Three quadratic regimes and a common exponential

Once the ternary signature has been obtained, its quadratic realization is defined by

$$\mathbb{A}_\tau = \mathbb{R}[X]/(X^2 + \tau), \quad \tau \in \{-1, 0, +1\}. \quad (2.5)$$

If J_τ is the class of X , then $J_\tau^2 = -\tau I$. The structure distinguishes the elliptic type $\tau = +1$, the parabolic type $\tau = 0$, and the hyperbolic type $\tau = -1$. The real realization does not generate the signature: it expresses the types already produced by oriented reduction.

Proposition 2.5 (Exponentiation of the three types). *In each algebra (2.5),*

$$\exp(tJ_\tau) = C_\tau(t)I + S_\tau(t)J_\tau, \quad (2.6)$$

where

$$C_\tau(t) = \sum_{n \geq 0} \frac{(-\tau)^n t^{2n}}{(2n)!}, \quad S_\tau(t) = \sum_{n \geq 0} \frac{(-\tau)^n t^{2n+1}}{(2n+1)!}.$$

The three realizations are

τ	J_τ^2	$\exp(tJ_\tau)$
+1	-I	$\cos t I + \sin t J_\tau$
0	0	$I + t J_\tau$
-1	I	$\cosh t I + \sinh t J_\tau$

Proof. Separating the even and odd terms of the exponential series and substituting $J_\tau^{2n} = (-\tau)^n I$ and $J_\tau^{2n+1} = (-\tau)^n J_\tau$ gives the formulas. For $\tau = 0$, all terms of degree at least two vanish. In the other two cases, the trigonometric or hyperbolic series are recovered. $\square$

The exponential is common to all three regimes: it is not assigned exclusively to the parabolic branch. The invariants

$$C_\tau(t)^2 + \tau S_\tau(t)^2 = 1$$

follow from $\exp(tJ_\tau)\exp(-tJ_\tau) = I$. The same algebraic principle preserves unity through realizations that differ in their quadratic type.

In the hyperbolic regime, $P_\pm = (I \pm J_{-1})/2$ are complementary projectors and

$$\exp(tJ_{-1}) = e^t P_+ + e^{-t} P_-.$$

Growth and contraction appear jointly in two eigendirections. In the elliptic regime, the parameter describes a rotation; in the parabolic regime, a nilpotent translation. The temporal interpretation adopted by HMT associates +1 with return with memory, 0 with the threshold, and -1 with bidirectional opening. This interpretation requires an ordering of the parametrization; it does not add a fourth algebraic type.

2.2.2 Balanced division and composition of lifts

Ternary reduction expresses an integer evaluation of APP through a visible coordinate and a prolongation coordinate. Its function is not merely to abbreviate the notation for an integer: it provides a composition rule determining how the quotient changes when evaluations are added or multiplied. The balanced choice also makes the compatibility of this rule with sign reversal explicit.

Definition 2.6 (Balanced ternary coordinates). For $n \in \mathbb{Z}$, let

$$q_3(n) = \left\lfloor \frac{n+1}{3} \right\rfloor, \quad r_3(n) = n - 3q_3(n).$$

The map

$$\mathcal{B} : \mathbb{Z} \longrightarrow \{-1, 0, +1\} \times \mathbb{Z}, \quad n \longmapsto (r_3(n), q_3(n))$$

is called the balanced ternary decomposition. Its inverse is $\mathcal{L}(r, c) = r + 3c$.

Indeed, $r_3(n) \in \{-1, 0, +1\}$. If two pairs represent the same integer, $r + 3c = s + 3d$, then $r - s$ is a multiple of 3 and belongs to the interval $[-2, 2]$; necessarily $r = s$ and $c = d$. The representation is therefore unique. Division can be repeated on $q_3(n)$, so that each level preserves the data needed to reconstruct the preceding one. The quotient is not a perturbation of the residue: it is its complementary coordinate.

The signature obtained from an evaluation F , with $F = S$ or $F = P$ in APP, is $r_3 \circ F$. When reduction acts on the phase cursor, the same balanced section produces the classes $\{1, 4, 7\}$, $\{2, 5, 8\}$, and $\{3, 6, 9\}$, with values +1, -1, 0. The TPK selector defined below uses these classes to determine the operating mode. Reading an evaluation and selecting a mode thus share the same ternary reduction, but have different domains and functions.

Proposition 2.7 (Arithmetic of balanced pairs). *Let $n = r + 3c$ and $m = s + 3d$, with $r, s \in \{-1, 0, +1\}$. Define*

$$\chi(r, s) = \frac{r + s - r_3(r + s)}{3}.$$

Integer operations are transported to the pairs by

$$(r, c) \boxplus (s, d) = (r_3(r + s), c + d + \chi(r, s)), \quad (2.7)$$

$$(r, c) \boxtimes (s, d) = (rs, rd + sc + 3cd). \quad (2.8)$$

The map $\mathcal{L}$ preserves both operations.

Proof. The equality $r + s = r_3(r + s) + 3\chi(r, s)$ gives

$$n + m = r_3(r + s) + 3(c + d + \chi(r, s)).$$

On the other hand,

$$nm = rs + 3(rd + sc + 3cd).$$

Since the product of two balanced representatives again belongs to $\{-1, 0, +1\}$, it requires no additional residue reduction. Uniqueness of the decomposition proves both formulas. $\square$

For example, $2 = (-1) + 3$ and $4 = 1 + 3$. Their sum and product are reconstructed as

$$(-1, 1) \boxplus (1, 1) = (0, 2), \quad (-1, 1) \boxtimes (1, 1) = (-1, 3).$$

The resulting evaluations are 6 and 8. Adding the quotients suffices in this additive example; the product involves the cross terms rd , sc , and $3cd$. The discrepancy between the two operations also resides in the prolongation coordinate, not only in the residue.

The term χ is a cocycle of the balanced section. If $r \oplus s = r_3(r + s)$, it satisfies

$$\chi(r, s) + \chi(r \oplus s, t) = \chi(s, t) + \chi(r, s \oplus t).$$

Both sides represent the quotient of the same sum of three integers. This identity ensures that regrouping the operations does not change the reconstructed evaluation. At the same time, projection onto the first component suppresses this accounting: $1 + 1 = -1 + 3$, whereas the residue reading retains only -1 . The information removed is determined, in this case, by $\chi(1, 1) = 1$.

The relation with modulus 9 preserves a second discrete scale. The map

$$\beta : \{-1, 0, +1\} \times \mathbb{F}_3 \longrightarrow \mathbb{Z}/9\mathbb{Z}, \quad (r, \bar{c}) \longmapsto r + 3\bar{c} \pmod{9}$$

is well defined and bijective. Changing the representative of $\bar{c}$ adds a multiple of 9; conversely, equality of images fixes r first, by reduction modulo 3, and then $\bar{c}$. Addition transported by this bijection contains the term $\chi(r, s)$ from (2.7), now reduced modulo 3. Thus the two ternary levels do not compose by independent coordinatewise addition: normalization of the first level modifies the second.

In particular, the classes 3, 6, and 0 of $\mathbb{Z}/9\mathbb{Z}$ have first component 0, but second components 1, 2, and 0, respectively. The distinction already observed between direct reduction and extraction of the ideal coordinate appears as a property of β . Preserving this coordinate prevents the zero residue sector from being artificially collapsed into a single position. Modulus 1000, used later to organize decimal blocks, serves a different positional-representation function; its quotient is preserved through the corresponding division. A change of base preserves the integer when it retains residue and quotient, whereas coincidence of residues in different bases does not by itself identify their transport states.

2.2.3 Inversion, residue class, and non-visible coordinates

Additive inversion lifts unambiguously:

$$\mathcal{B}(-n) = (-r_3(n), -q_3(n)).$$

In particular, 3, 0, and -3 have coordinates $(0, 1)$, $(0, 0)$, and $(0, -1)$. They share a zero signature but represent different evaluations. In a transition restricted to a quotient of maximum amplitude 1, these are the three possibilities of the visible fiber over zero; in a general integer lift, that fiber contains all pairs $(0, c)$, with $c \in \mathbb{Z}$.

The quotient coordinate recovers the evaluated integer, but does not by itself recover the trajectory that produced it. Two APP paths may share evaluation, residue, and quotient while differing in orientation, sequence of modes, or winding. Thus decomposition $\mathcal{B}$ is integrated into the transport state rather than replacing it. The arithmetic memory of a division and the memory of a trajectory are related coordinates, not synonyms.

For a signature sequence of length n , oriented reversal is

$$C_{\text{rev}}(\tau_1, \dots, \tau_n) = (-\tau_n, \dots, -\tau_1).$$

Applying it twice restores the sequence. The operation reverses the order of positions and the sign of each signature; it preserves the length and the zero positions in reversed order. In the adopted parametrization, it interchanges return with memory, associated with $+1$, and bidirectional opening, associated with -1 , while retaining the threshold 0. Transport of the remaining trajectory data is specified by the corresponding TPK involution.

This signature reversal must be distinguished from conjugation within a quadratic algebra. The former may change the type τ ; the latter, constructed below, acts within a type already fixed. This distinction prevents attributing an algebraic equivalence between different regimes to a sign reversal.

2.2.4 Quadratic product, conjugation, and algebraic norm

An element of $\mathbb{A}_\tau$ has a unique expression $aI + bJ_\tau$. The relation $J_\tau^2 = -\tau I$ determines its product:

$$(aI + bJ_\tau)(cI + dJ_\tau) = (ac - \tau bd)I + (ad + bc)J_\tau. \quad (2.9)$$

This is a real realization of the type already selected. The coordinates a, b are not APP residues, and the product (2.9) does not replace (2.8); it expresses another operation in the declared quadratic codomain.

Definition 2.8 (Conjugation and algebraic norm). Conjugation of type τ and its algebraic norm are

$$\overline{aI + bJ_\tau} = aI - bJ_\tau, \quad N_\tau(aI + bJ_\tau) = a^2 + \tau b^2.$$

Proposition 2.9 (Multiplicativity and invertibility). For $z, w \in \mathbb{A}_\tau$,

$$z\bar{z} = N_\tau(z)I, \quad N_\tau(zw) = N_\tau(z)N_\tau(w).$$

Moreover, z is invertible if and only if $N_\tau(z) \neq 0$, in which case

$$z^{-1} = \frac{\bar{z}}{N_\tau(z)}.$$

Proof. The product of $aI + bJ_\tau$ and $aI - bJ_\tau$ is $(a^2 + \tau b^2)I$. Conjugation is multiplicative and the algebra is commutative, so that $(zw)\bar{z}\bar{w} = (z\bar{z})(w\bar{w})$. If the norm does not vanish, the stated formula supplies the inverse. Conversely, if $zw = I$, multiplicativity implies $N_\tau(z)N_\tau(w) = 1$. $\square$

The quadratic form N_τ is positive definite only in the elliptic type. For $\tau = 0$, it equals a^2 , and the element J_0 is nonzero although its square and norm are zero. For $\tau = -1$, it equals $a^2 - b^2$; the nonzero

elements $I + J_{-1}$ and $I - J_{-1}$ have zero norm and their product vanishes. Here the algebraic norm is a multiplicative quadratic form: it does not presuppose a positive vector-space norm.

The three types admit faithful realizations by

$$aI + bJ_\tau \mapsto \begin{pmatrix} a & -\tau b \\ b & a \end{pmatrix}.$$

The determinant of this matrix is $N_\tau(aI + bJ_\tau)$. The complex structure, the dual-number structure, and the split real structure are thus represented by a single matrix expression. Each preserves its own law and zero divisors. The form of signature $(1, 1)$ in the last case admits a Lorentzian interpretation in dimension $1 + 1$; identification of a physical metric additionally requires a specific space, field, and transport rule.

2.2.5 Multiplicative conservation and composition of evolutions

The exponential already obtained belongs to the set of norm-one elements:

$$N_\tau(\exp(tJ_\tau)) = 1.$$

This conservation persists under composition of parameters. Indeed, all factors are functions of the same generator, and

$$\exp(tJ_\tau) \exp(sJ_\tau) = \exp((t + s)J_\tau).$$

Comparing the two components gives the addition laws

$$C_\tau(t + s) = C_\tau(t)C_\tau(s) - \tau S_\tau(t)S_\tau(s), \quad (2.10)$$

$$S_\tau(t + s) = S_\tau(t)C_\tau(s) + C_\tau(t)S_\tau(s). \quad (2.11)$$

Conjugation changes t to $-t$ and coincides with inversion within this subgroup. It does not change τ : it reverses an evolution of the chosen regime.

In the elliptic type, the preserved form is $a^2 + b^2$, and the exponential trajectory traverses the unit-norm circle. In the hyperbolic type, the eigencoordinates are $u = a + b$, $v = a - b$, with $uv = N_{-1}(z)$. The exponential gives

$$(u, v) = (e^t, e^{-t}), \quad uv = 1.$$

Growth of one coordinate is accompanied by contraction of the other. Conservation expresses a relation between the two coordinates; it does not assert that each remains constant.

In the parabolic type,

$$(I + tJ_0)(I + sJ_0) = I + (t + s)J_0, \quad (I + tJ_0)^{-1} = I - tJ_0.$$

The evolution does not stop because $J_0^2 = 0$. Nilpotency removes terms of order at least two, but preserves the linear term that transports the parameter. The visible symbol 0, the zero element of the algebra, and the nonzero generator J_0 are distinct. This distinction is necessary to describe the threshold without identifying it with an absence of state or dynamics.

Proposition 2.10 (Composition of affine coordinates). *In the chart where the scalar component does not vanish, represent a direction by $I + uJ_\tau$. If $1 - \tau uv \neq 0$, the direction of its product with $I + vJ_\tau$ has coordinate*

$$u \star_\tau v = \frac{u + v}{1 - \tau uv}.$$

Proof. We have

$$(I + uJ_\tau)(I + vJ_\tau) = (1 - \tau uv)I + (u + v)J_\tau.$$

Dividing by the scalar component gives the stated expression. $\square$

The affine chart preserves the ratio of components, not the scalar factor divided out. When $1 - \tau uv = 0$, the product must be examined before changing charts: it may have a well-defined direction with zero scalar component, or be the zero element in the presence of zero divisors. The composition law still belongs to the full algebra; a coordinate singularity does not automatically imply a singularity of the object. This formula will reappear in the elliptic regional reading. Its use requires retaining the denominator and, where applicable, the normalization factor.

2.2.6 Transport of type and subsequent realizations

The algebras $\mathbb{A}_{+1}$, $\mathbb{A}_0$, and $\mathbb{A}_{-1}$ are not isomorphic as unital real algebras. The first is a field; the second contains a nonzero nilpotent; the third has nontrivial idempotents and is isomorphic to $\mathbb{R} \times \mathbb{R}$. Thus a change of signature organizing a route cannot be interpreted as an undifferentiated isomorphism between the three realizations. The TPK preserves the regime index, the domains, and the effective operator acting at each transition.

The electronic application developed in Article I takes the pair of projections of the APP sheets and constructs on it a normalized commutator whose square is $-I$. The same block realizes an involution of square I and two nilpotent directions. This article reuses that operator result to type the complex structure of two-forms and the Cartan–Holst response; the preceding classification establishes why the three quadratic relations coexist within one matrix algebra, preserving their respective domains and actions.

In the exceptional prolongation, the Paley matrix reduced modulo 3 satisfies $A_W^2 = -I_6$. This relation endows the ternary space with a structure of dimension 3 over $\mathbb{F}_9$. Its connection to a real operator root is expressed through the integral presentation

$$\mathbb{Z}[u]/(u^2 + 1).$$

Base changes to $\mathbb{F}_3$ and $\mathbb{R}$ produce representations of the same relation. Each retains its characteristic, dimension, and incidence information. Developing the respective operators establishes the connection, not equality of their cardinalities.

Norm-one conservation consequently describes the quadratic realizations of evolutions. Conservation of quotients, orientation, and trajectory belongs to the transported arithmetic state. Both properties participate in the construction, with different functions: one determines the identities of the representations; the other makes it possible to compose, reconstruct, and prolong the operations that give rise to them.

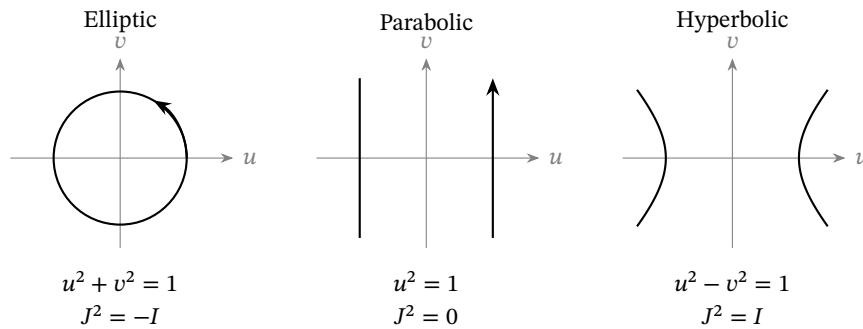


Figure 2: Quadratic realizations of TRIT. For $J_\tau = \begin{pmatrix} 0 & -\tau \\ 1 & 0 \end{pmatrix}$, the exponential preserves $u^2 + \tau v^2$. Parabolic degeneration preserves u and transports v linearly. The three diagrams represent orbits of quadratic forms; their geometric type corresponds to the algebraic relation of the generator.

2.3 TPK: Topological Phase Kernel

The dynamics uses the phase selector

$$M_{\text{ph}}(r) = \begin{cases} +1, & r \in \{1, 4, 7\}, \\ -1, & r \in \{2, 5, 8\}, \\ 0, & r \in \{3, 6, 9\}. \end{cases}$$

Its value determines which evaluation operates. Cardinal transport moves the corresponding cursor; the record preserves the previous value, the new quotient, the orientation, and the transition. Selection and transport are therefore different functions: a ternary sign is not a displacement on the torus.

We shall denote by $\hat{X}$ the space of arithmetic states with trajectory memory. An element contains the positions and directions of both cursors, the active mode, the TRIT signature, the phase, the residues and quotients of both evaluations, the carries, the constructed prefix, and the boundary data. When a refinement cylinder is introduced, it forms part of the state. This notation avoids replacing the object by a reduced version that retains only the visible symbol.

Definition 2.11 (TPK update). Let Sel_t be mode selection through M_{ph} , Tra_t the lifted cardinal transport, and Mem_t the record update. The transition is

$$\mathcal{U}_t = \text{Mem}_t \circ \text{Tra}_t \circ \text{Sel}_t. \quad (2.12)$$

Each application preserves the fields it does not modify and updates the remaining fields by Euclidean division and trajectory concatenation.

The notation (2.12) expresses the order of composition; it does not replace the effective rules. The following section specifies its finite two-cursor realization completely. Prolongation will then be obtained not by erasing its memory and repeating the first emission, but by transporting the prefixes and their compatible boundaries. Thus APP supplies the evaluations, TRIT distinguishes their regimes, and TPK composes their evolution.

2.3.1 Dynamic state, observables, and operational dependencies

The Topological Phase Kernel organizes the evolution of the two APP sheets with the orientation determined by TRIT. Its definition comprises the state on which it acts, the transition rules, and the relations that make prolongations compatible. Equation (2.12) acquires operational content when the coordinates received and modified by each factor are specified. Position belongs to the toroidal support; phase belongs to the calendar; orientation determines transport; quotients record the arithmetic evaluation before reduction. These coordinates are related through the transition and retain different mathematical functions.

The observable projection of a state is

$$q = (x_+, d_+, x_-, d_-, \theta) \in \mathcal{Q}_{\text{obs}} := (X_9 \times \mathcal{D})^2 \times \Theta_{108}, \quad \mathcal{D} = \{N, E, S, O\}, \quad \Theta_{108} = \mathbb{Z}/108\mathbb{Z}. \quad (2.13)$$

The subscripts distinguish the additive and multiplicative cursors. The fibers of complete states over q preserve the ordered trajectory, the integer evaluations of both sheets, their residues and quotients, and the orientation and boundary records. A precision prolongation adds the prefix and the compatible cylinder. Thus two states with the same projection q may retain different records.

Operational dependencies have three levels. At the local level, the selector activates a sheet, transport changes its position, and the update computes the new arithmetic coordinates. At the trajectory level, these transitions compose and retain the records of preceding steps. At the refinement level, extension relations determine which continuation preserves the predecessor's prefix, signature, and boundary. The finite calendar describes the phase of these operations; holonomy describes their composition on the history.

2.3.2 Tritic selection and lifted cardinal transport

Let $\bar{\theta} \in \{0, \dots, 107\}$ be the calendar representative. The operating phase and the dodecaphase sector are

$$r(\theta) = 1 + (\bar{\theta} \bmod 9), \quad \beta(\theta) = \left[\left[\frac{\bar{\theta}}{9} \right] \right]_{12}. \quad (2.14)$$

The previously defined selector may be written $\varepsilon_9 = M_{\text{ph}}$. Its value vector in the ordered chart of D_9 is

$$m_{\text{ph}} = (1, -1, 0, 1, -1, 0, 1, -1, 0).$$

The function $M_{\text{ph}} : D_9 \rightarrow \{-1, 0, +1\}$ and this vector are two presentations of the same selector, with their respective domains.

Define the activation indicators

$$a_+(\theta) = \mathbf{1}_{\{1,4,7\}}(r(\theta)), \quad a_\times(\theta) = \mathbf{1}_{\{2,5,8\}}(r(\theta)).$$

Cursor transport is then

$$x'_+ = x_+ + a_+(\theta)v_{d_+}, \quad x'_\times = x_\times + a_\times(\theta)v_{d_\times} \quad \text{in } X_9. \quad (2.15)$$

In phases 3, 6, 9, both positions remain unchanged; the neutral event forms part of the chronology and its record. Cardinal directions determine the vectors v_d , whereas the phase signature determines which vector is applied.

The integer lift preserves boundary crossing. If $s : X_9 \rightarrow D_9^2$ chooses positive representatives and $x' = x + av_d$, with $a \in \{0, 1\}$, define

$$b_d(x, a) = \frac{s(x) + av_d - s(x')}{9} \in \mathbb{Z}^2. \quad (2.16)$$

Divisibility by 9 follows from equality in X_9 . If $z \in \mathbb{Z}^2$ records previous crossings, the update is $z' = z + b_d(x, a)$. Thus

$$s(x') + 9z' = s(x) + 9z + av_d.$$

The integer transport data remain reconstructible after the position has been reduced on the torus.

Proposition 2.12 (Conservation of lifted displacement). *For a cardinal trajectory of n steps, with activation indicators a_t ,*

$$s(x_n) + 9z_n = s(x_0) + 9z_0 + \sum_{t=0}^{n-1} a_t v_{d_t}.$$

The same equality is preserved under concatenation of two compatible trajectories.

Proof. The one-step identity is (2.16). Summing it over the n steps cancels the intermediate coordinates. Splitting the summation interval into two segments produces exactly the concatenation law. For a closed trajectory, the difference $z_n - z_0$ is the previously defined winding number. $\square$

The same principle governs APP evaluations. If a sheet takes the integer value N_t , its record simultaneously preserves $N_t = R_t + 9q_t$. The difference between two steps satisfies

$$N_{t+1} - N_t = (R_{t+1} - R_t) + 9(q_{t+1} - q_t).$$

Spatial-crossing memory z_t and arithmetic quotient q_t have different definitions; preserving them jointly maintains the geometric and arithmetic provenance of every transition.

2.3.3 Two-cursor chronology and composite returns

The cardinal involution j interchanges N with S and E with O . After applying (2.15), both directions are reversed at the end of a block of 27 steps. Writing

$$h(\theta) = \mathbf{1}_{\{0,27,54,81\}}((\bar{\theta} + 1) \bmod 108),$$

the complete rule on the observable projection is

$$F_{\text{obs}}(q) = (x'_+, j^{h(\theta)}d_+, x'_\times, j^{h(\theta)}d_\times, \theta + 1). \quad (2.17)$$

The reading that forms the emission uses the positions before transport, according to the explicit recurrence in Section 5. The observable update and emission therefore compose in a fixed order.

Proposition 2.13 (Reversibility and period of the observable calendar). *The map F_{obs} is a permutation of $\mathcal{Q}_{\text{obs}}$ and satisfies $F_{\text{obs}}^{108} = I$. Within a block of 54 steps, positions and directions are restored; the coordinate in Θ_{108} advances by 54.*

Proof. Given the final state, the preceding phase is first recovered by subtracting one in Θ_{108} . That phase determines whether directions were reversed and which cursor moved. Applying the corresponding involution and subtracting the cardinal vector uniquely recovers the initial state.

Each interval of 27 steps contains nine activations of each cursor. The following block preserves the activations and reverses the directions, so the integer displacements cancel over 54 steps. The rule has period 54 in its position and orientation increments; the same cancellation holds for any calendar origin. After 108 steps, θ is also restored. $\square$

We reserve

$$\mathcal{K}_{27} = F_{\text{obs}}^{27} : \mathcal{Q}_{\text{obs}} \longrightarrow \mathcal{Q}_{\text{obs}}$$

for the iterate of 27 transitions. Its fourth iterate is the identity on the observable state defined in (2.13). Projecting history onto the calendar makes this finite check possible. The trajectory record, its evaluations, and the new prefixes remain in the dynamic state on which $\mathcal{U}_t$ acts; observable equality does not reinitialize them.

2.3.4 Affine memory transport and composition of states

The observable projection admits fibers of arithmetic and geometric data. Over a configuration q , we write a state in the form

$$x = (q, o, h, c, \sigma, C, b, \lambda).$$

Here o preserves orientation and sheet; h , residues and quotients; c , additive transport memory; σ , the boundary signature; C , the precision cylinder; b , its boundary label; and λ , the recorded trajectory. The signature is determined by a map

$$\text{Sig}_q : H_q \times C_q \times \text{Hist}_q \longrightarrow S_q, \quad \sigma = \text{Sig}_q(h, c, \lambda).$$

The symbols H_q , C_q , Hist_q , and S_q denote, respectively, the sets of residue–quotient data, the abelian transport-memory group, the recorded trajectories, and the boundary signatures. Realizations of this signature are specified component by component in the regional and terminal constructions. This dependence prevents treating the signature as a free parameter once the data producing it are fixed.

The elementary residue–quotient chart is explicit. For $n \in \mathbb{Z}$, let

$$r = 1 + ((n - 1) \bmod 9), \quad u = \frac{n - r}{9}.$$

Then $n = r + 9u$, with $r \in D_9$ and $u \in \mathbb{Z}$. Uniqueness follows because two representatives in D_9 congruent modulo 9 are equal. This chart supplies local integer reconstruction. The precision towers additionally incorporate the truncation maps and cylinders developed in 5 and 6.

Let $\gamma : q \rightarrow q'$ be an observable trajectory. Fiber transport is expressed through maps

$$H(\gamma) : H_q \rightarrow H_{q'}, \quad C(\gamma) : C_q \rightarrow C_{q'}, \quad \text{Hist}(\gamma) : \text{Hist}_q \rightarrow \text{Hist}_{q'}.$$

The maps respect identities and composition, and $C(\gamma)$ is an abelian-group homomorphism. The increment $k_{\text{tr}}(\gamma) \in C_{q'}$ satisfies

$$\begin{aligned} k_{\text{tr}}(\text{id}_q) &= 0, \\ k_{\text{tr}}(\gamma_2 \circ \gamma_1) &= k_{\text{tr}}(\gamma_2) + C(\gamma_2)k_{\text{tr}}(\gamma_1). \end{aligned} \tag{2.18}$$

Here k_{tr} denotes the transport cocycle; the rational coordinate κ of record K retains its notation in the dodecaphase-closure chapter.

Transportable coordinates are updated by

$$\begin{aligned} h' &= H(\gamma)h, & c' &= C(\gamma)c + k_{\text{tr}}(\gamma), \\ \lambda' &= \gamma \circ \lambda, & \sigma' &= \text{Sig}_{q'}(h', c', \lambda'). \end{aligned} \tag{2.19}$$

The second equation is an affine action: the linear term transports previous memory and the cocycle adds the increment produced by the trajectory. In the spatial-crossing realization for each cursor separately, $C(\gamma) = I$ and $k_{\text{tr}}(\gamma)$ is the sum of the vectors in (2.16). This concrete instance connects the abstract law with the cardinal operations already defined.

Proposition 2.14 (Compatibility of the update with concatenation). *Updating by (2.19) along γ_1 , followed by updating along γ_2 , coincides with updating along $\gamma_2 \circ \gamma_1$.*

Proof. For the affine coordinate, two updates give

$$c'' = C(\gamma_2)C(\gamma_1)c + C(\gamma_2)k_{\text{tr}}(\gamma_1) + k_{\text{tr}}(\gamma_2).$$

Functoriality of C and (2.18) turn this expression into $C(\gamma_2 \circ \gamma_1)c + k_{\text{tr}}(\gamma_2 \circ \gamma_1)$. The statement for h follows from composition of H , and that for λ from associativity of trajectories. The final signature agrees because it is evaluated on the same three reconstructed coordinates. $\square$

The cylinder and its boundary complete this law through the relations Ext_γ , defined in 2.3.7. Their elements determine which of the preceding updates preserve an admissible prolongation. States, together with the triples (γ, x, x') admitted by these relations, form a category over observable trajectories:

$$(\gamma_2, x', x'') \circ (\gamma_1, x, x') = (\gamma_2 \circ \gamma_1, x, x'').$$

The identity of x is (id_q, x, x) . Composition preserves arithmetic transport and boundary prolongability simultaneously. This is the structure in which returns with memory and successive refinements take place.

2.3.5 Dodecaphase record, phase, and accumulated memory

The 108 transitions are distributed over the ordered windows

$$E_m = \{9m - 8, \dots, 9m\}, \quad 1 \leq m \leq 12.$$

The record preserves the phase origin, orientation, and data produced during each window. Its map is

$$\mathcal{R}_{12}(x) = \left((E_m, \tau_m, \sigma_m, \kappa_m, \Lambda_m) \right)_{m=1}^{12},$$

where τ_m is the window's subroute, σ_m its orientation, κ_m the integer boundary increment, and Λ_m the local incidence record. We adopt the chart developed in 9.2; its window indices correspond to $m = \beta + 1$. The domain comprises compatible TPK histories, and the codomain their oriented temporal records. Composition of segments and their increments is governed by (2.19). By contrast, the group

C_{12} describes translation of the window index; Θ_{108} preserves the step calendar. This distinction specifies what is transformed and what is observed when a cycle is completed.

In coordinates $\theta = 9b + r$, with $0 \leq r < 9$, advancing 9 steps preserves r and increases b by one. Addition of phases introduces the quotient

$$c_0(r, r') = \left\lfloor \frac{r + r'}{9} \right\rfloor,$$

whose cocycle identity ensures associativity of composition. The finite calendar preserves $[b]_{12}$; a record of turns preserves $b \in \mathbb{Z}$, or $b \in \mathbb{N}_0$ for a history with fixed origin. Section 5 proves the nonsplit exact sequence (5.5). Its function is to describe the relation between these coordinates before they are used to construct the signed state.

The signed record receives phase and memory data through the composition developed in Section 9:

$$R_{\text{sgn}} = \Pi_H \circ \mathcal{E}_{108}^{90,120} \circ \mathcal{R}_{12}. \quad (2.20)$$

Its domain is the terminal state with memory; its output belongs to $\mathbb{Z}^{12}$. The reversible transformation between this signed state and K , together with the rational reading of K , is proved in the corresponding chapter. Record operations precede these transformations. Reconstructing a record from its cyclic differences verifies its recoverability; its dynamic provenance must be preserved through the events that produced it.

For the encoded trace (d_t) retained by the record, the event reader counts codes $\{2, 4, 6, 8\}$ in each window with orientation $(+, +, +, -, -, -, +, +, +, -, -, -)$. These codes and the cardinal symbols of $\mathcal{D}$ are different record coordinates. Chapter 9 develops this reader, the integral decomposition $1 + 5 + 6$, the congruences characterizing its image, and its exact inverse. The inversion proof establishes which data this chart preserves; the definition of $\mathcal{R}_{12}$ specifies the oriented history on which it acts.

2.3.6 Phase incidence and internal channel construction

The selector determines three subsets of D_9 :

$$H_+ = \{1, 4, 7\}, \quad H_- = \{2, 5, 8\}, \quad H_0 = \{3, 6, 9\}.$$

Let $H_{\text{act}} = H_+ \sqcup H_-$ be the set of active phases and $O_{\text{ph}} = D_9 \setminus \{9\}$ the chart preceding the marked return. Denote by $P_2(H_{\text{act}})$ the set of unordered pairs of active phases. Internal incidence is given by

$$\mathcal{F}_6^{\text{ph}} = H_{\text{act}} \times P_2(H_{\text{act}}), \quad \mathcal{F}_8^{\text{ph}} = O_{\text{ph}} \times P_2(H_{\text{act}}), \quad \Theta_{\text{ph}} = D_9 \times P_2(H_{\text{act}}). \quad (2.21)$$

The indices describe the number of positions in the first component. The construction uses the phases and calendar origin already fixed; subsequent exceptional realizations will transport these incidences to their own domains.

Proposition 2.15 (Cardinalities and oriented phase incidence). *The sets in (2.21) have cardinalities 90, 120, 135, respectively. If*

$$\partial_{\text{ph}} = \Theta_{\text{ph}} \times \{-1, +1\}, \quad E_{\text{ph}} = \partial_{\text{ph}} \sqcup \{\infty\}, \quad L_{\text{ph}} = H_0 \times P_2(H_{\text{act}}),$$

then

$$|\partial_{\text{ph}}| = 270, \quad |E_{\text{ph}}| = 271, \quad |L_{\text{ph}}| = 45, \quad |\partial_{\text{ph}} \sqcup L_{\text{ph}}| = 315.$$

Simultaneously translating the phase origin and marked return preserves all these cardinalities and inclusions.

Proof. There are six active phases, so $|P_2(H_{\text{act}})| = \binom{6}{2} = 15$. Multiplying by 6, 8, 9 gives 90, 120, 135. Orientation doubles 135; the return mark adds one element; the three neutral phases give $3 \cdot 15 = 45$. A calendar translation transports each factor by a bijection and carries the marked point to its image. Products and disjoint unions thus preserve the incidence. $\square$

Here the cardinality 271 has a phase-incidence realization. The cubic completion in Section 5 has its own construction of the corona 1000 – 729. The correspondence between the two realizations must preserve the maps and marks identifying them, in addition to their cardinality. Likewise, the selector ε_9 , the incidence (90, 120), and the extractor $\mathcal{E}_{108}^{90,120}$ retain the domains and operations just distinguished.

2.3.7 Trajectory reversal and conjugate prolongations

Bidirectionality begins with an operation on trajectories before scalar values are assigned to them. For $\gamma = (x_0; d_1, \dots, d_n)$, with endpoint x_n , define

$$\text{rev } \gamma = (x_n; jd_n, \dots, jd_1).$$

The reversed trajectory starts at the endpoint of the original. Composition satisfies

$$\text{rev}^2 \gamma = \gamma, \quad \text{rev}(\gamma_2 \circ \gamma_1) = \text{rev } \gamma_1 \circ \text{rev } \gamma_2,$$

and integer displacement changes sign. These identities follow by reversing the order of steps and replacing each cardinal vector with its opposite. For a loop, it follows that $\text{wind}(\text{rev } \gamma) = -\text{wind}(\gamma)$.

On records, reversal also requires transporting the order of components and the orientation of events. Route reversal and signature conjugation are distinguishable operations: their composition is specified when applied to a concrete record. In particular, the regional involution interchanging closure and propagation components preserves the self-scaling axis; its formula is given in Section 6.13. This regional operation acts on a different domain from cardinal trajectory reversal.

Extensions over a route $r : q \rightarrow q'$ are expressed by relations between fibers of compatible states. If $\mathcal{F}_q$ denotes the fiber of complete states over q , we require

$$\text{Ext}_r \subseteq \mathcal{F}_q \times \mathcal{F}_{q'}, \quad \Delta_{\mathcal{F}_q} \subseteq \text{Ext}_{\text{id}_q},$$

$$\text{Ext}_{r_2} \circ \text{Ext}_{r_1} \subseteq \text{Ext}_{r_2 \circ r_1}.$$

Their composition preserves the evaluations of both sheets, orientation, quotients, prefix, and boundary. Route reversal does not by itself imply that every extension relation is a bijection. Conjugate prolongation is defined on the domain where transport of these data satisfies the corresponding compatibilities.

2.3.8 Nonadic holonomy and continuity of refinement

A phase turn composes transitions on the state with memory:

$$\Gamma_{9,k} = \mathcal{U}_{k+8} \circ \dots \circ \mathcal{U}_k. \quad (2.22)$$

On an individualized history, this functional composition realizes the holonomy relation; on the full domain we retain $\Gamma_{9,k} \subseteq \widehat{X}_k \times \widehat{X}_{k+9}$, with the admissible prolongations developed later. The cyclic base has nine phases, while the fiber preserves trajectory, quotients, and boundary. Monodromy is the action of one turn of this base on the fibers; holonomy $\Gamma_{9,k}$ expresses its realization through the effective TPK transitions.

If g observes phase and M the number of retained turns,

$$g(\Gamma_{9,k}x) = g(x), \quad M(\Gamma_{9,k}x) = M(x) + 1.$$

The first identity expresses phase closure and the second identifies the new temporal depth. Coinductive prolongation also preserves its cylinders and prefixes. Restrictions between depths compose an inverse system; a compatible section gathers its antecedents without replacing them by the latest observation. This arithmetic of histories and its unity are developed in 5.8–5.10; the regions and their readers are constructed in Section 6.

Phase transfer on emissions already constructed has a different domain. Let $\mathcal{A}_+$ and $\mathcal{A}_\times$ be the additive and multiplicative signature alphabets, with cardinalities 18 and 26. Their product identifies the catalog $\mathcal{U}_6$. For $f : \mathcal{A}_+ \times \mathcal{A}_\times \rightarrow \mathbb{R}$, the sheet averages are

$$(E_+f)(s, e) = \frac{1}{18} \sum_{s' \in \mathcal{A}_+} f(s', e), \quad (E_\times f)(s, e) = \frac{1}{26} \sum_{e' \in \mathcal{A}_\times} f(s, e').$$

These emissions and their cardinalities are constructed through the recurrence in Section 5. Once they have been obtained, the transfer operator is defined on $\mathcal{U}_6 \times C_9$ by

$$P_+ = E_+, \quad P_- = E_\times, \quad P_0 = I, \quad \mathcal{K}_{\text{ph}}((u, r), (v, r+1)) = P_{M_{\text{ph}}(r)}(u, v),$$

with zero weight for all other phase transitions. The index r uses the positive representative of the cyclic class. The phase order $(+1, -1, 0)^3$ then produces the renewal

$$\mathcal{K}_{\text{ph}}^9|_r = E_+E_\times.$$

Indeed, both averages are idempotent, commute, and average different factors; their composition assigns weight $1/(18 \cdot 26) = 1/468$ to each emission. This equality characterizes a subsequent transfer observable. History transport remains given by (2.22), with its memory and boundary.

The resulting hierarchy establishes the function of the following chapters. Finite emission constructs the domain and regions; extension relations preserve their prefixes; the nonadic connection prolongs them; the dodecaphase record determines the data entering K and the closure of α . Numerical and incidence realizations receive this preceding structure. This order allows every consequence to be presented in its own chapter while preserving, from the formal kernel onwards, the objects that make it possible.

2.4 Accumulated memory and transport resolvents

Calendar return does not remove increments produced during the journey. To express this distinction in operator form, an accumulated trajectory-event record is constructed first, followed by its generating series. Reduction of memory variables will be performed on this transport, also preserving the source and output reader.

2.4.1 Oriented events and record update

Let (d_t) be the sequence of integer direction codes retained by a TPK trajectory. The code d_t and the cursor's cardinal direction are different coordinates; no identification of their alphabets is assumed. We number the windows starting from zero:

$$I_m = \{9m + 1, \dots, 9m + 9\}, \quad m \geq 0.$$

In the first cycle, I_m is window E_{m+1} of the dodecaphase record. Its orientation is

$$\sigma = (1, 1, 1, -1, -1, -1, 1, 1, 1, -1, -1, -1).$$

For a prolonged history, $\sigma_m \in \{-1, 1\}$ is the sign retained in the corresponding window. The signed event is

$$a_m = \sigma_m \sum_{t \in I_m} \mathbf{1}_{\{2,4,6,8\}}(d_t), \quad |a_m| \leq 9. \quad (2.23)$$

Each summand is incorporated when its transition occurs. This definition counts trace events; it does not select directions through a value that is to be obtained later.

Let $\mathbf{e}_0, \dots, \mathbf{e}_{11}$ be the basis of $\mathbb{Z}^{12}$ and $\mathbf{S}\mathbf{e}_j = \mathbf{e}_{j+1 \bmod 12}$. Write $R = S^{-1}$. The accumulated record in fixed coordinates satisfies

$$z_{m+1} = z_m + a_m \mathbf{e}_{m \bmod 12}.$$

A prefix with no accumulated history starts at $z_0 = 0$; at a different origin, z_0 retains the preceding record. The frame accompanying the calendar is $y_m = S^{-m}z_m$.

Proposition 2.16 (Update and return with memory). *The record in the moving frame satisfies*

$$y_{m+1} = Ry_m + \eta_m, \quad \eta_m = a_m R \mathbf{e}_0. \quad (2.24)$$

For every $n \geq 1$,

$$y_n = R^n y_0 + \sum_{j=0}^{n-1} R^{n-1-j} \eta_j. \quad (2.25)$$

In particular,

$$y_{12} = y_0 + \sum_{j=0}^{11} a_j \mathbf{e}_j. \quad (2.26)$$

Proof. Multiplying the update of z_m by $S^{-(m+1)}$ gives $y_{m+1} = S^{-1}y_m + a_m S^{-1} \mathbf{e}_0$, since $S^{-m} \mathbf{e}_{m \bmod 12} = \mathbf{e}_0$. Formula (2.25) follows by induction. For $n = 12$, $R^{12} = I$ and $R^{11-j} \eta_j = a_j R^{12-j} \mathbf{e}_0 = a_j \mathbf{e}_j$, which proves (2.26). $\square$

Thus twelve windows restore the frame and preserve the event vector. The count alone does not recover the order of all steps within each window; that order remains in the recorded trajectory.

To specify the balance, define

$$D_3 = I - S^3, \quad D_4 = I - S^4, \quad B = D_3^\top D_3 + D_4^\top D_4,$$

$$\mathcal{M}(y) = \frac{1}{2} y^\top B y, \quad q(y) = \sum_{j=0}^{11} y_j.$$

The first is a quadratic form of the record; the second records the sum of its coordinates. They are not identified here with physical energy or charge.

Proposition 2.17 (Event-induced balance). *Update (2.24) satisfies*

$$\begin{aligned} \mathcal{M}(y_{m+1}) - \mathcal{M}(y_m) &= a_m (B y_m)_0 + 2a_m^2, \\ q(y_{m+1}) - q(y_m) &= a_m. \end{aligned} \quad (2.27)$$

Proof. Expanding the products gives $B = 4I - S^3 - S^{-3} - S^4 - S^{-4}$; hence $R^\top B R = B$ and $B_{00} = 4$. Since $y_{m+1} = R(y_m + a_m \mathbf{e}_0)$, the quadratic difference is $a_m \mathbf{e}_0^\top B y_m + \frac{1}{2} a_m^2 B_{00}$. The coordinate sum is invariant under permutation R , giving the second equality. $\square$

2.4.2 Generating series and uniform memory control

The series preserves all increments, not only their sum at the end of a cycle. With a formal variable u , let

$$Y(u) = \sum_{m \geq 0} y_m u^m, \quad H_\eta(u) = \sum_{m \geq 0} \eta_m u^m.$$

Proposition 2.18 (Resolvent and tail bound). *The formal identity*

$$Y(u) = (I - uR)^{-1}(y_0 + uH_\eta(u)) \quad (2.28)$$

holds. Both series converge for $|u| < 1$. If $\ell : \mathbb{R}^{12} \rightarrow \mathbb{R}$ is linear, $L = \|\ell\|_{1 \rightarrow \mathbb{R}}$, $M_0 = \|y_0\|_1$, and $0 < r < 1$, then, for every integer $N \geq 0$ and $|u| \leq r$,

$$\left| \sum_{m > N} \ell(y_m) u^m \right| \leq L r^{N+1} \left(\frac{M_0 + 9(N+1)}{1-r} + \frac{9r}{(1-r)^2} \right). \quad (2.29)$$

The derivative series converges uniformly on every disk $|u| \leq r < 1$.

Proof. Multiplying (2.24) by u^{m+1} and summing gives $Y - y_0 = uRY + uH_\eta$. The constant term of $I - uR$ is invertible, so its formal inverse is $\sum_{k \geq 0} u^k R^k$. Moreover, R preserves the ℓ^1 norm and $\|\eta_m\|_1 \leq 9$, whence $\|y_m\|_1 \leq M_0 + 9m$. For the tail, sum the majorants $L(M_0 + 9m)r^m$, using

$$\sum_{m > N} r^m = \frac{r^{N+1}}{1-r}, \quad \sum_{m > N} m r^m = r^{N+1} \left(\frac{N+1}{1-r} + \frac{r}{(1-r)^2} \right).$$

Finally, $\sum_{m \geq 1} Lm(M_0 + 9m)r^{m-1} < \infty$ majorizes the derivative series and proves its uniform convergence. $\square$

The variable u counts windows of nine transitions. Identifying it with another reader's variable requires preserving that clock and the map relating the two readings.

2.4.3 Memory elimination: operator, source, and reader

On a domain where the update is fixed as $U : X \rightarrow X$, the free vector space $V = \mathbb{Q}^{(X)}$ on the states admits the operator $T\delta_x = \delta_{U(x)}$. The clock is included in x if the transition depends on it. This linear extension preserves distinct states even when they share an observable phase.

Consider a decomposition $V = V_0 \oplus V_1$, where V_1 is the auxiliary sector to be eliminated, and write

$$T = \begin{pmatrix} A & B_1 \\ C & D \end{pmatrix}, \quad v = \begin{pmatrix} v_0 \\ v_1 \end{pmatrix}, \quad \ell = (\ell_0, \ell_1).$$

Here $A : V_0 \rightarrow V_0$, $B_1 : V_1 \rightarrow V_0$, $C : V_0 \rightarrow V_1$, and $D : V_1 \rightarrow V_1$. The following identities are formal; they also hold wherever the resolvents converge.

Proposition 2.19 (Schur reduction with source and reader). *Let*

$$D_u = (I - uD)^{-1}, \quad \mathcal{F}(u) = I - uA - u^2 B_1 D_u C.$$

The retained component of $(I - uT)^{-1}v$ is

$$w_0 = \mathcal{F}(u)^{-1}(v_0 + uB_1 D_u v_1), \quad w_1 = D_u(v_1 + uCw_0). \quad (2.30)$$

Its complete reading satisfies

$$\begin{aligned} \ell(I - uT)^{-1}v &= (\ell_0 + u\ell_1 D_u C) \mathcal{F}(u)^{-1}(v_0 + uB_1 D_u v_1) \\ &\quad + \ell_1 D_u v_1. \end{aligned} \quad (2.31)$$

Proof. The two equations of $(I - uT)w = v$ are $w_0 = v_0 + uAw_0 + uB_1w_1$ and $w_1 = v_1 + uCw_0 + uDw_1$. Solving the second and substituting it into the first gives (2.30); applying ℓ_0 and ℓ_1 to the two components gives (2.31). All formal inverses exist because their constant terms are identities. $\square$

The term $u^2B_1D_uC = \sum_{k \geq 0} u^{k+2}B_1D^kC$ describes departure into memory, k steps within it, and a return. By contrast, $u\ell_1D_uC$ reads memory before that return. Thus reducing only the operator does not necessarily preserve the observation: the source and reader must also be transformed.

The distinction is visible in the cycle already constructed. If P retains $\mathbf{e}_0$, then $PRP = 0$, but

$$\langle \mathbf{e}_0, (I - uR)^{-1}\mathbf{e}_0 \rangle = \frac{1}{1 - u^{12}}, \quad \langle \mathbf{e}_{11}, (I - uR)^{-1}\mathbf{e}_0 \rangle = \frac{u}{1 - u^{12}}. \quad (2.32)$$

Indeed, $R^n\mathbf{e}_0 = \mathbf{e}_{-n \bmod 12}$: the two series collect, respectively, $n \equiv 0$ and $n \equiv 1 \pmod{12}$. The resolvent of the compression is I , whereas compression of the resolvent preserves the omitted returns. This example concerns the twelve-window record, not the entire TPK state.

2.4.4 Segment composition and coherence of reduction

Let three composable affine transports be $F_i(y) = R_iy + b_i$, with compatible domains and codomains. The memory produced by the complete journey is

$$F_3F_2F_1(y) = R_3R_2R_1y + b_3 + R_3b_2 + R_3R_2b_1. \quad (2.33)$$

The proof is successive substitution of the three affine formulas. Grouping the first two segments first, or the last two, produces the same expression; the factors transport each increment from its point of origin. In (2.24), $R_i = R$ and b_i is the event of its window.

Eliminating several layers preserves a similar coherence. To make it explicit, now decompose $V = V_0 \oplus V_1 \oplus V_2$ and write

$$T = \begin{pmatrix} A & B_1 & B_2 \\ C_1 & D_{11} & D_{12} \\ C_2 & D_{21} & D_{22} \end{pmatrix}, \quad Q_2 = (I - uD_{22})^{-1}.$$

Eliminating the third component leaves the four blocks

$$\begin{aligned} A' &= A + uB_2Q_2C_2, & B'_1 &= B_1 + uB_2Q_2D_{21}, \\ C'_1 &= C_1 + uD_{12}Q_2C_2, & D'_{11} &= D_{11} + uD_{12}Q_2D_{21}. \end{aligned} \quad (2.34)$$

Proposition 2.20 (Transitivity of elimination). *If $D_* = (D_{ij})_{i,j=1}^2$, the complement obtained in two stages coincides with that obtained by eliminating both memories together:*

$$\begin{aligned} I - uA' - u^2B'_1(I - uD'_{11})^{-1}C'_1 &= I - uA \\ &\quad - u^2(B_1 \ B_2)(I - uD_*)^{-1} \begin{pmatrix} C_1 \\ C_2 \end{pmatrix}. \end{aligned} \quad (2.35)$$

Proof. Solving the third equation of $(I - uT)w = v$ and substituting it into the first two produces (2.34). Eliminating the second component then gives the left-hand side of (2.35). Solving the two memory equations together gives the right-hand side, by Proposition 2.19. The formal solution is unique because $I - uT$ has constant term I . Both procedures also transform the source and reader according to (2.30)–(2.31). $\square$

Balance (2.27) describes record increments. The preceding identities allow them to be transported when memory is reused in dodecaphase closure; they do not by themselves determine additional analytic coefficients.

3 Joint construction of the continuum through compatible histories and prolongations

The arithmetic expressed by a normal form preserves a precise reading of the state that produces it. To situate that reading within HMT, it is also necessary to retain the histories, the sheets, and the operations acting upon them. This chapter brings together that common substrate. Its different constructions are obtained from the same tree of APP-TRIT-TPK executions; the expository divisions allow their properties to be studied without turning them into independent domains chosen by a subsequent application.

The starting point is the formal kernel presented above. An admissible emission contains the APP operation, the TRIT orientation, the change of sheet, the residue and the quotient, together with the effective TPK memory update. When only a residue is expressed, the execution history still belongs to the source object. Two executions with an identical final reading are not identified for that reason alone.

3.1 Admissible histories and memory by composition

Let $H_0 := \{*\}$. The level H_1 is the finite nonempty set of admissible APP-TRIT seeds, and, for $k \geq 1$, H_k is the set of enriched histories of length k admitting compatible TPK prolongations at every finite horizon. Its elements are constructed through successive emissions from the initial domain of the kernel. We write $\zeta\varepsilon$ for the prolongation of ζ by an admissible emission ε , and $\rho_k(\zeta\varepsilon) = \zeta$ for deletion of the final emission. The label produced by the emission is retained, even when different labels express the same scalar state. The space of complete histories is defined by

$$H_\infty^{\text{enr}} := \varprojlim_k (H_k, \rho_k).$$

The cofinal subsystem $X_N := H_{6N}$, with $X_0 = H_0$, coincides with the sextet clock used in the cardinal proof; by cofinality there is a canonical identification $H_\infty^{\text{enr}} \cong \varprojlim_N (X_N, \rho_{6(N+1), 6N})$.

On the two arithmetic sheets, the local data satisfy the exact divisions

$$\delta_\diamond = r_\diamond + 9q_\diamond, \quad 0 \leq r_\diamond < 9, \quad \diamond \in \{\Sigma, \Pi\}.$$

The oriented TRIT coordinate similarly admits an expression $\Delta = o_r + 3\kappa$, with the oriented representative declared by the kernel. Preservation of the quotient allows these representations to be composed without replacing an emission by its isolated residue.

In a memory chart, the update of an emission is written

$$U_\varepsilon(m) = C_\varepsilon m + \kappa_\varepsilon.$$

For a path $\alpha = (\varepsilon_1, \dots, \varepsilon_k)$, the effective operation is the ordered composition of those updates. Its linear part and its translation are

$$C_\alpha = C_{\varepsilon_k} \cdots C_{\varepsilon_1}, \quad \kappa_\alpha = \sum_{j=1}^k C_{\varepsilon_k} \cdots C_{\varepsilon_{j+1}} \kappa_{\varepsilon_j}.$$

The empty product in the last sum is the identity.

Proposition 3.1 (Affine compatibility of concatenation). *If α precedes β , then*

$$C_{\beta\alpha} = C_\beta C_\alpha, \quad \kappa_{\beta\alpha} = C_\beta \kappa_\alpha + \kappa_\beta, \quad U_{\beta\alpha} = U_\beta U_\alpha.$$

Proof. Substituting $U_\alpha(m)$ into U_β yields $C_\beta C_\alpha m + C_\beta \kappa_\alpha + \kappa_\beta$. Decomposition into linear part and translation gives the three identities. Associativity of composition proves that the result does not depend on an auxiliary subdivision of the path. $\square$

This cocycle identity is the precise content of the accumulated memory. The quantity transported by a second path depends on the chart update it performs; in general, it is not the untransported sum of two records.

The nonadic return preserves a second distinction. In the phase and turn coordinates, with $\bar{r} \in \{0, \dots, 8\}$, its representation is

$$\sigma_9(w, r) = \left(w + \left\lfloor \frac{\bar{r} + 1}{9} \right\rfloor, r + 1 \pmod{9} \right), \quad \sigma_9^9(w, r) = (w + 1, r).$$

Over nine steps, exactly one crossing from 8 to 0 occurs. The phase returns and the turn coordinate increases. The enriched holonomy also retains the memory updates accompanying that traversal.

3.2 Tree refinement and measure of histories

The initial measure is the canonical one from the census: $\mu(*) = 1$ and $\mu(c) = 1/|H_1|$ for each seed $c \in H_1$. In this subsection, each history has a finite positive number of prolongations. Deletion of emissions defines the refinement maps. Admissibility of the kernel provides the prolongations; when a residence uses a neutral emission, that emission explicitly provides a section of ρ_k . The existence of this section concerns the tree that includes that emission, not an arbitrary restriction of routes.

Let $d(\zeta) \geq 1$ be the number of admissible emissions at ζ . The APP order and the census of prolongations determine the canonical law $p_{\text{can}}(\varepsilon \mid \zeta) = 1/d(\zeta)$. Without additional parameters, this law defines

$$\mu(\zeta\varepsilon) = \frac{\mu(\zeta)}{d(\zeta)}.$$

A nonuniform weighting may replace p_{can} only when it has already been produced and added to the enriched record. The correlative constructor of this chapter uses p_{can} exclusively; therefore, its measure is a function of the APP-TRIT-TPK tree and not of an external choice.

Proposition 3.2 (Cylindrical compatibility and conditional expectation). *At every level,*

$$\sum_{\rho_k(y)=x} \mu(y) = \mu(x).$$

In the spaces $L^2(H_k, \mu_k)$, the maps

$$(J_k f)(y) = f(\rho_k(y)), \quad (E_k g)(x) = \sum_{\rho_k(y)=x} \frac{\mu(y)}{\mu(x)} g(y)$$

satisfy $E_k = J_k^$, $E_k J_k = I$, and $\|J_k f\| = \|f\|$.*

Proof. The first identity is the normalization of the local probabilities. Applying it to $|f(x)|^2$ yields the isometry. Grouping the sum by fibers of ρ_k yields the formula for E_k ; on a function constant on each fiber, that formula returns its value. $\square$

The compatible masses define a probability measure on the space of infinite histories: the cylinder of a finite history receives its mass $\mu(\zeta)$. The measure extends from the algebra of cylinders, and its uniqueness follows because they generate the σ -algebra. If the tree is finitely branching and every history has a prolongation, the inverse limit of its levels under deletion of emissions is nonempty. In the cylinder topology, every nonempty cylinder has positive measure.

This measure distinguishes histories converging to the same representation. The uniform measure on terminal values would be a different reading, since it would eliminate the multiplicities that the construction has just preserved.

3.2.1 Enriched readers and fiber memory

The difference between history and value admits an operational criterion. At a finite level H_k , let $q_k : H_k \rightarrow Z_k$ be a reader and let $M_k : H_k \rightarrow \mathcal{M}_k$ be the preserved record. The enriched reader is $\hat{q}_k = (q_k, M_k)$. Memory distinguishes states within a single fiber of the reader; its sufficiency is verified on the domain under consideration. This formulation incorporates into the article itself the complete development of the fiber information used below.

Proposition 3.3 (Recoverability at a finite level). *An inverse of $\hat{q}_k$ exists on its image if and only if $\hat{q}_k$ is injective. For a random variable Z taking values in H_k with distribution μ_k ,*

$$H(Z) = H(q_k(Z)) + H(Z | q_k(Z)).$$

If $\hat{q}_k$ is injective on the support of Z , then

$$H(Z | q_k(Z), M_k(Z)) = 0.$$

Here $H(Z) = -\sum_z \Pr(Z = z) \log \Pr(Z = z)$, with $0 \log 0 = 0$, is finite entropy in nats; conditional entropy is the average entropy of the conditional distributions.

Proof. An inverse recovers a unique antecedent, which requires injectivity; conversely, injectivity makes it possible to assign to each expressed pair its unique antecedent. Since $q_k(Z)$ is a deterministic function of Z , grouping the sum defining $H(Z)$ by fibers of q_k yields the chain rule. If the pair $(q_k(Z), M_k(Z))$ distinguishes the support, each corresponding conditional distribution is concentrated on a single state, and its entropy is zero. $\square$

Not publishing a coordinate and erasing the record that permits its recovery are distinct operations. This result does not identify entropy with the cardinality of the continuum or introduce a thermal interpretation: its domain, reader, and distribution are precisely those just specified.

3.2.2 Equivariance of the measure under tree symmetries

Proposition 3.4 (Joint transport of the tree and its measure). *Let $g = (g_k)_{k \geq 0}$ be a collection of bijections $g_k : H_k \rightarrow H_k$ that commutes with truncations, fixes the root, and bijectively transports admissible emissions while preserving their multiplicities. Then*

$$d(g_k \zeta) = d(\zeta), \quad (g_k)_* \mu_k = \mu_k, \quad (g_\infty)_* \mu = \mu.$$

More generally, if μ_ζ is the law of continuations of a history $\zeta \in H_k$, then

$$(g_\infty)_* \mu_\zeta = \mu_{g_k \zeta}.$$

Proof. The bijection of emissions preserves their number with multiplicities. The uniform seed distribution is likewise invariant. Each path and its image therefore receive the same product of local weights $1/d(\zeta)$. Equality of measures follows at every level and on every cylinder. Compatibility with truncations defines g_∞ , and uniqueness of the measure determined by cylinders proves equality in the limit. The same argument, beginning at ζ , proves the assertion about continuations. $\square$

This establishes the naturality of the tree and its measure: a coherent change of representation transports the weights and does not require choosing them anew. Preservation of emissions and multiplicities is an essential hypothesis of the proposition, not a datum added after the construction.

3.3 Transport by sheets and exact accounting of outputs

First consider the space with orthonormal basis formed by the histories at level k . The transport preserving the label of each prolongation is

$$\mathcal{U}_k e_\zeta = \sum_{\varepsilon \text{ admissible at } \zeta} \sqrt{p_{\text{can}}(\varepsilon \mid \zeta)} e_{\zeta\varepsilon}.$$

The successor fibers of distinct antecedents are disjoint. The normalization of p_{can} therefore proves $\mathcal{U}_k^* \mathcal{U}_k = I$. The representation by masses of histories is related to this one by the diagonal change of basis incorporating $\sqrt{\mu(\zeta)}$.

The sheet label is part of the complete record of a history. We write

$$\hat{\zeta} = ((\zeta_0, s_0), \dots, (\zeta_k, s_k)), \quad s_j \in \{\Sigma, \Pi\},$$

where s_j is the active sheet at step j . The prolongation adds the emission and the sheet s_{k+1} determined by the operational TPK selector, retaining the prefix in full, including the initial sheet. In the basis of these lifted histories,

$$\mathcal{U}_k e_{\hat{\zeta}} = \sum_{\varepsilon \text{ admissible at } \hat{\zeta}} \sqrt{p_{\text{can}}(\varepsilon \mid \hat{\zeta})} e_{\hat{\zeta}\varepsilon}, \quad R_k e_{\hat{\zeta}} = \mathbf{1}_{s_k=\Sigma} e_{\hat{\zeta}}, \quad Q_k = I - R_k.$$

Thus, R_k reads the current sheet, while the distinction between histories is maintained by their complete prefixes. Two initial histories that reach the same active sheet retain distinct successor fibers. The proof of isometry by disjoint collections of prolongations applies verbatim to this basis. The tritic orientation is another coordinate of the record: the value 0 preserves the threshold reading on each sheet and does not add a third sheet to this decomposition. We now separate sheet preservation and sheet exchange:

$$A_k = R_{k+1} \mathcal{U}_k R_k + Q_{k+1} \mathcal{U}_k Q_k, \quad \eta_k = Q_{k+1} \mathcal{U}_k R_k + R_{k+1} \mathcal{U}_k Q_k.$$

Proposition 3.5 (Gram identity by sheets). *For a linear transport U , whether or not it is isometric, the preceding decomposition satisfies*

$$A^* A + \eta^* \eta = P U^* U P + Q U^* U Q.$$

In particular, for $U = \mathcal{U}_k$,

$$A_k^* A_k + \eta_k^* \eta_k = I.$$

Proof. In $A^* A$ and $\eta^* \eta$, the mixed terms containing $R_{k+1} Q_{k+1}$ vanish. Summing the remaining terms produces $R_k U^* (R_{k+1} + Q_{k+1}) U R_k$, and its analogue with Q_k . The first identity follows from complementarity of the projectors; the second also uses $U^* U = I$. $\square$

The first identity is the general formula. Replacing its right-hand side by $U^* U$ requires the cross-blocks of that Gram matrix to vanish. Isometric preservation of history transport provides that condition here; it must not be attributed automatically to a subsequent compression.

Write $F_{0,0} = I$, $F_{0,n} = A_{n-1} \cdots A_0$, and $T_N = F_{0,N}^* F_{0,N}$.

Theorem 3.6 (Finite decomposition with terminal residue). *For each horizon N ,*

$$I = \sum_{n=0}^{N-1} F_{0,n}^* \eta_n^* \eta_n F_{0,n} + T_N.$$

The sequence T_N is positive and decreasing, and has a positive strong limit T_∞ . The sum of outputs converges strongly to $I - T_\infty$.

Proof. Proposition 3.5 gives $T_n - T_{n+1} = F_{0,n}^* \eta_n^* \eta_n F_{0,n}$. The telescoping sum proves the finite equality and monotonicity. For each vector, the quadratic forms decrease to a limit; polarization defines a positive bounded form and, by the representation theorem for bounded forms, an operator T_∞ . Convergence is strong because, for a positive operator $S \leq I$, $\|Sv\|^2 \leq \langle v, Sv \rangle$; this is applied to $S = T_n - T_\infty$. $\square$

The terminal residue is retained until its extinction in the domain under consideration is proved. This distinction prevents a phase return, by itself, from becoming an assumption that all histories are exhausted.

3.4 Oriented balance and cancellation under refinement

A history carries its accumulated orientation $o(h)$ and the counts of its visits to the two sheets. The corresponding local balance is

$$b(h) = o(h)(N_\Sigma(h) - N_\Pi(h)), \quad b^+ = \frac{|b| + b}{2}, \quad b^- = \frac{|b| - b}{2}.$$

When a reader expresses the mean over the prolongations, $b_c = Eb_f$ is defined. The orientation remains in the argument of b_f ; it is not removed before averaging.

Proposition 3.7 (Exact cancellation defect). *The quantity*

$$\kappa_{\text{can}} = \frac{E|b_f| - |Eb_f|}{2}$$

is nonnegative and satisfies

$$E(b_f^+) = (b_c)^+ + \kappa_{\text{can}}, \quad E(b_f^-) = (b_c)^- + \kappa_{\text{can}}.$$

Proof. The weighted triangle inequality gives $|Eb_f| \leq E|b_f|$. Writing the positive and negative parts as $(|b| \pm b)/2$ yields both equalities. $\square$

The information omitted when only the balance is expressed is the common cancellation of the opposing contributions. This quantity is determined from the same prolongations used by the preceding transport. The factor $1/9$ appears only when the fiber contains nine prolongations with equal weights; conditional expectation is the rule that remains valid for general branching and weightings.

3.5 Ternary incidence of the sheets and rectangular complex

The two sheets determine the module $V = \mathbb{F}_3 e_\Sigma \oplus \mathbb{F}_3 e_\Pi$. Its four projective directions are the lines generated by $e_\Sigma, e_\Pi, e_\Sigma + e_\Pi, e_\Sigma - e_\Pi$. They give rise to six unordered pairs. The eight nonzero vectors are their oriented representatives. Thus, the counts 4, 6, 8 belong to distinct operations on the same sheet module.

On the six pair positions, define

$$(Ba)_{\{L,L'\}} = a_L + a_{L'}, \quad s(q) = \sum_{\{L,L'\}} q_{\{L,L'\}}.$$

Each direction occurs in three pairs; hence $sB = 0$ in $\mathbb{F}_3$. The reader $W(q) = \mathbf{1}_{s(q)=0} - 1/3$ is invariant under $q \mapsto q + Ba$. On the uniform ambient space $\mathbb{F}_3^6$, its moments are

$$\mathbb{E}W = 0, \quad \mathbb{E}W^2 = \frac{2}{9}, \quad \mathbb{E}W^3 = \frac{2}{27}.$$

Indeed, s is surjective and each fiber contains 3^5 elements. The reader equals $2/3$ with probability $1/3$ and $-1/3$ with probability $2/3$. These means describe the uniform ambient space; the distribution induced by a subset of histories must be calculated with its own multiplicities, already retained in H_k .

To fix the ports without losing multiplicities, at depth k the labeled readings are retained: $\lambda_\Sigma(\zeta) = (\zeta, \Sigma)$ and $\lambda_\Pi(\zeta) = (\zeta, \Pi)$. Their sets are

$$\mathcal{P}_k^\Sigma = \{\lambda_\Sigma(\zeta) : \zeta \in H_k\}, \quad \mathcal{P}_k^\Pi = \{\lambda_\Pi(\zeta) : \zeta \in H_k\}.$$

They are finite and nonempty because H_k is. Values, quotients, and residues are expressed on these ports; equality of values does not identify their labels. The product $\mathcal{P}_k^\Sigma \times \mathcal{P}_k^\Pi$ is the domain of the rectangular reading, within which the support of the admissible incidences is recorded.

With this level fixed, write $I = \mathcal{P}_k^\Sigma$, $J = \mathcal{P}_k^\Pi$, and $A_m = \mathbb{Z}/3^m\mathbb{Z}$. The lexicographic APP order determines the minimal surviving branch and, within it, the reference ports i_0, j_0 . No external choice is added: changing coordinates produces a canonically isomorphic complex. The maps

$$\kappa(u, v)_{ij} = u_i - v_j, \quad (\delta w)_{ij} = w_{ij} - w_{i_0j_0} - w_{i_0j} + w_{i_0j_0}$$

define the following sequence, valid over the finite ring A_m :

$$0 \longrightarrow A_m \longrightarrow A_m^I \oplus A_m^J \xrightarrow{\kappa} A_m^{I \times J} \xrightarrow{\delta} A_m^{(I \setminus \{i_0\}) \times (J \setminus \{j_0\})} \longrightarrow 0.$$

Proposition 3.8 (Exactness of rectangular incidence). *The preceding sequence is exact, with its first arrow given by the constant diagonal.*

Proof. The equality $u_i - v_j = 0$ for every pair forces all coordinates of u and v to be the same constant. Direct substitution gives $\delta\kappa = 0$. If $\delta w = 0$, take $u_i = w_{i_0j_0}$ and $v_j = w_{i_0j_0} - w_{i_0j}$. Then $u_i - v_j = w_{ij}$. Finally, any matrix in the last term is prolonged with zero reference row and column; its image under δ is the initial matrix. The proof uses only additions and subtractions, and therefore does not require division by a unit of the ring. $\square$

Reduction from A_{m+1} to A_m , while keeping $k, \mathcal{P}_k^\Sigma, \mathcal{P}_k^\Pi, i_0, j_0$ fixed, commutes with both maps. The reconstruction formulas are the same at every modular depth. This is the arithmetic compatibility proved in this subsection. The prolongation of histories $k \rightarrow k+1$ preserves the prefix labels; its maps between sets of ports are data distinct from the change of ring $m \rightarrow m+1$.

Those maps are obtained from deletion of emissions:

$$\rho_k^\Sigma(\zeta\varepsilon, \Sigma) = (\zeta, \Sigma), \quad \rho_k^\Pi(\zeta\varepsilon, \Pi) = (\zeta, \Pi).$$

The minimal surviving branch of the APP order just constructed has compatible prolongations. Its two labels define compatible reference ports at every level. Coordinate transport is precomposition:

$$(L_k u)_{i'} = u_{\rho_k^\Sigma(i')}, \quad (L_k v)_{j'} = v_{\rho_k^\Pi(j')}, \quad (L_k w)_{i'j'} = w_{\rho_k^\Sigma(i'), \rho_k^\Pi(j')}.$$

For the last module in the sequence, the matrix is first prolonged by zero on the reference row and column, this same formula is applied, and the result is restricted to the new complement of the reference ports. This rule also defines L_k when a new port has the marked port as its antecedent.

Proposition 3.9 (Naturality of the rectangular complex under prolongation). *The preceding maps satisfy $L_k \kappa_k = \kappa_{k+1} L_k$ and $L_k \delta_k = \delta_{k+1} L_k$. They also commute with coefficient reduction $A_{m+1} \rightarrow A_m$ and compose in accordance with prefix deletion.*

Proof. The first identity follows from $u_{\rho_k^\Sigma(i')} - v_{\rho_k^\Pi(j')}$. In the second, each of the four terms defining δ is transported by the same precomposition. Compatibility of the marked ports identifies their rows and columns; when an antecedent is the marked port, the four terms cancel and agree with prolongation by zero. All formulas are sums and differences with integer coefficients, so they commute with modular reduction. The composition of precompositions is precomposition with the composite prefix, proving the final assertion. $\square$

3.6 Memory complex and compatibility of incidences

The same accumulated memory coordinate $c_y \in \mathbb{Z}$, at a residence y , defines a free integral complex. Its reduction to A_m uses the same integer record before reduction; therefore, successive depths are compatible. Its generators are f_y in degree 2, e_y, b_y, r_y in degree 1, and g_y in degree 0:

$$\partial f_y = 3c_y e_y + b_y, \quad \partial e_y = g_y, \quad \partial b_y = -3c_y g_y, \quad \partial r_y = 0.$$

The identity $\partial^2 = 0$ is verified on f_y by $3c_y g_y - 3c_y g_y = 0$, and is immediate on the other generators.

If a path $\alpha : y \rightarrow y'$ updates the memory, its transport in the complex sends the like-named generators to their new representatives, except for

$$\Psi_\alpha(b_y) = b_{y'} + 3(c_{y'} - c_y)e_{y'}.$$

Proposition 3.10 (Naturality of the memory complex). *The identities $\partial\Psi_\alpha = \Psi_\alpha\partial$ and $\Psi_{\beta\alpha} = \Psi_\beta\Psi_\alpha$ hold. The maps extend linearly over the chosen coefficient ring.*

Proof. On f_y , the image of its boundary is $3c_y e_{y'} + b_{y'} + 3(c_{y'} - c_y)e_{y'} = 3c_{y'} e_{y'} + b_{y'}$. On b_y , the boundary of its image is $-3c_{y'} g_{y'} + 3(c_{y'} - c_y)g_{y'} = -3c_y g_{y'}$. The other two cases are immediate. Under concatenation, the memory differences sum telescopically, proving the second identity. $\square$

The relation between two representatives of the same residence can also be written as an explicit boundary. For an integer coordinate v , reduced in the same ring when appropriate, let

$$z^- = r_y, \quad z^+ = r_y + 3c_y v e_y + v b_y, \quad h_* = -v f_y.$$

Then $\partial h_* = z^- - z^+$. Under a transport that also updates v , the correction $K_\alpha = (v_{y'} - v_y)f_{y'}$ satisfies $K_{\beta\alpha} = K_\beta + \Psi_\beta K_\alpha$. The symbols K_α in this local homotopy do not denote the dodecaphase seal K : they belong to another domain, and their function is specified by the preceding identity. Their function can also be checked in the equality

$$z_{y'}^+ - \Psi_\alpha z_y^+ = \partial K_\alpha.$$

Both sides equal $(v_{y'} - v_y)(3c_{y'} e_{y'} + b_{y'})$.

Proposition 3.11 (Homology and memory of the local complex). *Over $\mathbb{Z}$, or after reduction to A_m , one has*

$$H_0(\Gamma) = H_2(\Gamma) = 0, \quad H_1(\Gamma) \cong A_m[r_y]$$

in the modular realization, and $H_1(\Gamma) \cong \mathbb{Z}[r_y]$ in the integral realization.

Proof. Define the degree-1 homotopy by $h(g_y) = e_y$, $h(b_y) = f_y$, and zero on the other generators. Let p be the projection onto the summand generated by r_y , and ι its inclusion. Verification on each generator gives

$$\partial h + h\partial = I - \iota p.$$

In particular, on b_y the terms $3c_y e_y$ and $-3c_y e_y$ cancel. The complex deformation retracts onto the module generated by r_y , concentrated in degree 1, proving the assertion. $\square$

The transport Ψ_α is a shear of determinant 1. Thus, this complex explicitly preserves how c_y changes at chain level, but its local homology does not distinguish the values of that coordinate. The complete memory remains in the labels and transports of the source state; it is not recovered from H_1 alone. A memory-dependent numerical torsion requires the specified assembly and determinant reading, in addition to this local complex.

3.7 Assembly with terminal paths

For a finite horizon, the residences and their prolongations form a path category. Each residence ν is assigned the free module $W(\nu) = A_m[\text{Path}(\nu, \text{Term})]$, which retains each terminal continuation as a generator. An initial path acts on W by precomposition. The complex $\Gamma(\nu)$ of the preceding subsection acts covariantly through Ψ .

The assembly uses both transports in the same bigraded module:

$$\mathcal{B}_{q,r} = \bigoplus_{\nu_0 \rightarrow \dots \rightarrow \nu_q} W(\nu_q) \otimes_{A_m} \Gamma_r(\nu_0).$$

The normalized path complex is used: degenerate chains containing identities are omitted. The finite horizon then bounds the number of strict prolongations and the horizontal degree. The horizontal boundary deletes an intermediate residence by composing the adjacent arrows. At the first endpoint it transports Γ ; at the last it precomposes W . Alternating signs give the path boundary d_{bar} . The total differential is

$$D = d_{\text{bar}} + (-1)^q \partial_\Gamma.$$

The face identities cancel the terms of d_{bar}^2 pairwise. Naturality in Proposition 3.10 implies that d_{bar} commutes with ∂_Γ ; the factor $(-1)^q$ cancels the cross terms in the square. Therefore, $D^2 = 0$. Compatibility is proved on path generators and is not introduced by a label asserting membership in a complex.

Terminal evaluation also does not require choosing a particular history. For each continuation h , the effective composition of its updates is applied. The result retains the pair formed by h and its reading, with its multiplicity. If $k < \ell < N$, the continuations decompose as a disjoint union of prefixes up to ℓ and their continuations up to N . Proposition 3.1 proves that direct evaluation and evaluation of those two parts produce the same reading. This yields the terminal naturality square.

The path modules, incidences, memory transport, and their differentials arise from the same prolongations. Their assembly does not consist in adding to a state five coordinates that would be retained by definition: their relations are constructed through sums, boundaries, and effective compositions, and are verified on their generators.

3.8 Determinants, limit, and arithmetic reading

At each residence, the free local complex Γ admits its graded determinant line

$$\text{Det } \Gamma = \bigotimes_r \left(\bigwedge^{\text{rk } \Gamma_r} \Gamma_r \right)^{(-1)^r}.$$

The total assembly of the preceding subsection, in turn, has the modules $\text{Tot}_d \mathcal{B} = \bigoplus_{q+r=d} \mathcal{B}_{q,r}$. At finite horizon and over the same ring, they are free of finite rank and only finitely many of them are nonzero. Its line is

$$\text{Det}(\text{Tot } \mathcal{B}) = \bigotimes_{q,r} \left(\bigwedge^{\text{rk } \mathcal{B}_{q,r}} \mathcal{B}_{q,r} \right)^{(-1)^{q+r}}.$$

This formula follows from the direct-sum decomposition in each degree. It distinguishes the line of the local complex from that of the assembly with all paths; their bases are ordered by means of the retained labels. Modular reduction commutes with these exterior powers because the modules and their bases arise from the same free integral modules at fixed horizon. The line is defined by the modules and their degrees, both over $\mathbb{Z}$ and after reduction to A_m ; a negative exponent denotes the dual module. A nonzero numerical torsion additionally requires specifying the complex over a field, the bases, and the relevant homology identifications. To calculate Smith factors, one uses the integer matrix produced

before reduction, not an arbitrary lift of a modular matrix. An integer square block with nonzero determinant is invertible over $\mathbb{Q}$; its reduction is invertible over A_m only when that determinant is a unit, that is, when it is not divisible by 3. Refinement of memories and incidences makes it possible to formulate and verify the stability of the relevant invariant factors; these conditions do not follow from the existence of the determinant line.

Passage to the limit preserves the compatible projective systems of residues, the histories, and their multiplicities. The squares proved in the preceding subsections remain commutative because their maps are given by the same formulas at each level. For an Archimedean coordinate, the additional step consists in publishing nonempty cylinder intervals, compatible under inclusion and with diameter tending to zero. Their nested compact closures have a one-point intersection: existence follows from compactness and uniqueness from the diameter bound. That point is the value of the limit reading. If the intervals are half-open, the point may belong only to their closures; the boundary representation rule then determines its positional notation. The value does not replace the history that produced it. The other representation preserves the incidences of the same state. The two targets share their origin without thereby receiving the same information.

The coinductive regions of Section 6, the dodecaphase record of Section 9, and the exceptional incidence of Section 8 are expressed from that substrate with their respective maps. The Hilbert-space realization subsequently retains the phase labels and specifies its own unital inclusion; duality and the screens act on the codomains constructed there. The joint operations assembled here transmit histories, sheets, orientation, and memory to those representations. Identifying a specific functional with an external analytic form in turn requires writing its action on the generators and verifying their inner products: the conservation identities of the present chapter do not replace that composition.

3.9 Mathematical scope of the joint construction

The constructions of state, tree, and measure; the correlative operations; and the terminal assembly with naturality and limit are developed in full in this article. In this chapter their balance, incidence, and memory identities have been formulated and proved on the same generators. The five constructions thus remain operations of a single tree. Each specialization keeps its exact hypotheses visible: isometry of transport, canonical weighting law, treatment of the terminal residue, and nonsingularity of the determinant block, where applicable.

The preceding equations prove the memory cocycles, weighted cancellation, rectangular exactness, and compatibility of the memory complex at every finite horizon, as well as its compatible passage to the limit. A subsequent analytic identification must compose its operator on these generators and verify the additional properties of its codomain; it does not alter or replace the joint construction proved here.

3.10 Joint emergence of the five constructions

For $0 \leq k \leq N$, let

$$\rho_{N,k} := \rho_k \circ \rho_{k+1} \circ \cdots \circ \rho_{N-1}, \quad \rho_{k,k} := \text{id}_{H_k},$$

and, for a surviving history $\zeta \in H_k$, define

$$\text{Desc}_{k,N}(\zeta) := \{x \in H_N : \rho_{N,k}(x) = \zeta\}.$$

The finite tree $\mathcal{T}_{k,N}(\zeta)$ has as vertices

$$\prod_{j=k}^N \{y \in H_j : \rho_{j,k}(y) = \zeta\}$$

and as edges only the TPK prolongations $a : y \rightarrow z$ with $z \in \rho_j^{-1}(y)$. Therefore, neither its vertices nor its prolongations arise from an auxiliary tree.

For $x \in \text{Desc}_{k,N}(\zeta)$, the marked branch $\lambda_{\zeta,x}$ is the sequence

$$\zeta = x_k \longrightarrow x_{k+1} \longrightarrow \cdots \longrightarrow x_N = x$$

determined by the truncations of x . The common base record is

$$\begin{aligned} \mathcal{R}_{k,N}(\zeta; x) := & (\mathcal{T}_{k,N}(\zeta), \lambda_{\zeta,x}; \\ & (\varepsilon_a, s_a, o_a, (\delta_{\diamond,a}, r_{\diamond,a}, q_{\diamond,a})_{\diamond \in \{\Sigma, \Pi\}}, \text{Carr}_a, C_a, t_a, \text{Fr}_a, \text{Route}_a, \text{Mem}_a, \text{Surv}_a)_a; \\ & \text{src}, \text{tgt}, \circ, (\rho_j)_{k \leq j < N}), \end{aligned} \quad (\text{C1})$$

where

$$U_a(m_0) = C_a m_0 + t_a, \quad \delta_{\diamond,a} = r_{\diamond,a} + 9q_{\diamond,a}.$$

The symbol m_0 in this formula is a memory coordinate; it is not the modular depth m used below. The true base space is

$$\mathfrak{R}_{k,N}(\zeta) := \{\mathcal{R}_{k,N}(\zeta; x) : x \in \text{Desc}_{k,N}(\zeta)\}.$$

The marked branch is not an added input: it is the sequence of edges actually produced that ends at x . The remainder of the record simultaneously preserves the lateral fibers of the tree; therefore, the mark does not replace the branching with an isolated trajectory.

Set $R_m := \mathbb{Z}/3^m\mathbb{Z}$, and let $V_j(\zeta) := \text{Desc}_{k,j}(\zeta)$. The transport, the sheet projectors, and the terminal residue are restricted to the subtree by

$$\begin{aligned} \mathcal{U}_j^\zeta &:= \mathcal{U}_j \upharpoonright_{\ell^2(V_j(\zeta), \mu_j)}, & P_j^\zeta &:= P \upharpoonright_{V_j(\zeta)}, & Q_j^\zeta &:= I - P_j^\zeta, \\ A_j^\zeta &:= P_{j+1}^\zeta \mathcal{U}_j^\zeta P_j^\zeta + Q_{j+1}^\zeta \mathcal{U}_j^\zeta Q_j^\zeta, & \eta_j^\zeta &:= Q_{j+1}^\zeta \mathcal{U}_j^\zeta P_j^\zeta + P_{j+1}^\zeta \mathcal{U}_j^\zeta Q_j^\zeta. \end{aligned}$$

For $k \leq n \leq N$, let

$$F_{k,k}^\zeta := I, \quad F_{k,n}^\zeta := A_{n-1}^\zeta \cdots A_k^\zeta, \quad T_{k,n}^\zeta := (F_{k,n}^\zeta)^* F_{k,n}^\zeta.$$

The proper ports of the same subtree are

$$\mathcal{P}_j^\Sigma(\zeta) := \{(y, \Sigma) : y \in V_j(\zeta)\}, \quad \mathcal{P}_j^\Pi(\zeta) := \{(y, \Pi) : y \in V_j(\zeta)\};$$

we denote by $B_j^\zeta, \kappa_j^\zeta, \delta_j^\zeta$ the restrictions of the incidence and of the rectangular complex to those ports, and by L_j^ζ the precomposition induced by prefix deletion.

To define the path reader without abbreviations, let

$$W_\ell^\zeta(y) := R_m[\text{Path}_{\mathcal{T}_{k,\ell}(\zeta)}(y, \text{Term}_\ell)]$$

and set

$$\mathcal{D}_{q,r;k,\ell}^{(m)}[\mathcal{R}] := \bigoplus_{y_0 \rightarrow \cdots \rightarrow y_q \text{ in } \mathcal{T}_{k,\ell}(\zeta)} W_\ell^\zeta(y_q) \otimes_{R_m} (\Gamma_r(y_0) \otimes_{\mathbb{Z}} R_m), \quad \text{Tot}_d \mathcal{D} := \bigoplus_{q+r=d} \mathcal{D}_{q,r;k,\ell}^{(m)}[\mathcal{R}].$$

The formulas already proved produce, on each $\mathcal{R} \in \mathfrak{R}_{k,N}(\zeta)$, the five objects

$$\begin{aligned} \mathcal{F}_{\text{sp}}[\mathcal{R}] &:= (\ell^2(V_j(\zeta), \mu_j), \mathcal{U}_j^\zeta, A_j^\zeta, \eta_j^\zeta, T_{k,j}^\zeta)_{k \leq j \leq N}, \\ \mathcal{F}_{\text{sol}}[\mathcal{R}] &:= (b, \kappa_{\text{can}}, C_\alpha, \kappa_\alpha)_{\alpha \in \mathcal{T}_{k,N}(\zeta)}, \\ \mathcal{F}_{\text{gau}}^{(m)}[\mathcal{R}] &:= (\mathcal{P}_j^\Sigma(\zeta), \mathcal{P}_j^\Pi(\zeta), B_j^\zeta, \kappa_j^\zeta, \delta_j^\zeta; R_m)_{k \leq j \leq N}, \\ \mathcal{F}_{\text{coh}}^{(m)}[\mathcal{R}] &:= (\Gamma_y \otimes R_m, \partial, \Psi_\alpha, H_\alpha)_{y, \alpha \in \mathcal{T}_{k,N}(\zeta)}, \\ \mathcal{F}_{\text{det}}^{(m)}[\mathcal{R}] &:= (\text{Det}(\text{Tot } \mathcal{D}_{\cdot, \cdot; k, \ell}^{(m)}[\mathcal{R}]))_{k \leq \ell \leq N}. \end{aligned} \quad (\text{C2})$$

Here $\mathcal{D}_{\cdot, \cdot; k, \ell}^{(m)}[\mathcal{R}]$ is the normalized bicomplex of paths and memory from the preceding sections, restricted to the subtree of horizon ℓ and reduced over R_m . The last reader preserves the ordered

collection of determinant lines up to horizon N ; in this way, its truncation is a canonical projection and does not presuppose a nonexistent determinant factorization. The law μ is the canonical law of the census, and the ports are fixed by the lexicographically minimal branch of the APP order. Therefore, none of the five readers receives an external weighting or an external port.

For $T \in \mathfrak{T} := \{\text{sp}, \text{sol}, \text{gau}, \text{coh}, \text{det}\}$, define the typed total space

$$\tilde{\mathcal{F}}_{T;k,N}^{(m)}(\zeta) := \prod_{\mathcal{R} \in \mathfrak{R}_{k,N}(\zeta)} \{\mathcal{F}_T^{(m)}[\mathcal{R}]\}_{\mathcal{R}}, \quad p_T : \tilde{\mathcal{F}}_{T;k,N}^{(m)}(\zeta) \longrightarrow \mathfrak{R}_{k,N}(\zeta),$$

where $\mathcal{F}_{\text{sp}}^{(m)} := \mathcal{F}_{\text{sp}}$ and $\mathcal{F}_{\text{sol}}^{(m)} := \mathcal{F}_{\text{sol}}$; that is, the superscript m is inert for those two readers. The projection p_T forgets only the reader structure and preserves its source record. The correct fiber product is

$$\begin{aligned} \mathcal{J}_{k,N}^{(m)}(\zeta) := & \tilde{\mathcal{F}}_{\text{sp};k,N}^{(m)}(\zeta) \times_{\mathfrak{R}_{k,N}(\zeta)} \tilde{\mathcal{F}}_{\text{sol};k,N}^{(m)}(\zeta) \times_{\mathfrak{R}_{k,N}(\zeta)} \tilde{\mathcal{F}}_{\text{gau};k,N}^{(m)}(\zeta) \\ & \times_{\mathfrak{R}_{k,N}(\zeta)} \tilde{\mathcal{F}}_{\text{coh};k,N}^{(m)}(\zeta) \times_{\mathfrak{R}_{k,N}(\zeta)} \tilde{\mathcal{F}}_{\text{det};k,N}^{(m)}(\zeta). \end{aligned} \quad (\text{C3})$$

The equality of the five projections forces preservation of the same tree, the same edges, the same produced branch, and the same record. For each $\mathcal{R}$, the five constructions determine the element

$$s_{k,N}^{(m)}(\mathcal{R}) := (\mathcal{F}_T^{(m)}[\mathcal{R}])_{T \in \mathfrak{T}} \in \mathcal{J}_{k,N}^{(m)}(\zeta).$$

The existence of this element arises from the operations of (C2), not from the fiber-product notation.

Deletion of the last layer induces

$$\text{res}_{N+1,N}^{\mathfrak{R}}(\mathcal{R}_{k,N+1}(\zeta; x)) := \mathcal{R}_{k,N}(\zeta; \rho_N(x)).$$

The restriction of each operator to the retained generators defines

$$r_{N+1,N}^m : \mathcal{J}_{k,N+1}^{(m)}(\zeta) \longrightarrow \mathcal{J}_{k,N}^{(m)}(\zeta).$$

In the determinantal reader, $r_{N+1,N}^m$ deletes only the last coordinate of the collection of lines in (C2). The reduction $R_{m+1} \rightarrow R_m$ defines, by base change,

$$q_{m+1,m}^N : \mathcal{J}_{k,N}^{(m+1)}(\zeta) \longrightarrow \mathcal{J}_{k,N}^{(m)}(\zeta).$$

On each generator, the following identities hold:

$$\begin{aligned} r_{N+1,N}^m q_{m+1,m}^{N+1} &= q_{m+1,m}^N r_{N+1,N}^{m+1}, \\ \partial \Psi_\alpha &= \Psi_\alpha \partial, \quad L_j^\zeta \kappa_j^\zeta = \kappa_{j+1}^\zeta L_j^\zeta, \quad L_j^\zeta \delta_j^\zeta = \delta_{j+1}^\zeta L_j^\zeta, \end{aligned} \quad (\text{C4})$$

and the compositions of the maps r and q satisfy the functorial identities. The first equality holds because deleting the last layer and reducing the coefficients act on distinct coordinates. The other three are the naturality identities proved above. A single pro-system in (N, m) is obtained, not five juxtaposed limits.

At the root, define

$$\mathfrak{F}_{N,\bullet}^{(m)} := \mathcal{J}_{0,N}^{(m)}(*). \quad (\text{C5})$$

The joint discrete structure of the continuum is the limit of those correlative objects:

$$\boxed{\mathcal{C}_{\text{cont}}^{\text{disc}} := \lim_{\substack{\leftarrow \\ N \geq 1 \\ m \geq 1}} (\mathfrak{F}_{N,\bullet}^{(m)}, r_{N+1,N}^m, q_{m+1,m}^N).} \quad (\text{C6})$$

This is a definition by the pro-system generated by the full tree; it is not the image or the graph of a map that retains the input state as its first coordinate.

The records possess a produced terminal mark. Their compatible projections induce

$$\Pi : \mathcal{C}_{\text{cont}}^{\text{disc}} \longrightarrow H_\infty^{\text{enr}}.$$

Conversely, a complete history determines, through its already emitted prefixes, the compatible sections $s_{0,N}^{(m)}$. Thus, after having defined the continuum, one obtains

$$\text{Gen}_{\text{cont}} : H_{\infty}^{\text{enr}} \longrightarrow \mathcal{C}_{\text{cont}}^{\text{disc}}, \quad \Pi \circ \text{Gen}_{\text{cont}} = \text{id}_{H_{\infty}^{\text{enr}}}. \quad (\text{C7})$$

This section proves recoverability; it is not used to define $\mathcal{C}_{\text{cont}}^{\text{disc}}$.

Theorem 3.12 (Correlative emergence of the five constructions). *For every surviving history $\zeta \in H_k$, every $N > k$, every $m \geq 1$, and every $\mathcal{R} \in \mathfrak{R}_{k,N}(\zeta)$, the five constructions of (C2) are defined on the same record, and (C3) contains $s_{k,N}^{(m)}(\mathcal{R})$. Their truncations, modular reductions, and transports satisfy (C4). Furthermore, $\mathcal{C}_{\text{cont}}^{\text{disc}} \neq \emptyset$, the map Π is surjective and is split by Gen_{cont} . The five components are inseparable aspects of the same continuum, not five domains or five independent maps.*

Proof. The survival that defines H_k and the TPK prolongation law give $\rho_k^{-1}(\zeta) \neq \emptyset$ for each vertex. Because the fibers are finite, König's lemma yields $H_{\infty}^{\text{enr}} \neq \emptyset$.

For a fixed record $\mathcal{R}$, the transport of histories and sheets produces $\mathcal{F}_{\text{sp}}[\mathcal{R}]$; affine composition and the exact cancellation defect produce $\mathcal{F}_{\text{sol}}[\mathcal{R}]$; the incidences and the rectangular sequence produce $\mathcal{F}_{\text{gau}}^{(m)}[\mathcal{R}]$; the memory complex and its transport produce $\mathcal{F}_{\text{coh}}^{(m)}[\mathcal{R}]$; and the lines of the normalized path assembly produce $\mathcal{F}_{\text{det}}^{(m)}[\mathcal{R}]$. Each construction acts on the same generated edges, not on an assumed subdomain or on the abstract ambient space of 729 words.

The affine, transport, incidence, and memory compatibilities prove (C4). Truncating the determinant collection of (C2) commutes by definition with r ; its base change commutes with q . The sections $s_{0,N}^{(m)}$ are compatible and thus produce Gen_{cont} . The identity (C7) proves simultaneously the nonemptiness of the limit, the surjectivity of Π , and the recoverability of each history without introducing it as a defining coordinate.

The fiber, relation, number, orientation, carry, memory, boundary, and route remain in the common record; the multisection, the terminals, and the linear and probabilistic readers remain in the five correlative constructions. Therefore, passage to the limit preserves the closure of dependencies and does not disaggregate the continuum. $\square$

The subsequent recognition of the single continuum constructed through spectral, solenoidal, gauge, cohomological, or determinant language belongs to the subsequent conventional comparison. It neither selects its states nor intervenes in its APP–TRIT–TPK generation.

4 Topological Phase Kernel and generation of the seed catalogue

The two APP evaluations acquire generative efficacy once their activation, evaluation position, positional transport and the information preserved by each transition have been specified. The *Topological Phase Kernel*, abbreviated TPK, combines these operations into a dynamics on enriched states. Its definition comprises selection, transport, arithmetic updating and preservation of the record. The finite construction developed in this section provides the seeds on which the regional lifts and coinductive prolongation subsequently act.

The principal result of this stage is a generated and classified catalogue: the initial conditions of the two cursors produce integer emissions; their ternary reading determines oriented words; and the dihedral action organises these words into orbits. Each step preserves its preimages and the relations that permit recovery of the preceding description. The chain of cardinalities

$$104\,976 \longrightarrow 468 \longrightarrow 243 \subset 729, \quad 243 \longrightarrow 43,$$

abbreviates distinct objects. Each object is constructed and each count justified below, before any of them is used as data for regional selection.

4.1 Phase selection, transport and enriched state

4.1.1 Tritic selector of operative phase

The nonadic phase takes its values in $D_9 = \{1, \dots, 9\}$. The tritic selector is the function

$$M_{\text{ph}} : D_9 \longrightarrow \{-1, 0, +1\}, \quad M_{\text{ph}}(g) = \begin{cases} +1, & g \in \{1, 4, 7\}, \\ -1, & g \in \{2, 5, 8\}, \\ 0, & g \in \{3, 6, 9\}. \end{cases} \quad (4.1)$$

In the emitter realisation, the value $+1$ activates the additive evaluation, -1 activates the multiplicative evaluation, and 0 records the neutral event. The cardinal orientation of the cursor is another state coordinate. This distinction specifies both the operation being read and the direction in which its evaluation point is transported.

For transitions indexed by $t \geq 1$, the phase preceding the reading is

$$g_t = 1 + [(t-1)]_9, \quad \epsilon_t = M_{\text{ph}}(g_t), \quad (4.2)$$

where $[n]_b$ denotes the representative of n modulo b in $\{0, \dots, b-1\}$. A window of nine transitions contains the sequence $(+1, -1, 0, +1, -1, 0, +1, -1, 0)$.

4.1.2 Oriented cursors and cardinal transport

The indices $I_9 = \{0, \dots, 8\}$ are used for positions, and the positive labels $(a, b) = (i+1, j+1)$ for APP evaluations. A finite cursor is

$$c = (i, j; d) \in \mathcal{S} := I_9^2 \times \mathcal{D}, \quad \mathcal{D} = \{N, E, S, O\}.$$

The four directions act by

$$v_N = (-1, 0), \quad v_E = (0, 1), \quad v_S = (1, 0), \quad v_O = (0, -1).$$

The cardinal map corresponding to d is

$$T_d(i, j) = ([i + (v_d)_1]_9, [j + (v_d)_2]_9).$$

Thus M_{ph} and T_d have distinct domains, codomains and actions. Tritic activation selects one of the two cursors; cardinal transport changes its position in the direction already retained by that cursor.

The integer lift of the cursor is $\tilde{c} = (x, y; d) \in \mathbb{Z}^2 \times \mathcal{D}$, with finite position $([x]_9, [y]_9)$. Its winding counts are recovered from

$$x = 9[x/9] + [x]_9, \quad y = 9[y/9] + [y]_9. \quad (4.3)$$

These two divisions concern spatial transport. The division of an additive or multiplicative evaluation constitutes a distinct arithmetic record, which is preserved together with them.

4.1.3 Composition of the transition

The state of the finite realisation includes the two lifted cursors, their directions, the phase, the accumulators of the active window, the ordered list of blocks already emitted and the transition record. Each local record contains the preceding positions, the integer evaluations of both sheets, their residues and quotients, the active regime and the orientations. Prefix and boundary fields arising in a prolongation are incorporated into the same state with their compatibility rules.

The transition is ordered as

$$\mathcal{U}_t = \text{Mem}_t \circ \text{Tra}_t \circ \text{Sel}_t. \quad (4.4)$$

Selection determines the regime and preserves the sample taken before any displacement. Transport applies the active movement and the cardinal reversal prescribed by the schedule. Updating incorporates that sample into the accumulators and the record; at the end of a window, it emits the corresponding block. The composition therefore uses a sample taken before transport, although its definitive storage occurs at the end of the transition.

4.2 Initial conditions and reversible enumeration

The additive and multiplicative sheets have independent cursors. The domain of initial conditions is the ordered product

$$\Omega_{\text{seed}} = \mathcal{S}_+ \times \mathcal{S}_\times, \quad |\mathcal{S}| = 9^2 \cdot 4 = 324.$$

Consequently,

$$|\Omega_{\text{seed}}| = 324^2 = 104\,976. \quad (4.5)$$

The number 81 counts positions; 324 counts the oriented positions of one cursor; 104 976 counts ordered pairs of cursors. Choosing a position in APP retains only part of an initial condition of the emitter.

Fix the numbering

$$\text{ind}(N) = 0, \quad \text{ind}(E) = 1, \quad \text{ind}(S) = 2, \quad \text{ind}(O) = 3,$$

and define

$$\nu(i, j; d) = 4(9i + j) + \text{ind}(d). \quad (4.6)$$

Proposition 4.1 (Exact enumeration of the initial conditions). *The map ν is a bijection from $\mathcal{S}$ onto $\{0, \dots, 323\}$. The pair of indices is reversibly encoded by $324\nu_+ + \nu_\times \in \{0, \dots, 104\,975\}$.*

Proof. Division by four recovers

$$\text{ind}(d) = [\nu]_4, \quad j = [\lfloor \nu/4 \rfloor]_9, \quad i = \lfloor \nu/36 \rfloor.$$

The bounds on the domain ensure that these three quantities belong to their respective index sets and reconstruct ν through (4.6). For the pair, Euclidean division by 324 recovers the quotient ν_+ and the residue $\nu_\times$. $\square$

The enumeration traverses the complete domain before any regional grouping. The indices locate the preimages of an emission while keeping explicit the condition that produced each trajectory.

4.3 Sheet evaluation and arithmetic record

At a positive position (a, b) , the integer evaluations are $A_+(a, b) = a + b$ and $A_\times(a, b) = ab$. For $n > 0$, retain

$$\rho_9(n) = 1 + [n - 1]_9, \quad q_9(n) = \lfloor (n - 1)/9 \rfloor, \quad n = \rho_9(n) + 9q_9(n). \quad (4.7)$$

Additive activation incorporates $\rho_9(A_+)$ into the corresponding accumulator. Multiplicative activation uses an observable of the residue $\rho_9(A_\times)$, defined next.

4.3.1 Discrete exponent in the units modulo nine

The powers of 2 modulo 9 traverse

$$2^0, 2^1, \dots, 2^5 \equiv 1, 2, 4, 8, 7, 5 \pmod{9}, \quad 2^6 \equiv 1 \pmod{9}.$$

The discrete logarithm in this group of six units is represented by

$$\ell_9(1) = 0, \quad \ell_9(2) = 1, \quad \ell_9(4) = 2, \quad \ell_9(8) = 3, \quad \ell_9(7) = 4, \quad \ell_9(5) = 5. \quad (4.8)$$

The emission law extends the observable by $\ell_9(3) = \ell_9(6) = \ell_9(9) = 0$. To preserve its provenance, the recorded sample includes the original residue and the indicator of membership in the group of units. An observed value of zero can therefore originate from the unit 1 or from one of the three non-invertible residues, with its origin remaining recoverable.

4.3.2 Order of reading and movement

Evaluations are performed at the positions preceding the transition. If $\epsilon_t = +1$, the additive reading is accumulated and its cursor is displaced. If $\epsilon_t = -1$, the multiplicative discrete exponent is accumulated and the second cursor is displaced. If $\epsilon_t = 0$, the neutral counter is incremented and both positions are preserved.

At the end of steps 27 and 54, both directions are replaced by their opposites. The windows are concatenated with positions and orientations preserved. At the start of each window, only its three local accumulators are reset to zero; the preceding record remains available.

Proposition 4.2 (Inversion of the lifted transport). *Let $a, b \in \{0, 1\}$ be the activation and reversal indicators of a cursor. If opp interchanges $N \leftrightarrow S$ and $E \leftrightarrow O$, the transformation*

$$F_{a,b}(z, d) = (z + av_d, \text{opp}^b(d))$$

has inverse

$$F_{a,b}^{-1}(z', d') = (z' - av_{\text{opp}^b(d')}, \text{opp}^b(d')).$$

Spatial reduction modulo nine intertwines this transformation with the finite transport.

Proof. Applying opposition twice recovers the initial direction. Once that direction has been recovered, subtracting av_d reverses the displacement. For the spatial projection, it suffices to use $[z + av_d]_9 = [[z]_9 + av_d]_9$ componentwise. Directional reversal preserves this equality. $\square$

The phase determines the indicators used by the inverse. The spatial and arithmetic records can therefore be preserved jointly, including when the boundary of the representative I_9^2 is crossed.

4.4 Six windows and decimal block emission

4.4.1 Accumulation within a nonadic window

Window m , with $1 \leq m \leq 6$, comprises steps $9m - 8, \dots, 9m$. Let S_m be the sum of its active positive additive residues, E_m the sum of its active discrete exponents, and Z_m its number of neutral events.

Lemma 4.3 (Activation multiplicities). *Each window contains three additive activations, three multiplicative activations and three neutral activations. In particular, $Z_m = 3$.*

Proof. The phases of a window traverse exactly $1, \dots, 9$. Each of the three sets in (4.1) has cardinality three. $\square$

The six windows comprise 54 transitions and produce six blocks. This length belongs to the initial emitter defined here. Continuation in depth subsequently acts on the words and their compatible records.

4.4.2 Decimal encoding rule

The emission law specifies the digits

$$\begin{aligned} d_{m,1} &= [S_m]_{10}, \\ d_{m,2} &= [S_m + E_m]_{10}, \\ d_{m,3} &= [3S_m + 5E_m + 7Z_m]_{10}, \\ U_m &= 100d_{m,1} + 10d_{m,2} + d_{m,3}. \end{aligned} \quad (4.9)$$

The weights 100, 10, 1 are the hundreds, tens and units positions. The subscript 10 denotes the decimal residue, and each block satisfies $0 \leq U_m \leq 999$. The emitted word is the ordered sequence $\mathbf{U} = (U_1, \dots, U_6)$, with a fixed width of three digits per block.

The coefficients 3, 5, 7 constitute the explicit weights of this emission law. Once the rule has been fixed, each digit is calculated from the accumulators, and the results of this section follow from that rule. The number 1 appearing in its reduced form has a further specific provenance: $7Z_m = 21 \equiv 1 \pmod{10}$.

If $s_m = [S_m]_{10}$ and $e_m = [E_m]_{10}$, one obtains

$$G(s_m, e_m) = 100s_m + 10[s_m + e_m]_{10} + [3s_m + 5e_m + 1]_{10} = U_m. \quad (4.10)$$

Here the letter e_m denotes a component of the discrete-exponent signature. Its definition belongs to the finite emitter.

Proposition 4.4 (Recovery of the two signatures). *The map $G : \{0, \dots, 9\}^2 \rightarrow \{0, \dots, 999\}$ is injective. For a block $U = 100a + 10b + c$ in its image, the inverse is*

$$s = a, \quad e = [b - a]_{10}, \quad c = [3s + 5e + 1]_{10}.$$

Consequently, the word $\mathbf{U}$ determines the two six-component signatures $\mathbf{s}$ and $\mathbf{e}$.

Proof. The last two summands of (4.10) belong to $\{0, \dots, 99\}$, so the hundreds digit recovers s . The tens digit is $[s + e]_{10}$; subtracting s and reducing modulo ten recovers e . The units digit verifies the third congruence. The reconstruction applies independently to the six positions of the word. $\square$

The integer totals S_m, E_m are recovered from the recorded readings; the hundreds and tens digits recover their residues. For example, a total $S_m = 15$ produces $s_m = 5$. This distinction identifies precisely which information each level of encoding preserves.

4.4.3 Positional encoding of the six-block word

The six blocks admit a fixed-width integer encoding,

$$\mathcal{N}(\mathbf{U}) = \sum_{m=1}^6 U_m 1000^{6-m}, \quad 0 \leq \mathcal{N}(\mathbf{U}) < 1000^6. \quad (4.11)$$

The first block occupies the position of greatest weight, and the sixth the base-thousand units position. A block 058 occupies a complete position with integer value 58; its three-digit width belongs to the decimal presentation of that position.

Proposition 4.5 (Positional recovery of the emissions). *The encoding (4.11) determines the complete word by*

$$U_m = \left\lfloor \left\lfloor \frac{\mathcal{N}(\mathbf{U})}{1000^{6-m}} \right\rfloor \right\rfloor_{1000}, \quad 1 \leq m \leq 6.$$

Proof. On division by 1000^{6-m} , the preceding blocks yield integer multiples of 1000, block m yields U_m , and the contribution of the subsequent blocks belongs to $[0, 1)$. Taking the integer part eliminates the latter contribution, and the residue modulo 1000 eliminates the former. $\square$

This encoding preserves the six emissions. Its domain has ordered base-thousand positions, whereas the APP domain has spatial positions modulo nine. The composition retains both indices: m identifies an emission window, and (i, j) identifies the cell visited during a transition in that window.

4.5 Complete calculation of an emission beginning with 501

Fix the initial conditions

$$c_{+,0} = (0, 4; N), \quad c_{\times,0} = (2, 3; N), \quad \nu_+ = 16, \quad \nu_{\times} = 84.$$

The initial positive positions are $(1, 5)$ and $(3, 4)$. The preceding chronological index is $t = 0$, the first phase read is $g_1 = 1$, and the record is empty. These coordinates are the inputs to the calculation; the blocks are obtained by executing the recurrence.

Table 1: First window: preceding readings and active accumulation.

t	ϵ_t	position +	position $\times$	ΔS	ΔE	(S, E, Z)
1	+1	(1, 5)	(3, 4)	6	0	(6, 0, 0)
2	-1	(9, 5)	(3, 4)	0	0	(6, 0, 0)
3	0	(9, 5)	(2, 4)	0	0	(6, 0, 1)
4	+1	(9, 5)	(2, 4)	5	0	(11, 0, 1)
5	-1	(8, 5)	(2, 4)	0	3	(11, 3, 1)
6	0	(8, 5)	(1, 4)	0	0	(11, 3, 2)
7	+1	(8, 5)	(1, 4)	4	0	(15, 3, 2)
8	-1	(7, 5)	(1, 4)	0	2	(15, 5, 2)
9	0	(7, 5)	(9, 4)	0	0	(15, 5, 3)

Note. Positions are positive labels of the cells preceding movement. Neutral events increment only Z .

The three active additive evaluations are

$$1 + 5 = 6, \quad 9 + 5 = 14 = 5 + 9, \quad 8 + 5 = 13 = 4 + 9,$$

so $S_1 = 6 + 5 + 4 = 15$. The three active multiplicative evaluations are $3 \cdot 4 = 12$, $2 \cdot 4 = 8$ and $1 \cdot 4 = 4$. Their positive residues are 3, 8, 4, with observables 0, 3, 2; hence $E_1 = 5$. Finally, $Z_1 = 3$. The decimal rule gives

$$d_{1,1} = [15]_{10} = 5, \quad d_{1,2} = [15 + 5]_{10} = 0, \quad d_{1,3} = [45 + 25 + 21]_{10} = 1,$$

and one obtains $U_1 = 501$.

Table 2: The six windows of initial condition (16, 84).

m	steps	S_m	E_m	Z_m	s_m	U_m	$[-U_m]_3$
1	1–9	15	5	3	5	501	0
2	10–18	6	5	3	6	614	1
3	19–27	24	5	3	4	498	0
4	28–36	21	5	3	1	169	2
5	37–45	12	5	3	2	272	1
6	46–54	12	5	3	2	272	1

The second additive reading above has arithmetic quotient 1, whereas its lifted cursor is at $(-1, 4; N)$, with vertical winding count -1 . These are distinct data in the same record. Both participate in the complete recovery of the evaluation and its position.

The resulting word and its signatures are

$$\mathbf{U} = (501, 614, 498, 169, 272, 272),$$

$$\mathbf{s} = (5, 6, 4, 1, 2, 2), \quad \mathbf{e} = (5, 5, 5, 5, 5, 5).$$

Cardinal reversal after step 27 changes the trajectories of the final three windows. The repeated block 272 corresponds to two successive windows with their own positions and chronological record.

As a second case, the minimal initial condition $(\nu_+, \nu_-) = (0, 0)$ produces

$$\mathbf{U} = (252, 109, 252, 903, 850, 850), \quad \mathbf{s} = (2, 1, 2, 9, 8, 8), \quad \mathbf{e} = (3, 9, 3, 1, 7, 7).$$

In this case, both cursors return to their initial positions and orientations after 54 transitions. The nonadic phase completes six cycles, the chronological counter advances by 54 units, and the record contains 54 samples. The positional reading of the return and the recorded history describe distinct properties of the execution.

4.6 Factorisation of the finite census

The signatures $f_+(\nu) = \mathbf{s}$ and $f_-(\nu) = \mathbf{e}$ are calculated by executing the corresponding cursor through the six windows. The independence of positions and directions between the two cursors gives

$$T_{\text{em}}(\nu_+, \nu_-) = G^{(6)}(f_+(\nu_+), f_-(\nu_-)), \quad (4.12)$$

where $G^{(6)}$ applies (4.10) at each position. This factorisation first permits enumeration of $324 + 324$ trajectories and then combination of the distinct signatures, preserving the multiplicities of their preimages.

4.6.1 Images of the two signatures

The following tables include every signature. Its multiplicity is the number of initial conditions producing it; the minimal index locates one such condition. The complete preimage is defined by

$$f_{\sigma}^{-1}(a) = \{\nu \in \{0, \dots, 323\} : f_{\sigma}(\nu) = a\}.$$

The recurrence of the preceding sections determines every element of this set by a finite integer calculation.

Table 3: The eighteen additive signatures.

signature	multiplicity	$\nu_{\min}$	signature	multiplicity	$\nu_{\min}$
122564	18	17	564122	18	16
122898	18	24	645212	18	4
212645	18	5	654889	18	33
212988	18	0	889221	18	13
221456	18	29	889654	18	32
221889	18	12	898122	18	25
456221	18	28	898465	18	20
465898	18	21	988212	18	1
546988	18	9	988546	18	8

Table 4: The twenty-six multiplicative signatures.

signature	multiplicity	$\nu_{\min}$	signature	multiplicity	$\nu_{\min}$
000000	108	8	555555	24	84
177393	8	1	555717	8	14
177555	8	54	555771	8	18
177771	8	11	555933	8	24
339555	8	6	717555	8	12
339771	8	15	717717	8	21
339933	8	59	717933	8	19
393177	8	0	771177	8	9
393393	8	45	771339	8	13
393555	8	40	771555	8	16
555177	8	52	933339	8	57
555339	8	4	933555	8	26
555393	8	41	933717	8	17

Proposition 4.6 (Image and multiplicities of the emitter). *The image $\mathcal{U}_6 = \text{im } T_{\text{em}}$ contains 468 distinct words. Their preimages are distributed into 432 fibres of cardinality 144, 18 of cardinality 432 and 18 of cardinality 1944.*

Proof. Exhaustive evaluation of the algorithm in Section 4.9 on the indices $0, \dots, 323$ gives the two tables above. The first partition has 18 classes of cardinality 18; the second has 24 classes of cardinality 8, one of cardinality 24 and one of cardinality 108. The sums $18 \cdot 18 = 324$ and $24 \cdot 8 + 24 + 108 = 324$ verify the total mass of both partitions. Proposition 4.4 makes the coupling of their images injective; there are therefore $18 \cdot 26 = 468$ emissions. By (4.12), the preimage of a pair of signatures is the product of their two preimages. This gives $18 \cdot 24 = 432$ products of cardinality $18 \cdot 8 = 144$, 18 of cardinality $18 \cdot 24 = 432$, and 18 of cardinality $18 \cdot 108 = 1944$. Finally,

$$432 \cdot 144 + 18 \cdot 432 + 18 \cdot 1944 = 104\,976.$$

The enumerative part is an exact finite check of the recurrence; the factorisation explains why this count covers every pair. $\square$

The emission in Example 4.5 has a fibre of cardinality $18 \cdot 24 = 432$. The example with indices $(0, 0)$ has a fibre of cardinality $18 \cdot 8 = 144$. In both cases, the emitted word recovers the signatures, whereas identification of a particular initial condition uses its index or its trajectory record.

4.7 Oriented ternary reading and preservation of fibres

Oriented reduction acts separately on each block:

$$\Psi : \mathcal{U}_6 \longrightarrow \mathbb{F}_3^6, \quad \Psi(U_1, \dots, U_6) = ([-U_1]_3, \dots, [-U_6]_3). \quad (4.13)$$

Its sign fixes an orientation of the catalogue. The balanced identification $0 \mapsto 0, 1 \mapsto +1, 2 \mapsto -1$ is then applied componentwise.

Modulo nine organises APP evaluations and positions; residues modulo ten form each digit; base thousand combines three decimal positions; and modulo three determines the ternary signature of each block. These four uses have defined objects and functions. In particular, Ψ reads six blocks and produces six trits; a positional conversion of the integer formed by eighteen digits would be a different operation.

For the two conditions already calculated, one obtains

$$\begin{aligned} \Psi(501, 614, 498, 169, 272, 272) &= (0, 1, 0, 2, 1, 1), \\ \Psi(252, 109, 252, 903, 850, 850) &= (0, 2, 0, 0, 2, 2). \end{aligned}$$

The decimal emitter is defined before this reduction. The function of Ψ is to produce the ternary object on which orbital symmetry and regional selectors act. This function explains its position in the genealogy: it connects an already calculated emission with a structural classification.

The image $\mathcal{W}_6 := \Psi(\mathcal{U}_6)$ contains exactly 243 words within the ambient set of $3^6 = 729$ words. Enumeration by the final algorithm gives the histogram

$ \Psi^{-1}(w) $	1	2	3	4	5	6
number of words w	132	66	18	3	6	18.

(4.14)

The sums for the two readings are

$$132 + 66 + 18 + 3 + 6 + 18 = 243, \quad 132 + 2 \cdot 66 + 3 \cdot 18 + 4 \cdot 3 + 5 \cdot 6 + 6 \cdot 18 = 468.$$

The first counts ternary words; the second recovers the number of emissions distributed among their fibres.

Example 4.7 (Two emissions with the same ternary signature). The emissions

$$(498, 501, 614, 252, 252, 109), \quad (870, 810, 923, 252, 252, 109)$$

are distinct and both have image $(0, 0, 1, 0, 0, 2)$ under Ψ . Each has 144 initial conditions. The ternary word preserves a reading class; the emissions and their preimages distinguish the data that converge in that class.

Preservation of fibres is expressed concretely by

$$\mathcal{F}_w = \{\mathbf{U} \in \mathcal{U}_6 : \Psi(\mathbf{U}) = w\}, \quad \Omega_{\mathbf{U}} = T_{\text{em}}^{-1}(\mathbf{U}).$$

The complete preimage of w in the initial domain is therefore the disjoint union of the sets $\Omega_{\mathbf{U}}$, with $\mathbf{U} \in \mathcal{F}_w$. The enriched record can retain the emission, its signatures and the initial condition together with its ternary reading. The exact inverse then concerns the recorded datum with its type; the isolated six-trit word retains the precise scope of (4.13).

4.8 Dihedral action and the forty-three orbits

Define the following permutations on six coordinates:

$$r(u_1, u_2, u_3, u_4, u_5, u_6) = (u_3, u_1, u_2, u_5, u_6, u_4), \quad (4.15)$$

$$s(u_1, u_2, u_3, u_4, u_5, u_6) = (u_4, u_5, u_6, u_1, u_2, u_3). \quad (4.16)$$

The first rotates the two groups of three coordinates in conjugate directions; the second interchanges the groups. Their composition satisfies $r^3 = s^2 = 1$ and $srs = r^{-1}$. They therefore generate an action of the six-element dihedral group D_3 .

Proposition 4.8 (Equivariance and stability of the catalogue). *Reduction by Ψ commutes with the action of D_3 . The sets $\mathcal{U}_6$ and $\mathcal{W}_6$ are stable under its generators.*

Proof. The equality $\Psi(g\mathbf{U}) = g\Psi(\mathbf{U})$ holds at every coordinate because g permutes positions and Ψ applies the same reduction at each position. Stability of the catalogue is a further check: generating its 468 elements by (4.12), one verifies that $r\mathbf{U}$ and $s\mathbf{U}$ again belong to that set. The algorithm in Section 4.9 performs these membership checks. Equivariance then transfers stability to the ternary image. $\square$

Proposition 4.9 (Complete orbital decomposition). *The set $\mathcal{W}_6$ decomposes into 43 orbits: 38 of cardinality six and 5 of cardinality three.*

Proof. For each $w \in \mathcal{W}_6$, form

$$\mathcal{O}(w) = \{w, rw, r^2w, sw, srw, sr^2w\}.$$

Choose the lexicographically least word in this set as representative. Two words have the same representative exactly when they belong to the same orbit. The algorithm generates the list of representatives in Table 5; their cardinalities sum to $38 \cdot 6 + 5 \cdot 3 = 243$. Since every word in the catalogue participates in this operation and the catalogue is stable, the list covers the entire image. $\square$

Table 5: Minimal representatives and cardinalities of the forty-three orbits.

representative	cardinality	representative	cardinality	representative	cardinality
001002	6	002112	6	012222	6
001011	6	002211	6	021022	6
001012	6	002220	6	021121	6
001021	6	011021	6	021202	6
001022	6	011112	6	022022	3
001100	3	011120	6	022112	6
001101	6	011201	6	022121	6
001110	6	011211	6	022202	3
001112	6	011220	6	022211	6
001202	6	011222	6	022222	6
001210	6	012112	6	112112	3
001211	6	012121	6	112211	3
001220	6	012202	6	112222	6
001222	6	012210	6		
002012	6	012220	6		

The orbits inherit integer invariants from their fibres. For each word, $n(w) = |\mathcal{F}_w|$ counts emissions, whereas

$$M(w) = \sum_{\mathbf{U} \in \mathcal{F}_w} |\Omega_{\mathbf{U}}|$$

counts initial conditions. The symbol census $c(w) = (n_0, n_1, n_2)$ records the number of occurrences of each ternary class. Together with the orientation and the support of the trajectories, these data provide the predicates for subsequent regional selection. The classification thus preserves more information than the form of a representative word.

4.9 Self-contained enumeration and verification algorithm

The enumeration uses integers, finite lists and the operations already defined. The following procedure specifies its execution; σ takes the values $+$ and $\times$.

1. For each $v = 0, \dots, 323$, recover $(i, j; d)$ using the inverse of (4.6). Initialise an empty signature list.
2. For each window $m = 1, \dots, 6$, initialise the accumulator $A = 0$. Traverse $t = 9m - 8, \dots, 9m$, calculating ϵ_t by (4.2).
3. If $\sigma = +$ and $\epsilon_t = +1$, add $\rho_9(i + j + 2)$ to A and update $(i, j) \leftarrow T_d(i, j)$. If $\sigma = \times$ and $\epsilon_t = -1$, add $\ell_9(\rho_9((i + 1)(j + 1)))$ and perform the same transport.
4. At the end of $t = 27$ or $t = 54$, replace d by its opposite. At the end of the window, append $[A]_{10}$ to the signature list. Preserve the position and direction for the next window.
5. Group the 324 indices according to equality of their six-component lists. These two groupings give f_+ , $f_\times$ and all their preimages.
6. For each pair of distinct signatures, apply $G^{(6)}$. Assign to the emission the product multiplicity of the two preimages. Verify signature recovery in each block and the sum of multiplicities 104 976.
7. Apply Ψ to each emission and group by equality of words. Count the 243 values and the multiplicities in (4.14).
8. Apply r, s to each emission and verify membership in the catalogue. For each ternary word, form its six dihedral images, remove repetitions and record its lexicographic minimum. Count the 43 orbits and the two sizes in Table 5.

The execution terminates: the first stage evaluates 648 trajectories of 54 transitions; the second combines 468 pairs of signatures; the third groups a set of 243 words. An additional direct check can execute the 104 976 pairs and compare the result with the factorisation (4.12); its scope is the same finite domain.

4.10 Preservation of the record and the function of the seeds

Let $\text{Sample}(q_t)$ be the sample preceding the transition, and $\mathcal{M}_t$ the accumulated record. The memory update is

$$\mathcal{M}_{t+1} = \mathcal{M}_t \parallel \text{Sample}(q_t), \quad (4.17)$$

where $\parallel$ denotes ordered concatenation. Induction gives

$$\mathcal{M}_n = \mathcal{M}_0 \parallel (\text{Sample}(q_0), \dots, \text{Sample}(q_{n-1})).$$

The accumulators are reconstructed by summing the active contributions of each segment of nine records. Their quotients, the winding counts of the cursors and the orientations remain associated with the same transitions. Execution determines that record through the recurrence; storage preserves the specific provenance of the readings.

At the end of this stage, the complete domain of initial conditions, decimal emission, its two recoverable signatures, the ternary image, its fibres and its orbital classification have been constructed. The counts 104 976, 468, 243, 729 and 43 now have a domain and a mechanism of derivation. Regional selections receive these structures with their invariants, and the lifts act on oriented words with preserved provenance. This continuity makes the finite catalogue a constructive antecedent of the coinductive development.

5 Development of the formal kernel: initial conditions, geometry, and refinement

The eighty-one-position support does not coincide with the domain of initial conditions for the dynamics. An initial condition must also specify an orientation, and the additive and multiplicative sheets

have independent cursors. This distinction allows exhaustive reconstruction of the emission without selecting a numerical region in advance.

5.1 Initial states and the emission recurrence

For enumeration we use indices $I_9 = \{0, \dots, 8\}$, related to positive labels by $i \mapsto i + 1$. Let

$$\mathcal{S} = I_9^2 \times \{N, E, S, O\}, \quad \Omega_{\text{seed}} = \mathcal{S}_+ \times \mathcal{S}_\times.$$

Each initial condition $\omega = (s_+, s_\times)$ determines two positions and two directions. The domain comprises all ordered pairs, so

$$|\mathcal{S}| = 324, \quad |\Omega_{\text{seed}}| = 324^2 = 104\,976. \quad (5.1)$$

The independent annex enumerates all these conditions; the body presents the law transforming them. The rows of an emission table are fibers of this initial-condition domain.

The finite realization uses six windows of nine transitions. At each instant t , the phase signature is $M_{\text{ph}}(1 + ((t - 1) \bmod 9))$. If its value is $+1$, the positive residue $\Sigma(i + 1, j + 1) = \rho_9(i + j + 2)$ of the first cursor is read and that cursor is then moved; if its value is -1 , the positive residue $\Pi(i + 1, j + 1) = \rho_9((i + 1)(j + 1))$ of the second cursor is read and the second cursor is then moved. Zero increments the neutral counter. At the end of transitions twenty-seven and fifty-four, cardinal orientations are reversed. Integer evaluations and their quotients are retained in the record, but the emitter uses the residue readings specified here.

In the unit sector, the multiplicative reader uses the base-two discrete logarithm in $(\mathbb{Z}/9\mathbb{Z})^\times$:

$$\ell_9(1) = 0, \quad \ell_9(2) = 1, \quad \ell_9(4) = 2, \quad \ell_9(8) = 3, \quad \ell_9(7) = 4, \quad \ell_9(5) = 5.$$

For finite emission it is extended by zero to the non-unit sector. This extension is an observable, not an identification of the elements 3, 6, 9 with the multiplicative identity. The sheet and its membership in the radical sector remain in the trajectory record.

Let S_m be the sum of the three positive additive residues Σ in window m , E_m the sum of the three discrete exponents of the multiplicative residues Π , and $Z_m = 3$ the number of neutral events. Thus S_m does not sum the integer evaluations $S = i + j$, whose quotients remain in the record. With $[a]_{10} \in \{0, \dots, 9\}$, the decimal emission rule is

$$\begin{aligned} d_{m,1} &= [S_m]_{10}, \\ d_{m,2} &= [S_m + E_m]_{10}, \\ d_{m,3} &= [3S_m + 5E_m + 7Z_m]_{10}, \\ U_m &= 100d_{m,1} + 10d_{m,2} + d_{m,3}. \end{aligned} \quad (5.2)$$

The integer coefficients and calendar of this recurrence are explicit parts of the declared emission law. Values of π, φ, e or α are not received. Distinguishing an emission rule from a constant recognized from its output prevents the genealogy from being hidden in a decimal table.

If $s_m = [S_m]_{10}$ and $e_m = [E_m]_{10}$, the same rule is written

$$U_m = 100s_m + 10[s_m + e_m]_{10} + [3s_m + 5e_m + 1]_{10}. \quad (5.3)$$

The final one is the residue of $7Z_m = 21$ modulo ten. The oriented ternary reduction of the catalog adopts the convention

$$\Psi(U_1, \dots, U_6) = ([-U_1]_3, \dots, [-U_6]_3). \quad (5.4)$$

The sign of this chart is explicit: changing it modifies the words and requires transport of subsequent orientations. The balanced identification of each component of $\mathbb{F}_3$ with TRIT is performed after (5.4).

Proposition 5.1 (Injective factorization of emission). *The word $U = (U_1, \dots, U_6)$ determines both signatures $s = (s_1, \dots, s_6)$ and $e = (e_1, \dots, e_6)$. The emission image has $18 \cdot 26 = 468$ elements. The ternary projection has 243 distinct values in the ambient set of 729 words.*

Proof. The hundreds digit of U_m recovers s_m . If b_m is its tens digit, then $e_m = [b_m - s_m]_{10}$. The units digit checks the third congruence of (5.3). Hence coupling is injective on the signature product.

Enumerating the 324 conditions for one cursor through the preceding recurrence produces eighteen additive signatures, each with eighteen preimages, and twenty-six multiplicative signatures, of which twenty-four have eight preimages, one has twenty-four, and one has one hundred and eight. The annex preserves these signatures and their initial conditions, allowing the finite enumeration to be repeated without a reference numerical catalog. Injectivity of the coupling gives $18 \cdot 26$ emissions and multiplicities

$$\begin{array}{ccc} 432 & \text{fibers of } 144, & 18 \text{ fibers of } 432, & 18 \text{ fibers of } 1944. \\ 18 \cdot 24 & & & \end{array}$$

The sum is $432 \cdot 144 + 18 \cdot 432 + 18 \cdot 1944 = 104\,976$. Applying (5.4) to the 468 emissions produces exactly 243 words. This last number is the cardinality of the image, not of the space $\mathbb{F}_3^6$. $\square$

This factorization explains the difference between finite exhaustiveness and unlimited depth. Domain (5.1) is complete for the declared support and emitter; it does not contain every prefix of every prolongation in advance. Depth belongs to the system of compatible extensions constructed over those conditions.

5.2 Calendar closure and memory advance

The calendar of one hundred and eight transitions consists of twelve nonadic windows. Its group structure is more precise than an informal product of two periods:

$$0 \longrightarrow C_{12} \xrightarrow{\iota} C_{108} \xrightarrow{p} C_9 \longrightarrow 0, \quad \iota([b]) = [9b], \quad p([t]) = [t]_9. \quad (5.5)$$

If $t = 9b + r$ with $0 \leq r < 9$, adding two phases produces the carry

$$(b, r) + (b', r') = \left(b + b' + \left\lfloor \frac{r + r'}{9} \right\rfloor, [r + r']_9 \right),$$

where the first coordinate is reduced modulo twelve. Composition preserves the term linking the two coordinates.

Proposition 5.2 (Nonsplitting of the calendar). *Sequence (5.5) is exact and admits no section that is a group homomorphism.*

Proof. The kernel of reduction modulo nine is the subgroup of multiples of nine, the image of ι . If the extension split, commutativity would give $C_{108} \cong C_{12} \times C_9$. The first group has exponent 108, whereas the second has exponent $\text{lcm}(12, 9) = 36$, a contradiction. $\square$

Nine transitions restore the nonadic phase and advance one window; one hundred and eight restore the finite calendar. Neither fact requires erasing the trajectory or restoring a prolongation prefix. Nonadic holonomy Γ_9 denotes transport through one turn on states with memory, so that

$$g(\Gamma_9 x) = g(x), \quad M(\Gamma_9 x) = M(x) + 1. \quad (5.6)$$

Phase equality and memory increment are compatible and constitute two observations of the same transport.

5.2.1 Temporal parametrization of compatibility and tritic orientation

The temporal interpretation of TRIT is established on transitions of the complete state. Let $x \xrightarrow{\text{adm}} y$ be the admissibility relation determined by TPK operations on $\hat{X}$. A trajectory parametrized by a discretely ordered set $\mathcal{T}$ is a map

$$\gamma : \mathcal{T} \longrightarrow \hat{X}, \quad \gamma(t) \xrightarrow{\text{adm}} \gamma(t^+),$$

where t^+ is the successor of t . The compatibility relation precedes the choice of parameter: the order allows its transitions to be expressed as an evolution. When the term section is used, the projection $p : E \rightarrow \mathcal{T}$ and condition $p \circ \gamma = \text{id}_{\mathcal{T}}$ are also declared. The state space, temporal order, and transported information are thereby specified.

The signature τ selects a quadratic regime and admits a temporal reading relative to this parametrization. Type +1, with $J_{+1}^2 = -I$, represents return and memory; type 0, with $J_0^2 = 0$, represents the transition threshold; type -1, with $J_{-1}^2 = I$, represents opening in two eigendirections. In the model's temporal terminology, these functions correspond, respectively, to past, present, and future. The correspondence concerns the operational position of a transition within a history: retained antecedent, update, and admissible prolongation.

This interpretation acquires content through three distinct operations. Restriction of a trajectory recovers the prefix already constructed. The update $\mathcal{U}_t$ composes selection, transport, and recording in the observed transition. Prolongation considers continuations that preserve the boundary conditions of that prefix. If $\mathcal{P}_n(x)$ denotes the set of admissible continuations of length n from x , deleting the last step defines

$$r_n : \mathcal{P}_{n+1}(x) \longrightarrow \mathcal{P}_n(x).$$

An unlimited continuation is a collection $(\gamma_n)_{n \geq 1}$ satisfying $r_n(\gamma_{n+1}) = \gamma_n$. The inverse system thus preserves the restrictions each depth imposes on preceding depths. Existence of a compatible collection is established in the corresponding prolongation domain; merely specifying the sets $\mathcal{P}_n(x)$ does not replace that proof.

Bidirectionality therefore combines reconstruction of antecedents with compatibility of continuations. On the signature sequence, the involution C_{rev} reverses order and changes the sign of each component. The trajectory is also subject to the transformations of position, sheet, and boundary specified by transport. The reversed signature is an observation of the conjugate trajectory, not a replacement for its other coordinates. The threshold remains distinguished: the visible value 0 retains its transition function even when its quotient or historical record is nonzero.

The relation with exponential geometries is equally precise. In each fixed regime, $\exp(tJ_\tau)$ composes parameters and preserves the algebraic norm. Elliptic closure belongs to that local representation. Phase return of Γ_0 , by contrast, acts on the trajectory with memory, according to (5.6). A trajectory can recover its phase while retaining a new antecedent: historical data have changed although the phase observation is the same. The parabolic threshold retains a linear displacement; hyperbolic opening separates two eigendirections of expansion and contraction. These three realizations provide composition laws, whereas the TPK determines their selection and effective concatenation.

The aperiodic suspension in Section 6 will make this distinction explicit through a finite phase factor, an irrational rotation, and a memory degree. The link between TRIT and temporality is thus expressed by operations on trajectories and their quadratic representations. Its content exceeds a classification of signs: it preserves the regime, the direction of transport, and the relation between antecedent, transition, and continuation.

The monodromic representation of this return is developed in Section 6.10. There, the holonomy of the state is distinguished from its representation on phase and counter. The symbolic suspension in Section 6.12 defines the aperiodic time-crystal model, and Section 6.14 composes this temporality with the three Euler realizations.

5.3 Observable compression and orthogonal conservation

An isometric realization provides a quantitative expression of what a reduced observation loses. Let $U : \mathcal{H} \rightarrow \mathcal{H}$ be an isometric realization of transport through one turn, $U^*U = I$, and $J : \mathcal{H}_{\text{vis}} \rightarrow \mathcal{H}$ an isometric inclusion. Define

$$T = J^*UJ, \quad D = (I - JJ^*)UJ.$$

Then

$$UJ = JT + D, \quad J^*D = 0, \quad I - T^*T = D^*D. \quad (5.7)$$

The last equality follows by taking the adjoint product of the first and using orthogonality. Contraction of the visible component does not by itself constitute destruction of information in the total state.

Proposition 5.3 (Finite-horizon conservation). *If $I - T^*T = D^*D$, then, for every $n \geq 1$,*

$$I = \sum_{r=0}^{n-1} T^{*r} D^* D T^r + T^{*n} T^n. \quad (5.8)$$

Proof. Multiplying $D^*D = I - T^*T$ by T^{*r} and T^r , then summing, produces a telescoping sum. The interior terms cancel, leaving $I - T^{*n} T^n$. $\square$

The final summand preserves the terminal state. Removing it before justifying a limit would alter the equality. This distinction is the operational content of conservation of unity under compression: the norm is distributed among the visible component, memory, and terminal, rather than being replaced by a single residue.

5.4 Local development: commutation, Lorentzian signature, and discrete curvature

The adjacent diagonal block with equal diagonals modulo nine is unique. Indeed,

$$k^2 \equiv (k+1)^2 \pmod{9} \iff 2k+1 \equiv 0 \pmod{9}$$

gives $k = 4$. Its visible matrix is

$$B_c = \begin{pmatrix} 7 & 2 \\ 2 & 7 \end{pmatrix} = 7I + 2R, \quad R = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad R^2 = I.$$

With $P_{\pm} = (I \pm R)/2$, one obtains $B_c = 9P_+ + 5P_-$. The unity of stochastic normalization and conservation of an indefinite form are two realizations of this block:

$$\frac{B_c}{9} = P_+ + \frac{5}{9}P_-, \quad \frac{B_c}{\sqrt{45}} = \exp(\eta R), \quad \eta = \frac{1}{2} \log \frac{9}{5}.$$

For $Q = \text{diag}(1, -1)$, the second matrix satisfies $B_c^T Q B_c = 45Q$. This yields a local realization in $SO^+(1, 1)$; no prior introduction of four-dimensional spacetime is required to prove it.

Uniform averaging over the classes $C_1 = \{1, 4, 7\}$, $C_2 = \{2, 5, 8\}$, and $C_0 = \{3, 6, 9\}$ gives

$$M_{\Pi} = \begin{pmatrix} 4 & 5 & 6 \\ 5 & 4 & 6 \\ 6 & 6 & 9 \end{pmatrix}, \quad M_{\Sigma} = \begin{pmatrix} 5 & 6 & 4 \\ 6 & 4 & 5 \\ 4 & 5 & 6 \end{pmatrix}.$$

The characteristic polynomial of M_{Π} is

$$(\lambda + 1)((\lambda - 9)^2 - 72),$$

so its quadratic form has signature $(2, 1)$. The local spectral gap $9 - 5 = 4$, the coefficient 2, the aggregate difference $51 - 45 = 6$, and the integrated defect $9 \cdot 6 = 54$ describe different levels of the same arithmetic comparison. They are not interchangeable indices of a single operator.

Discrete curvature can be described directly before this spectral realization. In residues, let $S(x, y) = x + y$, $P(x, y) = xy$, and let Δ_x, Δ_y be forward differences. Link variables are differences of potentials, $A_x^T = \Delta_x T$, $A_y^T = \Delta_y T$, and we set

$$F(T_x, T_y) = \Delta_x \Delta_y T_y - \Delta_y \Delta_x T_x.$$

Since $\Delta_x S = \Delta_y S = 1$, $\Delta_x P = y$, and $\Delta_y P = x$,

$$F(S, S) = F(P, P) = 0, \quad F(S, P) = +1, \quad F(P, S) = -1 \pmod{9}. \quad (5.9)$$

Sum and product, taken separately, induce flat connections; their ordered mixture changes the sign of plaquette curvature. Summing over a rectangle of sides a, b cancels interior edges, and the oriented boundary sum equals $\pm ab$ modulo nine. An exact relation thus appears between composition order and boundary orientation.

5.5 Cubic completion and two notions of boundary

Expanding the cubic support of side nine to that of side ten admits an exact positional stratification:

$$10^3 = 9^3 + 3 \cdot 9^2 + 3 \cdot 9 + 1 = 729 + 243 + 27 + 1. \quad (5.10)$$

The first term corresponds to three interior coordinates; the second to one new coordinate; the third to two; and the last to the vertex with three new coordinates. The expansion corona contains 271 positions. By contrast, the internal boundary of the discrete cube of side nine contains $9^3 - 7^3 = 386$. The difference between these counts is a difference of domain, not a conflict of values.

Dividing (5.10) by 1000 gives a normalized partition of unity. Cube incidence distinguishes six faces, twelve edges, and eight vertices. The connection to Steiner hexads and octads is realized through the combinatorial maps of Section 11, which preserve the cardinalities and incidence of each domain. This completion fixes the relation between bases 729 and 1000 here; the numerical generation in Section 6, the electronic coordinate $\rho_E = \varphi/\pi^2$ in (6.6), and the exceptional prolongation subsequently form three typed outputs, while the gravitational construction reuses the action sections derived from that genealogy in Section 12.

5.5.1 Stratification of the completion and restoration of the apex

The completion admits a set-theoretic description making its operation explicit. Let E_9 be the set of nine residue classes, represented positively by $1, \dots, 9$, and let $*$ be an additional element. The symbol $*$ differs from the zero class, already represented by 9. Consequently, extending E_9 to $\hat{E} = E_9 \sqcup \{*\}$ increases the cardinality from nine to ten without changing the internal modular identification. The cubic completion is $\hat{C} = \hat{E}^3$, and the initial cube is $C = E_9^3$.

Definition 5.4 (Strata of combinatorial codimension). For $0 \leq r \leq 3$, define

$$S_r = \{(x_1, x_2, x_3) \in \hat{E}^3 : \#\{j : x_j = *\} = r\}.$$

The apex of the completion is $a_* = (*, *, *)$, and the punctured completion is $\hat{C}^\circ = \hat{C} \setminus \{a_*\}$.

Proposition 5.5 (Exact decomposition of the completion). *The strata are disjoint and satisfy*

$$\hat{C} = S_0 \sqcup S_1 \sqcup S_2 \sqcup S_3, \quad |S_r| = \binom{3}{r} 9^{3-r}, \quad (5.11)$$

$$|\hat{C}^\circ| = 729 + 243 + 27 = 999, \quad |\hat{C}| = 999 + 1 = 1000. \quad (5.12)$$

The punctured expansion contains 270 elements in addition to the initial cube; the complete expansion contains 271.

Proof. An element of S_r is obtained by choosing the r coordinates containing $*$, in $\binom{3}{r}$ ways, and assigning one of the nine classes of E_9 to each remaining coordinate. Classification by the number of additional coordinates is exhaustive and its classes are disjoint. The sole element of S_3 is a_* . Removing it subtracts one from the total cardinality; restoring it recovers the complete Cartesian product. $\square$

The expression *holographic extreme and mean ratio*, abbreviated EMRH in the corpus, here denotes this articulation of the initial body, expansion strata, and unit restoration. Its immediate content is the exact partition of a finite product and its normalization. The number 999 represents a definite object, the punctured completion; 1000 represents its closure through restoration of the apex. The added unit therefore has an explicit incidence location.

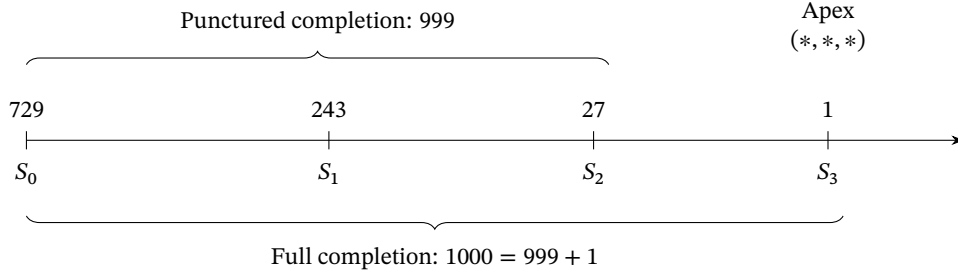


Figure 3: Strata of the cubic completion. The arrangement represents the set-theoretic decomposition, not a metric scale. The 243 elements of S_1 form three coordinate strata of 81 elements; the 27 elements of S_2 form three strata of 9. The apex completes the partition.

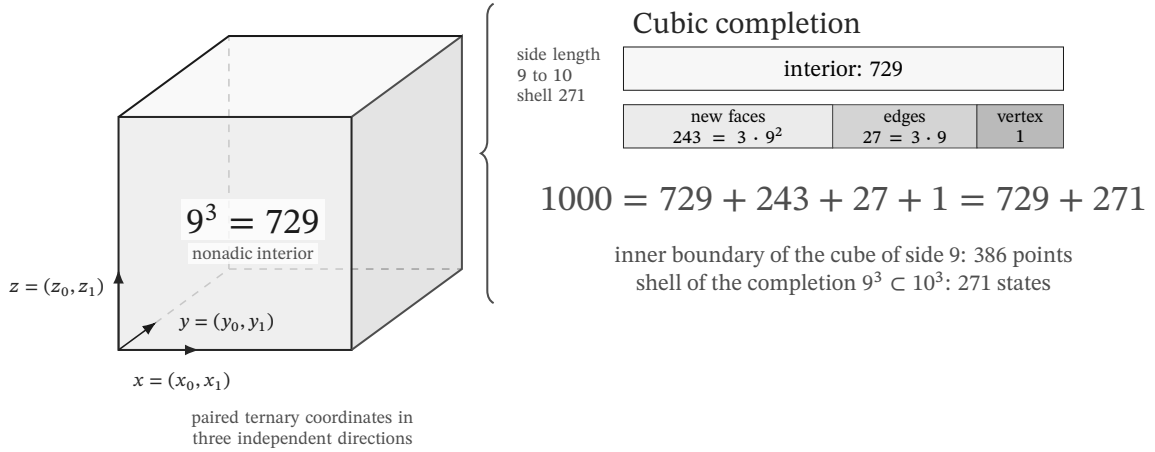


Figure 4: Cubic representation of the expansion from nine to ten classes per coordinate. Strata S_1, S_2, S_3 locate the added incidences on faces, edges, and the apex; their cardinalities form the 271-element shell. The 386-element boundary belongs to the initial cube. This geometric perspective complements the partition in Figure 3; graphical spacing is schematic.

5.5.2 Conservation of unity and dimensional compatibility

Normalization is not limited to dividing an isolated identity. In dimension d , the product $\hat{E}^d$ carries the uniform measure $\mu_d(\{x\}) = 10^{-d}$. The coordinate projection $p_d : \hat{E}^{d+1} \rightarrow \hat{E}^d$ deletes the last coordinate.

Proposition 5.6 (Projective compatibility of normalized measures). *For every $d \geq 1$,*

$$(p_d)_* \mu_{d+1} = \mu_d, \quad \mu_d(\hat{E}^d) = 1.$$

The mass of the stratum with r additional coordinates is $\binom{d}{r} 9^{d-r} / 10^d$, and the masses of all strata sum to one.

Proof. Each point of $\hat{E}^d$ has ten preimages. Their total mass is $10 \cdot 10^{-(d+1)} = 10^{-d}$. Equality for any subset follows by summing over its points. The stratified formula follows from the same count as (5.11); the binomial $(9 + 1)^d$ gives total normalization. $\square$

In dimension three, removing the apex leaves mass 999/1000, and restoring it adds 1/1000. Renormalizing before restoration would define a different measure: the distinction is operationally relevant. In particular, conservation of probability mass, conservation of transport records, and thermodynamic dissipation concern different objects. The proposition proves the first and provides the compatible support for studying the second; the third requires a specified dynamic and energetic realization.

The internal geometric boundary also remains distinct. On the Cartesian representative $\{1, \dots, 9\}^3$, removing $\{2, \dots, 8\}^3$ leaves 386 points. This boundary belongs to the initial cube, whereas $\hat{C} \setminus C$ belongs to its expansion. Cubic completion, base-thousand positional expansion, and exceptional-incidence reading share support data; their respective maps determine which information from these data each representation uses.

5.6 Cubic incidence and Lorentzian realization

Cubic completion supplies a relation between interior and boundary. Its geometry can be developed through incidence of the six faces with the eight vertices. Label faces by (i, ε) , $i \in \{1, 2, 3\}$, $\varepsilon \in \{+1, -1\}$, and vertices by $v \in \{+1, -1\}^3$. The incidence matrix is

$$M_{(i,\varepsilon),v} = \mathbf{1}_{\{v_i=\varepsilon\}}.$$

Let $u = (1, \dots, 1)^\top \in \mathbb{R}^6$, let R_F denote face opposition, and let $P_u = uu^\top/6$, $P_- = (I - R_F)/2$.

Proposition 5.7 (Four-dimensional incidence quotient). *The incidence Gram matrix satisfies*

$$MM^\top = 12P_u + 4P_-.$$

Its eigenvalues 12, 4, 0 have multiplicities 1, 3, 2. Consequently,

$$Q_\square = \mathbb{R}^6 / \ker(MM^\top) \simeq \mathbb{R}u \oplus E_-, \quad E_- = \text{im } P_-.$$

Proof. A face contains four vertices; two opposite faces have empty intersection, and two distinct non-opposite faces share two vertices. The matrix $12P_u + 4P_-$ has exactly these entries. The uniform subspace has dimension one; differences between opposite faces generate a three-dimensional space orthogonal to it. Its complement has dimension two and belongs to the kernel. $\square$

The decomposition $1 + 3$ comes from incidence. The temporal choice is declared through the bilinear form

$$B_\square([x], [y]) = \langle x, (P_- - P_u)y \rangle.$$

It is well defined on the quotient because both projectors annihilate the kernel. In the basis

$$t = [u]/\sqrt{6}, \quad d_i = [e_{i,+} - e_{i,-}]/\sqrt{2},$$

its matrix is $\text{diag}(-1, 1, 1, 1)$. The domain, decomposition, and sign convention producing the Minkowski realization are thus explicit. The positive Gram matrix determines the quotient; the temporal assignment determines the indefinite form on it.

The operator

$$W_\square = \sqrt{6}P_u + \sqrt{2}P_-, \quad W_\square e_{i,\varepsilon} = t + \varepsilon d_i$$

sends the six face labels to null directions, since $B_\square(t + \varepsilon d_i, t + \varepsilon d_i) = -1 + 1 = 0$. Direction reversal and face opposition thus acquire a concrete geometric realization.

5.6.1 Soldering, reversal, and subdivision

The cellular realization uses an invertible transport $U_m(a)$, preserving the Lorentzian form, between the endpoint fibers of an oriented dual edge a , a face label $\lambda_m(a)$, and an orientation $\sigma_m(a)$. With $\ell_m = \ell_0/9^m$, the soldering map is

$$\beta_m(a) = \ell_m \sigma_m(a) W_{\square, s(a)} e_{\lambda_m(a)}.$$

Compatible reversal satisfies

$$\beta_m(\bar{a}) = -U_m(a) \beta_m(a).$$

Identify the source fibers at two consecutive levels through the declared refinement isometry. If the nine subsegments $a_1, \dots, a_9$, at level $m+1$, preserve label and orientation when transported to that common fiber, then

$$\beta_m(a) = \sum_{j=1}^9 U_{m+1}(a_1 \cdots a_{j-1})^{-1} \beta_{m+1}(a_j). \quad (5.13)$$

Each transported term equals $\ell_m/9$ times the same null direction, proving the identity. The compatibility hypothesis determines which subdivisions realize this refinement. The equation brings together direction, scale, and transport, rather than limiting geometric correspondence to a dimension count.

With orientation and the Lorentzian form also fixed, Hodge duality on $\Lambda^2 Q_{\square}^*$ satisfies $\star^2 = -I$. In dimension four, the sign is $(-1)^{2(4-2)+1} = -1$. This yields a real complex structure on two-forms and, after complexification, the eigenspaces with eigenvalues $+i$ and $-i$. The electronic application in Article I constructs its persistent section on the pair of APP sheets and fixes the polarity entering the electromagnetic realization. In this article, the declared orientation and Lorentzian form transport that structure to the gravitational domain: Hodge duality organizes the components of the current, and Section 26 incorporates them into the algebraic Cartan–Holst equation.

5.7 Compatibility of modular and positional reductions

The block-decimal base enters through Euclidean division preserving residue and quotient simultaneously:

$$n = 1000q + r, \quad 0 \leq r < 1000.$$

Since $1000 \equiv 1 \pmod{9}$ and $1000 \equiv 1 \pmod{3}$,

$$n \equiv q + r \pmod{9}, \quad n \equiv q + r \pmod{3}.$$

Reduction of a block therefore requires the quotient's contribution to recover the integer's class. The numbers 0 and 1000 have the same remainder modulo 1000, but different classes modulo 9 and 3. This is an operational reason for preserving memory through a change of resolution.

Ternary refinement and determination of a decimal prefix can be coordinated through a single integer remainder. If $N/3^L$ is a cylinder endpoint and $D/1000^r$ a decimal endpoint, define

$$A = 1000^r N - D 3^L.$$

On adding six ternary symbols, $N' = 729N + \nu$, $L' = L + 6$, with $0 \leq \nu < 729$. If the decimal prefix remains fixed, one obtains

$$A' = 729A + \nu 1000^r.$$

If a decimal block δ is also incorporated, then $D' = 1000D + \delta$, $r' = r + 1$, and

$$A' = 729000A + \nu 1000^{r+1} - 729\delta 3^L. \quad (5.14)$$

Both identities follow by substitution of the definitions. Comparing endpoints is thus reduced to integer operations retaining each prolongation's contribution. This compatibility makes it possible to refine a region before evaluating its real coordinate.

5.8 Number as a compatible section and value as a reading

Distinguishing state from observation specifies what object is prolonged as resolution increases. At depth n , let X_n be the set of admissible states produced by APP-TRIT-TPK. Their data include position, sum and product sheets, residues and quotients, regime, orientation, carry, phase, route, boundary, memory, and survival. These are not independent coordinates: earlier arithmetic evaluations and transports impose their relations. A prolongation preserves these relations in each antecedent, even while adding new steps and memory data. This is the meaning of the arithmetic of histories: the tritic selection and cardinal transport of Section 2.3.2, the memory composition of Section 2.3.4, and the nonadic holonomy of Section 2.3.8 act on a single prolonged state.

Let $r_{m,n} : X_m \rightarrow X_n$, for $m \geq n$, be depth restrictions, with $r_{n,n} = \text{id}$ and $r_{\ell,n} = r_{m,n} \circ r_{\ell,m}$. Each restriction preserves the prefix and the information already determined in it.

Definition 5.8 (Compatible numerical section). An HMT number is a compatible history of the state system:

$$x = (x_n)_{n \geq 0} \in X_\infty := \varprojlim_n X_n, \quad r_{m,n}(x_m) = x_n. \quad (5.15)$$

A reader is an additional operation on a specified domain of these histories. A digit is a finite observation; a scalar value is an evaluation of the reader. The pair consisting of a section and its reader constitutes a read presentation, not a redefinition of the section's identity.

On the Archimedean domain X_∞^{Arch} , the reader associates a closed cylinder with each prefix, preserves nesting, and makes its diameter tend to zero. It thus determines

$$\text{ev}_\mathbb{R} : X_\infty^{\text{Arch}} \longrightarrow \mathbb{R}, \quad \{\text{ev}_\mathbb{R}(x)\} = \bigcap_n I_n(x). \quad (5.16)$$

The concrete construction of these cylinders and the determination of their boundaries are developed in Section 6. Definition (5.15) does not by itself assert that the image of (5.16) is the entire line: a coverage claim requires its own theorem. Nor does equality of values establish equality of histories without an injectivity result. This separation allows precision to be discussed as depth while preserving the distinction between the arithmetic antecedent and its scalar realization.

5.9 Composition of histories and memory multiplier

Fix a sector of states at depth n and a category $\mathcal{C}$ whose arrows are its admissible histories. Composition is written $vu = v \circ u$: first traverse u , then v . Elementary steps come from TPK selection, transport, and updating. A local tritic fiber admits the notation $\mathbb{R}[J_x]/(J_x^2 + \tau(x)1_x)$, where $\tau(x) \in \{+1, 0, -1\}$. Regime and orientation belong to the transported state; a transition between regimes does not amount to identifying their quadratic algebras.

The following formulation specifies an algebraic sector of this composition. For each object x , a unital associative real algebra A_x , possibly noncommutative, is given. For each $u : x \rightarrow y$, a unital homomorphism $\rho_u : A_x \rightarrow A_y$ is given. Assume a strict action:

$$\rho_{\text{id}_x} = \text{id}_{A_x}, \quad \rho_v \rho_u = \rho_{vu}. \quad (5.17)$$

For each composable pair $u : x \rightarrow y$, $v : y \rightarrow z$, fix an invertible central multiplier $\Omega(v, u) \in Z(A_z)^\times$. Require the normalization $\Omega(\text{id}_y, u) = \Omega(u, \text{id}_x) = 1_y$ and, for every composable triple, the identity

$$\rho_w(\Omega(v, u))\Omega(w, vu) = \Omega(w, v)\Omega(wv, u). \quad (5.18)$$

These are explicit hypotheses on transports and their coefficients, not a consequence of calling the dynamics holonomic. Applying this presentation to a TPK sector requires identifying its ρ_u and $\Omega(v, u)$. Transitions expressed through bimodules require preserving that structure and are not replaced by (5.17).

The space of finite sums

$$\mathcal{H}(\mathcal{C}, A, \rho, \Omega) = \bigoplus_{u \in \text{Mor}(\mathcal{C})} A_{t(u)} T_u$$

receives the bilinear product

$$(aT_v)(bT_u) = a\rho_v(b)\Omega(v, u)T_{vu}, \quad (5.19)$$

when the arrows are composable, and zero otherwise. Here $t(u)$ denotes the terminal state. Addition gathers histories with coefficients; multiplication concatenates histories while preserving the record multiplier.

Proposition 5.9 (Sectorial associativity with a central multiplier). *Under (5.17) and (5.18), product (5.19) is associative. If the object set is finite, its identity is $\sum_x 1_x T_{\text{id}_x}$.*

Proof. For $u : x \rightarrow y, v : y \rightarrow z, w : z \rightarrow t$, let $a \in A_y, b \in A_z, c \in A_t$. The two coefficients of T_{wvu} are

$$\begin{aligned} ((cT_w)(bT_v))(aT_u) &: c\rho_w(b)\Omega(w, v)\rho_{wv}(a)\Omega(wv, u), \\ (cT_w)((bT_v)(aT_u)) &: c\rho_w(b)\rho_w\rho_v(a)\rho_w(\Omega(v, u))\Omega(w, vu). \end{aligned}$$

The strict action identifies $\rho_w\rho_v(a)$ with $\rho_{wv}(a)$. Centrality of $\Omega(w, v)$ allows it to move to the right of that factor; (5.18) then equates the remaining products. The coefficients a, b, c are not permuted. If the triple is not composable, both parenthesizations are zero because of their initial and terminal states. Bilinearity completes the proof. Local identities and normalization of Ω verify the stated identity element. $\square$

Calendar carry provides an effective example. In the category with one object and arrows $\mathcal{C}_9$, take coefficients $A = \mathbb{R}[z, z^{-1}]$, the identity action, and

$$\Omega(b, a) = z^{c_0(a, b)}, \quad c_0(a, b) = \left\lfloor \frac{a+b}{9} \right\rfloor, \quad 0 \leq a, b < 9.$$

The two carry sums of a triple are $\lfloor (a+b+c)/9 \rfloor$; hence (5.18) holds. The formal variable z records the turn without assigning it a numerical value. Subsequently imposing $z^{12} = 1$ recovers the dodecaphase record of calendar (5.5); retaining z without that relation distinguishes successive turns. The example realizes this calendar's memory without identifying it with all TPK memory.

5.10 Double variance of refinement and conservation of unity

Restricting states prolongs observables in the opposite direction. Consider the function algebras, with pointwise product, $D_n = \text{Fun}(X_n, \mathbb{R})$. Precomposition defines

$$r_{m,n}^* : D_n \longrightarrow D_m, \quad r_{m,n}^* f = f \circ r_{m,n}. \quad (5.20)$$

These are unital morphisms; they are also injective when the corresponding restrictions are surjective. The direct limit $D_{\text{cyl}} = \varinjlim_n (D_n, r_{n+1,n}^*)$ gathers finite-depth observations. This algebra is not identified here with the whole route algebra: prolonging transport operators also requires their compatibility with refinement.

Lemma 5.10 (Identity and naturality of evaluation). *If e_x is the indicator of $x \in X_n$, then*

$$r_{m,n}^* e_x = \sum_{y: r_{m,n}(y)=x} e_y, \quad r_{m,n}^* 1_n = 1_m, \quad (5.21)$$

and, for every $f \in D_n, y \in X_m$,

$$\langle r_{m,n}^* f, y \rangle = \langle f, r_{m,n} y \rangle.$$

Each section $x \in X_\infty$ determines a well-defined evaluation $D_{\text{cyl}} \rightarrow \mathbb{R}$, given by $[f] \mapsto f(x_n)$ for $f \in D_n$.

Proof. The fibers of $r_{m,n}$ partition X_m . Their indicators are orthogonal and sum to 1_m , proving (5.21). Naturality is the definition of precomposition. Finally, $(r_{m,n}^* f)(x_m) = f(x_n)$, so two compatible representatives give the same evaluation in the direct limit. $\square$

The global identity is the constant function one on all possibilities; an individual section selects a compatible history. Preserving the former does not mean immobilizing the latter. Refinement may increase the number of extensions and the depth of each history while preserving the identity and all evaluations of its antecedents.

5.11 Phase return, Weyl representation, and two readings

Law (5.6) acquires an operator realization in the bilateral suspension constructed in Section 6. There, T is one-step advance, V_9 observes the nonadic phase, W_δ the rotation phase, and D_M the memory degree. Writing $\hat{\Gamma}_9 = T^9$ for the turn in this representation, the Weyl relations (6.13) imply, on finitely supported vectors,

$$\begin{aligned} V_9 \hat{\Gamma}_9 &= \hat{\Gamma}_9 V_9, & W_\delta \hat{\Gamma}_9 &= e^{2\pi i 9\delta} \hat{\Gamma}_9 W_\delta, \\ [D_M, \hat{\Gamma}_9] &= \hat{\Gamma}_9. \end{aligned} \tag{5.22}$$

The first equality follows from $\zeta_9^9 = 1$; the second from nine applications of the covariance of W_δ ; the third expresses the unit memory increment. The finite phase returns while irrational rotation and memory distinguish the next state. Characters and parameter δ are realized subsequently from the numerical construction specified there: they do not select the seeds retrospectively. This representation describes suspension observables and translations; it does not replace all TPK coordinates or operators.

The double reading extends the same distinction between history and observable. On its Archimedean domain, (5.16) applies; on the calibrated incidence domain, the maps Q_n of Section 11.2 preserve words and boundary frames. The equality proved there, $r_{n,m} Q_n = Q_m p_{n,m}$, determines their prolongation Q_∞ . On the common domain of both readings, $\text{ev}_\mathbb{R}(x)$ and $Q_\infty(x)$ can be considered jointly: the former retains the value; the latter, the transported incidence preserved by its codomain. The section precedes both. This articulation makes it possible to pass from the arithmetic of histories to constants and exceptional reading without turning their realizations into new initial data.

6 Coinductive generation of π, φ, e to arbitrary depth

The six-block emission is a finite stage of the construction, not a complete decimal expansion. Its function is to produce regions with multiplicity, orientation, and transport rules. Prolongation must preserve these data and determine a compatible collection of prefixes of arbitrary length. The word *generation* designates this operation; *genealogy* designates the reconstruction of its dependencies. Both notions enter the discussion, but perform distinct functions.

6.1 Orbits, regions, and characters of closure, propagation, and self-scaling

On a word $w = (w_1, \dots, w_6)$, we define

$$r(w) = (w_3, w_1, w_2, w_5, w_6, w_4), \quad s(w) = (w_4, w_5, w_6, w_1, w_2, w_3).$$

We have $r^3 = s^2 = I$ and $srs = r^{-1}$; these permutations therefore realize the dihedral group D_3 . The action preserves the catalog and multiplicities and commutes with Ψ . The quotient of the 243 visible words has 43 orbits: thirty-eight of cardinality six and five of cardinality three.

The closure class is recognized by a fiber predicate: its six orientations have five emissions per word, total multiplicity 1008, and distribution $432 + 4 \cdot 144$. There is a single orbit with these properties in the catalog. To specify the other two selections, we write $f(w) = \#\Psi^{-1}(w)$, $m(w) = \sum_{U \in \Psi^{-1}(w)} \text{mult}(U)$, and $\nu(w) = (n_0, n_1, n_2)$. The additive diagonal class of an emission is $d(U) = [i + j]_9$ in indices from zero to eight; its additive signature preserves this class and the orientation *ES* or *NO*. Let $\mathcal{D}(\mathcal{O})$ be the set of these classes for all emissions over an orbit.

Among the orbits of size six, the joint predicate

$$f(w) = 1, \quad m(w) = 144 \quad (w \in \mathcal{O}), \quad \mathcal{D}(\mathcal{O}) = \{0, 3, 6\}$$

leaves exactly six orbits. In this sector, the census $\nu = (2, 3, 1)$, equal to that of closure, selects propagation uniquely; the census $\nu = (2, 2, 2)$ selects self-scaling uniquely. The diagonal support cannot be omitted: without it there are four balanced singleton-fiber orbits of multiplicity 144, not one. Finally, the existence of an emission with $d = 6$ and orientation *ES* selects a representative of each orbit and leads to

$$w^{\text{cl}} = 010211, \quad w^{\text{pr}} = 201101, \quad w^{\text{au}} = 121200. \quad (6.1)$$

These symbols first designate regions of the catalog. Scalar identification will be proved through their readers; they are not chosen because they coincide with digits of a known constant.

The closure word 010211 has the following five preimages:

Six-block emission	Multiplicative signature	Role	Multiplicity
501, 614, 498, 169, 272, 272	555555	transversal	432
810, 923, 870, 169, 272, 272	339555	positive orientation	144
810, 983, 810, 169, 272, 272	393555	positive orientation	144
870, 923, 810, 169, 272, 272	933555	positive orientation	144
870, 923, 810, 418, 674, 521	933717	oriented return	144

The multiplicity 432 and constant signature 555555 characterize the transversal region. They do not identify the central position (5, 5) of APP. Confusing a six-window signature with a position of the support would eliminate the distinction between numerical region and electronic structure.

The microfiber also has a precise bipartite morphology. Each emission is a product of an additive-cursor class and a multiplicative class. The three positive regions share the same additive class of eighteen conditions; together they contain three multiplicative classes of eight conditions each. The transversal region has a product 18×24 ; the three positive regions together, another 18×24 ; and the return, a product 18×8 . The decomposition by emissions has five parts, whereas the decomposition by connected components of the bipartite graph has three. Both preserve the same 1008 seeds without reducing form to a cardinality.

6.2 Finite transport and the prolongation boundary

The endomorphisms L_0, L_1 of the ternary-word space act before Archimedean realization. Their determination begins with transitions of the arithmetic state, not with the prescription of two matrices. If U_t is the composite step of selection, transport, and updating, the observed block satisfies

$$b_j^\chi = \Pi_6(x_j^\chi), \quad x_{j+1}^\chi = U_{9j+9} \cdots U_{9j+1}(x_j^\chi), \quad \chi \in \{\text{cl}, \text{pr}, \text{au}\}.$$

Six transitions are formed for each regime: two consecutive transitions in each of the three regions. The pairs of source and target matrices obtained in the record are

X_0	Y_0	X_1	Y_1
010211	012222	010211	002111
012222	010211	002111	110221
201101	121221	102011	012222
121221	102011	012222	102011
121200	112202	121020	010210
112202	121020	010210	010200

Each six-digit string represents a row in $\mathbb{F}_3^6$. Since $\det X_0 = 1$ and $\det X_1 = -1$ in $\mathbb{F}_3$, the equations $X_a L_a = Y_a$ uniquely determine $L_a = X_a^{-1} Y_a$. In that row-vector chart, the result is

$$L_0 = \begin{pmatrix} 2 & 2 & 2 & 1 & 2 & 1 \\ 2 & 1 & 2 & 2 & 1 & 1 \\ 1 & 0 & 0 & 1 & 0 & 2 \\ 0 & 0 & 1 & 1 & 1 & 0 \\ 2 & 2 & 2 & 0 & 2 & 1 \\ 2 & 1 & 2 & 1 & 0 & 0 \end{pmatrix}, \quad L_1 = \begin{pmatrix} 2 & 1 & 2 & 0 & 2 & 2 \\ 1 & 2 & 1 & 1 & 2 & 0 \\ 1 & 2 & 1 & 1 & 1 & 2 \\ 0 & 1 & 0 & 0 & 2 & 1 \\ 2 & 0 & 2 & 1 & 0 & 1 \\ 0 & 2 & 2 & 2 & 1 & 1 \end{pmatrix} \pmod{3}.$$

The calendar (L_0, L_0, L_1, L_1) produces four new blocks of six components. Their concatenations form

$$w_6 \longrightarrow w_{12} \longrightarrow w_{18} \longrightarrow w_{24} \longrightarrow w_{30}.$$

The subscripts of w indicate accumulated length. The boundary R_{36} preserves the terminal structure and opens the prolongation relation; it is not renamed as a final periodic block that could be repeated indefinitely.

The identity $L = X^{-1}Y$ proves uniqueness once the transitions have been fixed; it does not replace the production of those transitions by U_t . The provenance of their record and the examination of the sixteen binary calendars are documented in the technical supplement. The indicated calendar is the only one of those sixteen that jointly reproduces the three sequences of five blocks.

6.3 Five closure regions and angular compensation

The closure region contains a transversal component and four companions. The symmetric reader acts on the space of its five positions through

$$P_5 = \frac{1}{5} \mathbf{1} \mathbf{1}^\top, \quad \mathbf{1} = (1, 1, 1, 1, 1)^\top.$$

The invariance $P_5 \lambda = \lambda$ and conservation $\mathbf{1}^\top \lambda = 1$ force $\lambda = \mathbf{1}/5$. This normalization distributes unity among reader positions; it does not replace the seed multiplicities 432, 144, 144, 144, 144 by uniform frequencies. The reader's elliptic chart realizes each coordinate through $1 + \lambda_j J$; its slope is λ_j , so the primitive slope is $1/5$. This makes explicit the rule passing from a normalized coordinate to a slope, instead of identifying the two notions without a map. The reader composes the four companion positions by the law

$$a \oplus b = \frac{a + b}{1 - ab}. \quad (6.2)$$

This law is obtained by multiplying $(1 + aJ)(1 + bJ)$ in the regime $J^2 = -I$ and dividing by its scalar component. Slope composition is therefore a realization of the quadratic structure of TRIT, not an ordinary sum of four fractions.

Proposition 6.1 (Closure compensator). *The composition of four slopes $1/5$ is $120/119$. The condition of return to slope one determines a unique positive compensator of the form $1/q$, with $q > 1$, and gives $q = 239$.*

Proof. The first doubling gives $(1/5) \oplus (1/5) = 5/12$; the second gives $(5/12) \oplus (5/12) = 120/119$. The condition

$$\frac{120/119 - 1/q}{1 + (120/119)/q} = 1$$

is equivalent to $(120 - 119)q = 120 + 119$, hence $q = 239$. $\square$

The resulting closure character has realization

$$\Pi_{\text{cl}} = 4 \left(4 \arctan \frac{1}{5} - \arctan \frac{1}{239} \right). \quad (6.3)$$

Machin's angular identity recognizes $\Pi_{\text{cl}} = \pi$. The meaning of the construction is precise: the slope and number of compositions are declared by the penta-fiber reader; the compensation equation forces 239. Once this character has been produced, its evaluation by alternating series does not select the seed orbit anew.

For $0 < x < 1$, the alternating sums

$$A_n(x) = \sum_{j=0}^n \frac{(-1)^j x^{2j+1}}{2j+1}$$

enclose $\arctan x$ between two consecutive truncations, with difference $x^{2n+3}/(2n+3)$. Applied to the two slopes in (6.3), they provide rational endpoints with explicit error. The numerical value is thus calculated as a consequence of the exact character, without introducing a target prefix.

6.4 Propagation reader and normalization of unity

The propagation region is realized through a formal collection $P_t \in \mathcal{A}[[t]]$, where $\mathcal{A}$ is a unital commutative algebra over $\mathbb{Q}$. Its composition satisfies $P_{t+s} = P_t P_s$, with unit $P_0 = 1$, and oriented elementary transport fixes the scalar generator $P'_0 = +1$. The condition fixes its sign, not only its magnitude. Writing

$$P_t = \sum_{n \geq 0} a_n t^n,$$

normalization determines $a_0 = a_1 = 1$. Comparing the coefficient of $s^n t$ in $P_{s+t} = P_s P_t$ gives $(n+1)a_{n+1} = a_n a_1$; equivalently, $P'_t = P_t$. Thus,

$$(n+1)a_{n+1} = a_n, \quad a_n = \frac{1}{n!}. \quad (6.4)$$

The character at one unit of the parameter is $\Pi_{\text{pr}} = P_1$, subsequently recognized as e .

The normalization of the generator matters. Without that datum, the semigroup law would admit $P_t = e^{ct}$ for different c ; the condition fixing the parameter unit in the reader is therefore not omitted. This condition precedes decimal comparison and belongs to the exposition of the operational rule, not to the choice of an experimental value.

Proposition 6.2 (Rational propagation bounds). *For $n \geq 1$, let $E_n = \sum_{j=0}^n 1/j!$. Then*

$$E_n < \Pi_{\text{pr}} < E_n + \frac{n+2}{(n+1)(n+1)!}.$$

Proof. The first omitted term is $1/(n+1)!$. Each subsequent ratio is at most $1/(n+2)$. The geometric upper bound for the tail is

$$\frac{1}{(n+1)!} \frac{1}{1 - 1/(n+2)} = \frac{n+2}{(n+1)(n+1)!}.$$

The strict inequality follows from the decrease in successive ratios. $\square$

6.5 Modal incidence and generation of self-scaling

The tritic selector produces the cycle $(+1, -1, 0)$. In the modes Σ, Π , the rule is $+1 : m \mapsto \Sigma, -1 : m \mapsto \Pi, 0 : m \mapsto m$. With cyclic closure, the observed mode is (Σ, Π, Π) . Its edges are $\Sigma \rightarrow \Pi, \Pi \rightarrow \Pi$, and $\Pi \rightarrow \Sigma$. Its incidence, in that order of modes, is therefore

$$F_{av} = \begin{pmatrix} 0 & 1 \\ 1 & 1 \end{pmatrix}. \quad (6.5)$$

The four entries are obtained from the calendar, not from an equation previously supplied with the value of φ . Repetition of the calendar gives counts $3F_{av}, 9F_{av}, 36F_{av}$ at lengths nine, twenty-seven, and one hundred eight. These counts are not the powers of the matrix.

Theorem 6.3 (Positive self-scaling character). *The incidence (6.5) has a unique strictly positive eigenray. When normalized as $(1, x)^T$, its coordinate x satisfies $x^2 - x - 1 = 0$, $x > 0$, and is $\varphi = (1 + \sqrt{5})/2$.*

Proof. The eigenvalue equation gives $x = \lambda$ and $1 + x = \lambda x$. Eliminating λ produces the polynomial. Only one of its two roots is positive. Since F_{av}^2 has all entries positive, the positive ray is unique and simple. The radical expression recognizes the coordinate obtained; it does not participate in producing F_{av} . $\square$

With $F_0 = 0, F_1 = 1, F_{n+1} = F_n + F_{n-1}$, the incidence induces, for $n \geq 1$,

$$F_{av}^n = \begin{pmatrix} F_{n-1} & F_n \\ F_n & F_{n+1} \end{pmatrix}.$$

Consecutive ratios F_{n+1}/F_n alternate around φ . The determinant identity

$$F_{n+1}^2 - F_n F_{n+2} = (-1)^n$$

gives an exact width $1/(F_n F_{n+1})$ between two consecutive ratios. Their growth makes this rational bracket arbitrarily small.

The measure on all compatible histories does not coincide with the mode frequency of the three-event calendar. The chronological frequency is $(1/3, 2/3)$. The measure of maximal entropy of the symbolic envelope defined by F_{av} is constructed from its eigenvectors and has weight ratio φ^{-2} . The latter enters the areal–volumetric reading; it is not obtained by simply substituting one counting frequency for another.

After the characters have been generated, the two orientations of the reader

$$R(x, y) = \frac{x}{y^2}$$

provide

$$\rho_A = \frac{\pi}{\varphi^2}, \quad \rho_E = \frac{\varphi}{\pi^2}, \quad \varphi^3 \rho_A^2 \rho_E = 1. \quad (6.6)$$

The electronic application in Article I incorporates the second expression, $\rho_E = \varphi/\pi^2$, into the exponent of its central section. The first, $\rho_A = \pi/\varphi^2$, is also obtained as the limit of the marked reading $\pi F_{n-1}/F_{n+1}$. Exchanging the arguments defines the bidirectional reparameterization of (6.6); in the gravitational construction, ρ_A organizes the areal reader and ρ_E preserves the electronic provenance of the action sections, without altering the coefficients that each application determines in its own domain.

6.6 Exponential realizations of the generated coordinates

The regional coordinates already determined allow recognition of concrete realizations of the TRIT quadratic collection. The order is substantive: construction of the regions precedes evaluation of an exponential at parameters π or $2 \log \varphi$. These expressions identify the role of the coordinates in operators constructed from APP and TPK; they do not participate in selecting the prefixes that produce them.

6.6.1 Ternary cycle, nilpotent transport, and sheet incidence

Multiplication $a(r) = 4r$ in $\mathbb{Z}/9\mathbb{Z}$ has order 3. On the units it determines the orbits $\{1, 4, 7\}$ and $\{2, 8, 5\}$. In the augmentation module of either orbit, consisting of integer combinations whose coefficients sum to zero, the basis $(e_r - e_{4r}, e_{4r} - e_{7r})$ gives

$$C = \begin{pmatrix} 0 & -1 \\ 1 & -1 \end{pmatrix}, \quad C^2 + C + I = 0.$$

The restricted form has Gram matrix $\begin{pmatrix} 2 & -1 \\ -1 & 2 \end{pmatrix}$. On residue evaluations modulo 9, the action distinguishes the sheets: $S(ax, ay) = aS(x, y)$, whereas $P(ax, ay) = a^{-1}P(x, y)$. The cyclic realization therefore preserves a contragredient orientation between sum and product. Quotients of the integer lift are retained in the corresponding arithmetic record.

Elementary parabolic transport is represented by

$$N = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}.$$

The areal–volumetric incidence already constructed in (6.5) gives

$$F_{av} = \begin{pmatrix} 0 & 1 \\ 1 & 1 \end{pmatrix}, \quad A = F_{av}^2 = \begin{pmatrix} 1 & 1 \\ 1 & 2 \end{pmatrix}.$$

The three operators act on declared real spaces: the augmentation module tensored with $\mathbb{R}$, the nilpotent-transport plane, and the modal-incidence plane. Sharing a matrix dimension does not identify these spaces or their coordinates.

Proposition 6.4 (Concrete generators of the three types). *The operators*

$$J_e = \frac{2C + I}{\sqrt{3}}, \quad J_p = N, \quad J_h = \frac{2A - 3I}{\sqrt{5}}$$

satisfy, respectively,

$$J_e^2 = -I, \quad J_p^2 = 0, \quad J_h^2 = I.$$

Proof. The relation for C implies $(2C + I)^2 = -3I$. Direct multiplication gives $N^2 = 0$. Finally, A has trace 3 and determinant 1, so $A^2 - 3A + I = 0$ and $(2A - 3I)^2 = 5I$. $\square$

Proposition 6.5 (Three evaluations of the exponential identity). *In the preceding realizations,*

$$\boxed{\exp(\pi J_e) + I = 0, \quad \exp(J_p) = I + J_p, \quad \exp(2 \log \varphi J_h) = F_{av}^2.} \quad (6.7)$$

Proof. The elliptic specialization of the tritic exponential gives $\cos \pi I + \sin \pi J_e = -I$. Nilpotency removes exactly the terms of degree at least 2 in the second expression. For the third, $\varphi^2 = \varphi + 1$ implies

$$\cosh(2 \log \varphi) = \frac{3}{2}, \quad \sinh(2 \log \varphi) = \frac{\sqrt{5}}{2}.$$

Thus the exponential equals $\frac{3}{2}I + \frac{\sqrt{5}}{2}J_h = A$. $\square$

The first equality represents the elliptic half-turn; the complete turn satisfies $\exp(2\pi J_e) = I$. The second retains a nonzero linear term: the parabolic regime expresses transport, not absence of evolution. The third realizes a unimodular hyperbolic advance. Its eigenvalues are φ^2 and φ^{-2} ; one direction expands and the other contracts while preserving the determinant. The three evaluations use different parameters of one common exponential collection.

In this collection, e enters through the scalar evaluation $\exp(I) = eI$ and the law $\exp(sI)\exp(tI) = \exp((s+t)I)$. Its role is not restricted to the parabolic type. The coordinate e , previously obtained from its region, is recognized here as the unit evaluation of the exponential; this identity does not replace its coinductive genealogy.

6.6.2 Polynomial resolution of the regimes

The three realizations admit a compact presentation on the direct sum of their spaces. Define

$$\mathbb{J} = J_e \oplus J_p \oplus J_h, \quad Q = \mathbb{J}^2.$$

The preceding relations imply

$$\mathbb{J}^6 = \mathbb{J}^2, \quad Q^3 = Q.$$

Using Q keeps this operator square distinct from the dodecaphase record K .

Proposition 6.6 (Polynomial regime projectors). *The maps*

$$p_e = \frac{Q^2 - Q}{2}, \quad p_p = I - Q^2, \quad p_h = \frac{Q^2 + Q}{2}$$

are mutually orthogonal idempotents, sum to I , and recover the three summands of the representation.

Proof. The three polynomials take values $(1, 0, 0)$, $(0, 1, 0)$, and $(0, 0, 1)$ at the points $-1, 0, +1$, respectively. The differences between their squares and themselves, as well as their cross products, are multiples of $X^3 - X$. Substituting Q , which annihilates that polynomial, proves the identities. On the three blocks, Q acts as $-I, 0, I$. $\square$

One thus obtains

$$\begin{aligned} \exp(t\mathbb{J}) &= p_e(\cos tI + \sin t\mathbb{J}) + p_p(I + t\mathbb{J}) \\ &\quad + p_h(\cosh tI + \sinh t\mathbb{J}). \end{aligned} \tag{6.8}$$

Each summand preserves its quadratic relation, and conjugation reverses its evolution. The total representation gathers the three regimes, but does not by itself define a transition operator between them. Such a transition requires the TPK transport maps with their domains, orientation, and memory. The exponential articulation allows recognition of their types without replacing those dynamics by an identification of fibers.

6.7 Compatible cylinders and arbitrary depth

The realization of a ternary block $b = (b_1, \dots, b_6)$ is the integer

$$\nu(b) = \sum_{j=1}^6 b_j 3^{6-j} \in \{0, \dots, 728\}.$$

A sequence of blocks defines integer prefixes by $N_{n+1} = 729N_n + \nu(b_{n+1})$, starting at $N_0 = m$, where m is the integer part determined by the reader. The preceding brackets place the closure, propagation, and self-scaling characters in $(3, 4)$, $(2, 3)$, and $(1, 2)$, respectively; their initial values are therefore $m = 3, 2, 1$. Subsequent blocks refine the fractional part. The corresponding cylinder is

$$I_n = \left[\frac{N_n}{729^n}, \frac{N_n + 1}{729^n} \right], \quad |I_n| = 729^{-n}. \tag{6.9}$$

The half-open convention is used to select a representative when a value falls on a boundary. For passage to the limit, the closures of these intervals are used and the two equivalent boundary expansions are identified. This specification prevents the assumption that every intersection of nested half-open intervals is automatically nonempty.

Theorem 6.7 (Determination of scalar realization). *A compatible block history determines a unique limit of the left endpoints of (6.9). If an exact character has rational brackets of arbitrarily small width and is separated from the boundary of each decimal cylinder examined, each decimal prefix of finite length is determined after a finite computation. The prefixes thus obtained are mutually compatible.*

Proof. Extending a block preserves the preceding prefix by Euclidean division. The closed cylinders are nested and their diameter tends to zero; their left endpoints are Cauchy and have a unique limit. For a fixed decimal length, a value separated from the boundaries of the partition has positive distance from them. A sufficiently narrow bracket lies within a single cell and determines its prefix. Uniqueness of the cell and nesting of the partitions prove compatibility under truncation. At points with a terminating expansion, the exact boundary decision and its representation convention must be added; that decision does not follow from numerical width alone. $\square$

The closure, propagation, and self-scaling characters have the preceding brackets. After their recognition, the irrationality of π, e, φ guarantees their separation from the rational boundaries of any finite decimal precision. Arbitrary depth means *for every finite horizon there exists a computation determining its prefix*; it does not require a finite machine to store an infinite number of digits all at once. The mathematical object is the complete compatible collection, and an execution produces a finite projection of it.

6.8 Specular involution and synchronization of the regions

The nonadic connection transports the three characters jointly. In the ninth-phase chart, the self-scaling axis and the ternary sum of closure and propagation remain fixed under specular exchange. Helical inversion changes the direction of transport; the mirror of the two channels changes their relative orientation. They are different involutions, although both act on the same system of trajectories.

Coincidences in the census illustrate this coordination:

Phase	N_π	N_e	N_φ	Equality of counts
4	207	207	122	$N_\pi = N_e$
5	39	151	151	$N_e = N_\varphi$
6	110	110	69	$N_\pi = N_e$
7	81	29	81	$N_\varphi = N_\pi$
8	59	59	59	triple synchronization
9	43	43	43	triple synchronization

These integers count compatible coordinates of the regions. They do not assert equality among π, e , and φ as real numbers. If

$$\partial(a, b, c) = (a - b, b - c, c - a),$$

its image belongs to the lattice $A_2 = \{(u, v, w) \in \mathbb{Z}^3 : u + v + w = 0\}$. At phases eight and nine the relative component vanishes, but the diagonal component changes from (59, 59, 59) to (43, 43, 43). Relative synchronization coexists with a shift of -16 per channel. The closure leading to α must preserve both pieces of information: relative coincidence and the evolution of memory.

6.9 Sturmian helical suspension

The relation between bases nine and ten of the completion defines the dimensionless parameter

$$\delta = \frac{\log(10/9)}{\log 10}. \quad (6.10)$$

It is not obtained from a target physical constant. It is the relative logarithmic scale of two bases already present in the construction. Its use as a symbolic reading of refinement has an exact consequence.

Lemma 6.8 (Irrationality of the slope). *The number δ is irrational.*

Proof. If $\delta = p/q$, $q > 0$, then $(10/9)^q = 10^p$ and $9^q = 10^{q-p}$. The factorization of the left-hand side contains the prime three and that of the right-hand side only the primes two and five. Equality is impossible. $\square$

The lower mechanical word, with intercept θ , is

$$z_n(\theta) = \lfloor (n+1)\delta + \theta \rfloor - \lfloor n\delta + \theta \rfloor \in \{0, 1\}.$$

It is obtained by observing the rotation $\theta \mapsto \theta + \delta \pmod{1}$ through the interval $[1 - \delta, 1)$. The event sequence has density δ , and the sum over any window of length L equals $\lfloor L\delta \rfloor$ or $\lceil L\delta \rceil$. Since $21 < 1/\delta < 22$, the gaps between events are twenty-one or twenty-two. The word indicating which intervals are short is another mechanical word of slope $\sigma = 22 - 1/\delta$; its events are separated by six or seven primary intervals. The second scale is induced by the first and is not a second free parameter.

6.9.1 Capacity index, vacancies, and induction of the tritic regime

The vacancy sequence is linked to refinement through an increment identity. To exhibit it, fix the phase origin $\theta = 0$. Let

$$\rho = \log_{10} 9 = 1 - \delta, \quad \mathcal{C}_t = \lfloor (t+1)\rho \rfloor \quad (t \geq 0).$$

The index $\mathcal{C}_t$ expresses the decimal capacity of refinement in the normalization comparing a ternary block of 6 positions, with 729 possibilities, and a decimal block of 3 positions, with 1000 possibilities. Indeed, $\log_{1000} 729 = \log_{10} 9 = \rho$. The letter K is reserved for the dodecaphase record constructed later.

Proposition 6.9 (Complementarity of capacity and vacancy). *For $t \geq 1$,*

$$\mathcal{C}_t - \mathcal{C}_{t-1} = 1 - z_t(0). \quad (6.11)$$

Consequently, if $0 \leq a < b$,

$$\mathcal{C}_b - \mathcal{C}_a + \sum_{t=a+1}^b z_t(0) = b - a.$$

Proof. Irrationality of δ gives

$$\mathcal{C}_t = t + 1 - \lfloor (t+1)\delta \rfloor = t - \lfloor (t+1)\delta \rfloor.$$

Subtracting consecutive terms produces (6.11). Summation over the interval telescopes. $\square$

Each transition increases the block counter by one. If $z_t = 0$, capacity also increases; if $z_t = 1$, capacity remains unchanged while a transition is incorporated into the history. Here a vacancy is a capacity-retention event, with a determined position and antecedent. The preceding balance preserves length exactly between acquired capacity and recorded vacancies. The observed phase $g_t^{\text{cap}} = \lfloor \mathcal{C}_t \rfloor_9$ satisfies

$$g_t^{\text{cap}} - g_{t-1}^{\text{cap}} \equiv 1 - z_t \pmod{9}.$$

This capacity phase and chronological phase $[t]_9$ are two different observations. Local capacity retention and a complete holonomy turn have distinct mechanisms, although both allow the phase observation to be preserved while the state changes.

The relation with TRIT can be followed in the vacancy positions themselves. For $j \geq 1$, define

$$p_j = \left\lfloor \frac{j}{\delta} \right\rfloor, \quad v_j = \mathcal{C}_{p_j}, \quad h_j = p_{j+1} - p_j.$$

The inequality $p_j \delta < j < (p_j + 1)\delta$ identifies p_j as the position of event j . Since $0 < \delta < 1$, it also gives $\lfloor (p_j + 1)\delta \rfloor = j$, and hence

$$v_j = p_j - j, \quad v_{j+1} - v_j = h_j - 1.$$

If $s_j = 22 - h_j$, then $s_j \in \{0, 1\}$ and

$$[v_{j+1}]_3 = [v_j - s_j]_3. \quad (6.12)$$

An interval of length 22 preserves the class; one of length 21 shifts it by one in the negative direction. The canonical selector is applied to the positive phase $\bar{r}_j = 1 + [v_j]_9$:

$$\tau_j = M_{\text{ph}}(\bar{r}_j).$$

A tritic regime is thus determined by the capacity index retained at each event, in addition to the vacancy's binary encoding.

The positions of the short intervals form the induced mechanical coding of slope $22 - 1/\delta$. Their separations 6 and 7 determine the lengths of plateaus of the class $[v_j]_3$, once the origin and endpoint convention are fixed. The first classes are 2 for 6 events, 1 for 7, and 0 for 7; the first segment of 6 is the initial restriction of a plateau. Their signatures traverse 0, -1 , $+1$. Equations (6.11) and (6.12) make explicit the link between refinement, vacancies, and regime selection. The quadratic reading of TRIT subsequently realizes these types through their respective generators; the temporal event and its geometric representation thus retain an identifiable dependence.

Inverting the monotone index supplies another useful observation:

$$\mathcal{T}(k) = \min\{t : \mathcal{C}_t = k\} = \left\lfloor \frac{k}{\rho} \right\rfloor - 1 \quad (k \geq 1).$$

If $\sigma = \rho^{-1} - 1$, the number of blocks needed to gain m units of capacity is

$$\mathcal{T}(k + m) - \mathcal{T}(k) = m + \lfloor (k + m)\sigma \rfloor - \lfloor k\sigma \rfloor \in \{m + \lfloor m\sigma \rfloor, m + \lceil m\sigma \rceil\}.$$

In particular, 9 units of capacity require 9 or 10 blocks. The period of a finite phase remains exact, whereas return time measured by the other index admits two values. This distinction belongs to refinement dynamics and is not an a posteriori correction of its numerical values.

Theorem 6.10 (Complexity of phase readings). *For $m \geq 1$, the mechanical word has factor complexity $p(m) = m + 1$. When the phase $n \bmod 9$ is retained, the complexity is $p_9(m) = 9(m + 1)$; when only its ternary class is retained, it is $p_3(m) = 3(m + 1)$. All three topological entropies are zero.*

Proof. The predecessors of the two endpoints of the coding interval form, for factors of length m , $m + 1$ distinct points on the circle: coincidence between shifted endpoints eliminates duplications and irrationality prevents additional coincidences. These points determine $m + 1$ distinct intervals of constant itinerary. Every irrational orbit visits each interval, whence $p(m) = m + 1$.

For a fixed phase modulo nine, positions of that phase follow a rotation of step 9δ , also irrational. By density, all factors occur in every phase, and the first coordinate distinguishes their nine copies. The same argument with step 3δ gives three copies after ternary reduction. Finally, $m^{-1} \log(c(m + 1)) \rightarrow 0$ for $c = 1, 3, 9$. $\square$

The phase product $C_9 \times \mathbb{T}$, translated by $(1, \delta)$, has characters of frequency

$$\frac{a}{9} + k\delta \pmod{1}, \quad a \in \mathbb{Z}/9\mathbb{Z}, k \in \mathbb{Z}.$$

The spectral module is not $\log 10$: it is the additive module generated by $1/9$ and δ , modulo the integers. The factor $\log 10$ belongs to the normalization in (6.10).

An operator presentation makes memory explicit. In $\ell^2(\mathbb{Z})$, let $T|n\rangle = |n+1\rangle$, $V_9|n\rangle = \zeta_9^n|n\rangle$, $W_\delta|n\rangle = e^{2\pi i n \delta}|n\rangle$, and $D_M|n\rangle = \lfloor n/9 \rfloor |n\rangle$. On finitely supported vectors,

$$V_9 T = \zeta_9 T V_9, \quad W_\delta T = e^{2\pi i \delta} T W_\delta, \quad [D_M, T^9] = T^9. \quad (6.13)$$

The first two relations are Weyl covariances; the third expresses that a nonadic turn increases the memory degree. The base rotation is abelian, whereas the algebra of observables and translations is not. The inversion $|n\rangle \mapsto |-n\rangle$ conjugates advance with retreat. This realization provides an aperiodic helical suspension with finite phase return, not a periodic word of length nine.

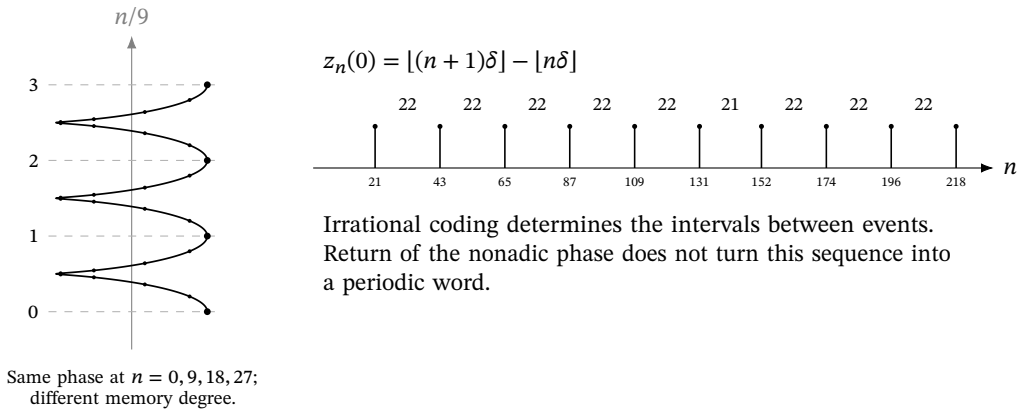


Figure 5: Complementary representations of transport. Left: projection of the helix $(\cos(2\pi n/9), \sin(2\pi n/9), n/9)$: advancing nine steps recovers the phase and increases the height. Right: the first ten events of the lower word for $\delta = \log_{10}(10/9)$, with exact separations 21 or 22. The helical representation introduces no additional physical metric.

For this count we fix the intercept $\theta = 0$ and the accumulated endpoints $(N_0, \dots, N_4) = (0, 729, 972, 999, 1000)$, corresponding to the cubic order $(729, 243, 27, 1)$. The lower counts are $\lfloor N_j \delta \rfloor - \lfloor N_{j-1} \delta \rfloor$, and give $(33, 11, 1, 0)$. The upper counts are $\lceil N_j \delta \rceil - \lceil N_{j-1} \delta \rceil$, with the same initial boundary zero, and give $(34, 11, 1, 0)$. These values depend on the intercept and the ordering of the strata; they are not asserted for an arbitrary initial phase. Their totals are forty-five and forty-six. The increment corresponds to a single boundary unit, located in the first stratum of this ordered partition. The numbers thirty-four and eleven thus appear in a single count preceding any chart of units; identifying them with physical exponents additionally requires the corresponding dimensional derivation.

The term *aperiodic crystal* is used here for this symbolic order, with its complexity, spectrum, and return specified. Zero topological entropy does not by itself amount to zero physical dissipation: a thermodynamic realization requires a defined energy dynamics and erasure operation. Likewise, this Sturmian reading does not replace all TPK operators or by itself prove periodicity or aperiodicity of the complete output of α . Its function is to make explicit an order structure not described by the simple return of the calendar.

6.10 Nonadic connection, holonomy, and monodromy representation

The regional structure requires distinguishing the transport producing a history, observation of its phase, and representation of its returns. APP supplies the two arithmetic evaluations with remainder and quotient; TRIT determines regime and orientation; the Topological Phase Kernel composes selection, transport, and update. The initial conditions of cardinality 104 976, their catalog of 468 emissions, and the 243 visible words in the ambient space of 729 elements belong to different stages of that construction. The chain

$$w_6 \longrightarrow w_{12} \longrightarrow w_{18} \longrightarrow w_{24} \longrightarrow w_{30} \longrightarrow R_{36} \longrightarrow G_9$$

leads to a joint connection over the three regions. Its nine phases are positions of a single transport, and the regional coordinates π, φ, e , together with the dodecaphasic determination of α , arise from the compatible readings constructed in the dodecaphasic record of Section 9, in the coinductive prolongation of Section 6, and in the two routes to α assembled in Section 10.3.

At prolongation index k , a state chart preserves

$$\hat{x}_k = (B_k, C_k, p_k, c_k, \Sigma_k, W_k, g_k, m_k).$$

Here B_k collects the three regional blocks, C_k is the cylinder, p_k the prefix, c_k the carry record, Σ_k the boundary signature, and W_k the incidence information. The phase is $g_k = 1 + [(k - 1)]_9$; m_k counts complete turns. The reduction $\beta_k = [m_k]_{12}$ preserves the dodecaphasic position, whereas the integer m_k distinguishes its successive turns.

Definition 6.11 (Transport through one nonadic turn). Let $\mathcal{R}_k \subseteq \hat{X}_k \times \hat{X}_{k+1}$ be the admissible extension relations of the TPK. The realization of holonomy at level k is the relational composition

$$\Gamma_{9,k} = \mathcal{R}_{k+8} \circ \cdots \circ \mathcal{R}_k.$$

For $s \geq 1$, its prolongation through s turns is

$$\Gamma_{9s,k} = \Gamma_{9,k+9(s-1)} \circ \cdots \circ \Gamma_{9,k}.$$

The relational formulation preserves admissible continuations. On an individualized history, each arrow leads to the state actually prolonged; the isolated visible block may still admit more than one continuation. This difference is precisely the function of memory and boundary conditions in determining the trajectory.

Proposition 6.12 (Iteration of return and accumulated memory). *In every admissible composition of s turns,*

$$g_{k+9s} = g_k, \quad m_{k+9s} = m_k + s, \quad \beta_{k+9s} = \beta_k + s \pmod{12}.$$

Proof. The one-turn rule preserves the phase and increments the counter by one. Applying it to the already transported state repeats exactly these two operations. Induction gives the first two equalities; reducing the second modulo 12 gives the third. $\square$

The counter thus provides an integer witness of the difference between turns. After 12 turns the pair of finite phases is restored, but the counter increases by 12. The nonsplit extension (5.5) expresses the composition of the calendar through its cocycle; the trajectory also preserves the integer lift of that datum. Preservation here means transport of prior contributions, not immobility of all coordinates.

6.10.1 Bilateral representation of the return index

The geometric realization of the cyclic graph admits an especially simple operator representation. Here we number the phases from 0 to 8, shifting the origin relative to g_k , and write $k = 9m + r$, with $0 \leq r < 9$, on the integer line covering the phase cycle. On $\mathcal{H}_{\text{ind}} = \ell^2(\mathbb{Z})$, the basis $|k\rangle = |r, m\rangle$ allows the unit shift to be expressed as

$$T|r, m\rangle = \begin{cases} |r+1, m\rangle, & r < 8, \\ |0, m+1\rangle, & r = 8. \end{cases}$$

The inverse decreases r by one when $r > 0$, and sends $|0, m\rangle$ to $|8, m-1\rangle$. Passing through the phase origin therefore incorporates the exact unit of quotient.

Proposition 6.13 (Representation of cycle monodromy). *Let $S = T^9$. Then*

$$S|r, m\rangle = |r, m+1\rangle, \quad \mathbb{Z} \longrightarrow \mathcal{U}(\mathcal{H}_{\text{ind}}), \quad s \longmapsto S^s$$

is a faithful unitary representation of the group of turns of the cycle.

Proof. Nine shifts preserve the remainder of division by 9 and increment the quotient. The powers satisfy $S^{s+t} = S^s S^t$, and $S^s|r, m\rangle = |r, m+s\rangle$ distinguishes every integer s . A permutation of an orthonormal basis is unitary. $\square$

This representation corresponds to the covering of the cycle, whose group of turns is $\mathbb{Z}$, and represents the phase with its counter. The TPK state also contains the trajectory coordinates already described. The bilateral extension admits negative indices as coordinates of the covering; a trajectory beginning at an initial condition uses its admissible half-axis. An operator inverse of the index is thus available without turning every extension relation into a bijection of the full state.

With $D_M|k\rangle = [k/9]|k\rangle$, the identity $[D_M, S] = S$ expresses the memory increment on the dense domain of finitely supported vectors. If $\mathcal{I}|k\rangle = |-k\rangle$ and P_0 projects onto indices divisible by 9, one also obtains

$$\mathcal{I}T\mathcal{I} = T^{-1}, \quad \mathcal{I}D_M\mathcal{I} = -D_M - (I - P_0). \quad (6.14)$$

Indeed, for $k = 9m + r$, $[-k/9] = -m$ if $r = 0$, and equals $-m - 1$ if $r > 0$. Inversion thus preserves the boundary term of Euclidean division.

6.11 Block clock and capacity clock

The index of a refinement map and the number of available decimal positions obey related but different laws. To express them, reserve n for the ternary-block index and define

$$\rho = \log_{1000} 729 = \log_{10} 9, \quad \delta = 1 - \rho, \quad \mathcal{C}_n = \lfloor (n+1)\rho \rfloor, \quad n \geq 0.$$

Here $\mathcal{C}_n$ denotes capacity, not the dodecaphasic record K . At the canonical phase origin, the preceding mechanical word provides

$$Z_n = \sum_{j=0}^n z_j(0) = \lfloor (n+1)\delta \rfloor.$$

Proposition 6.14 (Exact balance between capacity and vacancies). *For $n \geq 0$, and in the second identity for $n \geq 1$,*

$$\mathcal{C}_n = n - Z_n, \quad \mathcal{C}_n - \mathcal{C}_{n-1} = 1 - z_n(0).$$

Proof. Irrationality of δ implies $\lfloor (n+1)\delta \rfloor = \lfloor (n+1)\delta \rfloor + 1$. Then

$$\lfloor (n+1)(1-\delta) \rfloor = n+1 - \lfloor (n+1)\delta \rfloor = n - Z_n.$$

Subtracting two consecutive identities concludes the proof. $\square$

An event $z_n = 1$ adds a block to the history while preserving capacity $\mathcal{C}_n$; an event $z_n = 0$ increments both counters. Over any interval, capacity increments and vacancies sum exactly to its length. Reducing modulo q , for $q = 9$ or $q = 108$, gives

$$[\mathcal{C}_n]_q = [n]_q - [Z_n]_q.$$

The uniform block phase and capacity phase are related through the integrated memory of vacancies. In particular, a capacity hold and a complete turn of the connection have distinct mechanisms, although both may preserve a phase observation.

Monotone inversion of the clock gives a first-arrival time:

$$T_{\text{cap}}(a) = \min\{n : \mathcal{C}_n = a\} = \left\lceil \frac{a}{\rho} \right\rceil - 1, \quad a \geq 1.$$

With $\sigma = \rho^{-1} - 1$, the time needed to gain b units of capacity is

$$T_{\text{cap}}(a + b) - T_{\text{cap}}(a) = b + [(a + b)\sigma] - [a\sigma].$$

The difference of ceilings takes one of the two integers adjacent to $b\sigma$. This yields 9 or 10 blocks to gain 9 capacity units, and 113 or 114 to gain 108. These are inverse-clock times measured from a first arrival; the turn counter is always interpreted in the index to which it belongs.

6.12 Symbolic suspension and aperiodic time crystal

Rotation coding combines phase periodicity, local recurrence, and global aperiodicity. Let $\mathcal{X}_\delta \subseteq \{0, 1\}^{\mathbb{Z}}$ be the closure, in the product topology, of shifts of the bilateral word $z_n(\theta)$. Denote by σ_{sh} the shift of its sequences. Retaining a nonadic phase, the step is

$$F(r, z) = ([r + 1]_9, \sigma_{\text{sh}} z), \quad (r, z) \in C_9 \times \mathcal{X}_\delta.$$

Definition 6.15 (Symbolic aperiodic time crystal). The unit-roof suspension of the preceding system is the space

$$\mathcal{S}_\delta = ((C_9 \times \mathcal{X}_\delta) \times \mathbb{R}) / \sim, \quad (y, t + 1) \sim (Fy, t),$$

with flow $\Phi_s[y, t] = [y, t + s]$. This model of temporal order, together with its vacancy coding and turn-counter representation, is called a *symbolic aperiodic time crystal*.

The adjective temporal designates the action of the flow and its return section; aperiodicity characterizes its itineraries. The frequency of ones in each sequence is δ : the uniform discrepancy bound of less than one, obtained by telescoping over any window, persists under closure. A periodic sequence would have rational frequency. Therefore, $\mathcal{X}_\delta$ contains no periodic orbits. Nor does the suspension have closed orbits: a return of the flow would require an integer time and periodicity of the itinerary.

Finite words do recur. Length- m itineraries come from a partition of the circle by $m + 1$ points; every interval is visited recurrently by irrational rotation. Even when times are restricted to a fixed phase, the step 9δ remains irrational. This is why every word appears in all 9 phases and explains the already established complexities

$$p(m) = m + 1, \quad p_9(m) = 9(m + 1), \quad p_3(m) = 3(m + 1).$$

Recurrence of local configurations coexists with the absence of a global period. Its topological entropy is zero because these complexities grow linearly.

The realization through observables V_δ, W_δ and the shift T expresses the same order through the covariances (6.13). The frequencies of the uniform clock belong to $\frac{1}{9}\mathbb{Z} + \delta\mathbb{Z}$, modulo 1. The capacity clock additionally preserves the relation $[\mathcal{C}_n]_9 = [n]_9 - [Z_n]_9$; it is not replaced by the uniform phase when returns are interpreted. The term *time crystal* is thus defined through a mathematical dynamical system. Its physical realization requires specifying an energetic dynamics, its temporal observables, and its stability, data distinct from the present symbolic construction.

6.13 Specular symmetry of the regions and dual realization

Specular exchange acts on the regional block $B = (u^\pi, u^e, u^\varphi) \in (\mathbb{F}_3^6)^3$ by

$$\mathfrak{c}B = (u^e, u^\pi, u^\varphi).$$

Its fixed axis and aggregate are, respectively, u^φ and $s = u^\pi + u^e$. This fixedness concerns the involution in a fiber; both coordinates may evolve during nonadic transport. With $q = u^\pi + u^e + u^\varphi$, $a = u^\pi + u^e - u^\varphi$, and $d = u^\pi - u^e$, one recovers

$$u^\varphi = 2(q - a), \quad s = q - u^\varphi, \quad u^\pi = 2(s + d), \quad u^e = 2(s - d).$$

The equalities hold componentwise in $\mathbb{F}_3$, where $2^{-1} = 2$. The mirror preserves q, a, s and reverses d ; this antisymmetric coordinate individualizes the allocation that the aggregate preserves without distinguishing. Time-index inversion $\mathcal{J}$ and the channel involution $\mathfrak{c}$ therefore act on different data.

Phase return has an explicit witness:

$$B_1 = (012222 \mid 121221 \mid 112202),$$

$$B_{10} = (101222 \mid 112221 \mid 112002).$$

Both positions have phase 1; the block, axis, and memory have changed. Likewise, census synchronization in phases 8 and 9 annihilates their relative differences, while the diagonal component passes from (59, 59, 59) to (43, 43, 43). Equality of a relative observable does not exhaust the state determining it.

The dual realization preserves this architecture. In section 11.2, the same prefix determines nested cylinders on one side and a sequence of encoded words with their frames on the other. The first map passes to the limit by contraction of diameters; the second satisfies $r_{n,m}Q_n = Q_m p_{n,m}$ and defines Q_∞ . Figure 6 anticipates these two maps. The subsequent incidence construction produces codes, lattices, and the Moonshine realization in their own domains; the automorphism group acts on that realization, not on the digits through an implicit identification.

6.14 Temporal transport and tritic exponential composition

The Euler relations in section 6.6 describe local realizations of the three regimes. For a fixed signature τ , the generator satisfies $J_\tau^2 = -\tau I$; algebraic conjugation sends J_τ to $-J_\tau$. Consequently,

$$\exp(sJ_\tau) \exp(tJ_\tau) = \exp((s+t)J_\tau), \quad \overline{\exp(tJ_\tau)} = \exp(-tJ_\tau),$$

and the product of an evolution with its conjugate is the unit. In the elliptic type it is realized through $\cos^2 t + \sin^2 t = 1$; in the hyperbolic type, through $\cosh^2 t - \sinh^2 t = 1$; in the parabolic type, $(I + tN)(I - tN) = I$, since $N^2 = 0$. The same conservation law admits three quadratic expressions.

The elliptic realization closes a turn; the nilpotent one preserves a linear displacement; the hyperbolic one separates expansion and contraction with unit determinant. At the regional parameters already generated,

$$\exp(\pi J_e) = -I, \quad \exp(J_p) = I + J_p, \quad \exp(2 \log \varphi J_h) = F_{\text{av}}^2.$$

The evaluation $\exp(I) = eI$ recognizes the unit of the common exponential law. The number e is not assigned exclusively to the parabolic regime.

Transporting a local realization through an isomorphism U preserves these relations: if $J' = UJU^{-1}$, the exponential series gives $U \exp(tJ) = \exp(tJ')U$. The identity transports the regime and its norm. A transition between different quadratic types, by contrast, belongs to TPK sheet and regime selection; it is not a conjugation changing the value of J^2 . The temporal product therefore preserves the order of transitions and the information at their endpoints.

This articulation distinguishes the closure of a local representation from the holonomy of a history. One may have $\exp(2\pi J_e) = I$ at the same time as $S|r, m\rangle = |r, m+1\rangle$. The first equality restores the regime's orientation; the second records another turn. The discrete structure preserves both: the local return and the accumulated provenance of the state realizing it.

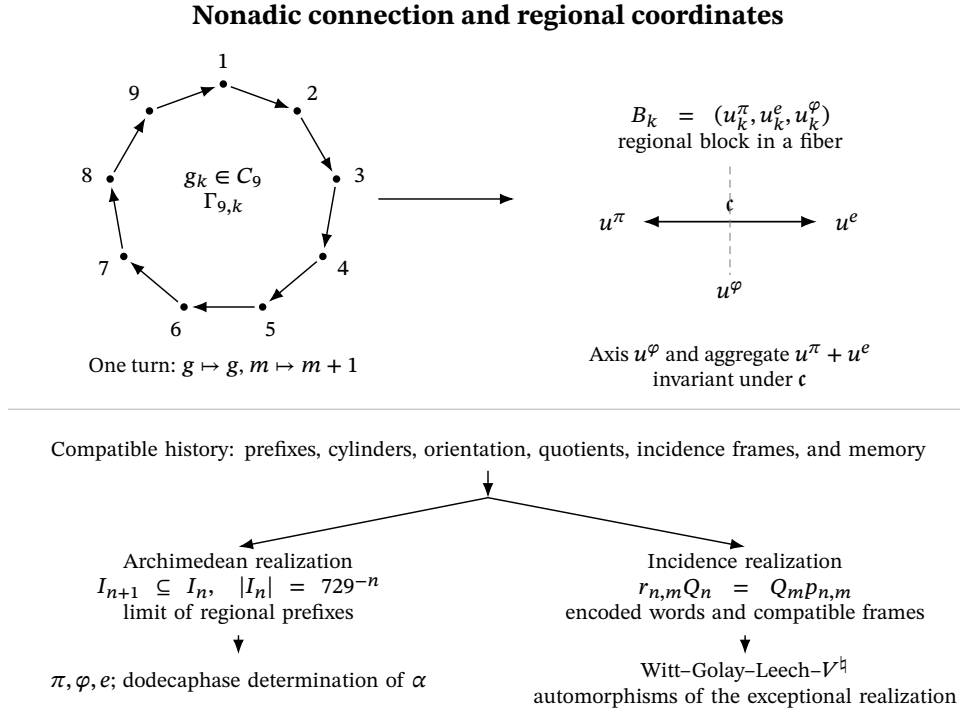


Figure 6: Nine phases of a connection and two realizations of a compatible history. The cycle represents phase; each turn increments the counter. The regional diagram represents the involution ς , which interchanges closure and propagation and fixes the self-scaling axis in a fiber. This invariance does not require the axis to remain constant during transport. The two lower maps use the same prefix: one determines its numerical limit, while the other preserves words and incidence frames. The maps of the second realization are specified in Section 11.2; the exceptional construction is developed afterwards.

6.15 Multiplicative irreducibility and return periods

The distinction between structure, evaluation, and phase also appears in the multiplicative sector. APP's local product contains both residue and quotient; iterating it constructs normal forms and factorizations before their decimal periods are examined. This development places the expression *prime mode* in a precise domain and relates irreducibility to the local APP operations defined in Section 2.1 and to the nonadic holonomy constructed in Section 6.10.

Let

$$\mathcal{W}_9 = \{\emptyset\} \sqcup \bigsqcup_{\ell \geq 1} \{1, \dots, 9\}^\ell.$$

Sequences are ordered from the least significant digit. The evaluation $\text{ev}_9(u) = \sum_j u_j 9^j$, with $\text{ev}_9(\emptyset) = 0$, is bijective onto the nonnegative integers: for a positive integer, division of $n - 1$ by 9 uniquely determines the first digit and continuation. This is bijective nonadic numeration; it retains the boundary representative 9.

Addition $\boxplus_\#$ is obtained by applying APP's additive sheet to the digits at each position and propagating their quotients. If there is only one digit, it is the provisional residue; a nonzero incoming quotient is normalized through a second additive operation. The sum of quotients is transported to the next position. To multiply a sequence by a digit d , the multiplicative cell produces $u_j d = 9q_j + r_j$, and the incoming quotient c_j is incorporated by

$$(w_j, c_{j+1}) = \begin{cases} (r_j, q_j), & c_j = 0, \\ (R^+(r_j, c_j), q_j + q^+(r_j, c_j)), & c_j > 0. \end{cases}$$

Start with $c_0 = 0$ and incorporate the final quotient when nonzero. During addition, $c_j \leq 2$; during multiplication by a digit, $c_j \leq 9$. Local operations thus receive arguments within their domain. Denote this second procedure by $d \odot_\# u$, with $d \odot_\# \emptyset = \emptyset$.

For $v = (v_0, \dots, v_{m-1})$, the complete product is constructed through the recurrence

$$a_m = \emptyset, \quad a_j = (9 \odot_{\#} a_{j+1}) \boxplus_{\#} (v_j \odot_{\#} u), \quad u \boxtimes_{\#} v = a_0.$$

The rule composes both local sheets. Its evaluation satisfies

$$\text{ev}_9(u \boxtimes_{\#} v) = \text{ev}_9(u) \text{ev}_9(v). \quad (6.15)$$

Indeed, each transition preserves the partial sum plus the pending quotient weighted by the power of 9. The recurrence then gives $\text{ev}_9(a_j) = 9 \text{ev}_9(a_{j+1}) + v_j \text{ev}_9(u)$, whose iteration proves (6.15). Uniqueness of normal form transports associativity and commutativity to the constructed product; its identity is (1).

Definition 6.16 (Reduced multiplicative coproduct). On the finitely supported vector space generated by nonempty normal forms, define

$$\tilde{\Delta}_{\#} e_w = \sum_{\substack{u \boxtimes_{\#} v = w \\ u, v \neq (1)}} e_u \otimes e_v.$$

A non-unit normal form is irreducible when its image under $\tilde{\Delta}_{\#}$ is zero.

Factorization fibers are finite by (6.15). In the completion $\mathcal{H}_{\#} = \ell^2(\mathcal{W}_9 \setminus \{\emptyset\})$, images of distinct forms have disjoint supports. If $d_{\#}(w)$ counts ordered non-unit factorizations, the domain of the closure is

$$\text{Dom}(\overline{\tilde{\Delta}_{\#}}) = \left\{ x \in \mathcal{H}_{\#} : \sum_w d_{\#}(w) |x_w|^2 < \infty \right\}.$$

Indeed, the squared norm of the image is this weighted sum. Hence the operator

$$A_{\#} = (\overline{\tilde{\Delta}_{\#}})^* \overline{\tilde{\Delta}_{\#}}$$

is diagonal, positive, and self-adjoint: its eigenvalue at e_w is the number of ordered non-unit factorizations of w . The projection

$$P_{\text{irr}} = (I - |e_{(1)}\rangle\langle e_{(1)}|) \mathbf{1}_{\{0\}}(A_{\#})$$

selects exactly the irreducible forms. Through multiplicative evaluation, these correspond to ordinary primes. Selection depends on factorization structure and keeps the unit state separate.

For a value $n \geq 2$ coprime to 10, decimal division induces the residue permutation $r \mapsto 10r \pmod{n}$. Return of residue 1 has period

$$\tau_n = \text{ord}_n(10).$$

If $n = \prod_p p^{a_p}$, the Chinese remainder theorem gives

$$\tau_n = \text{lcm}_{p^{a_p} \parallel n} \text{ord}_{p^{a_p}}(10).$$

Composite return synchronizes the returns of prime powers. For a prime $p \notin \{2, 3, 5\}$,

$$\tau_p \mid p - 1, \quad M_p = \frac{10^{\tau_p} - 1}{p} \in \mathbb{N}, \quad M_p \equiv 0 \pmod{9}.$$

The last congruence expresses nonadic closure of decimal repetition: $10^{\tau_p} - 1$ is a multiple of 9, and p is invertible modulo 9. The same congruence holds for every n coprime to 90. In particular, 7, 13, 77, 91, 143 have decimal period 6, although only the first two are irreducible. The coproduct and the period therefore record different properties of the same constructed value.

This differentiation specifies the relation with aperiodic refinement. The vacancy sequence determines increments of the capacity index; factorization determines multiplicative modes; decimal division determines their return times. The nonadic representative preserves phase closure, while quotient and factorization retain information that this closure does not distinguish. The connection to the treatise's spectral constructions resides in these operators and their intertwining. Proof of their subsequent analytic results retains its own development; it is not replaced by periodicity of a residue observation.

6.15.1 Vacancy discrepancy and Dirichlet readings

The connection also admits an exact analytic expression. Retain the same coding $z_n = z_n(0)$, for $n \geq 1$, and consider two subsequent observables: the series over all indices and the product over indices selected by irreducibility. For $\operatorname{Re} s > 1$,

$$Z_{\text{vac}}^+(s) = \sum_{n \geq 1} \frac{z_n}{n^s}, \quad Z_{\text{vac}}^\times(s) = \prod_p (1 - p^{-s})^{-z_p}.$$

Both converge absolutely in this half-plane. Their common factor is density δ , but their discrepancies differ.

Proposition 6.17 (Separation of vacancy density). *We have*

$$Z_{\text{vac}}^+(s) = \delta \zeta(s) + H_{\text{vac}}(s), \quad H_{\text{vac}} \in \mathcal{O}(\operatorname{Re} s > 0).$$

In $\operatorname{Re} s > 1$, with logarithms defined by absolutely convergent Euler series,

$$Z_{\text{vac}}^\times(s) = \zeta(s)^\delta G_{\text{vac}}(s), \quad \log G_{\text{vac}}(s) = \sum_p (z_p - \delta) \log(1 - p^{-s})^{-1}.$$

Proof. The centered partial sums satisfy

$$A_N := \sum_{n=1}^N (z_n - \delta) = \lfloor (N+1)\delta \rfloor - N\delta, \quad |A_N| < 1.$$

Abel summation represents the centered term by $s \int_1^\infty A_{\lfloor x \rfloor} x^{-s-1} dx$. The integral converges locally uniformly in $\operatorname{Re} s > 0$, proving holomorphy. The second identity follows by separating $z_p = \delta + (z_p - \delta)$ in the logarithm of the product, within its domain of absolute convergence. $\square$

The decomposition reveals what the reading on primes adds: it examines the same vacancy sequence on a subset determined by factorization. Controlling discrepancy over all integers gives the first holomorphic continuation; the second observable preserves its own discrepancy on primes. These are two applications of the same discrete structure, with explicit analytic domains. This articulation explains their relevance to the arithmetic kernel without transferring the proof of the treatise's terminal results to this article.

7 Integral compatibility of regional generation, cylindrical refinement, and incidence

Regional prolongation produces ordered sequences of blocks together with their reading depths. The Archimedean projection and ternary incidence are calculated on these same sequences. The connection between the two realizations can be formulated entirely through integer operations: selection by rational intervals, Euclidean division, lifting of six trits, carry reading, and the Witt transformation. This formulation makes it possible to trace each coefficient from its generating block to its subsequent use.

The construction uses the three regional readers already obtained from the APP-TRIT-TPK state. We denote them by the indices $c \in \{P, E, F\}$, corresponding, respectively, to the closure, propagation, and autoscaling regions. Their real realization will be written as v_c . The subsequent identifications $v_P = \pi_{\text{HMT}}$, $v_E = e_{\text{HMT}}$, and $v_F = \varphi_{\text{HMT}}$ preserve this causal order. The operation that emits a block inspects the rational intervals of the regional reader; the real value enters the proof of correctness of that operation.

7.1 Positional emission through rational separation of cells

For each region, two rational sequences $\ell_c(j), u_c(j)$, constructed by its reader, are available and satisfy

$$\ell_c(j) \leq v_c \leq u_c(j).$$

The cell-separation property proved for these readers states that, for each integer $s > 0$, there exists J such that, for every $j \geq J$,

$$\lfloor s\ell_c(j) \rfloor = \lfloor su_c(j) \rfloor - 1 = \lfloor sv_c \rfloor. \quad (7.1)$$

Avoidance of rational boundaries by the regional values is the hypothesis of the preceding theorems that guarantees this separation. Here we use their operational conclusion together with the readers and intervals that produce it.

Definition 7.1 (Separation index and emitted prefix). For $s > 0$, let

$$j_c(s) = \min\{j \in \mathbb{N} : \lfloor s\ell_c(j) \rfloor = \lfloor su_c(j) \rfloor - 1\}.$$

For an integer base $b \geq 2$ and a depth $n \geq 0$, define

$$P_c(b, n) = \lfloor b^n \ell_c(j_c(b^n)) \rfloor. \quad (7.2)$$

The prefix contains the integer part and the first n positions in base b ; consequently, $P_c(b, 0)$ is the regional integer part.

Theorem 7.2 (Termination and correctness of emission). *The index $j_c(s)$ exists for every $s > 0$. The emitted prefixes satisfy*

$$P_c(b, n) = \lfloor b^n v_c \rfloor, \quad P_c(b, n) \leq b^n v_c < P_c(b, n) + 1. \quad (7.3)$$

For each scale s , a single sufficiency index $J(s)$ can also be chosen that is valid simultaneously for all three regions.

Proof. The eventual equality in (7.1) makes the set defining $j_c(s)$ nonempty. Its minimum exists by the well-ordering of $\mathbb{N}$. At any index where the equality holds, the rational lower bound gives $\lfloor s\ell_c(j) \rfloor \leq \lfloor sv_c \rfloor$. The upper bound, together with avoidance of a cell boundary, gives $\lfloor sv_c \rfloor \leq \lfloor su_c(j) \rfloor - 1$. When the two integer endpoints coincide, the intervening integer coincides with them. This proves (7.3). For the joint statement, it suffices to take the maximum of the three sufficiency indices obtained at the same scale. The joint index controls all three emissions simultaneously, while each region retains its own intervals. $\square$

Proposition 7.3 (Exact compatibility of prefixes). *For $n, m \geq 0$,*

$$\left\lfloor \frac{P_c(b, n+m)}{b^m} \right\rfloor = P_c(b, n). \quad (7.4)$$

In particular, the next block $d_c(b, n) = P_c(b, n+1) \bmod b$ satisfies

$$P_c(b, n+1) = bP_c(b, n) + d_c(b, n), \quad 0 \leq d_c(b, n) < b. \quad (7.5)$$

Proof. Write $b^n v_c = N + \theta$, with $N = P_c(b, n)$ and $0 \leq \theta < 1$. Then $P_c(b, n+m) = b^m N + \lfloor b^m \theta \rfloor$, and the second summand belongs to $\{0, \dots, b^m - 1\}$. Euclidean division by b^m recovers N . The specialization $m = 1$ yields (7.5) and its residual bound. $\square$

7.2 Regional blocks of six trits at arbitrary depth

The integer equality $729 = 3^6$ determines the grouping of six ternary positions into a block. For $t \geq 0$, define

$$u_c(t) = P_c(729, t+1) \bmod 729, \quad B_{c,j}(t) = \left\lfloor \frac{u_c(t)}{3^{5-j}} \right\rfloor \bmod 3, \quad 0 \leq j < 6. \quad (7.6)$$

Thus $B_c(t)$ is an ordered word of six trits, and

$$u_c(t) = \sum_{j=0}^5 B_{c,j}(t) 3^{5-j}. \quad (7.7)$$

The word and its integer lift are mutually recoverable representations of the same emitted block.

Theorem 7.4 (Stability of extraction at any depth). *For every n ,*

$$P_c(729, n) = P_c(3, 6n).$$

If $t < n$, then

$$u_c(t) = \left\lfloor \frac{P_c(729, n)}{729^{n-t-1}} \right\rfloor \bmod 729. \quad (7.8)$$

Consequently, every extension of the prefix preserves all previously emitted blocks and all signatures calculated from them.

Proof. The first equality follows because both prefixes are separated at the same scale, $729^n = 3^{6n}$, and from Theorem 7.2. For the second, apply (7.4) at depths n and $t+1$, and then take the residue modulo 729. The ternary extraction in (7.6) is deterministic. Any explicit function of those eighteen coordinates therefore retains the same result when the prefix is extended. $\square$

The construction is quantified over $t \in \mathbb{N}$. A table materializing one hundred blocks is a finite evaluation of this construction; formula (7.8) determines its compatibility with every subsequent extension exactly.

Example 7.5 (First joint block). Evaluation of the selected regional readers gives the words

$$B_P(0) = 010211, \quad B_E(0) = 201101, \quad B_F(0) = 121200.$$

Their lifting by (7.7) produces

$$u_P(0) = 103, \quad u_E(0) = 523, \quad u_F(0) = 450.$$

For example, $201101_3 = 2 \cdot 243 + 27 + 9 + 1 = 523$. All six positions, including leading zeros, are retained in the band representation. The leading zero in 010211 has a defined position even though the decimal notation for 103 does not contain it.

7.3 Transition registers and reconstruction of linear lifts

Over $\mathbb{F}_3$, collect two consecutive emissions from each region in the matrix

$$R(t) = \begin{pmatrix} B_P(t) \\ B_P(t+1) \\ B_E(t) \\ B_E(t+1) \\ B_F(t) \\ B_F(t+1) \end{pmatrix}. \quad (7.9)$$

This definition fixes both the regional order and the temporal order. The first four registers are

$$X_0 = R(0), \quad Y_0 = R(1), \quad X_1 = R(2), \quad Y_1 = R(3).$$

Their values, obtained by extraction from the regional prefixes, are

$$\begin{aligned} X_0 &= \begin{pmatrix} 0 & 1 & 0 & 2 & 1 & 1 \\ 0 & 1 & 2 & 2 & 2 & 2 \\ 2 & 0 & 1 & 1 & 0 & 1 \\ 1 & 2 & 1 & 2 & 2 & 1 \\ 1 & 2 & 1 & 2 & 0 & 0 \\ 1 & 1 & 2 & 2 & 0 & 2 \end{pmatrix}, & Y_0 &= \begin{pmatrix} 0 & 1 & 2 & 2 & 2 & 2 \\ 0 & 1 & 0 & 2 & 1 & 1 \\ 1 & 2 & 1 & 2 & 2 & 1 \\ 1 & 0 & 2 & 0 & 1 & 1 \\ 1 & 1 & 2 & 2 & 0 & 2 \\ 1 & 2 & 1 & 0 & 2 & 0 \end{pmatrix}, \\ X_1 &= \begin{pmatrix} 0 & 1 & 0 & 2 & 1 & 1 \\ 0 & 0 & 2 & 1 & 1 & 1 \\ 1 & 0 & 2 & 0 & 1 & 1 \\ 0 & 1 & 2 & 2 & 2 & 2 \\ 1 & 2 & 1 & 0 & 2 & 0 \\ 0 & 1 & 0 & 2 & 1 & 0 \end{pmatrix}, & Y_1 &= \begin{pmatrix} 0 & 0 & 2 & 1 & 1 & 1 \\ 1 & 1 & 0 & 2 & 2 & 1 \\ 0 & 1 & 2 & 2 & 2 & 2 \\ 1 & 0 & 2 & 0 & 1 & 1 \\ 0 & 1 & 0 & 2 & 1 & 0 \\ 0 & 1 & 0 & 2 & 0 & 0 \end{pmatrix}. \end{aligned}$$

Theorem 7.6 (Unique reconstruction of the two initial links). *The equations $X_0 L_0 = Y_0$ and $X_1 L_1 = Y_1$ each have a unique solution in $M_6(\mathbb{F}_3)$. These solutions are*

$$\begin{aligned} L_0 &= \begin{pmatrix} 2 & 2 & 2 & 1 & 2 & 1 \\ 2 & 1 & 2 & 2 & 1 & 1 \\ 1 & 0 & 0 & 1 & 0 & 2 \\ 0 & 0 & 1 & 1 & 1 & 0 \\ 2 & 2 & 2 & 0 & 2 & 1 \\ 2 & 1 & 2 & 1 & 0 & 0 \end{pmatrix}, & L_1 &= \begin{pmatrix} 2 & 1 & 2 & 0 & 2 & 2 \\ 1 & 2 & 1 & 1 & 2 & 0 \\ 1 & 2 & 1 & 1 & 1 & 2 \\ 0 & 1 & 0 & 0 & 2 & 1 \\ 2 & 0 & 2 & 1 & 0 & 1 \\ 0 & 2 & 2 & 2 & 1 & 1 \end{pmatrix}. \end{aligned}$$

Both matrices are invertible.

Proof. The matrices

$$\begin{aligned} A_0 &= \begin{pmatrix} 1 & 1 & 2 & 0 & 1 & 2 \\ 0 & 0 & 1 & 0 & 0 & 1 \\ 2 & 1 & 2 & 1 & 0 & 1 \\ 0 & 2 & 0 & 1 & 0 & 2 \\ 2 & 1 & 1 & 0 & 2 & 2 \\ 2 & 1 & 1 & 1 & 1 & 2 \end{pmatrix}, & A_1 &= \begin{pmatrix} 1 & 0 & 1 & 2 & 0 & 0 \\ 0 & 0 & 1 & 1 & 2 & 2 \\ 0 & 1 & 2 & 0 & 1 & 1 \\ 1 & 1 & 0 & 2 & 0 & 0 \\ 1 & 1 & 2 & 1 & 1 & 2 \\ 1 & 0 & 0 & 0 & 0 & 2 \end{pmatrix} \end{aligned}$$

satisfy $A_i X_i = X_i A_i = I$, by multiplication in $\mathbb{F}_3$. The solutions are therefore forced by $L_i = A_i Y_i$; multiplication produces the two displayed matrices. Their inverses are

$$L_0^{-1} = \begin{pmatrix} 1 & 2 & 0 & 2 & 0 & 2 \\ 2 & 0 & 2 & 2 & 0 & 0 \\ 2 & 1 & 2 & 0 & 2 & 0 \\ 1 & 0 & 0 & 0 & 2 & 0 \\ 0 & 2 & 1 & 1 & 2 & 0 \\ 2 & 2 & 2 & 2 & 2 & 2 \end{pmatrix}, \quad L_1^{-1} = \begin{pmatrix} 2 & 0 & 2 & 1 & 2 & 1 \\ 2 & 1 & 0 & 0 & 2 & 0 \\ 2 & 0 & 2 & 2 & 0 & 2 \\ 2 & 0 & 0 & 1 & 1 & 0 \\ 1 & 1 & 1 & 1 & 1 & 0 \\ 2 & 0 & 1 & 2 & 2 & 0 \end{pmatrix}.$$

The bilateral verification $L_i L_i^{-1} = L_i^{-1} L_i = I$ proves exact recovery of each transformed word. All products use residues 0, 1, 2 with their interpretation in $\mathbb{F}_3$. $\square$

The causal reading of this result is precise: the regional emissions produce $R(t)$; the registers $R(0), \dots, R(3)$ determine the two lifts above; their inverses recover the transformed words. Continuity for every t belongs to the coinductive readers and Theorem 7.4. The matrix reconstruction proved here has the two links specified in its statement as its domain.

7.4 Simultaneous refinement in ternary and decimal bases

A ternary emission of six trits and a decimal emission of three digits are readings in different bases. To preserve their relationship, record the state

$$s = (n, k, N, D),$$

where N is a prefix at depth n in base 729, and D is a prefix at depth k in base 1000. Their Archimedean cylinders are

$$I_{729}(s) = \left[\frac{N}{729^n}, \frac{N+1}{729^n} \right), \quad I_{1000}(s) = \left[\frac{D}{1000^k}, \frac{D+1}{1000^k} \right). \quad (7.10)$$

The half-open upper endpoint assigns each value to a unique cell at a given base and depth.

Definition 7.7 (Integer compatibility defects). Define

$$A(s) = N1000^k - D729^n, \quad (7.11)$$

$$B(s) = (N+1)1000^k - D729^n = A(s) + 1000^k. \quad (7.12)$$

The state is compatible when

$$A(s) < 729^n, \quad B(s) > 0. \quad (7.13)$$

Lemma 7.8 (Integer criterion for intersection). *The cylinders in (7.10) have a nonempty intersection if and only if (7.13) holds.*

Proof. Two half-open intervals $[a, b)$ and $[c, d)$, with $a < b$ and $c < d$, intersect exactly when $a < d$ and $c < b$. Applying this condition to (7.10) and multiplying by the positive denominators gives $N1000^k < (D+1)729^n$ and $D729^n < (N+1)1000^k$. These are the two inequalities in (7.13). $\square$

Definition 7.9 (Update of depths and prefixes). For $0 \leq u < 729$ and $0 \leq d < 1000$, define

$$T_u(n, k, N, D) = (n+1, k, 729N+u, D),$$

$$T_{u,d}(n, k, N, D) = (n+1, k+1, 729N+u, 1000D+d).$$

The first operation refines only the ternary cylinder. The second refines both readings and preserves both depths in the state.

Proposition 7.10 (Exact boundary recurrences). *The updates above satisfy*

$$A(T_u s) = 729A(s) + u1000^k, \quad (7.14)$$

$$A(T_{u,d} s) = 729000A(s) + u1000^{k+1} - d729^{n+1}. \quad (7.15)$$

Proof. For the first identity,

$$(729N + u)1000^k - D729^{n+1} = 729(N1000^k - D729^n) + u1000^k.$$

For the second,

$$\begin{aligned} (729N + u)1000^{k+1} - (1000D + d)729^{n+1} \\ = 729000(N1000^k - D729^n) + u1000^{k+1} - d729^{n+1}. \end{aligned}$$

Both are integer identities preceding any real approximation. $\square$

Theorem 7.11 (Refinement of the generated regional cylinders). *Let*

$$s_c(n, k) = (n, k, P_c(729, n), P_c(1000, k)).$$

Then $s_c(n, k)$ is compatible for all n, k . Moreover,

$$\begin{aligned} T_{u_c(n)}s_c(n, k) &= s_c(n + 1, k), \\ T_{u_c(n), d_c(1000, k)}s_c(n, k) &= s_c(n + 1, k + 1). \end{aligned}$$

Euclidean division of the refined prefixes recovers the preceding prefixes exactly. For every $a, b \geq 0$,

$$\frac{P_c(729, n + a) - P_c(729, n + a) \bmod 729^a}{729^a} = P_c(729, n),$$

and the analogous identity holds in base 1000.

Proof. Theorem 7.2 places v_c simultaneously in both intervals in (7.10); the intersection criterion gives compatibility. The update identities are (7.5) in each base. Recovery follows from (7.4). Thus the defect recurrence and prefix recovery are consequences of the same update. $\square$

Example 7.12 (First pair of regional cylinders). For $n = k = 1$, evaluation of the three regions gives

c	N	D	A	B
P	2290	3141	211	1211
E	1981	2718	-422	578
F	1179	1618	-522	478

In all three rows, $A < 729$ and $B > 0$. The first prefix column decomposes as $2290 = 3 \cdot 729 + 103$, $1981 = 2 \cdot 729 + 523$, and $1179 = 729 + 450$: exactly the previously emitted six-trit blocks reappear. The different signs of A express the relative positions of the two lower endpoints without altering the regional value's membership in both intervals.

7.5 Selection of the next block by the regional reader

Compatibility of two intervals expresses a relationship between readings. Determining the next block additionally uses the rational interval produced by the region. This operation can be formulated without discarding the depth history.

Definition 7.13 (Admissibility of regional prolongation). For fixed c, n , let $s = 729^{n+1}$ and $j = j_c(s)$. A residue $u \in \{0, \dots, 728\}$ is admissible when

$$[s\ell_c(j)] \leq 729P_c(729, n) + u \leq [su_c(j)] - 1. \quad (7.16)$$

Theorem 7.14 (Uniqueness of the prolongation compatible with the reader). *Condition (7.16) is equivalent to $u = u_c(n)$. There is therefore exactly one admissible prolongation of each region at each depth.*

Proof. At the separation index, both integer endpoints in (7.16) coincide with $P_c(729, n + 1)$. The condition is therefore equivalent to $729P_c(729, n) + u = P_c(729, n + 1)$. By (7.5), the right-hand side is $729P_c(729, n) + u_c(n)$. Cancelling the common term gives equality of the residues. Conversely, this equality satisfies the condition. $\square$

The effective dependence of this selection is fully specified: selected region, rational reader, depth, separation index, preceding prefix, and next residue. The defect A summarizes a boundary relationship; the state (n, k, N, D) and the regional intervals retain the data used by the update.

7.6 Ternary signature, carry, Witt incidence, and reading clock

At a fixed depth t , write $B_{c,j} = B_{c,j}(t)$. For each column j , calculate the following integers and residues:

$$S_j = B_{P,j} + B_{E,j} + B_{F,j}, \quad q_j = S_j \bmod 3, \quad c_j = \lfloor S_j/3 \rfloor, \quad (7.17)$$

$$a_j = (B_{P,j} + B_{E,j} - B_{F,j}) \bmod 3, \quad w_j = \sum_c \mathbf{1}_{B_{c,j} \neq 0}, \quad \bar{w}_j = w_j \bmod 3. \quad (7.18)$$

The residue of the axial expression is taken in the integers. An implementation using natural numbers calculates it as $(B_{P,j} + B_{E,j} + 3 - B_{F,j}) \bmod 3$, which preserves that residue exactly.

The ternary incidence matrix used by the reader is

$$W = \begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 \\ 1 & 0 & 1 & 1 & 2 & 2 \\ 1 & 1 & 0 & 2 & 1 & 2 \\ 1 & 1 & 2 & 0 & 2 & 1 \\ 1 & 2 & 1 & 2 & 0 & 1 \\ 1 & 2 & 2 & 1 & 1 & 0 \end{pmatrix}. \quad (7.19)$$

Two residual charges are preserved for each region:

$$v_c^{\text{res}} = \sum_j B_{c,j} \bmod 3, \quad v_c^W = \sum_j (B_c W)_j \bmod 3. \quad (7.20)$$

The reader's complete signature is $\mathcal{S}(B) = (q, a, c, w, \bar{w}, v^{\text{res}}, v^W)$. Its structure separately preserves the column residue, carry, axial contrast, support weight, and the two charge readings.

Theorem 7.15 (Exact recovery and bounds of the signature). *For each column and each region,*

$$S_j = q_j + 3c_j, \quad 0 \leq q_j, a_j < 3, \quad 0 \leq c_j \leq 2, \quad (7.21)$$

$$B_{F,j} = 2(q_j + 3 - a_j) \bmod 3, \quad 0 \leq w_j \leq 3, \quad 0 \leq \bar{w}_j < 3. \quad (7.22)$$

Moreover,

$$v_c^W = \left(\sum_j ((B_c W)_j \bmod 3) \right) \bmod 3. \quad (7.23)$$

The stated recoveries determine each column's total and its autoscaling coordinate; the regional register additionally preserves the three ordered bands.

Proof. Identity (7.21) is Euclidean division of S_j by 3. Each summand of S_j lies between 0 and 2, so $0 \leq S_j \leq 6$ and $c_j \leq 2$. In $\mathbb{F}_3$, subtracting the two congruences defining q_j and a_j gives $q_j - a_j = 2B_{F,j}$. Since the inverse of 2 is 2, this yields (7.22); the summand 3 permits an expression using natural numbers. The bounds on w_j follow by summing three indicators. Finally, reducing an integer sum modulo 3 is equivalent to reducing each summand and then reducing their sum, which proves (7.23). $\square$

Example 7.16 (Signature of the first joint block). The three initial bands produce

$$\begin{aligned} q &= (0, 0, 2, 2, 1, 2), & a &= (1, 2, 0, 1, 1, 2), \\ c &= (1, 1, 0, 1, 0, 0), & w &= (2, 2, 2, 3, 1, 2), \\ v^{\text{res}} &= (2, 2, 0), & v^W &= (2, 1, 1). \end{aligned}$$

The fourth column has total $2 + 1 + 2 = 5$; its signature records $q_3 = 2$ and $c_3 = 1$, so $q_3 + 3c_3 = 5$. Axial recovery gives $2(2 + 3 - 1) \bmod 3 = 2$, precisely the autoscaling coordinate. The Witt images, reduced componentwise, are

$$(2, 0, 2, 1, 1, 2), \quad (0, 0, 0, 2, 0, 2), \quad (2, 1, 1, 2, 1, 0).$$

Their residual sums by row yield the three charges $(2, 1, 1)$.

Definition 7.17 (Integer clock of decimal reading). The transition index t and decimal representation depth are related by

$$\kappa(t) = \max\{k \in \mathbb{N} : 1000^k \leq 3^{6(t+1)}\}. \quad (7.24)$$

Proposition 7.18 (Characterization and monotonicity of the clock). *The integer $\kappa(t)$ is uniquely determined by*

$$1000^{\kappa(t)} \leq 729^{t+1} < 1000^{\kappa(t)+1}. \quad (7.25)$$

The function κ is monotone and satisfies $\kappa(20) = \kappa(21) = 20$.

Proof. The powers of 1000 are strictly increasing and unbounded; there is therefore a unique interval between consecutive powers containing 729^{t+1} . Monotonicity of the latter expression proves that of κ . The final two evaluations are the integer comparisons

$$1000^{20} \leq 729^{21} < 1000^{21}, \quad 1000^{20} \leq 729^{22} < 1000^{21}.$$

These comparisons use exact integer powers. □

Equality of decimal depth at two distinct transitions explains why both indices must be retained: the reading clock can remain constant while the ternary band advances and its signature is updated. Chronological information resides in the ordered register of transitions.

7.7 Composition theorem for the block, cylinder, and signature

Theorem 7.19 (Joint compatibility of regional readings). *For each depth n , there exists a unique triple of blocks (u_P, u_E, u_F) satisfying regional admissibility (7.16) in all three regions. This triple is $(u_P(n), u_E(n), u_F(n))$. Simultaneously:*

1. *each block updates its cylinder through recurrences (7.14) and (7.15);*
2. *its six trits produce the signature $S(B_P(n), B_E(n), B_F(n))$;*
3. *the column totals are recovered as $q + 3c$, and the Witt charges are obtained from the same bands through (7.23);*
4. *every deeper prefix returns exactly these blocks, truncated cylinders, and signatures.*

Proof. Apply Theorem 7.14 to each of the three regional indices. Coordinatewise uniqueness gives joint uniqueness. Substitute those same residues into the operations $T_u, T_{u,d}$, which allows Theorem 7.11 to be applied. The deterministic application of (7.6) produces the bands used by (7.17)–(7.20). The recovery identities are those of Theorem 7.15. Finally, Theorem 7.4 and truncation in each base prove stability at greater depth. At every step the identity of the emitted block has been preserved; neither reading requires the selection of a different block. □

The conclusion is an operational compatibility between cylindrical contraction and preservation of incidence. Both use the same generated register, retain the depths, and admit integer recovery. This result is the explicit form of the connection

$$\text{emitted regional block} \longrightarrow \begin{cases} \text{cylinder and integer boundary,} \\ \text{band, signature, carry, and Witt charge.} \end{cases}$$

The subsequent development of incidence and fields uses the second reading while preserving its common provenance with the first.

7.8 Calendar census and simultaneous compatibility of the regional biographies

The preceding rows are assembled, without yet applying any decimal reader, into the three block biographies produced by the transitions:

$$\begin{aligned} B^{\text{cl}} &= 010211|012222|010211|002111|110221, \\ B^{\text{pr}} &= 201101|121221|102011|012222|102011, \\ B^{\text{au}} &= 121200|112202|121020|010210|010200. \end{aligned} \quad (7.26)$$

Indeed, the consecutive pairs of the first two segments are the rows of X_0, Y_0 , and those of the last two are the rows of X_1, Y_1 . Thus, the biographies are another representation of the record of twelve transitions, not a collection of conventional prefixes.

To verify the operative word of the calendar, let $c = c_1c_2c_3c_4 \in \{0, 1\}^4$ and, for each regime $\chi \in \{\text{cl}, \text{pr}, \text{au}\}$, define

$$u_0^\chi = b_0^\chi, \quad u_j^\chi = u_{j-1}^\chi L_{c_j} \quad (1 \leq j \leq 4).$$

To condense the census, denote by

$$N_{\text{ar}}(c) := \#\{(j, \chi) : u_j^\chi = b_j^\chi, 1 \leq j \leq 4\}, \quad N_{\text{bio}}(c) := \#\{\chi : (u_0^\chi, \dots, u_4^\chi) = B^\chi\}.$$

Exact multiplication in $\mathbb{F}_3$ for the sixteen calendars gives:

c	$N_{\text{ar}}(c)$	$N_{\text{bio}}(c)$
0000, 0001	6	0
0010	9	0
0011	12	3
0100, 0101, 0110, 0111	3	0
1000, 1001, 1010, 1011	0	0
1100, 1101, 1110, 1111	0	0

(D)

The table exhausts the $2^4 = 16$ possibilities, and each of its entries is obtained by four multiplications of a row by one of the two displayed matrices. Consequently, 0011 is the unique calendar that simultaneously reconstructs the three complete biographies. In operator notation,

$$(L_0, L_0, L_1, L_1) \quad (7.27)$$

is not an independent prescription: it is the unique binary reading compatible with the twelve edges already produced by U_t .

The calendar (7.27) produces four new six-component blocks. Their concatenations form

$$w_6 \longrightarrow w_{12} \longrightarrow w_{18} \longrightarrow w_{24} \longrightarrow w_{30}.$$

The subscripts of w indicate cumulative length. The boundary R_{36} preserves the terminal structure and opens the prolongation relation; it is not renamed as a final periodic block that could be repeated indefinitely.

The identity $L = X^{-1}Y$ proves uniqueness once the transitions are fixed; it does not replace their production by U_t . Conversely, (D) proves within the article the uniqueness of the ordering of the four transports. The causal chain is thereby closed: the APP census produces the initial regions, TRIT fixes regime and orientation, TPK produces the transition record, and that record determines L_0, L_1 and 0011 before any Archimedean realization.

8 Regional determination of the dodecaphasic descriptor and selection of the coupling register

The determination of the coupling register combines two constructions that should be developed in their actual order. The first produces the inputs to the selector from the regional representations of the TPK: six-trit bands, column signatures, a conversion calendar, an oriented descriptor, and a functional panel. The second operates on these inputs and on the terminal eight-block repertoire, enumerates its admissible orderings, and selects a unique register through two simultaneous conditions: exceptional incidence and reading heterotypy. The value of the register appears at the end of this composition.

The present development presupposes the regional representations obtained in the preceding chapters from APP-TRIT-TPK. Its purpose is to make explicit every intermediate operation leading from those representations to the finite selector. The realization of the same register through signed incidences, Hadamard inversion, and the representation of the fine-structure constant are then incorporated. The three constructions retain their respective domains: selection of the register, recovery from its channels, and arithmetic evaluation of its representations.

8.1 Regional bands, encoding, and integer signatures

8.1.1 Domains and encoding of the regional bands

The channel order is fixed as

$$\chi \in \{\pi, e, \varphi\}, \quad B_\chi(t) \in \mathbb{F}_3^6, \quad t \in \mathbb{N}.$$

The channel symbols identify the regions already generated. Each $B_\chi(t)$ retains its place in its sequence of representations; the index t counts ternary blocks. The natural-number evaluation of a band $b = (b_0, \dots, b_5)$ is

$$\text{enc}_6(b) = 243b_0 + 81b_1 + 27b_2 + 9b_3 + 3b_4 + b_5, \quad 0 \leq \text{enc}_6(b) < 729. \quad (8.1)$$

Its fixed-width inverse is

$$\text{dig}_j(n) = \left\lfloor \frac{n}{3^{5-j}} \right\rfloor \bmod 3, \quad 0 \leq j < 6.$$

Leading zeros are thus retained: the words 002001 and 2001 could share an abbreviated integer notation, but the type of the former contains six positions. All subsequent concatenations use this width.

Within the hundred-block window, the six-hundred-trit representation of channel χ determines an integer $P_\chi^{(600)}$. Extraction is performed by

$$b_\chi(t) = \left\lfloor \frac{P_\chi^{(600)}}{729^{99-t}} \right\rfloor \bmod 729, \quad 0 \leq t < 100, \quad B_\chi(t) = \text{enc}_6^{-1}(b_\chi(t)). \quad (8.2)$$

The six-hundred-trit integer is a representation of the regional producer; the hundred-row table is the evaluation of (8.2). This separation allows a historical table to be replaced by its producer without changing the selector that consumes it.

8.1.2 Residues, quotients, occupancy, and orientation

For each column j and time t , introduce

$$T_j(t) = B_\pi(t)_j + B_e(t)_j + B_\varphi(t)_j, \quad (8.3)$$

$$q_j(t) = T_j(t) \bmod 3, \quad c_j(t) = \lfloor T_j(t)/3 \rfloor, \quad (8.4)$$

$$a_j(t) = (B_\pi(t)_j + B_e(t)_j - B_\varphi(t)_j) \bmod 3, \quad (8.5)$$

$$w_j(t) = \mathbf{1}_{B_\pi(t)_j \neq 0} + \mathbf{1}_{B_e(t)_j \neq 0} + \mathbf{1}_{B_\varphi(t)_j \neq 0}, \quad \bar{w}_j(t) = w_j(t) \bmod 3. \quad (8.6)$$

The subtraction in the third expression is performed in the integers and then reduced. In an implementation over natural numbers, the equivalent expression is $(B_\pi + B_e + 3 - B_\varphi) \bmod 3$; truncated subtraction would change the reader.

Proposition 8.1 (Reconstruction of the column sum). *For all admissible bands,*

$$T_j(t) = q_j(t) + 3c_j(t), \quad 0 \leq q_j(t) < 3, \quad 0 \leq c_j(t) \leq 2, \quad 0 \leq w_j(t) \leq 3.$$

Proof. Each column sums three representatives from $\{0, 1, 2\}$, so that $0 \leq T_j(t) \leq 6$. Euclidean division by three yields the stated residue and quotient. Occupancy counts three binary conditions; its bound follows directly from that definition. $\square$

This identity preserves the quotient of the local sum. Reconstructing the three individual bands requires the bands themselves or an additional reader that distinguishes them: a column sum has fewer coordinates than the triple that produces it. Accordingly, the temporal row used below jointly retains the three bands and the four fields $q, a, c, \bar{w}$.

8.2 Conversion clock and first zero advance

8.2.1 Ternary capacity and decimal depth

A representation of $t + 1$ bands contains $6(t + 1)$ trits. The associated conversion calendar is defined by the integer

$$d_t = \max\{k \in \mathbb{N} : 1000^k \leq 3^{6(t+1)}\} = \max\{k \in \mathbb{N} : 1000^k \leq 729^{t+1}\}. \quad (8.7)$$

The definition compares integer capacities. The counter t and the depth d_t record different quantities: a transition adds one six-trit band; the clock counts complete powers of the decimal base one thousand contained in the accumulated ternary capacity.

Lemma 8.2 (Integer characterization of the clock). *For $t, k \in \mathbb{N}$,*

$$d_t = k \iff 1000^k \leq 729^{t+1} < 1000^{k+1}.$$

Moreover, $d_t \leq d_{t+1}$.

Proof. The powers of one thousand form a strictly increasing, unbounded sequence. The power 729^{t+1} lies between two consecutive terms; the lower exponent is precisely the maximum in (8.7). The inequality $729^{t+1} < 729^{t+2}$ implies monotonicity of the maximum. $\square$

Proposition 8.3 (First zero advance). *The least time satisfying $d_{t+1} = d_t$ is*

$$t_* = 20.$$

In particular,

$$d_0 = 0, \quad d_{19} = 19, \quad d_{20} = 20, \quad d_{21} = 20, \quad d_{22} = 21.$$

Proof. The exact integer comparisons

$$1000^{20} \leq 729^{21}, \quad 729^{22} < 1000^{21}, \quad 1000^{21} \leq 729^{23} < 1000^{22}$$

give the values at times twenty, twenty-one, and twenty-two. For $0 \leq t \leq 20$, the expression $729^{t+1}/1000^t = 729(729/1000)^t$ is decreasing and its value at twenty is greater than or equal to one. Hence, $1000^t \leq 729^{t+1} < 1000^{t+1}$ and $d_t = t$. No time less than twenty can satisfy $d_{t+1} = d_t$; at twenty it does hold. All comparisons used involve integer powers. $\square$

The number twenty is therefore an output of the minimality criterion. The rule selecting the instant is “the first zero advance of the clock”; the algorithm and its proof can use this rule without receiving twenty as an external parameter.

8.2.2 Nonadic return and temporal antecedent

The length of the phase return determines

$$r = t_* - 9 = 11. \quad (8.8)$$

The descriptor queries are made at $t_* + 1$, $r + 1$, and r . This distribution preserves the relation between the pause instant, the nonadic translation, and the immediately associated phase reading. Times eleven, twelve, and twenty-one designate positions in the regional calendar; their temporal difference remains recorded even when two times produce the same decimal depth.

8.3 Oriented construction of the twenty-four-trit descriptor

8.3.1 Opposite half-band defects

Take the oriented defects of the regional chart

$$\delta_+ = (0, 0, 0, 1, 1, 1), \quad \delta_- = (0, 0, 0, 2, 2, 2) \in \mathbb{F}_3^6.$$

Their opposition is verified componentwise:

$$\delta_+ + \delta_- = 0 \quad \text{in } \mathbb{F}_3^6.$$

The representative 2 realizes -1 in the ternary chart. Concatenating the three-position tails in negative-positive order produces the word 222111. Adding the defects and concatenating their tails are different operations: the former has codomain $\mathbb{F}_3^6$, whereas the latter preserves the order of two segments of length three.

8.3.2 Dual charge and phase selection

In the regional chart, the six-coordinate transformation uses

$$A_{\text{reg}} = \begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 \\ 1 & 0 & 1 & 1 & 2 & 2 \\ 1 & 1 & 0 & 2 & 1 & 2 \\ 1 & 1 & 2 & 0 & 2 & 1 \\ 1 & 2 & 1 & 2 & 0 & 1 \\ 1 & 2 & 2 & 1 & 1 & 0 \end{pmatrix}.$$

With row vectors, the dual charge of $b \in \mathbb{F}_3^6$ is the sum of the coordinates of bA_{reg} . The integer row sums of A_{reg} are $(5, 7, 7, 7, 7, 7)$; this gives the local expression

$$\eta^\vee(b) = 2b_0 + b_1 + b_2 + b_3 + b_4 + b_5 \pmod{3}. \quad (8.9)$$

The reading phase at instant r is $\theta_r = (r - 1) \bmod 3$. For each channel, define

$$\varepsilon_\chi(r) = \begin{cases} 1, & \eta^\vee(B_\chi(r)) = \theta_r, \\ 2, & \eta^\vee(B_\chi(r)) \neq \theta_r. \end{cases}$$

The phase-charge representation is

$$D(r) = \varepsilon_\pi(r)B_\pi(r) + \varepsilon_e(r)B_e(r) + \varepsilon_\varphi(r)B_\varphi(r) \quad \text{in } \mathbb{F}_3^6. \quad (8.10)$$

The coefficient is determined by an equality of charges before evaluating the word that will result from the combination.

8.3.3 Composition of the descriptor

Let $\parallel$ denote the concatenation of fixed-width words. The complete construction is

$$\Omega_{24} = (B_e(t_* + 1) + \delta_-) \parallel (\delta_-|_{3,4,5} \parallel \delta_+|_{3,4,5}) \parallel (B_\pi(r + 1) + B_e(r + 1)) \parallel D(r). \quad (8.11)$$

Each length-six summand is evaluated in $\mathbb{F}_3^6$; the final concatenation has length $6 + 6 + 6 + 6 = 24$. This construction distinguishes the descriptor Ω_{24} from the regional stages w_{24}^χ : the two objects share a length, but their producers and their places in the composition differ.

Proposition 8.4 (Regional evaluation of the descriptor). *The regional representations and the first-zero-advance criterion determine*

$$\Omega_{24} = 020022\ 222111\ 211201\ 021101, \quad [\Omega_{24}]_3 = 66241910521.$$

Proof. The required bands, preserving the order (π, e, φ) , are

t	$B_\pi(t)$	$B_e(t)$	$B_\varphi(t)$
11	022212	110001	100021
12	012012	202222	000120
21	221022	020100	211021

and come from extraction (8.2). At time twenty-one, $020100 + 000222 = 020022$ in $\mathbb{F}_3^6$. The defect tails provide 222111. At time twelve, $012012 + 202222 = 211201$.

At time eleven, the dual charges are 0, 1, 2, respectively; the phase is $(11 - 1) \bmod 3 = 1$. The resulting coefficients are (2, 1, 2) and

$$2(022212) + (110001) + 2(100021) = 021101 \quad \text{in } \mathbb{F}_3^6.$$

Concatenating these four results gives the stated word. Its integer evaluation is obtained by Horner's method in base three, or by three successive concatenations in base 729. $\square$

The preceding table serves a verification purpose: each of its entries is obtained from the regional producer. Construction rule (8.11) makes it possible to reproduce the descriptor without introducing the final twenty-four-trit string as an input.

8.4 Functional panel and terminal repertoire

8.4.1 Nine regional positions

The functional panel retains the first three bands of each of the three channels, in the declared regional order:

$$\mathcal{P}_9 = (b_\pi(0), b_\pi(1), b_\pi(2), b_e(0), b_e(1), b_e(2), b_\varphi(0), b_\varphi(1), b_\varphi(2)). \quad (8.12)$$

Its evaluation is

$$\mathcal{P}_9 = (103, 161, 103, 523, 457, 301, 450, 398, 438).$$

The nine positions retain their regional provenance, although the value 103 occurs twice. The panel has nine ordered incidences and eight distinct values. The subsequent enumeration uses the incidences; candidate deduplication takes place at a later stage.

8.4.2 Unordered eight-block repertoire

The terminal repertoire involved in this construction is

$$\mathcal{S}_{\text{term}} = \{002001, 020111, 200110, 121012, 010122, 001001, 221110, 222110\} \subset \mathbb{F}_3^6. \quad (8.13)$$

Its eight elements are distinct. The integer encoding, sorted by value solely to fix a computational representation, is

$$\{28, 55, 98, 175, 437, 498, 687, 714\}.$$

The documentary provenance of this repertoire is the terminal segmentation of the seventy-two-trit word and its position-wise recovery through incidences of the phase-charge and affine readers. The selector developed below receives the repertoire without its historical ordering and tests its 8! orderings.

The exact formulation of the reassembled implementation maintains an explicit interface here: the regional producers materially replace the descriptor, the panel, and the temporal rows; repertoire (8.13) retains its earlier documentary producer. This distinction identifies where each input enters. Membership of eight words in a temporal reader repertoire, selection of those eight words, and determination of their terminal order are three different statements. The terminal-selection proof establishes the third for the declared repertoire.

Documentary recovery by position can be expressed exactly. Let $\mathcal{I}$ be the phase-charge incidence register and let $\mathcal{J}$ be the register of the affine extension. Each incidence retains the position ℓ in the seventy-two-trit word and the emitted word u . For $5 \leq \ell \leq 12$, form

$$\mathcal{S}_\ell = \{u : (\ell, u) \in \mathcal{I} \cup \mathcal{J}\}.$$

The recovery procedure checks that all eight positions occur and that each $\mathcal{S}_\ell$ is a singleton. If s_ℓ is its unique element, the output delivered to the selector is

$$\mathcal{S}_{\text{term}} = \{s_5, s_6, \dots, s_{12}\}.$$

The singleton proof for each position preserves the coherence of the historical witnesses; the subsequent export retains the eight words and removes only the historical ordering. The selector reconstructs the order through complete enumeration and the terminal conditions, instead of receiving it from the provenance register.

8.4.3 Temporal incidences of the terminal repertoire and recovery by position

The representation of the terminal repertoire uses the same regional blocks, the same carries, and the same calendar that produce the initial twenty-four-trit descriptor. Two operations in this composition should be made explicit separately. The first recovers eight words from recorded temporal incidences. The second presents those eight words to the dodecaphasic selector as an unordered set. The selector re-establishes the order through its incidence, cylinder, and closure conditions. The distinction preserves the provenance witnesses without turning the order of a historical segmentation into an input to the selector.

Let $B_t = (b_{t,0}, b_{t,1}, b_{t,2}) \in (\mathbb{F}_3^6)^3$ be the regional boundary already generated at time t , with $0 \leq t < 100$. In the chart used by the historical register, the incidence matrix is

$$A_{\text{hist}} = \begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 \\ 1 & 0 & 1 & 1 & 2 & 2 \\ 1 & 1 & 0 & 2 & 1 & 2 \\ 1 & 1 & 2 & 0 & 2 & 1 \\ 1 & 2 & 1 & 2 & 0 & 1 \\ 1 & 2 & 2 & 1 & 1 & 0 \end{pmatrix}.$$

The words are regarded as row vectors. Their two charges and the chronological phase are

$$r_{t,i} = \sum_{j=1}^6 b_{t,i,j}, \quad d_{t,i} = \sum_{j=1}^6 (b_{t,i} A_{\text{hist}})_j, \quad \vartheta_t = t - 1 \pmod{3}.$$

All operations in these three expressions take place in $\mathbb{F}_3$. The historical subscript identifies a specific chart of the Witt incidence; it keeps the coordinate transformation explicit when another presentation of that same incidence is used.

For $c \in \{r_t, d_t\}$, $\delta \in \mathbb{F}_3$ and $\epsilon \in \{1, -1\}$, define

$$\eta_i(c, \delta, \epsilon) = \begin{cases} \epsilon, & c_i = \vartheta_t + \delta, \\ -\epsilon, & c_i \neq \vartheta_t + \delta, \end{cases} \quad L_{t,c,\delta,\epsilon} = \sum_{i=0}^2 \eta_i(c, \delta, \epsilon) b_{t,i}.$$

The orientation -1 is written as 2 when printing the ternary coefficients. Alongside these twelve comparative readers per boundary, the register has the affine reader

$$L_t^{\text{af}} = \sum_{i=0}^2 (d_{t,i} + \vartheta_t) b_{t,i}.$$

Adding the phase in the affine reader and comparing charges in the preceding reader are different operations, both acting on data already produced by the TPK boundary.

t	$b_{t,0}$	$b_{t,1}$	$b_{t,2}$	r_t	d_t
8	200121	212212	110110	011	202
11	022212	110001	100021	001	012
30	210112	110202	211110	100	012
34	121022	000011	112110	220	021
37	120121	011202	112221	100	201
49	110212	200212	100122	110	201
58	111121	022121	122201	122	220
67	010012	001220	011120	122	122
75	220000	020111	100002	120	021
81	122201	020202	201122	202	001
90	022112	022110	121010	202	200

Table 6: Temporal boundaries realizing the printed incidences of the terminal repertoire. Each six-digit entry preserves the order of the six coordinates of the regional boundary.

The following table displays all comparative incidences of the register on the twelve blocks of the segmentation under consideration. The letters r and d denote the visible charge and the dual charge, respectively. Column j retains the position index of the historical witness; the operation delivering the data to the selector forgets this index after forming the set of words.

The tenth block has an affine witness. At $t = 30$, $d_{30} = (0, 1, 2)$ and $\vartheta_{30} = 2$, whence

$$(d_{30,0} + \vartheta_{30}, d_{30,1} + \vartheta_{30}, d_{30,2} + \vartheta_{30}) = (2, 0, 1)$$

and, using the bands in Table 6,

$$L_{30}^{\text{af}} = 2(210112) + 0(110202) + (211110) = (001001) \quad \text{in } \mathbb{F}_3^6.$$

The equality holds coordinatewise: before reduction modulo three, the sum vector is $(6, 3, 1, 3, 3, 4)$.

Proposition 8.5 (Complete recovery of the repertoire through incidences). *Consider the comparative-incidence register in Table 7, augmented by the affine witness $(j, t, L) = (10, 30, 001001)$. For each position $5 \leq j \leq 12$, the set of output words of its incidences is a singleton. Extracting these eight words and subsequently forgetting their order produces exactly*

$$\mathcal{S}_{\text{term}} = \{002001, 020111, 200110, 121012, 010122, 001001, 221110, 222110\}.$$

Their encoding as base-three integers is

$$\{28, 55, 98, 175, 437, 498, 687, 714\}.$$

j	t	charge	δ	ϵ	$(\eta_0\eta_1\eta_2)$	$L_{t,c,\delta,\epsilon}$
4	11	d	0	1	212	021101
5	67	d	1	-1	211	002001
5	67	d	2	1	211	002001
5	67	r	1	-1	211	002001
5	67	r	2	1	211	002001
6	81	d	0	1	222	020111
6	81	r	2	1	222	020111
7	34	r	1	-1	111	200110
7	75	d	1	-1	211	200110
7	75	r	2	-1	211	200110
8	90	r	0	1	121	121012
8	90	r	1	-1	121	121012
9	49	d	0	-1	121	010122
11	11	d	2	-1	211	221110
11	37	d	0	-1	121	221110
12	8	d	0	-1	111	222110
12	8	r	1	-1	111	222110
12	58	d	1	-1	111	222110
12	58	r	0	-1	111	222110

Table 7: Complete phase-charge incidences on the terminal segmentation within the hundred boundaries of the register. Word matches between rows retain their times, orientations, and charge types.

Proof. The incidence multiplicities at positions $j = 5, 6, 7, 8, 9, 10, 11, 12$ are, respectively,

$$(4, 2, 3, 2, 1, 1, 2, 4).$$

At each position, all rows print the same word. Each row is verified by substituting its three bands and three coefficients into the formula for L ; the tenth case has been calculated explicitly through L_{30}^{af} . For example, at $t = 67$ and with coefficients $(2, 1, 1)$, one obtains

$$2(010012) + (001220) + (011120) = (002001) \pmod{3}.$$

The eight results are distinct as words of length six. Their encoding $\sum_{k=1}^6 w_k 3^{6-k}$ gives, in position order, $(55, 175, 498, 437, 98, 28, 714, 687)$. Forgetting the order yields the stated set of integers. Thus, extraction preserves each complete word, including its zero positions, and delivers eight distinct elements to the selector. $\square$

The preceding proof is an exact recovery from identified incidences. Its input comprises the incidence register with its positions and witnesses. The subsequent orbital selection receives only $\mathcal{S}_{\text{term}}$ and the initial regional descriptor; it traverses the $8!$ orderings and determines their compatibility with the functional boundaries and the oriented incidences of the dodecaphasic wheel. Both operations are thus presented contiguously, with explicit domains: temporal provenance of the eight words and constructive determination of their order.

The comparative table contains the nineteen incidences that its formula produces on the twelve-block segmentation over the hundred declared times. This finite exhaustiveness is checked by traversing $100 \cdot 2 \cdot 3 \cdot 2 = 1200$ evaluations. The affine reader extends coverage of the terminal positions from seven to eight. This count characterizes the local reader repertoire just defined; the other TPK operations retain the domains and constructive functions established in their respective derivations.

8.5 Comparative readers, affine reader, and temporal incidence

8.5.1 Witt transformation of the selector

The selector uses the six-coordinate chart

$$A_{\text{ter}} = \begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 \\ 1 & 0 & 1 & 2 & 2 & 1 \\ 1 & 1 & 0 & 1 & 2 & 2 \\ 1 & 2 & 1 & 0 & 1 & 2 \\ 1 & 2 & 2 & 1 & 0 & 1 \\ 1 & 1 & 2 & 2 & 1 & 0 \end{pmatrix}.$$

Define $W(b) = bA_{\text{ter}}$ in $\mathbb{F}_3^6$, together with the charges

$$\eta(b) = \sum_{j=0}^5 b_j \pmod{3}, \quad \eta_{\text{ter}}^{\vee}(b) = \eta(W(b)).$$

The descriptor chart and the selector chart are related by the permutation $\sigma = (0, 1, 2, 4, 5, 3)$, written as a list of images:

$$(A_{\text{ter}})_{ij} = (A_{\text{reg}})_{\sigma(i), \sigma(j)}. \quad (8.14)$$

Lemma 8.6 (Compatibility of the charge functional). *For every band b ,*

$$\eta_{\text{ter}}^{\vee}(b) = \eta^{\vee}(b).$$

Proof. The integer sum of each corresponding row of the two matrices agrees: it is five in the first row and seven in the other five. By distributivity,

$$\sum_j \sum_i b_i (A_{\text{ter}})_{ij} = \sum_i b_i \sum_j (A_{\text{ter}})_{ij} = \sum_i b_i \sum_j (A_{\text{reg}})_{ij}.$$

Reduction modulo three gives the equality of charges. Matrix identity (8.14) is verified by substituting its thirty-six entries and also preserves the coordinate change. $\square$

Equality of the total charge functional allows its representations to be compared. The vector transformations remain identified by their matrices and the explicit permutation; the preceding proof preserves that coordinate information.

8.5.2 Twelve comparative readers per instant

Define the temporal phase

$$\theta(t) = (t + 2) \pmod{3}.$$

This expression realizes $(t - 1) \pmod{3}$ also at $t = 0$, where the phase equals two. For each charge type $\nu \in \{\eta, \eta_{\text{ter}}^{\vee}\}$, offset $o \in \{0, 1, 2\}$ and orientation $s \in \{1, 2\}$, take

$$\epsilon_{\chi}^{\nu, o, s}(t) = s \begin{cases} 1, & \nu(B_{\chi}(t)) = \theta(t) + o \pmod{3}, \\ 2, & \nu(B_{\chi}(t)) \neq \theta(t) + o \pmod{3}. \end{cases}$$

The corresponding comparative word is

$$C_{\nu, o, s}(t) = \sum_{\chi \in \{\pi, e, \varphi\}} \epsilon_{\chi}^{\nu, o, s}(t) B_{\chi}(t) \quad \text{in } \mathbb{F}_3^6. \quad (8.15)$$

The parameters generate twelve labelled representations per row. Two labels may produce the same word; the temporal witnesses and labels remain available in the complete calendar.

8.5.3 Affine phase–charge reader

The affine reader uses coefficients

$$a_{\chi}^{\text{af}}(t) = \eta_{\text{ter}}^{\vee}(B_{\chi}(t)) + \theta(t) \pmod{3},$$

and produces

$$F(t) = \sum_{\chi \in \{\pi, e, \varphi\}} a_{\chi}^{\text{af}}(t) B_{\chi}(t) \text{ in } \mathbb{F}_3^6. \quad (8.16)$$

The comparative rule distinguishes agreement and difference of charges; the affine rule adds charge and phase before combining the bands. This operational difference is what heterotypic selection uses.

8.5.4 Word–residue incidence

Let R be the cyclic rotation of the six positions of a band. In the row at time t , form the residual repertoire

$$\mathcal{R}(t) = \{R^j v : v \in \{q(t), a(t), c(t), \bar{w}(t)\}, 0 \leq j < 6\}.$$

The temporal incidence relations are

$$\mathcal{A}_{\text{cmp}} = \{(C_{\nu, o, s}(t), u) : 0 \leq t < 100, u \in \mathcal{R}(t), \nu, o, s \text{ admissible}\}, \quad (8.17)$$

$$\mathcal{A}_{\text{af}} = \{(F(t), u) : 0 \leq t < 100, u \in \mathcal{R}(t)\}. \quad (8.18)$$

Each pair relates a read word to a residue of the same row. Time is quantified jointly across both coordinates of the pair. In the calendar used, times $0, \dots, 99$ are distinct; identifying the same row and identifying the same time are therefore equivalent operations.

Proposition 8.7 (Existential compression of the temporal repertoire). *The function*

$$M(v, u) = \mathbf{1}_{(v, u) \in \mathcal{A}_{\text{cmp}}} + 2\mathbf{1}_{(v, u) \in \mathcal{A}_{\text{af}}} \in \{0, 1, 2, 3\}$$

preserves exactly the existence of each reading type for the pair (v, u) .

Proof. The first bit equals one exactly when there exist a time, a residual field, a rotation, and a comparative label that produce the pair. The second bit expresses the corresponding condition for the affine reader. Union by bitwise disjunction preserves each existence condition as all rows are traversed. A pair may admit both types, in which case the mask equals three. $\square$

The function M is an existence reader. The witness times, their multiplicities, and the individual labels remain in (8.18) and in the calendar that produces them. The finite implementation uses M to decide heterotypy, whereas the genealogical archive retains the richer provenance data.

8.6 Enumeration of cylinders and decimal candidates

8.6.1 Concatenation of seventy-two and seventy-eight trits

For an ordering $\tau = (s_1, \dots, s_8)$ of $\mathcal{S}_{\text{term}}$, define

$$P_{72}(\tau) = [\Omega_{24} \| s_1 \| \cdots \| s_8 \|]_3.$$

Recursively, start from $p_0 = [\Omega_{24}]_3$ and compute $p_i = 729p_{i-1} + [s_i]_3$ for $1 \leq i \leq 8$. This recurrence preserves the leading zeros of each block through explicit multiplication by 729. For each position b of the functional panel,

$$P_8(\tau, b) = 729P_{72}(\tau) + b.$$

The lengths are $24 + 8 \cdot 6 = 72$ and $72 + 6 = 78$, respectively. The corresponding ternary cylinder is

$$I_{\tau, b} = \left[\frac{P_8(\tau, b)}{3^{78}}, \frac{P_8(\tau, b) + 1}{3^{78}} \right). \quad (8.19)$$

8.6.2 Exact intersection with the twelve-triad grid

The decimal reading uses twelve triads, that is, the grid $N/10^{36}$. Write $D = 10^{36}$ and $Q = 3^{78}$ for brevity.

Lemma 8.8 (Integer endpoints of the cylinder). *For an integer prefix p of length seventy-eight, the numerators of the grid points belonging to its cylinder are*

$$\left\{ N \in \mathbb{Z} : \left\lceil \frac{pD}{Q} \right\rceil \leq N < \left\lceil \frac{(p+1)D}{Q} \right\rceil \right\}. \quad (8.20)$$

Each cylinder contains at most one point of this grid.

Proof. The condition $N/D \in [p/Q, (p+1)/Q]$ is equivalent to $pD/Q \leq N < (p+1)D/Q$. For integer N , the first inequality is equivalent to $\lceil pD/Q \rceil \leq N$; the second is equivalent to $N < \lceil (p+1)D/Q \rceil$. The interval length in coordinate N is $D/Q < 1$, since $10^{36} < 3^{78}$. It can therefore contain at most one integer. $\square$

Ceilings are evaluated by integer division: $\lceil a/Q \rceil = \lfloor (a+Q-1)/Q \rfloor$ for $a \geq 0$. The entire enumeration uses exact integers; the choice of boundary points is fixed by the half-open right endpoint.

First retain the indexed list of incidences

$$\mathcal{C}_{\text{ind}} = \{(\tau, j, N) : N/D \in I_{\tau, \mathcal{P}_9(j)}\}, \quad 1 \leq j \leq 9,$$

and then obtain the set of numerators

$$\mathcal{C} = \{N : \exists \tau, j (\tau, j, N) \in \mathcal{C}_{\text{ind}}\}.$$

Passing to $\mathcal{C}$ removes repetitions of a numerator, retained in $\mathcal{C}_{\text{ind}}$ together with their provenance.

Lemma 8.9 (Recovery of the ordering from the cylinder point). *If $(\tau, j, N) \in \mathcal{C}_{\text{ind}}$, block number i of the seventy-eight-trit word is recovered by*

$$\left\lfloor \frac{N 729^i}{D} \right\rfloor \bmod 729, \quad 1 \leq i \leq 13.$$

In particular, blocks five through twelve recover τ .

Proof. Membership in the cylinder means $p \leq QN/D < p+1$, where $p = P_8(\tau, \mathcal{P}_9(j))$. Hence, $\lfloor QN/D \rfloor = p$. Extraction of the earlier blocks agrees with successive division of the prefix p by powers of 729: truncating a positional expansion before extracting a block preserves that block. The displayed formula performs exactly this extraction. The first four blocks form Ω_{24} and the next eight are the ordering τ . $\square$

Proposition 8.10 (Exact candidate count). *For Ω_{24} , $\mathcal{S}_{\text{term}}$ and $\mathcal{P}_9$ defined above,*

$$\# \text{Ord}(\mathcal{S}_{\text{term}}) = 40320, \quad \# \mathcal{C}_{\text{ind}} = 21902, \quad \# \mathcal{C} = 19446.$$

Proof. The eight blocks are distinct, so their orderings are the $8! = 40320$ permutations. For each permutation and each of the nine panel positions, compute $p = P_8(\tau, b)$, apply endpoints (8.20), and add the unique integer when the integer interval is nonempty. The sum of the cardinalities of these intervals is 21902. The union of their integers contains 19446 distinct elements. The exhaustive evaluation of these operations is certified by the theorems `indexed_cylinder_count` and `candidate_count` of the finite selector. The algorithm performs exactly the enumeration described: permutations, nine panel incidences, bounds by integer division, and duplicate removal. $\square$

The numbers 21902 and 19446 count different objects. The former counts indexed incidences containing a decimal point; the latter counts distinct numerators. The total number of cylinder queries is $40320 \cdot 9 = 362880$, including cylinders whose intersection with the grid is empty.

8.7 Exceptional incidence and heterotypy condition

8.7.1 Exceptional face produced by the panel

A band is lifted to twelve coordinates by

$$\Gamma(b) = (b, W(b)) \in \mathbb{F}_3^{12}.$$

The support of a word is the set of positions of its nonzero coordinates, with human-readable indices $1, \dots, 12$. The first regional bands of the panel produce

$$\mathcal{B} = \text{supp } \Gamma(B_\pi(0)) \cap \text{supp } \Gamma(B_e(0)) = \{4, 6, 9, 10\}. \quad (8.21)$$

The operation uses the regional words 103 and 523 and the selector's Witt matrix. The face is computed before querying the candidate coordinates.

For $N \in \mathcal{C}$, its twelve triads are

$$K_i(N) = \left\lfloor \frac{N}{1000^{12-i}} \right\rfloor \bmod 1000, \quad 1 \leq i \leq 12. \quad (8.22)$$

The crown support is

$$\mathcal{H}(N) = \{i : K_i(N) \geq 729\}, \quad \mathcal{C}_{\mathcal{B}} = \{N \in \mathcal{C} : \mathcal{H}(N) = \mathcal{B}\}.$$

The boundary $729 = 3^6$ relates the six-trit band to the decimal triad. Exhaustive evaluation of the equality of supports gives

$$\#\mathcal{C}_{\mathcal{B}} = 79. \quad (8.23)$$

The support preserves positions, while the complete register also preserves the twelve magnitudes. The next stage uses this positional information together with relations between magnitudes.

8.7.2 Ternary words and oriented differences of the candidate

The i th ternary band of the point N/D is obtained by

$$T_i(N) = \left\lfloor \frac{N 729^i}{D} \right\rfloor \bmod 729, \quad 1 \leq i \leq 13. \quad (8.24)$$

The oriented edge $a \rightarrow b$ has residue

$$R_{a,b}(N) = (K_a(N) - K_b(N)) \bmod 729. \quad (8.25)$$

Subtraction is performed in $\mathbb{Z}$ before reduction. For this edge, the pair

$$(\text{enc}_6^{-1}(T_b(N)), \text{enc}_6^{-1}(R_{a,b}(N))).$$

is queried in relations (8.18). It is therefore a joint query of the destination word and the oriented difference of the triads.

Write $\text{Cmp}(a, b; N)$ when the pair belongs to $\mathcal{A}_{\text{cmp}}$ and $\text{Af}(a, b; N)$ when it belongs to $\mathcal{A}_{\text{af}}$.

8.7.3 Determination of the transverse orbit

The twenty-four-trit descriptor determines the first three decimal triads before the terminal filtrations. Indeed, its cylinder satisfies the exact inclusion

$$\left[\frac{66241910521}{3^{24}}, \frac{66241910522}{3^{24}} \right) \subseteq \left[\frac{234543140}{10^9}, \frac{234543141}{10^9} \right).$$

The two endpoint inequalities are verified by integer cross-products. Thus, every candidate shares the three reference positions 1, 2, 3 and their representations (234, 543, 140).

The step-four translation on $\mathbb{Z}/12\mathbb{Z}$ has the four orbits

$$(1, 5, 9), \quad (2, 6, 10), \quad (3, 7, 11), \quad (4, 8, 12).$$

The first three each contain one of the reference positions; the fourth is the only one disjoint from $\{1, 2, 3\}$. The oriented forest of steps three and four is fixed by the edges

$$\begin{aligned} 1 \rightarrow 4, \quad 1 \rightarrow 5, \quad 2 \rightarrow 6, \\ 3 \rightarrow 7, \quad 4 \rightarrow 8, \quad 5 \rightarrow 9, \\ 6 \rightarrow 10, \quad 7 \rightarrow 11, \quad 8 \rightarrow 12. \end{aligned}$$

Its first edge has step three and the remaining edges have step four. The two internal edges of the distinguished orbit are precisely $4 \rightarrow 8$ and $8 \rightarrow 12$. The location of the reading condition is thus fixed by the geometry of the positions and by the initial prefix, before examining the candidates of the exceptional face.

8.7.4 Terminal heterotypy condition

The terminal condition is

$$\begin{aligned} \mathcal{J}(N) = [\text{Cmp}(4, 8; N) \wedge \text{Af}(8, 12; N)] \\ \vee [\text{Af}(4, 8; N) \wedge \text{Cmp}(8, 12; N)]. \end{aligned} \quad (8.26)$$

Both assignments of types to the two edges are admitted. The selection requires the existence of a realization with distinct types; the individual labels and their times remain available in the incidence witnesses.

Theorem 8.11 (Terminal selection of the coupling register). *On the candidate domain produced by Ω_{24} , $\mathcal{S}_{\text{term}}$, $\mathcal{P}_3$ and the hundred-row regional calendar, there exists a unique numerator simultaneously satisfying exceptional incidence and heterotypy:*

$$\{N \in \mathcal{C}_{\mathcal{B}} : \mathcal{J}(N)\} = \{234543140729659824621058914794146601\}.$$

Its twelve-triad register is

$$K = (234, 543, 140, 729, 659, 824, 621, 058, 914, 794, 146, 601). \quad (8.27)$$

Proof. The preceding count produces the 79 elements of $\mathcal{C}_{\mathcal{B}}$. For each one, its twelve coordinates are extracted by (8.22); the destination bands T_8 , T_{12} and residues $R_{4,8}$, $R_{8,12}$ are computed; the two masks of the temporal repertoire are queried; and disjunction (8.26) is applied. The resulting list contains exactly the stated integer, according to the exhaustive evaluation certified by `terminal_selection`. Equality of the list with a singleton implies existence and uniqueness in the declared domain. Base-one-thousand extraction from that integer gives (8.27).

The selection procedure receives the enumerated producers and rules. The final integer appears in the statement of the verified equality after selection has been executed. This logical position allows the result to be checked without using it as a selection criterion. $\square$

8.7.5 Arithmetic witness of membership in a cylinder

One of the incidences of the selected numerator uses the ordering

$$(002001, 020111, 200110, 121012, 010122, 001001, 221110, 222110)$$

and the boundary $438 = 121020_3$. For this incidence,

$$P_2 = 5283881584882673136514102926090957,$$

$$P_{78} = 3851949675379468716518781033120308091.$$

The integer-endpoint formula yields

$$\left\lceil \frac{P_{78} 10^{36}}{3^{78}} \right\rceil = 234543140729659824621058914794146601,$$

$$\left\lceil \frac{(P_{78} + 1) 10^{36}}{3^{78}} \right\rceil = 234543140729659824621058914794146602.$$

The unique integer in the interval is therefore N_K . The thirteen ternary bands of its decimal point are

020022	222111	211201	021101
002001	020111	200110	121012
010122	001001	221110	222110
121020			

This calculation exhibits a concrete witness of membership in the enumerated domain. Uniqueness of the register follows from the exhaustive examination of all candidates, in addition to this particular witness.

8.8 Regional composition and continuity of the realizations

8.8.1 Composition of the regional generators

Let $\text{Sel}(\Omega, \mathcal{S}, \mathcal{P}, \mathcal{N})$ be the selector defined by the enumeration and the two preceding filters. The regional producers satisfy the equalities

$$\Omega_{\text{reg}} = \Omega_{24}, \quad \mathcal{P}_{\text{reg}} = \mathcal{P}_9, \quad \mathcal{N}_{\text{reg}} = \mathcal{N}_{100},$$

where $\mathcal{N}_{\text{reg}}$ is computed by band extraction and signature reading. By congruence of functions,

$$\begin{aligned} \text{Sel}(\Omega_{\text{reg}}, \mathcal{S}_{\text{term}}, \mathcal{P}_{\text{reg}}, \mathcal{N}_{\text{reg}}) \\ = \text{Sel}(\Omega_{24}, \mathcal{S}_{\text{term}}, \mathcal{P}_9, \mathcal{N}_{100}) = \{N_K\}. \end{aligned} \tag{8.28}$$

The equality replaces three documentary interfaces with their actual regional construction. The terminal repertoire retains the provenance specified in (8.13).

The register is then formed by the twelve Euclidean divisions in (8.22). Each coordinate belongs to $\{0, \dots, 999\}$ and the length is twelve by construction. The implementation that takes the first element of the selected list is justified by equality with the singleton: the default value of a computational operation on empty lists never enters this result.

8.8.2 Relation to signed incidences and inversion

The present construction provides a prospective selection in an explicit finite domain. The subsequent development of the dodecaphasic register produces its oriented channels from the enriched terminal state and proves an integral inversion through Hadamard. That inversion recovers a register from compatible channels; here a register is selected from its regional and incidence-based inputs. Equality of their results allows the two realizations to be composed while preserving the origin of each operation.

Uniqueness of selection and uniqueness of inversion are distinct properties. The former quantifies over the candidates constructed in this chapter. The latter quantifies over the registers sharing a compatible signature. Their conjunction justifies transporting the selected register to the signed realization and recovering it from that realization, without turning a retrospective recovery into the producer of the register.

8.8.3 Relation between the terminal counts

The sequence

$$40320 \longrightarrow 21902 \longrightarrow 19446 \longrightarrow 79 \longrightarrow 1$$

records, successively, orderings, indexed incidences with the grid, distinct numerators, candidates of the exceptional face, and heterotypic survivors. The arrow notation indicates the order of the operations; the second number counts a different class of objects from the first.

The regional-catalogue count $196 \cdot 197 \cdot 196 \longrightarrow 331 \longrightarrow 2 \longrightarrow 1$, developed in its corresponding place, starts from another population: regional blocks from three catalogues and terminal conditions of Gram, occupancy, distance, and orientation. Both counts retain their domains. Their convergence to the same representation is expressed through the identification of their readers and outputs; the intermediate numbers describe different operations and remain documented separately.

8.8.4 Regional composition of the fine-structure constant

The output K subsequently enters the ordered regional composition and the base-one-thousand carry. Its later use preserves the twelve positions and the reading boundary. Coupling to the analytic root and representations at arbitrary depth uses that same output, together with the regional coordinates already generated and the rules of the analytic chart. This dependency prepares the chapter devoted to the two correlated realizations of the fine-structure constant.

The scope of this chapter is precise: production of the descriptor and its regional inputs, determination of the finite domain, unique selection on that domain, and material composition with the producers. The analytic recurrence at all orders and the compatibility of infinite carries are developed subsequently with their own proofs.

8.9 Executable certification and traceability of the operations

The formal development preserves the correspondence between the mathematical definitions and their executable realizations. The signature and clock proofs are found in `N69RegionalSignature.lean`; band extraction and reproduction of the hundred rows, in `GeneratedN69Rows.lean`; the minimality criterion and construction of Ω_{24} , in `RegionalW24.lean`. Compatibility of the two Witt matrices is assembled in `WittReaderCompatibility.lean`.

The temporal readers, rotations, oriented residues, and heterotypic condition are formalized in `TerminalReaders.lean`. Enumeration and counts are formalized in `TerminalOrbitalSelection.lean`. The regional panel is linked through `RegionalPanelBridge.lean`, and composition (8.28) is expressed in `SelectedFromRegionalInputs.lean`.

The finite enumeration checks use Lean’s native evaluation; their reports record `Lean.ofReduceBool`, in addition to the library’s customary logical axioms. The regional reconstruction of the descriptor uses kernel proofs and ordinary reduction. This distinction describes the verification mechanisms actually employed. The compilation receipts attest to their declared modules and dependencies; the mathematical proofs retain the domains, producers, and hypotheses made explicit in the body of the chapter.

8.10 Complete census of cylinders and selection of the terminal boundary

The terminal readout jointly uses two kinds of information: compatibility of the regional cylinders and an incidence profile with labelled channels and positions. This section constructs the complete catalogues, enumerates their triples, and proves the effect of each reader condition. The starting point consists of the regional outputs already produced in Section 6, not values introduced to define the APP-TRIT-TPK transport. Reduction to a matrix belongs to a subsequent readout of these outputs and preserves the order of the three channels.

The result is a census theorem with an explicit starting structure. The profile of Gram products, weights, distances, and marginals is declared as a datum of the terminal reader. The exhaustiveness of the enumeration proves what this profile selects on the specified catalogues; it does not, by itself, prove that the profile is a universal law or that it can be reconstructed from the unique triple that ultimately satisfies it. This distinction allows the complete finite proof to be incorporated without reversing its genealogy.

8.10.1 From regional outputs to ternary catalogues

Let $x_\pi, x_e, x_\varphi \in [0, 1)$ be the fractional parts of the three regional coordinates already constructed. We use their expressed cylinders of one thousand decimal places,

$$J_\chi^{(1000)} = \left[\frac{d_\chi}{10^{1000}}, \frac{d_\chi + 1}{10^{1000}} \right), \quad \chi \in \{\pi, e, \varphi\}.$$

The integers d_χ are finite output readouts. A length of one thousand is sufficient for the integers that follow, as verified from the two bounds of each interval; no infinite depth is assumed. We define

$$p = 30, \quad r = 1050, \quad L = p + r = 1080, \quad k = 171, \quad D = 3^L, \quad Q = 1000^k, \quad (8.29)$$

and

$$P_\chi = \lfloor 3^{30} x_\chi \rfloor, \quad T_\chi = \lfloor Q x_\chi \rfloor.$$

Here k counts decimal triads and does not denote the dodecaphasic register K . We have $1000^{171} \leq 3^{1080} < 1000^{172}$.

Lemma 8.12 (Stable extraction of an integer). *For $d, m, h \in \mathbb{Z}$, with $m, h > 0$, the value of $\lfloor mx \rfloor$ is constant on $[d/h, (d+1)/h)$ if and only if*

$$\left\lfloor \frac{md}{h} \right\rfloor = \left\lfloor \frac{m(d+1) - 1}{h} \right\rfloor. \quad (8.30)$$

When this holds, either side gives that value.

Proof. The minimum of mx is attained at the left endpoint. Its supremum is $m(d+1)/h$ and is not attained. The largest integer value that $\lfloor mx \rfloor$ can take is $\lceil m(d+1)/h \rceil - 1$, which equals the second side because the numerator and denominator are integers. Equality of the endpoints of the range of values proves the assertion. $\square$

The stability test holds in all three channels for $m = 3^{30}, Q, 3^{1080}$. The last choice will be used solely for a subsequent comparison, not to select the profile. The first thirty recovered positions are precisely the five six-trit stages of the regional transport:

Channel	Thirty-trit prefix	P_χ
π	01021101222201021100211110221	29152671743887
e	201101121221102011012222102011	147887858824447
φ	121200112202121020010210010200	127247717616687

The agreement preserves the prolongation already constructed; the decimal readout is not used to replace its transport matrices retrospectively.

For a fixed channel, a tail of r trits is represented by $0 \leq R < 3^r$. Its cylinder and the cylinder of the first k triads are

$$I_{\chi,R} = \left[\frac{P_\chi 3^r + R}{D}, \frac{P_\chi 3^r + R + 1}{D} \right), \quad J_{\chi,T} = \left[\frac{T_\chi}{Q}, \frac{T_\chi + 1}{Q} \right). \quad (8.31)$$

The catalogue includes all tails for which these two cylinders have a nonempty intersection.

Proposition 8.13 (Catalogue by exact intersection). *The compatible tails are the integers in the interval $[a_\chi, b_\chi]$, where*

$$a_\chi = \max\left(0, \left\lfloor \frac{T_\chi D}{Q} \right\rfloor - P_\chi 3^r\right), \quad (8.32)$$

$$b_\chi = \min\left(3^r - 1, \left\lfloor \frac{(T_\chi + 1)D - 1}{Q} \right\rfloor - P_\chi 3^r\right). \quad (8.33)$$

For the specified regional cylinders, no clipping applies and we obtain

χ	$a_\chi \bmod 729$	$b_\chi \bmod 729$	$b_\chi - a_\chi + 1$
π	67	262	196
e	584	51	197
φ	117	312	196

(8.34)

Proof. Write $N = P_\chi 3^r + R$. The two half-open intervals in (8.31) intersect exactly when

$$NQ < (T_\chi + 1)D, \quad (N + 1)Q > T_\chi D.$$

The second inequality is equivalent to $N \geq \lfloor T_\chi D/Q \rfloor$; the first, to $N \leq \lfloor ((T_\chi + 1)D - 1)/Q \rfloor$. Subtracting $P_\chi 3^r$ and retaining $0 \leq R < 3^r$ proves the formulae. Evaluation by integer division of the T_χ extracted using (8.30) gives the table. Since $r \geq 6$, $P_\chi 3^r \equiv 0 \pmod{729}$, so the residue of the first endpoint can also be obtained directly from $\lfloor T_\chi D/Q \rfloor \bmod 729$. $\square$

The last six trits of a tail form the word $\nu_6(R \bmod 729)$, where ν_6 denotes the ternary representation of length six, including its leading zeros. By the table, each catalogue contains fewer than 729 consecutive elements, so this reduction is injective on it. Thus, the three boundary catalogues are completely described, without choosing the final triples:

$$\begin{aligned} \mathcal{B}_\pi &= \{\nu_6(67 + i) : 0 \leq i < 196\}, \\ \mathcal{B}_e &= \{\nu_6((584 + j) \bmod 729) : 0 \leq j < 197\}, \\ \mathcal{B}_\varphi &= \{\nu_6(117 + \ell) : 0 \leq \ell < 196\}. \end{aligned} \quad (8.35)$$

Requiring the left endpoint of $I_{\chi,R}$ to belong to $J_{\chi,T}$ would impose a different condition: it would lose the first cylinder that intersects the boundary and would give 195, 196, 195. The census uses the intersection of the complete cylinders, in accordance with its definition.

8.10.2 Declared profile and compatibility indicators

For $u, v \in \mathbb{F}_3^6$, let $g(u, v) = \sum u_i v_i \in \mathbb{F}_3$, $n(u) = g(u, u)$, $w(u) = \#\{i : u_i \neq 0\}$, and $h(u, v) = \#\{i : u_i \neq v_i\}$. The last two functions take integer values; the first two take values in $\mathbb{F}_3$. The reader's local profile, in the order (π, e, φ) , is

$$\begin{aligned} (g_{\pi e}, g_{\pi \varphi}, g_{e \varphi}) &= (2, 2, 0), & (n_\pi, n_e, n_\varphi) &= (0, 0, 0), \\ (w_\pi, w_e, w_\varphi) &= (3, 3, 3), & (h_{\pi e}, h_{\pi \varphi}, h_{e \varphi}) &= (2, 5, 3). \end{aligned} \quad (8.36)$$

The words are ordered as in (8.35). We use $[E] \in \{0, 1\}$ for the indicator of a condition E . For each pair of channels χ, ψ , the rectangular matrix $A_{\chi\psi}$ has entry $[g(u, v) = g_{\chi\psi}]$. The matrix $A_{\chi\psi}^h$ multiplies that entry by $[h(u, v) = h_{\chi\psi}]$. Let D_χ^n be the diagonal matrix of $[n(u) = 0]$, and D_χ^{nw} that of $[n(u) = 0][w(u) = 3]$.

Theorem 8.14 (Exhaustive enumeration by finite sums). *The product of the three catalogues contains 7 567 952 triples. The successive conditions in (8.36) leave*

$$7\,567\,952 \longrightarrow 275\,794 \longrightarrow 6235 \longrightarrow 3127 \longrightarrow 331. \quad (8.37)$$

These four counts are, respectively,

$$\begin{aligned} N_g &= \text{tr}(A_{\pi e} A_{e\varphi} A_{\varphi\pi}), \\ N_{gn} &= \text{tr}(D_{\pi}^n A_{\pi e} D_e^n A_{e\varphi} D_{\varphi}^n A_{\varphi\pi}), \\ N_{gnw} &= \text{tr}(D_{\pi}^{nw} A_{\pi e} D_e^{nw} A_{e\varphi} D_{\varphi}^{nw} A_{\varphi\pi}), \\ N_{gnwh} &= \text{tr}(D_{\pi}^{nw} A_{\pi e}^h D_e^{nw} A_{e\varphi}^h D_{\varphi}^{nw} A_{\varphi\pi}^h). \end{aligned} \quad (8.38)$$

All products and traces in this formula are calculated over the integers, not over the field with three elements.

Proof. The size of the product is $196 \cdot 197 \cdot 196$. When a trace in (8.38) is expanded, its indices run through exactly one triple of words. The summand equals one if it satisfies all the represented conditions and zero otherwise. Each triple is counted once. The diagonal matrices add the individual conditions; the rectangular factors add the pairwise conditions. This proves that the traces are precisely the stated cardinalities.

To make their evaluation explicit, fix the indices i, j of the first two channels and form the sets of indices ℓ of the third that satisfy each pairwise condition. Their indicators are intersected, the relevant individual indicators are added, and the cardinality is summed. The final sum runs over $0 \leq i < 196, 0 \leq j < 197$. Substitution of the words in (8.35) and the profile (8.36) gives, in that order, 275794, 6235, 3127, 331. The supplement records the four contributions of each of the 196 indices i and the complete list of 331 triples; its program evaluates these sums with exact binary indicators and checks each survivor by scalar evaluation. No sample or tolerance is involved. $\square$

8.10.3 Labelled incidence and orientation charge

For a surviving triple, let $B \in \mathbb{F}_3^{3 \times 6}$ be its matrix of ordered rows. The second profile requires

$$\text{rank}_{\mathbb{F}_3} B = 3, \quad w_{\text{col}}(B) = (1, 1, 2, 2, 3, 0), \quad q_{\partial}(B) = (1, 2, 2, 1, 2, 0), \quad (8.39)$$

where w_{col} counts the nonzero elements in each column and q_{∂} sums its entries modulo three. This is a profile of labelled columns; permuting them without transporting the profile defines a different reader. Denote by $\mathcal{F}$ the set of 331 triples from the preceding theorem.

Proposition 8.15 (Reduction of the census to the terminal pair). *The sums of indicators over $\mathcal{F}$, successively imposing rank, column weights, and column sums, are*

$$\sum_{B \in \mathcal{F}} [\text{rank } B = 3] = 331, \quad N_{\text{rango,pesos}} = 18, \quad N_{\text{rango,pesos,sumas}} = 2. \quad (8.40)$$

With catalogue indices starting at one, the two final triples are (120, 197, 157) and (129, 197, 148), and their matrices are

$$B_0 = \begin{pmatrix} 0 & 2 & 0 & 2 & 2 & 0 \\ 0 & 0 & 1 & 2 & 2 & 0 \\ 1 & 0 & 1 & 0 & 1 & 0 \end{pmatrix}, \quad B_1 = \begin{pmatrix} 0 & 2 & 1 & 0 & 2 & 0 \\ 0 & 0 & 1 & 2 & 2 & 0 \\ 1 & 0 & 0 & 2 & 1 & 0 \end{pmatrix}.$$

Proof. For each triple in the already enumerated set, the rank is computed by elimination over $\mathbb{F}_3$, the nonzero elements of the six columns are counted, and their entries are summed modulo three. The products of the indicators in (8.39) give (8.40). Substituting the two final indices into (8.35) gives the displayed matrices. The complete list of survivors and the exact filters in the supplement allow the elimination to be reproduced without presupposing these two matrices as the domain. $\square$

The orientation $\sigma = (1, 1, -1) = (1, 1, 2) \in \mathbb{F}_3^3$ selects the line $\langle \sigma \rangle = \{(0, 0, 0), (1, 1, 2), (2, 2, 1)\}$. The row charges of B_0, B_1 are (0, 2, 0) and (2, 2, 1); only the latter belongs to this line. This recovers the last step of the selector in Section 8.15, now preceded by the construction and enumeration of its complete domain. The stable readout of $[3^{1080} x_{\chi}]$, performed after the filters, gives the boundaries 021020, 001220, 100210, with the same ranks (129, 197, 148). This subsequent comparison does not define the profile or enter the sums of indicators.

8.10.4 Preserved information and scope of the result

The procedure preserves the thirty-trit prefix, the compatible tail, the regional order, the position of each trit, and the orientation. Reduction modulo 729 preserves the identity of each candidate here because the interval of tails has length less than 729; this injectivity is not asserted for an arbitrary history. The census of 331 also proves that the local profile alone does not select a single matrix: the labelled marginals and the charge supply additional effective conditions.

Thus, a complete finite enumeration and its selector have been recovered on output cylinders, with an explicit readout profile. Construction of the signed register additionally uses its memory reader and its channels, as specified below; the visible matrix is not identified with that register by equality of the number of components. Nor does this count prove the universality of the profile, generate the regional coordinates anew, or decide a claim about α or about the complete continuum.

9 Dodecaphasic record and rational phase constant

The generation of regional coordinates does not exhaust the information in the APP-TRIT-TPK state. One Archimedean reading determines a value through compatible cylinders; another reading preserves the opposition of the sheets, the orientation of the route, and the record of the incidences traversed. The dodecaphasic record belongs to this second reading before entering the closure of alpha. It is therefore not defined by subtracting a physical constant from a combination of previously known values. It is reconstructed from a signed coordinate of the same arithmetic state with trajectory memory that produces the regional coordinates.

The distinction is necessary even when two objects have twelve coordinates. A twelve-block word from the catalog, a list of twelve oriented events, a signed integer vector, and a base-thousand dodecaphasic record are different objects. Comparing them requires the maps that relate them. Equality of length does not provide those maps, and the loss of signs is not a mere simplification of notation.

This chapter maintains the causal cut fixed in the preceding chapters: the nine APP positions on the horizontal and vertical axes produce the arithmetic sheets; TRIT types regime and orientation; TPK selects, transports, updates, and prolongs, preserving remainder, quotient, carry, sheet, route, boundary, and memory. The coordinates used here are obtained from the joint discrete structure of the continuum. The chain $w_6 \rightarrow w_{12} \rightarrow w_{18} \rightarrow w_{24} \rightarrow w_{30} \rightarrow R_{36} \rightarrow G_9$ is not replaced by a finite calendar. Nor is the order $U_t \rightarrow \Gamma_9 \rightarrow K_{\text{ph}}$ reversed: the last map is a reduced-memory reading, not the generator of the signed record.

Reversible coordinatization and incidence determination. The result organizing this chapter is the equivalence between three coordinatizations of a record previously produced by APP-TRIT-TPK. The Hadamard transformation relates the signed record to the integer record; the difference extractor relates the latter to its transverse variations and its uniform component:

$$U \xrightleftharpoons[\mathcal{H}_{12}]{\mathcal{H}_{12}/4} K \xrightleftharpoons{(D_3, D_4, Q)} (D_3 K, D_4 K, Q(K)).$$

The first equivalence is established on the integral sublattice $\mathcal{L}_H$; the second, on the image of the extractor. With base 1000, the 12 positions, and the phase origin fixed, the evaluation $K \mapsto \kappa_{\text{per}}$ adds a fourth reversible coordinatization: multiplication by $1000^{12} - 1$ followed by Euclidean division recovers the complete record. The proofs in §9.7, §9.8, and §9.10 establish these equivalences in their respective domains.

Two compositions subsequently act on this record. The positional composition combines its components with the closure, propagation, and self-scaling coordinates, and its normalization determines α_A . The incidence composition applies the cubic threshold and yields

$$K \mapsto B_K = \{j : K_j \geq 729\} = \{4, 6, 9, 10\}.$$

The complete record allows both compositions to be reconstructed. The support B_K specifically preserves membership in the class selected by the threshold. The identity $B_K = H_e \cap P_{\pi}$ in (11.3) links this integer extraction to the incidence of the regional codes. The negative support of their balance with K completes the flag $B_K \subset H_{\alpha}^-$ developed in the exceptional extension. Thus, the function of K is defined through an invertible transformation and precise subsequent maps; the additional information on depth and transport remains in the state from which the record originates.

9.1 Cyclic extension and memory grading

The observable calendar contains one hundred eight positions, organized into twelve nonadic windows. With positive indices, let

$$E_m = \{9(m-1) + 1, \dots, 9m\}, \quad 1 \leq m \leq 12.$$

The ordered support $E_1 \sqcup \cdots \sqcup E_{12}$ preserves position and window membership. The group $C_{108} = \mathbb{Z}/108\mathbb{Z}$, by contrast, preserves the law of cyclic translation. For the representative $0 \leq \tilde{\theta} < 108$, the coordinates

$$\beta(\theta) = \left[\left[\frac{\tilde{\theta}}{9} \right] \right]_{12}, \quad r(\theta) = 1 + (\tilde{\theta} \bmod 9)$$

separate the dodecaphasic sector and the interior nonadic position. Neither by itself preserves the total number of turns.

Proposition 9.1 (Calendar carry). *The maps*

$$\iota : C_{12} \longrightarrow C_{108}, \quad [b] \longmapsto [9b], \quad p : C_{108} \longrightarrow C_9, \quad [\theta] \longmapsto [\theta]_9$$

form a nonsplit exact sequence

$$0 \longrightarrow C_{12} \xrightarrow{\iota} C_{108} \xrightarrow{p} C_9 \longrightarrow 0.$$

For the set-theoretic section choosing the representatives $0, \dots, 8$, the defect of additivity is the carry

$$c(r, r') = \left[\left[\frac{\tilde{r} + \tilde{r}'}{9} \right] \right]_{12}.$$

Proof. The kernel of p consists of the multiples of nine and equals the image of ι . Reduction is surjective, and multiplication by nine is injective on the indicated domain. If the sequence split, the middle group would be isomorphic to $C_{12} \times C_9$, whose exponent is $\text{lcm}(12, 9) = 36$; the exponent of C_{108} is 108. This contradiction proves nonsplitting. Finally, Euclidean division of $\tilde{r} + \tilde{r}'$ by nine gives the displayed carry and verifies the section identity. $\square$

The result explains a difference that will recur in alpha: returning to the phase is not equivalent to returning to the state. If q_{mem} counts nonadic turns, the calendar retains only $\beta = [q_{\text{mem}}]_{12}$. After one hundred eight steps, β returns, but q_{mem} increases by twelve. The record also preserves the cylinder, the incidence ledger, and vertical memory. Closing a finite chart does not turn that history into a memoryless cycle.

9.2 The oriented record and the integral 1 + 5 + 6 chart

For a history produced by TPK, let τ_m be the subroute over E_m , σ_m its orientation, κ_m the boundary carry, and Λ_m the local incidence ledger. The record is

$$\mathcal{R}_{12}(x) = ((E_m, \tau_m, \sigma_m, \kappa_m, \Lambda_m))_{m=1}^{12}.$$

Its domain is the space of arithmetic histories with trajectory memory, not the set of phase labels. Projection onto the calendar forgets precisely some of the data required by the signed reader.

An initial chart of the record can be written through oriented events. The digital reader fixes the set of event codes $\{2, 4, 6, 8\}$ and the sector orientation

$$\Sigma_{12} = (+, +, +, -, -, -, +, +, +, -, -, -).$$

If d_t is the symbol of the digital record at position t , one obtains

$$e_i = \Sigma_{12}(i+1) \# \{t \in E_{i+1} : d_t \in \{2, 4, 6, 8\}\}, \quad 0 \leq i < 12.$$

The formula keeps the nonnegative event count separate from orientation. The digital symbol, cardinal direction, and TRIT sign retain their respective domains. The vector e , whose components are signed local counts, is not yet the vector U appearing below either.

Opposite positions determine, for $0 \leq i < 6$,

$$p_i = e_i + e_{i+6}, \quad a_i = e_i - e_{i+6}.$$

With

$$s_C = \sum_{i=0}^5 p_i, \quad M_i = p_i - p_5 \quad (0 \leq i < 5),$$

the integral chart is defined by

$$\mathcal{L}_{12}(e) = (s_C, M_0, \dots, M_4, a_0, \dots, a_5).$$

The 1+5+6 decomposition is not an arbitrary allocation of coordinates: one coordinate preserves charge, five compare pairs relative to a reference pair, and six preserve the opposition of the semicoronas.

Lemma 9.2 (Inversion of the integral chart). *An integer tuple $(s_C, M_0, \dots, M_4, a_0, \dots, a_5)$ belongs to the image of $\mathcal{L}_{12}$ if and only if*

$$s_C - \sum_{i=0}^4 M_i \equiv 0 \pmod{6}, \quad p_i \equiv a_i \pmod{2} \quad (0 \leq i < 6),$$

where

$$p_5 = \frac{s_C - \sum_{i=0}^4 M_i}{6}, \quad p_i = p_5 + M_i \quad (i < 5).$$

In that case its preimage is unique and is given by

$$e_i = \frac{p_i + a_i}{2}, \quad e_{i+6} = \frac{p_i - a_i}{2}.$$

Proof. The sum of the five margins is $s_C - 6p_5$, yielding division by six. Once the p_i are reconstructed, each pair of equations $p_i = e_i + e_{i+6}$, $a_i = e_i - e_{i+6}$ has the indicated solution. This solution is integral exactly under the parity congruence. All operations are inverted, so no residual freedom remains. $\square$

This reversibility shows what information is preserved; it does not authorize removing the rest of the ledger. The full record has a larger domain than this count chart. Extracting the channels feeding U uses that full record and does not follow from the list (e_i) through an identification of names.

9.3 Incidence evaluation preceding the dodecaphasic record

Recovering a record from its cyclic differences and total charge is distinct from evaluating those coordinates on the incidence ledger. The first problem is solved through the transverse extractor of Proposition 9.8. For the second, a historical presentation of the record provides three integer counts in each window. It is useful to make their composition explicit and determine exactly the information their evaluation requires.

Let $\mathcal{L}$ be a previously generated ledger of visits and oriented traces. Each visit preserves the route, APP position, sheet, instant, multiplicity, and boundary incidence. Arithmetic evaluation produces the integer and its decomposition

$$n_t = r_t + 9q_t, \quad 1 \leq r_t \leq 9.$$

The integer n_t is computed from the sum or product sheet; its remainder is not received as an independent value. The distinction between corona incidence and interior incidence remains explicit for visits with remainder 9.

In window E_m , the count presentation considers

$$A_m = \sum_{v \in E_m} w(v) \mathbf{1}_{\{1,3,5,7\}}(r(v)), \quad (9.1)$$

$$C_m = \sum_{j \geq 0} (\nu_{120,m}^-(j) - \nu_{120,m}^+(j)), \quad (9.2)$$

$$V_m = \sum_{v \in E_m} w(v) \mathbf{1}_{\{9\}}(r(v)) \epsilon_{\partial}(v). \quad (9.3)$$

Here $w(v)$ is the incidence multiplicity, and $\varepsilon_\partial(v) = 1$ on the corona, -1 in the interior. The traces counted by $v^\pm$ must be enumerated through the corresponding ledger rule. The integer sum requires finite support or an alternative construction determining its meaning. The number of chronological incidences and the reduced length $120(1 + 3j)$ have different types; their relation requires the reader's length unit.

Define $[x]_{10} \in \{0, \dots, 9\}$ as the Euclidean representative and retain the quotients

$$x = [x]_{10} + 10\lfloor x/10 \rfloor.$$

The incidence block is

$$z_m = 100[A_m]_{10} + 10[C_m]_{10} + [V_m]_{10}. \quad (9.4)$$

Composition yields the map

$$\mathcal{E}_{\text{cnt}}(\mathcal{L}) = \left((z_m - z_{m+3})_{m=1}^{12}, (z_m - z_{m+4})_{m=1}^{12}, \sum_{m=1}^{12} z_m \right). \quad (9.5)$$

Indices are read cyclically modulo 12. In this subsection, $(D_j z)_m = z_m - z_{m+j}$, for $j = 3, 4$, and $Q(z) = \sum_{m=1}^{12} z_m$; thus the preceding composition is $(D_3 z, D_4 z, Q(z))$. This convention is retained in §9.8, where complete integral reconstruction is proved. This formula produces its coordinates from the counts. It does not use K , U , α , or the expected transverse coordinates as arguments. Its identification with $\mathcal{E}_{108}^{90,120}$ requires composition with the effective collection and incidences of the terminal state; it does not follow solely from invertibility of the transverse system.

Proposition 9.3 (Sufficient and necessary arithmetic information). *Let $N, N' \in \mathbb{Z}^{12 \times 3}$, ordered by the counts (A, C, V) . For the map defined by (9.4)–(9.5),*

$$\mathcal{E}_{\text{cnt}}(N) = \mathcal{E}_{\text{cnt}}(N') \iff N - N' \in 10\mathbb{Z}^{36}.$$

In particular, the 36 sector residues determine exactly the 25-coordinate output. Preservation of quotients and provenance constitutes additional information in the record.

Proof. Equality modulo 10 produces the same blocks and therefore the same transverse coordinates. Conversely, if the transverse coordinates agree, $D_3(z - z') = D_4(z - z') = 0$. Steps 3 and 4 generate the cycle of length 12, so $z - z'$ is constant. Equality of total charges annihilates that constant. Finally, the decimal representation of an integer between 0 and 999 has three unique digits. Thus the three residues of every window are recovered. The statement is an algebraic identity for all N, N' , not an inference obtained from finite tests. $\square$

9.4 Conjugation of counts and boundary terms

Let $\rho(m) = 13 - m$. Consider an incidence conjugation that reverses the window order, preserves the arithmetic and boundary classes of visits, and exchanges the trace channels. Then

$$(A_m^\dagger, C_m^\dagger, V_m^\dagger) = (A_{\rho(m)}, -C_{\rho(m)}, V_{\rho(m)}).$$

If $c_m = [C_m]_{10}$, decimal reduction produces the identity

$$z_m^\dagger = z_{\rho(m)} + 100\mathbf{1}_{\{c_{\rho(m)} \neq 0\}} - 20c_{\rho(m)}. \quad (9.6)$$

Indeed, $[-C]_{10} = 0$ when $[C]_{10} = 0$, and $[-C]_{10} = 10 - [C]_{10}$ otherwise. The formula preserves exactly this difference. Negating the channel is not equivalent to negating the entire decimal block.

Applying this identity to a concrete TPK involution requires checking its action on incidences. For an occupation reader evaluated before advancement, reversing a route $\gamma = (x_0, \dots, x_n)$ satisfies

$$\sum_{k=0}^{n-1} f(x_{n-k}) = \sum_{k=0}^{n-1} f(x_k) + f(x_n) - f(x_0).$$

Endpoint terms are incorporated before reduction modulo 10. On closed routes they vanish; in open windows they can alter the residues. This correction, already present in the bidirectional transport development, specifies what the connection between route and counts must preserve.

Proposition 9.4 (Necessity of the record's nonperiodic information). *Suppose the observable of a fixed route configuration satisfies $o(t + 54) = o(t)$, multiplicity is preserved under that return, and the same local reader acts in windows of 9 instants. Then its counts and blocks satisfy*

$$(A, C, V)_{m+6} = (A, C, V)_m, \quad z_{m+6} = z_m.$$

Proof. Translation $t \mapsto t + 54$ bijectively sends E_m to E_{m+6} , preserving each counted incidence and its multiplicity. Equality passes to the sum and decimal reduction. $\square$

Thus a presentation with 12 distinct blocks requires the reader to preserve some information that does not factor through that period-54 observable: incidence selection, effective orientation, endpoints, multiplicity, or memory. The proposition delimits an inadmissible state reduction; it does not assert that the full dynamics has period 54 or itself determine its terminal reader.

9.5 Two readings of the terminal state

Denote by x_{term} the state produced by the composition of selection, transport, and update up to the terminal cut. The operator expression is

$$x_{\text{term}} = (\mathcal{U}_{108} \circ \cdots \circ \mathcal{U}_1)(x_{\text{seed}}), \quad \mathcal{U}_t = \text{Upd}_t \circ \text{Tra}_t \circ \text{Sel}_t.$$

Its two relevant readings are

$$x_{\text{term}} \xrightarrow{(\pi_{\text{term}}, R_{\text{sgn}})} (B_{\text{term}}, U).$$

The matrix B_{term} retains the visible ternary reading; the vector U preserves another coordinate of the same history. No arrow $B_{\text{term}} \rightarrow U$ is inserted between them.

The terminal selector defined by the composition $\mathcal{U}_{108} \circ \cdots \circ \mathcal{U}_1$ and evaluated by the incidence ledger of Subsection 9.3 operates on catalogs of sizes 196, 197, and 196. The local signatures reduce their $196 \cdot 197 \cdot 196 = 7\,567\,952$ triples to 331, and the column margin $q_{\partial} = (1, 2, 2, 1, 2, 0)$ leaves two matrices:

$$B_0 = \begin{pmatrix} 0 & 2 & 0 & 2 & 2 & 0 \\ 0 & 0 & 1 & 2 & 2 & 0 \\ 1 & 0 & 1 & 0 & 1 & 0 \end{pmatrix}, \quad B_1 = \begin{pmatrix} 0 & 2 & 1 & 0 & 2 & 0 \\ 0 & 0 & 1 & 2 & 2 & 0 \\ 1 & 0 & 0 & 2 & 1 & 0 \end{pmatrix}.$$

The row orientation is $\sigma = (1, 1, -1) = (1, 1, 2)$ in $\mathbb{F}_3^3$. The row charges of the candidates are $(0, 2, 0)$ and $(2, 2, 1)$. Only the second belongs to $\langle \sigma \rangle = \{(0, 0, 0), (1, 1, 2), (2, 2, 1)\}$; hence $B_{\text{term}} = B_1$. This check proves the final step of the selector from the declared census. It does not make the column margin a consequence of the row charge, nor replace the preceding enumeration with inspection of these two matrices.

The signed reader has the typed composition

$$R_{\text{sgn}} = \Pi_H \circ \mathcal{E}_{108}^{90,120} \circ \mathcal{R}_{12} : \mathcal{X}_{\text{TPK}}^{\text{term,mem}} \longrightarrow \mathbb{Z}^{12}. \quad (9.7)$$

The channel record determines

$$\mathcal{E}_{108}^{90,120}(\mathcal{R}_{12}(x_{\text{term}})) = (b^{90}, b^{120}, Q_{\text{TPK}}),$$

with coordinates

$$\begin{aligned} b^{90} &= (-495, -116, -684, 108, 601, -90, \\ &\quad -173, -88, 313, 560, -397, 461), \\ b^{120} &= (-425, -281, -481, 671, -255, 30, \\ &\quad 475, -543, 680, 251, 6, -128), \\ Q_{\text{TPK}} &= 6263. \end{aligned}$$

These quantities are used as coordinates of the preceding record, not as parameters fitted through alpha. The reconstruction proved in the following subsections starts exactly at this typed output. The preceding evaluation is developed in §9.3: visits and traces determine the counts, which produce the blocks through (9.4) and the differences and charge through (9.5). Proposition 9.3 exactly characterizes the arithmetic information in that evaluation; subsection 9.4 preserves its boundary terms under conjugation. Identification of the effective incidence collection of this terminal state has the scope specified there. The terminal domain, the incidence evaluation, the boundary conjugation, and the arithmetic characterization producing the typed output are thus identified within the article. The harmonic reconstruction below begins with that output and preserves the preceding evaluation.

9.6 From the channel record to the signed vector

The harmonic reader Π_H can be written without using either K or alpha. To avoid confusing its orbital sums with the alpha blocks, we denote them by s_1, s_2, s_3 . Let

$$B_r = \sum_{j=0}^3 b_{r+3j}^{120}, \quad d_{r,j} = b_{r+3j}^{90}, \quad 1 \leq r \leq 3,$$

and

$$s_1 = \frac{Q_{\text{TPK}} + 2B_1 + B_2}{3}, \quad s_2 = s_1 - B_1, \quad s_3 = s_2 - B_2. \quad (9.8)$$

Division by three belongs to the three-orbit reader. Its integrality is checked on compatible record outputs; it is not assumed for an arbitrary element of $\mathbb{Z}^{25}$.

The three signed blocks are

$$U^{(r)} = (s_r, d_{r,1} - d_{r,3}, d_{r,0} + d_{r,2}, d_{r,0} - d_{r,2}). \quad (9.9)$$

Integer substitution gives

$$(B_1, B_2, B_3) = (972, -1073, 101), \quad (s_1, s_2, s_3) = (2378, 1406, 2479),$$

and therefore

$$\begin{aligned} U^{(1)} &= (2378, -452, -668, -322), \\ U^{(2)} &= (1406, 998, -204, -28), \\ U^{(3)} &= (2479, -551, -371, -997). \end{aligned}$$

Interleaving by columns yields

$$U = (2378, 1406, 2479, -452, 998, -551, -668, -204, -371, -322, -28, -997). \quad (9.10)$$

The presence of negative values and components greater than 999 shows that U is not a base-thousand digit word. Nor is it $U_6 \parallel U_6$, a catalog concatenation of period six. The difference concerns domain and preserved information, not just values.

The constructive order so far is

$$x_{\text{term}} \longrightarrow \mathcal{R}_{12} \longrightarrow (b^{90}, b^{120}, Q_{\text{TPK}}) \longrightarrow U.$$

The inversion below produces the dodecaphasic record. The subsequent ability to recover the channels from the dodecaphasic record is a reversibility check; it does not reverse this causal order.

9.7 Hadamard transformation by orbits and integer reconstruction

On the twelve-position cycle, step three determines three orbits:

$$(1, 4, 7, 10), \quad (2, 5, 8, 11), \quad (3, 6, 9, 12).$$

The separation into four sectors of each orbit is what, once origin and orientation are fixed, admits the angular reading of ninety degrees. The transformation does not act on three consecutive four-element sets in the natural order.

Let

$$H_4 = \begin{pmatrix} 1 & 1 & 1 & 1 \\ 1 & 1 & -1 & -1 \\ 1 & -1 & 1 & -1 \\ 1 & -1 & -1 & 1 \end{pmatrix}.$$

If P_{orb} reorders according to the three indicated orbits, define

$$\mathcal{H}_{12} = P_{\text{orb}}^{-1}(H_4 \oplus H_4 \oplus H_4)P_{\text{orb}}.$$

Lemma 9.5 (Hadamard inversion and integrality). *We have*

$$H_4^2 = 4I_4, \quad H_4^{-1} = \frac{1}{4}H_4, \quad \det H_4 = -16.$$

For $u \in \mathbb{Z}^4$, the preimage $H_4^{-1}u$ is integral if and only if $H_4u \equiv 0 \pmod{4}$.

Proof. The rows are orthogonal and have squared norm four; the matrix is symmetric. This gives $H_4^2 = 4I_4$ and the inverse. Integer elimination, or expansion of the determinant, gives the negative sign and value -16 . The integrality condition is exactly divisibility by four of all four coordinates of H_4u . $\square$

Definition 9.6 (Integral dodecaphasic transformation). The sublattice of signed records and its integral coordinatization are

$$\mathcal{L}_H = \{u \in \mathbb{Z}^{12} : \mathcal{H}_{12}u \equiv 0 \pmod{4}\}, \quad \mathcal{K}(u) = \frac{1}{4}\mathcal{H}_{12}u.$$

The map $\mathcal{K} : \mathcal{L}_H \rightarrow \mathbb{Z}^{12}$ is an isomorphism of abelian groups, with inverse $k \mapsto \mathcal{H}_{12}k$. The canonical record K is the image of the signed record generated by the preceding operations, with phase origin and orientation fixed.

Proof. The congruence defines exactly the integrality of $\mathcal{K}(u)$. The identity $\mathcal{H}_{12}^2 = 4I$ proves the two inverse compositions. In particular, every $k \in \mathbb{Z}^{12}$ has preimage $\mathcal{H}_{12}k \in \mathcal{L}_H$. $\square$

Proposition 9.7 (Canonical integral record). *The inversion $K^{(r)} = H_4U^{(r)}/4$ of the blocks in (9.9) produces*

$$\begin{aligned} K^{(1)} &= (234, 729, 621, 794), \\ K^{(2)} &= (543, 659, 58, 146), \\ K^{(3)} &= (140, 824, 914, 601). \end{aligned}$$

In the natural order, the dodecaphasic record is

$$K = (234, 543, 140, 729, 659, 824, 621, 058, 914, 794, 146, 601). \quad (9.11)$$

It is the unique rational preimage of U , and belongs to $\{0, \dots, 999\}^{12}$.

Proof. The three products before division are

$$H_4 U^{(1)} = (936, 2916, 2484, 3176),$$

$$H_4 U^{(2)} = (2172, 2636, 232, 584),$$

$$H_4 U^{(3)} = (560, 3296, 3656, 2404).$$

All coordinates are divisible by four. Dividing and undoing the orbit permutation gives (9.11). Uniqueness follows from rational invertibility, not from a search among candidates close to a physical value. $\square$

Integrality can also be formulated in lattice terms. The determinantal divisors of H_4 are 1, 2, 4, 16: the entries have greatest common divisor one; the order-two minors are even, with at least one of absolute value two; the order-three minors are multiples of four, with at least one of absolute value four. Hence

$$\text{SNF}(H_4) = \text{diag}(1, 2, 2, 4),$$

and for the global transformation,

$$\text{coker } \mathcal{H}_{12} \cong (\mathbb{Z}/2\mathbb{Z})^6 \oplus (\mathbb{Z}/4\mathbb{Z})^3.$$

The index of the image is $16^3 = 4096$. Thus not every signed integer vector admits an integer preimage; the vector obtained satisfies the precise congruences required by inversion. The factor $1/4$ is forced by the matrix and does not have the status of a free coefficient.

9.8 Cyclic differences and the uniform component

With cyclic indices modulo twelve, let

$$(D_3 z)_m = z_m - z_{m+3}, \quad (D_4 z)_m = z_m - z_{m+4}, \quad Q(z) = \sum_{m=1}^{12} z_m.$$

The first operator compares sectors separated by three positions; the second compares sectors separated by four. The channel names 90 and 120 refer here to these separations in the oriented cycle, not to an identification with a physical operator from another domain.

Proposition 9.8 (Completeness of the transverse system). *The operator*

$$\mathcal{D} = (D_3, D_4, Q) : \mathbb{Z}^{12} \longrightarrow \mathbb{Z}^{25}$$

has rank twelve and nonzero invariant factors

$$\text{diag}(1, \dots, 1, 12),$$

with eleven units. In particular, its outputs determine a unique dodecaphasic record.

Proof. If $D_3 z = D_4 z = 0$, the vector is constant along both steps. Since 3 and 4 generate the cycle modulo twelve, $\ker D_3 \cap \ker D_4 = \langle \mathbf{1} \rangle$. Total charge eliminates this last freedom because $Q(\mathbf{1}) = 12$.

For the integral claim, the rows of (D_3, D_4) are oriented incidences of a connected graph. A spanning tree provides eleven unit invariant factors. Every order-twelve minor must contain the row Q . Adding the first eleven columns to the last makes the latter zero in the incidence rows and twelve in the Q row. All maximal minors are divisible by twelve, and the tree produces one of absolute value exactly twelve. The last factor is therefore twelve. $\square$

Checking (9.11) gives

$$D_3 K = b^{90}, \quad D_4 K = b^{120}, \quad Q(K) = 6263.$$

Formulas (9.8) and (9.9) in turn reconstruct U from these differences. This closes a triangle of exact representations of the same object: transverse record, signed vector, and dodecaphasic record. Equality of results does not collapse their codomains into one; it proves that the declared maps recover the relevant information.

9.9 Integral compatibility and determination of the transverse record

The domain of this construction is the representation of the record of an APP-TRIT-TPK history. Evaluation of record events, compatibility of its channels, and inversion of its coordinates are distinguished. The following result gives a complete integral criterion for the second level and an explicit factorization connecting the first to it. The proof uses the record operators and establishes their relations for every element of the indicated integer domains.

9.9.1 Characterization by adjacent increments

The indices belong to $\mathbb{Z}/12\mathbb{Z}$. Let

$$(Sz)_i = z_{i+1}, \quad D_3 = I - S^3, \quad D_4 = I - S^4, \quad Q(z) = \sum_{i=0}^{11} z_i.$$

Retain the transverse extractor

$$\mathcal{D} : \mathbb{Z}^{12} \longrightarrow \mathbb{Z}^{25}, \quad k \longmapsto (D_3k, D_4k, Q(k)).$$

For a triple of integer coordinates (b, c, q) , write

$$w_i = c_i - b_{i+1}, \quad P_0 = 0, \quad P_i = \sum_{j=0}^{i-1} w_j \quad (1 \leq i \leq 12), \quad T(w) = \sum_{i=0}^{11} P_i. \quad (9.12)$$

Theorem 9.9 (Integral image of the transverse extractor). *A triple $(b, c, q) \in \mathbb{Z}^{25}$ belongs to $\text{im } \mathcal{D}$ if and only if it satisfies*

$$b_i = w_i + w_{i+1} + w_{i+2} \quad (0 \leq i < 12), \quad (9.13)$$

$$\sum_{i=0}^{11} w_i = 0, \quad (9.14)$$

$$q + T(w) \equiv 0 \pmod{12}. \quad (9.15)$$

Its unique integer preimage is

$$k_i = \frac{q + T(w)}{12} - P_i, \quad 0 \leq i < 12. \quad (9.16)$$

The twelve equations of (9.13) and equation (9.14) are linearly independent over $\mathbb{Q}$.

Proof. For an output of $\mathcal{D}$, one has

$$c_i - b_{i+1} = (k_i - k_{i+4}) - (k_{i+1} - k_{i+4}) = k_i - k_{i+1}.$$

The sum of three increments gives b_i ; their full cyclic sum is zero. Moreover, $P_i = k_0 - k_i$, so $q = 12k_0 - T(w)$. All conditions and the inverse formula follow.

Conversely, (9.15) makes formula (9.16) integral; (9.14) allows its cyclic prolongation. Its adjacent differences are w_i . Thus, $D_3k = b$ and

$$(D_4k)_i = w_i + w_{i+1} + w_{i+2} + w_{i+3} = w_i + b_{i+1} = c_i.$$

Summing the inverse gives $Q(k) = q$. The same formula proves uniqueness. For independence, the invertible integer change $(b, c) \mapsto (b, w = c - Sb)$ turns the first twelve relations into $b = (I + S + S^2)w$. Each solves for a different coordinate of b . The condition $\sum w_i = 0$ is an additional independent relation. $\square$

The rational image has dimension twelve. Its integral structure adds a congruence of order twelve: this is the constructive manifestation of the final Smith factor of $\mathcal{D}$. The eleven increments $w_0, \dots, w_{10}$ and the charge q are sufficient coordinates, with $w_{11} = -\sum_{i=0}^{10} w_i$ and congruence (9.15). The original twenty-five coordinates preserve these same twelve degrees of freedom through thirteen relations.

9.9.2 Commutativity with the harmonic reading

Let $\mathcal{H}_{12}$ be the record's Hadamard transformation, with orbits $(r, r+3, r+6, r+9)$ for $r = 0, 1, 2$. Its block is

$$H_4 = \begin{pmatrix} 1 & 1 & 1 & 1 \\ 1 & 1 & -1 & -1 \\ 1 & -1 & 1 & -1 \\ 1 & -1 & -1 & 1 \end{pmatrix}, \quad H_4^2 = 4I_4.$$

For compatible (b, c, q) , define

$$B_r = \sum_{j=0}^3 c_{r+3j}, \quad A_0 = \frac{q + 2B_0 + B_1}{3}, \quad A_1 = A_0 - B_0, \quad A_2 = A_1 - B_1.$$

The reader Π_H of (9.9) has blocks

$$(\Pi_H(b, c, q))^{(r)} = (A_r, b_{r+3} - b_{r+9}, b_r + b_{r+6}, b_r - b_{r+6}).$$

Proposition 9.10 (Agreement of the two reconstructions). *On $\text{im } \mathcal{D}$,*

$$\Pi_H \mathcal{D} = \mathcal{H}_{12}, \quad \mathcal{D}^{-1} = \frac{1}{4} \mathcal{H}_{12} \Pi_H.$$

The image of Π_H restricted to that domain is exactly

$$\mathcal{L}_H = \{u \in \mathbb{Z}^{12} : \mathcal{H}_{12}u \equiv 0 \pmod{4}\}.$$

Proof. For $k = \mathcal{D}^{-1}(b, c, q)$, let $a_r = \sum_{j=0}^3 k_{r+3j}$. Shifting four positions sends each orbit to the next, so $B_r = a_r - a_{r+1}$. Together with $q = a_0 + a_1 + a_2$, these identities give $A_r = a_r$. On an orbit, write (x_0, x_1, x_2, x_3) for the coordinates of k and $d_j = x_j - x_{j+1}$. Then

$$d_1 - d_3 = x_0 + x_1 - x_2 - x_3, \quad d_0 + d_2 = x_0 - x_1 + x_2 - x_3, \quad d_0 - d_2 = x_0 - x_1 - x_2 + x_3.$$

These are the remaining three rows of $H_4 k^{(r)}$. Inversion and the integral characterization follow from $\mathcal{H}_{12}^2 = 4I$. $\square$

The condition $\Pi_H(b, c, q) \in \mathcal{L}_H$ controls only the harmonic output. Membership of the twenty-five coordinates in $\text{im } \mathcal{D}$ additionally requires (9.13)–(9.15). For example, adding the vector $e_0 - e_3$ to c preserves the three orbital sums and leaves Π_H unchanged. Starting from $(0, 0, 0)$, one thus obtains a triple with $\Pi_H = 0$ that violates (9.13). This is an exact negative control of channel compatibility, independent of any terminal evaluation.

9.9.3 Determination and transport on the record's orbits

Theorem 9.9 determines k once its channels and charge have been produced. As a condition on outputs not yet individualized, the same image is a group isomorphic to $\mathbb{Z}^{12}$. Fixing only a charge q , the solutions form an affine class of

$$L_0 = \{v \in \mathbb{Z}^{12} : \sum v_i = 0\} = \bigoplus_{i=0}^{10} \mathbb{Z}(e_i - e_{11}).$$

For example, $1001 + e_0$ and $1001 + e_1$ are two different records of charge 1201, with components in $\{0, \dots, 999\}$. Each satisfies every image condition and every Hadamard congruence through its own channels. These witnesses compare compatibility equations; they are not attributed the status of terminal HMT histories.

Proposition 9.11 (Cyclic covariance and stabilizer of the record). *Let ρ be window translation on a record domain $\mathcal{R}$, and S the preceding translation on $\mathbb{Z}^{12}$. On the orbit of $R \in \mathcal{R}$, a covariant reader is determined by a vector*

$$k \in \text{Fix}(S^p),$$

where $p \mid 12$ is the least period of R under ρ . The charge condition $Q(k) = q$ admits integer solutions if and only if $12/p$ divides q ; when they exist, they form an affine class of rank $p - 1$. In particular, on a free orbit, covariance and charge leave eleven degrees of freedom.

Proof. The rule $F(\rho^j R) = S^j k$ is well defined exactly when $S^p k = k$. Such vectors repeat a word of length p exactly $12/p$ times. Their charge is $\frac{12}{p} \sum_{i=0}^{p-1} k_i$. Divisibility and rank follow from this formula. Channel covariance follows from $D_a S = S D_a$ and $Q S = Q$. On $\mathcal{L}_H$, the corresponding action is $\hat{S}_H = \mathcal{H}_{12} S \mathcal{H}_{12}/4$; by Proposition 9.10, both reconstructions intertwine these actions. $\square$

If the domain carries a marked origin, the action transporting that mark as well must be used. The proposition concerns this declared cyclic action; it does not assume periodicity of the enriched history. Naturality under truncations after the 108 cutoff is obtained, for any already defined reader F_{108} , through $F_N = F_{108} \circ \tau_{N,108}$. This naturality preserves the initial evaluation rule; it does not by itself choose one of the possible rules on the first one hundred eight events.

9.9.4 Event evaluation and composition of contributions

The preceding characterization provides a factorization of extraction through two effective readings of the record:

$$\omega : \mathcal{R} \longrightarrow \{w \in \mathbb{Z}^{12} : \sum w_i = 0\}, \quad q_{\text{reg}} : \mathcal{R} \longrightarrow \mathbb{Z}, \quad q_{\text{reg}}(R) + T(\omega(R)) \equiv 0 \pmod{12}. \quad (9.17)$$

These maps must be evaluated on the events, orientation, incidence, and memory produced by APP-TRIT-TPK. Once their actions on these data are given, the composition

$$R \longmapsto (w, q) \longmapsto ((I + S + S^2)w, (I + S + S^2 + S^3)w, q) \xrightarrow{\Pi_H} u \xrightarrow{\mathcal{H}_{12}/4} k$$

is determined and satisfies all inverse checks. The formula $w = c - Sb$ also allows connection to a producer already delivering both channels: it is a check on its output, prior to any comparison with the terminal witness.

Proposition 9.12 (Accumulation of contributions and prospective uniqueness). *Suppose the initial state and each event a_t of the record produce, through previously fixed rules, contributions $(\delta w_t, \delta q_t)$ with*

$$\sum_i \delta w_{t,i} = 0, \quad \delta q_t + T(\delta w_t) \equiv 0 \pmod{12}.$$

With compatible initial conditions $(w^{(0)}, q^{(0)})$, the update

$$w^{(t+1)} = w^{(t)} + \delta w_t, \quad q^{(t+1)} = q^{(t)} + \delta q_t$$

produces a unique record at each stage. Its increment is

$$\delta k_{t,i} = \frac{\delta q_t + T(\delta w_t)}{12} - P_i(\delta w_t).$$

The construction respects concatenation of records, and the reading of a prefix remains unchanged under subsequent prolongations.

Proof. The compatibility conditions are preserved under addition. Formula (9.16) is additive in (w, q) and produces the indicated increment. Induction fixes each state successively. The sum over two segments is the concatenated sum, and a prefix receives exactly the same summands before and after prolonging the history. $\square$

The proposition provides a sufficient criterion, not a requirement of linearity for every TPK reader. If events are transported by nontrivial actions, their contributions can be expressed in the terminal chart through the corresponding affine cocycle. The cocycle equation then preserves the same independence from the partition of the trajectory. In both cases, event rules and the initial state constitute the constructive data that must come from the ledger: choosing them to reproduce an expected vector would replace that provenance with another task.

The result of this development is the exact determination condition (9.17), its proved inversion, and its transport identities. Its scope is the transverse record; the remaining HMT constructions retain their own domains and proofs.

9.10 Two rational readings: κ_{36} and κ_{per}

Once K has been constructed, the base-thousand chart permits two distinct operations. The first reads a single turn; the second prolongs the dodecaphasic record periodically. Let $B = 1000$, and define its concatenation integer

$$N_K = \sum_{m=1}^{12} K_m B^{12-m} = 234543140729659824621058914794146601. \quad (9.18)$$

The leading zeros of a block, such as that of 058, are part of its three-digit representation. They do not change its integer value, but allow the twelve blocks to be concatenated unambiguously.

Definition 9.13 (Finite and periodic readings of the dodecaphasic record). The one-turn reading is

$$\kappa_{36} = \sum_{m=1}^{12} K_m B^{-m} = \frac{N_K}{B^{12}}.$$

The periodic prolongation is

$$K_j^{\text{per}} = K_{1+((j-1) \bmod 12)}, \quad \kappa_{\text{per}} = \sum_{j \geq 1} K_j^{\text{per}} B^{-j}.$$

Proposition 9.14 (Exact value and period of the prolonged dodecaphasic record). *We have*

$$\kappa_{\text{per}} = \frac{N_K}{B^{12} - 1}, \quad \kappa_{\text{per}} - \kappa_{36} = \frac{N_K}{B^{12}(B^{12} - 1)} = B^{-12} \kappa_{\text{per}}. \quad (9.19)$$

The infinite word of the dodecaphasic record has least period twelve in base thousand and least period thirty-six in base ten.

Proof. The sum of the first turn is $N_K B^{-12}$; each new turn multiplies its contribution by B^{-12} . The geometric series yields $N_K/(B^{12} - 1)$, and subtraction gives the second identity.

The twelve components of K are distinct. A proper period of the cycle would identify two of them; the least block period is therefore twelve. The decimal expansion is canonical, since the dodecaphasic record has no tail of 999 blocks. If its least decimal period were d , it would divide 36. Grouping digits in threes would produce a block period $d/\gcd(d, 3)$. No proper divisor of 36 gives twelve through this operation. Hence the least decimal period is 36. $\square$

Both readings are rational, but they are not the same rational number. The first terminates after twelve blocks; the second repeats the turn. Moreover, $0 < \kappa_{\text{per}} - \kappa_{36} < B^{-12}$. Their closeness does not permit substituting one for the other in an exact identity. In particular, the finite alpha closure uses κ_{36} , whereas the sum of the prolonged section uses κ_{per} .

9.11 Reversibility of the rational constant and phase change

The rational constant κ_{per} determines the entire record K when the base, length, and reading origin are retained. Its numerical and structural functions are related by an exact inversion.

Proposition 9.15 (Integral recovery and cyclic transformation). *Let $I(K) = \sum_{j=1}^{12} K_j 1000^{12-j}$. Then*

$$I(K) = (1000^{12} - 1)\kappa_{\text{per}}.$$

The integer expansion of $I(K)$ into twelve base-1000 blocks recovers every K_j , including the leading zeros in each block. If $\rho(K) = (K_2, \dots, K_{12}, K_1)$, then

$$\kappa(\rho K) = 1000\kappa(K) - K_1. \quad (9.20)$$

Proof. The first identity inverts the definition of periodic evaluation. Iterated Euclidean division by 1000, with twelve fixed positions, recovers the vector. Furthermore,

$$I(\rho K) = 1000I(K) - K_1(1000^{12} - 1).$$

Dividing by $1000^{12} - 1$ yields (9.20). $\square$

Recovery therefore composes $\kappa \mapsto K \mapsto U$ and the already constructed transformations D_3K, D_4K, Q . Rational evaluation preserves the signed record and the cyclic differences that depend on it. The total depth and transport operators of a history remain in their own coordinates: the domain of the preceding inversion is the integral dodecaphasic record.

Write $\kappa_j = \kappa(\rho^j K)$, with indices modulo 12. The same identity gives

$$K_{j+1} = 1000\kappa_j - \kappa_{j+1}, \quad \sum_{j=0}^{11} \kappa_j = \frac{\sum_{j=1}^{12} K_j}{999} = \frac{6263}{999}.$$

Phase acts exactly on the constant: its value is fixed in the canonical reading, while its rotations follow an affine law. The orbit sum is invariant under the choice of origin.

The subsequent incidence also has an expression in these coordinates. The cubic threshold determines

$$B_K = \{j + 1 : 1000\kappa_j - \kappa_{j+1} \geq 729\} = \{4, 6, 9, 10\}.$$

The tetrad is recovered from the rational orbit because the latter first recovers the integer components. The signed support associated with α additionally uses its regional records and boundary normalization. The relationship between coordinate and incidence is thus expressed through identifiable operations.

This is an operational definition of the function of K : it provides a reversible integral coordinatization of the oriented record, whose periodic evaluation and incidence functions participate in subsequent constructions. The value, transformation, and domain are preserved together.

9.12 Periodicity and cokernel of the transport operator

The periodicity just proved belongs to the dodecaphasic record. It does not imply periodicity of the regional coordinates, the closure carries, or alpha. It is useful to clarify this separation through a second, purely integral quotient.

Let S be the cyclic shift of twelve coordinates, with fixed orientation, and let $b \geq 2$ be an integer base. The periodic carry operator is $\partial_b = S - bI$. If a periodic identity is written

$$A = P + E - \Phi - K + \partial_b C,$$

then the transformations

$$K \mapsto K + \partial_b h, \quad C \mapsto C + h$$

preserve its right-hand side. This invariance compares representatives of a carry class; it does not authorize changing the dodecaphasic record already selected by the record.

Proposition 9.16 (Quotient of periodic carry). *With the preceding orientation,*

$$\text{coker}(S - bI) \cong \mathbb{Z}/(b^{12} - 1)\mathbb{Z}.$$

Proof. The cyclic module is presented as $\mathbb{Z}[t]/(t^{12} - 1)$. Imposing $S = bI$ is equivalent to imposing $t = b$ and leaves the relation $b^{12} - 1 = 0$. Evaluating the coefficients at b , with the chosen concatenation order, represents the resulting class. $\square$

The denominator $B^{12} - 1$ of κ_{per} and this quotient arise from the same cyclic length, but their objects remain different: one is a rational evaluation and the other a quotient of a carry module. The finite alpha closure will be an oriented chain with a right boundary. Its prolongation will transport a compatible remote boundary, not a carry identified by decree between the first and thirteenth positions.

The next chapter is thus prepared. The dodecaphasic record has been fixed before solving the alpha recurrence; its Hadamard reconstruction is integral and unique; its transverse differences recover the record; and its two rational evaluations have separate domains and uses. None of these facts uses alpha as an input, and none yet determines the arithmetic type of the output of the complete closure.

10 Derivation of the fine-structure constant

In this development, the fine-structure constant receives an arithmetic determination that brings together the three regional coordinates and the integral dodecaphase record. The operation combines their blocks with fixed orientation and transports the quotients of positional normalization. The resulting parameter expresses compatibility of that oriented difference with cylinder prolongation. Its value and the incidence data used in its construction will remain related throughout the exceptional development.

The internal order is important. The coordinates π , e , and φ , already constructed in the preceding section, are prior coordinates of the same arithmetic state with trajectory memory and may legitimately act as inputs to a subsequent reader. They are not conventional constants injected to select alpha. Correlated generation of the four coordinates shares origin and coinductive compatibility without requiring that all four components be evaluated in a single step.

The two representations act in distinct operator codomains. The first realizes an integer closure by Euclidean division in base 1000, preserves the tail, and produces the complete section α_{HMT} . The analytic reading has two strata that must be distinguished. Its ninth-order jet, constructed by the vacancy resolvent $E_{21/22}$, provides a finite characterization, isolates a simple root, and determines an exact involution on that fibre. The subsequent analytic completion is a correlated representation of the same state: it calculates all its coefficients from the regional and terminal coordinates already expressed, proves uniform convergence, and constructs rational intervals compatible with the integer cylinders. It is not presented as an independent selector of alpha. The final section proves equality of the two representations and explicitly states their common dependency.

Nonuniqueness of a tail chosen from the bare jet prevents inferring the completion from the nine isolated coefficients. The complete rule displayed below removes that freedom once the enriched state is fixed; it introduces no conventional constant and does not alter the causal order of the dodecaphase generation of alpha. These results remain within the complete HMT state on which the continuum is constructed and the cardinal proof is formulated; their conventional digits are not used as retrospective selectors.

10.1 First route: positional determination through integer closure

10.1.1 Domain and orientation of integer closure

Let $P_j, E_j, \Phi_j \in \{0, \dots, 999\}$ be the canonical blocks of the three regional coordinates, and prolong the dodecaphase record by

$$K_j = K_{1+((j-1) \bmod 12)}.$$

The channel orientation is $(1, 1, -1)$. The contribution of the dodecaphase record is subtracted according to the already typed dodecaphase closure. For $B = 1000$, the local operation distinguishes the signed balance

$$Z_j := P_j + E_j - \Phi_j - K_j$$

from the sum incorporating the incoming quotient. Normalization is

$$s_j = Z_j + c_{j+1}, \quad A_j = s_j - B \left\lfloor \frac{s_j}{B} \right\rfloor, \quad c_j = \left\lfloor \frac{s_j}{B} \right\rfloor. \quad (10.1)$$

Thus $A_j \in \{0, \dots, B-1\}$, even when s_j is negative. The carry entering from the right belongs to the domain of the operation; it cannot be omitted because the table is printed from left to right.

Definition 10.1 (Finite-depth closure). For a depth N and an integer boundary t , the map $\mathcal{C}_N$ receives $(P_{1:N}, E_{1:N}, \Phi_{1:N}, K_{1:N}, c_{N+1} = t)$ and applies (10.1) in the order $N, N-1, \dots, 1$. Its output is the pair of words

$$(A_{1:N}, c_{1:N+1}).$$

Euclidean division has no residual sign choice: the remainder is fixed in $\{0, \dots, 999\}$, and the quotient is determined. For example, $-431 = 569 - 1000$ and $-1200 = 800 - 2 \cdot 1000$. The negative carries of closure are not anomalies; they are the integer information needed to maintain the oriented equality.

10.1.2 Dodecaphase window and carry conditions

The first twelve triads obtained are

$$\begin{aligned} P &= (141, 592, 653, 589, 793, 238, 462, 643, 383, 279, 502, 884), \\ E &= (718, 281, 828, 459, 045, 235, 360, 287, 471, 352, 662, 497), \\ \Phi &= (618, 033, 988, 749, 894, 848, 204, 586, 834, 365, 638, 117). \end{aligned}$$

In the closure channel, the five regions determining coordinate P preserve their distinct antecedents. Sharing this reading does not identify them as regions or histories.

With boundary $c_{13} = 0$, the recurrence produces the following table. Each input, incoming carry, total before normalization, obtained block, and outgoing carry is shown. The table is a reproducible integer computation, not a list of blocks selected for their resemblance to a target.

j	P_j	E_j	Φ_j	K_j	c_{j+1}	s_j	A_j	c_j
1	141	718	618	234	0	7	007	0
2	592	281	033	543	0	297	297	0
3	653	828	988	140	-1	352	352	0
4	589	459	749	729	-1	-431	569	-1
5	793	045	894	659	-2	-717	283	-1
6	238	235	848	824	-1	-1200	800	-2
7	462	360	204	621	0	-3	997	-1
8	643	287	586	058	-1	285	285	0
9	383	471	834	914	-1	-895	105	-1
10	279	352	365	794	0	-528	472	-1
11	502	662	638	146	0	380	380	0
12	884	497	117	601	0	663	663	0

Table 8: Initial closure with zero right boundary. The dependency propagates from the last row to the first.

The carry word is

$$(c_1, \dots, c_{13}) = (0, 0, 0, -1, -1, -2, -1, 0, -1, -1, 0, 0, 0),$$

and the window reading is the rational number

$$\alpha_{12}^{(0)} = \frac{7297352569283800997285105472380663}{10^{36}} = 0.007297352569283800997285105472380663. \quad (10.2)$$

The superscript recalls the chosen boundary. This window is exact in its finite domain; it is not simply identified with the truncation of the infinite section. The difference between the two readings will become visible when the carry of the remote tail is transported.

10.1.3 Telescoping and reversibility in a boundary fiber

The local form of the equality is

$$P_j + E_j - \Phi_j - K_j - A_j = Bc_j - c_{j+1}. \quad (10.3)$$

The carry term is a link between adjacent positions. Eliminating it coordinate by coordinate gives a false equality. Only an evaluation preserving its endpoints allows it to telescope.

Proposition 10.2 (Integer identity at arbitrary depth). *For $\iota_N(z) = \sum_{j=1}^N z_j B^{N-j}$, every execution of $\mathcal{C}_N$ satisfies*

$$\iota_N(P) + \iota_N(E) - \iota_N(\Phi) - \iota_N(K) - \iota_N(A) = B^N c_1 - c_{N+1}. \quad (10.4)$$

Proof. Multiplying (10.3) by B^{N-j} and summing produces on the right-hand side

$$\sum_{j=1}^N (Bc_j - c_{j+1})B^{N-j} = B^N c_1 - c_{N+1}.$$

Each interior carry appears twice with opposite signs; only the two endpoints survive. $\square$

For Table 8, both endpoints are zero. If p_{12}, e_{12}, f_{12} are the three truncated fractional sums,

$$\alpha_{12}^{(0)} = p_{12} + e_{12} - f_{12} - \kappa_{36}.$$

The dodecaphase-record constant appearing here is κ_{36} , not κ_{per} . The domain of this identity contains truncations and a finite boundary; its form does not authorize erasing both conditions when passing to the complete section.

Local inversion is exact as well:

$$K_j = P_j + E_j - \Phi_j + c_{j+1} - A_j - Bc_j.$$

Given P, E, Φ , blocks A , and the complete carries, it recovers the dodecaphase record. Concatenation additionally recovers

$$\iota_N(K) = \iota_N(P) + \iota_N(E) - \iota_N(\Phi) - \iota_N(A) - B^N c_1 + c_{N+1}.$$

This reversibility is a property of the fiber with fixed data and boundaries. It must not be turned into a new causal recipe starting from alpha and manufacturing the record afterward. The direction of construction remains signed record, dodecaphase record, closure, and output.

10.1.4 Prolongation and canonical normalization of the complete section

Prolongation preserves the periodic dodecaphase record and receives the complete regional sequences. Let

$$p = \sum_{j \geq 1} P_j B^{-j}, \quad e_0 = \sum_{j \geq 1} E_j B^{-j}, \quad f = \sum_{j \geq 1} \Phi_j B^{-j}.$$

Each series is a realization of the corresponding internal character. The integer parts already obtained are 3, 2, and 1, so $p = \pi - 3$, $e_0 = e - 2$, and $f = \varphi - 1$.

Proposition 10.3 (Existence and uniqueness of complete normalization). *The series*

$$y = p + e_0 - f - \kappa_{\text{per}} = \sum_{j \geq 1} (P_j + E_j - \Phi_j - K_j) B^{-j} \quad (10.5)$$

converges absolutely. On the domain of the obtained blocks, $0 < y < 10^{-2}$. There is a unique canonical expansion (A_j) of y , with no eventually constant 999 tail, and a unique bounded integer sequence (c_j) satisfying (10.3) with $c_1 = 0$. These carries belong to the invariant set $\mathcal{C} = \{-2, -1, 0, 1, 2\}$.

Proof. The coefficients of the series are bounded in absolute value by $2(B - 1)$, so a geometric series dominates it. If $Z_j = P_j + E_j - \Phi_j - K_j$ and $R_j(Z) = \sum_{r \geq 0} Z_{j+r} B^{-r-1}$, each tail is the difference of two sums of two canonical tails and belongs to $(-2, 2)$. The first coefficient is $Z_1 = 141 + 718 - 618 - 234 = 7$; hence $y = (7 + R_2(Z))/1000 \in (0.005, 0.009)$.

Let (A_j) be the canonical expansion of y . Define

$$c_j = R_j(Z) - R_j(A).$$

Multiplying equality of the two series by B^{j-1} and separating their first $j - 1$ terms gives

$$c_j = \sum_{r=1}^{j-1} (A_r - Z_r) B^{j-1-r} \in \mathbb{Z},$$

and $c_1 = 0$. Furthermore, $R_j(A) \in [0, 1)$, so $-3 < c_j < 2$, placing all carries in $\mathcal{C}$. The tail-shift identities give $Bc_j = Z_j - A_j + c_{j+1}$, which is the required recurrence.

Conversely, any bounded solution with canonical output and $c_1 = 0$ telescopes in the limit to the same series y . The canonical expansion is unique, and once it is fixed, the tail formula determines all carries. Finally, if a finite execution is applied with $c_{j+1} \in \mathcal{C}$, then $-2000 \leq s_j \leq 2000$, so its Euclidean quotient again belongs to $\mathcal{C}$. $\square$

The proposition makes explicit the normalization of the subsequent reader: it does not replace generation of the three sequences with a conventional series introduced as data. Once those sequences have been produced by the coinductive state, their combination and the normalization boundary are determined without consulting alpha.

The finite procedure preserved in the corpus propagates the five boundaries of $\mathcal{C}$ and retains only the prefix common to all of them. If two depths have coalesced before a boundary, the deterministic right-to-left recurrence gives the same blocks to its left. This is compatibility with truncation. A boundary is not chosen because it gives favorable digits, nor is all-depth compatibility inferred from observing a long run.

The initial comparison shows why this precaution matters. Stabilized prolongation gives $c_{13}^\infty = -1$, whereas the isolated window imposed $c_{13} = 0$. Thus,

$$884 + 497 - 117 - 601 - 1 = 662$$

in the last block of the first turn. The first eleven triads do not change, but the prolongation begins

$$(007, 297, 352, 569, 283, 800, 997, 285, 105, 472, 380, 662, 999, \dots).$$

An occurrence of block 999 does not violate the canonical convention: what is excluded is an eventually constant 999 tail. The zero-boundary window ends in 663; the truncation of the complete section ends in 662. Both statements describe different and exact operations. They must not be merged into a single projection notation.

10.1.5 Operative definition and complete additive equation

Definition 10.4 (Alpha coordinate of closure). The integer route determines the coordinate

$$\alpha_A = \sum_{j \geq 1} A_j B^{-j} = p + e_0 - f - \kappa_{\text{per}}, \quad (10.6)$$

where (A_j, c_j) is the compatible canonical normalization of (10.1). In the scalar realization of the HMT coordinates,

$$\boxed{\alpha_A = \pi + e - \varphi - 4 - \kappa_{\text{per}}.} \quad (10.7)$$

The integer four is not calibrated: it is $3 + 2 - 1$, the combination of the integer parts of the three coordinates. Base thousand arises from the cubic chart, and period twelve from the dodecaphase record. The signs are those of the oriented channel and closure, and the remote boundary is that imposed by the compatible section. These are the effective origins of the coefficients of the equation.

At this level, alpha can be described as the scalar coordinate of the closure defect between the three regional realizations and the rational dodecaphase return, computed with compatible carry. The term *defect* denotes a determinate algebraic difference, not an arbitrary loss or an imperfection of the system. This description does not suppress the genealogy: without the regions, signed record, orientation, and carry law, the scalar equality would not be a sufficient exposition of the construction.

Three objects that alpha notation may accompany in subsequent readers must also be kept separate: A is a word of blocks; the negative pre-carry support is obtained from the signs of the totals before normalization; and an exceptional radial vector belongs to an incidence codomain. None is interchangeable with the real value α_A .

Signed balance and sum with incoming quotient. It is useful to distinguish the two consecutive stages of positional division by means of a common notation. The balance before quotient transport is

$$Z_j := P_j + E_j - \Phi_j - K_j^{\text{per}}.$$

The variable s_j in (10.1) denotes the sum that already includes the incoming contribution, so that

$$s_j = Z_j + c_{j+1}, \quad A_j = Z_j + c_{j+1} - 1000c_j.$$

In (11.2), the letter s_j locally denotes the balance before that contribution, namely Z_j in the present notation. The two formulas therefore express the same Euclidean division with differently named stages. The incidence configuration H_α^- uses the signs of the balance Z_j in the initial window; the determination of the digits also uses the linking quotients c_{j+1} and c_j . This distinction jointly preserves the signed support and the normalization of the expansion.

Exact dependence on the truncation boundary. The regional coordinates and the dodecaphasic record originate in the common APP-TRIT-TPK generation. Once produced, their positional composition has an explicit dependence on reading depth. Let $B = 1000$,

$$Z_j = P_j + E_j - \Phi_j - K_j^{\text{per}}, \quad y_N = \sum_{j=1}^N Z_j B^{-j}, \quad R_{N+1}(Z) = \sum_{r \geq 0} Z_{N+1+r} B^{-r-1}.$$

The complete series in (10.5) satisfies exactly

$$\alpha_A = y_N + B^{-N} R_{N+1}(Z), \quad -2 < R_{N+1}(Z) < 2.$$

This identity combines the contribution of the window with that of its continuation, both expressed in the same coordinates.

Proposition 10.5 (Boundary transport and canonical cylinder). *Let $A_j^{(N,t)}$ be the output of finite positional division with $c_{N+1} = t \in \mathcal{C} = \{-2, -1, 0, 1, 2\}$, and let $a_N^{(t)} = \sum_{j=1}^N A_j^{(N,t)} B^{-j}$. For the records in the domain under consideration,*

$$\alpha_A - a_N^{(t)} = B^{-N}(R_{N+1}(Z) - t), \quad |\alpha_A - a_N^{(t)}| < 4B^{-N}. \quad (10.8)$$

The quotient of the complete continuation is $c_{N+1}^\infty = \lfloor R_{N+1}(Z) \rfloor$. With this boundary,

$$0 \leq \alpha_A - a_N^{(c_{N+1}^\infty)} < B^{-N},$$

so that the resulting window is the canonical prefix of order N .

Proof. Telescoping (10.3) gives $a_N^{(t)} = y_N + tB^{-N} - c_1$. Since $Z_1 = 7$, each incoming state belongs to $\mathcal{C}$, and the first dividend lies between 5 and 9, we have $c_1 = 0$. Substituting this equality into the series decomposition proves (10.8). The bound follows from $R_{N+1}(Z) \in (-2, 2)$ and $t \in \mathcal{C}$.

For complete canonical normalization, $c_{N+1}^\infty = R_{N+1}(Z) - R_{N+1}(A)$ and $R_{N+1}(A) \in [0, 1)$. The integrality of the quotient implies $c_{N+1}^\infty = \lfloor R_{N+1}(Z) \rfloor$. The error formula then becomes $\alpha_A - a_N^{(c_{N+1}^\infty)} = B^{-N} R_{N+1}(A)$, proving membership in the canonical cylinder. $\square$

The proposition identifies the meaning of the right boundary: it is the integer part of the signed tail expressed at the scale of the next position. Computation over the five boundaries preserves this information as a finite collection of possibilities until the blocks on the left stabilize. Quantitative contraction follows from depth and tail transport; its object is the normalization of the coordinates already generated.

Graded domain of the analytic resolvent. The analytic composition exhibited in §10.2.2 acts on $\text{span}\{e_7, e_8, e_9\}$ and satisfies $Ne_9 = 0$. Its evaluation yields exactly $T_{7;9}$; degree 9 is the final component of this domain. The identity $N^3 = 0$ explains why the resolvent is expressed as three summands. To describe a continuation to higher degrees, one must additionally specify the transport between successive graded domains and its action on the initial component of each domain. Local nilpotence and global compatibility thus perform different mathematical functions.

10.1.6 Constructive determination and the function of the dodecaphasic record

The preceding definition makes it possible to specify what it means to determine α from the structure. The domain of the operation contains four records of known provenance: the three regional sequences and the integral phase record. Normalization additionally incorporates a boundary condition compatible with continuation. Once these data are fixed, Euclidean division determines each remainder and its quotient. Proposition 10.3 extends this determination to the complete system. Independence from a target physical value is therefore a property of the construction order: the value compared at the end does not participate in selecting any argument of this map.

This causal independence must be distinguished from algebraic independence between numbers. The former is established by identifying the domain, operations, and provenance of the coefficients; the latter would be a statement about polynomial relations. The result used here is the former. The construction determines a coordinate through records produced by APP-TRIT-TPK and subsequently permits verification of the relations it satisfies. This order makes it possible to use an already generated coordinate in a later operation without turning it into an external parameter.

The function of the record K is more precise than that of a numerical correction added to a formula. Its twelve positions originate in the reversible transformation of the signed record. The equation $U = \mathcal{H}_{12}K$ preserves the possibility of returning to that record, while the transverse differences and total sum allow another reconstruction. In periodic evaluation, the order of the positions determines the positional weight of each component. The same set of components, arranged in a different order, generally produces a different rational coordinate. The phase origin and orientation therefore form part of the definition of the number occurring in (10.7).

Alpha normalization composes this information with the regions. Its local identity distinguishes two successive operations. First, the oriented balance of blocks is formed; then that balance is represented by a canonical remainder and a quotient linking adjacent positions. The first operation preserves the sign of the balance and allows a support to be extracted subsequently. The second allows prefixes of the real coordinate to be formed. Both readings remain related because the state preserves the data before and after normalization. The signed support and the positional expansion are thus different realizations of an explicit arithmetic composition.

The telescoping identity expresses the global content of this composition. The interior quotients cancel in pairs when a prefix is evaluated, while the endpoint terms remain in the equality. This cancellation is a property of the weighted sum; it neither removes the quotients from the state nor allows them to be discarded before the sum is performed. The right boundary of a window receives the contribution of the continuation. For this reason, evaluating an isolated window and restricting a continued section are different operations. The change of the final block from 663 to 662, proved above, provides an explicit realization of this difference without altering the already stabilized blocks to its left.

In this sense, bidirectionality has a concrete arithmetic formulation. Generation provides compatible continuations; normalization transmits the integer information at their boundary from positions of lower weight towards those of higher weight. The equation preserves the contribution of both directions through its indices and boundary conditions. Reversibility in the fixed fibre is verified by reconstructing K from the other records and the complete quotients. This inversion certifies the consistency of the original operation; the genealogical direction remains the one that first produces the record and then the alpha coordinate.

The second determination has complementary content. In it, the immediate object is a vacancy balance, and the parameter appears as its root in an isolating domain. The positional route provides a canonical normalization of records; the analytic route studies the vanishing of an operator. Presenting both requires precisely this difference in function to be retained. Their relationships with the common state must be expressed through the refinement maps and the comparison of their complete realizations, as developed in the corresponding subsection. Agreement at a finite approximation constitutes a localized check; the identification of the complete constructions has the domain and scope of its own statement.

Finally, this organization fixes alpha's place in the article. Its equation composes the regional generation of Section 6 with the record K of Section 9. The electronic application in Article I subsequently incorporates α_{HMT} as a factor in its central section; the present article receives the action sections (12.9) from that application and uses them in gravitational normalization. The exceptional prolongation of Section 11, by contrast, receives the record, its supports, and the preserved incidence. Both continuations share a provenance and perform different operations: evaluating a central section and constructing a boundary configuration. Their internal maps keep visible the complete structure that determines the scalar coordinate.

10.2 Analytic reading: finite witness and correlated representation

10.2.1 Transport invariants and graded resolvent

The finite analytic witness receives invariants of the oriented state, not the output α_A . Its sixth-order truncation is

$$P_6(x) = 2\pi x - \frac{7}{4}x^2 + \frac{x^3}{2\pi} + \frac{x^4}{20} - \frac{2x^5}{21} - \frac{x^6}{46}, \quad (10.9)$$

and the vacancy condition uses

$$\delta = \log_{10} \frac{10}{9}.$$

The ratio 10/9 compares the decimal chart with the nine APP positions of the nonadic construction. Its logarithm belongs to this subsequent compactification reading; it is not a target value of alpha.

The genealogy of the coefficients must be preserved in full, but with its exact logical strength. Recent developments bring together a marked signature of the state. The synchronization matrix and associated census are

$$D_{\text{syn}} = \begin{pmatrix} 85 & 112 \\ 41 & 52 \end{pmatrix}, \quad N_9 = 43.$$

From its determinant one extracts

$$a = -\frac{\det D_{\text{syn}}}{N_9} = \frac{172}{43} = 4.$$

The clock induced on the short gaps has secondary spectrum $\{6, 7\}$; its upper endpoint gives $b = 7$. The tritic cardinality $m = 3$ determines

$$t_* = mb = 21, \quad k_* = mb - 1 = 20.$$

The vacancy reading in the thousand chart gives

$$v_- = [1000\delta] = 45, \quad v_+ = [1000\delta] = 46.$$

These integers are typed antecedents of the truncation's evaluation. They are not obtained from its root or from a metrological approximation to it.

The lower branch additionally coincides with the determinant of the APP block

$$B_c = \begin{pmatrix} 7 & 2 \\ 2 & 7 \end{pmatrix}, \quad \det B_c = 45.$$

The transverse rank of the preceding chapter's record gives $r_{\text{dod}} = 11$, and the phase semicrown provides $f_{\text{ph}} = 8\binom{6}{2} = 120$. In particular, $45r_{\text{dod}} = 495$.

With $\omega = 2\pi$, the signature

$$\mathcal{I}_9 = (\omega, m, a, b, k_*, t_*, v_-, v_+, f_{\text{ph}}, r_{\text{dod}})$$

feeds the graded reader

$$\begin{aligned} \mathfrak{G}_9(\mathcal{I}_9)(x) = & \omega x - \frac{b}{a}x^2 + \omega^{-1}x^3 + \frac{x^4}{k_*} - \frac{2x^5}{t_*} - \frac{x^6}{v_+} \\ & - \frac{x^7}{f_{\text{ph}}} + \frac{x^8}{v_-} + \frac{2x^9}{v_-r_{\text{dod}}}. \end{aligned} \quad (10.10)$$

Substitution recovers the coefficients of P_6 and its ninth-order prolongation. The pair 45/46 crosses the separation between the two: the upper branch occupies degree six, and the lower reappears in degrees eight and nine.

The factorization of (10.10) is formulated for a state preserving origin, channel order, and orientation. Naturality with respect to morphisms preserving those data is not equivalent to uniqueness of every imaginable reader after they are forgotten. In the marked domain, the preceding formula fixes each coefficient through $(\omega, m, a, b, k_*, t_*, v_-, v_+, f_{\text{ph}}, r_{\text{dod}})$; therefore two readers that preserve the signature and apply the same graded evaluation agree term by term through degree nine. This is the uniqueness used below and requires no additional documentary reference.

10.2.2 Nilpotent resolvent in degrees 7, 8, and 9

The first prolongation admits an especially explicit finite operator construction. On the space with basis e_7, e_8, e_9 , let

$$Ne_7 = -\frac{8}{3}e_8, \quad Ne_8 = \frac{2}{11}e_9, \quad Ne_9 = 0, \quad v_7 = -\frac{1}{120}e_7.$$

The symbol N here denotes a nilpotent operator, not a truncation depth. Its domain and grading are fixed before the polynomial is evaluated at a number.

Lemma 10.6 (Exact nilpotent prolongation). *One has $N^3 = 0$ and*

$$(I - N)^{-1}v_7 = -\frac{e_7}{120} + \frac{e_8}{45} + \frac{2e_9}{495}.$$

Evaluation $e_n \mapsto x^n$ produces

$$T_{7:9}(x) = -\frac{x^7}{120} + \frac{x^8}{45} + \frac{2x^9}{495}. \quad (10.11)$$

Proof. The operator raises degree and vanishes on the last basis vector, so its cube is zero. The inverse is the finite resolvent $I + N + N^2$. Moreover,

$$Nv_7 = \frac{1}{45}e_8, \quad N^2v_7 = \frac{2}{495}e_9.$$

Summing the three terms and evaluating gives the formula. $\square$

Here the denominators 120, 45, and 495 are not decimal fits: they arise from the semicrown, APP determinant, and its product with the dodecaphase rank. The transfers $-8/3$ and $2/11$, together with the sign and degree of the seed, belong to the typed reader. Preserving only cycle symmetry and replacing this operator by another symmetric operator does not necessarily preserve the coefficients.

10.2.3 Multiplicity-one root of the ninth-order truncation

Define

$$P_9 = P_6 + T_{7:9}, \quad E_{21/22}^{(9)}(x) = P_9(x) - \delta, \quad I_\alpha = (0, 10^{-2}).$$

This finite operator has its own existence and uniqueness result.

To isolate its root by rational arithmetic, let

$$A_n(t) = \sum_{j=0}^n \frac{(-1)^j t^{2j+1}}{2j+1}, \quad L_n(t) = 2 \sum_{j=0}^n \frac{t^{2j+1}}{2j+1} \quad (0 < t < 1).$$

The alternating-series criterion and the geometric sum of the tail give

$$A_{2n+1}(t) < \arctan t < A_{2n}(t), \quad (10.12)$$

$$L_n(t) < \log \frac{1+t}{1-t} < L_n(t) + \frac{2t^{2n+3}}{(2n+3)(1-t^2)}. \quad (10.13)$$

We apply (10.12) to Machin's identity $\pi = 16 \arctan(1/5) - 4 \arctan(1/239)$, and (10.13) to

$$\log(10/9) = \log \frac{1+1/19}{1-1/19}, \quad \log 10 = \log(10/9) + 2 \log \frac{1+1/2}{1-1/2}.$$

We write

$$\delta := \frac{\log(10/9)}{\log 10}.$$

With truncation orders 80, 20, 40, 120, applied respectively to the preceding four series, rational intervals are obtained for π and δ , both of width less than 10^{-70} . No value of alpha is consulted in these bounds.

Proposition 10.7 (Multiplicity-one root of the truncation). *The operator $E_{21/22}^{(9)}$ has a unique root ξ_9 in I_α , and that root is simple. It also has the rational isolation*

$$\frac{729735256928380099}{10^{20}} < \xi_9 < \frac{729735256928380100}{10^{20}}. \quad (10.14)$$

Proof. The internal-coordinate bounds $3.14 < \pi < 3.15$ and $0.045 < \delta < 0.046$ suffice for existence. At zero, $E_{21/22}^{(9)}(0) = -\delta < 0$. At 10^{-2} , the linear contribution minus the negative terms exceeds 0.0626, so the operator value is positive.

For $0 \leq x \leq 10^{-2}$, discarding the positive terms of the derivative gives

$$(E_{21/22}^{(9)})'(x) \geq 2\pi - \frac{7}{2} \cdot 10^{-2} - \frac{10}{21} \cdot 10^{-8} - \frac{6}{46} \cdot 10^{-10} - \frac{7}{120} \cdot 10^{-12} > 6.24.$$

Continuity gives existence; the derivative bound gives uniqueness and simplicity. For the left endpoint l and right endpoint u of (10.14), rational substitution of the bounds (10.12)–(10.13) into the polynomial gives, with directed rounding,

$$-4.6 \cdot 10^{-20} < E_{21/22}^{(9)}(l) < -4.5 \cdot 10^{-20} < 0, \quad 0 < 1.6 \cdot 10^{-20} < E_{21/22}^{(9)}(u) < 1.8 \cdot 10^{-20}.$$

The monotonicity already proved therefore places the unique root between the two endpoints. The interval is verified after constructing the operator and does not enter the selection of its coefficients. $\square$

The result concerns ξ_9 . It does not authorize writing $\xi_9 = \alpha_A$ as an exact equality of complete numbers. A truncation root may share a prefix with the root of the prolonged operator and differ at a later position. Simplicity allows control of root movement under perturbations; it does not annul those perturbations.

10.3 Compatibility of the dodecaphase and analytic representations of alpha

Compatibility is established at every depth. The reversible dodecaphase representation and the analytic representation of the alpha coordinate $E_{21/22}$ are two internal representations with distinct codomains, produced from the same APP-TRIT-TPK enriched state. The jet $E_{21/22}^{(9)}$ is an exact finite witness, but does not by itself determine the tail. The completion constructed below does not follow from the bare jet: it receives the shared coordinate of the state and expresses its analytic image through an explicit recurrence. It therefore does not constitute an independent causal selection; it constitutes a typed analytic representation of the same generated point.

This distinction preserves the causal order

$$\text{APP} \longrightarrow \text{TRIT} \longrightarrow \text{TPK} \longrightarrow \mathcal{X}_\alpha^{\text{enr}} \longrightarrow (P, E, \Phi, U_{12}^{\text{sgn}}, K) \longrightarrow \alpha_{\text{HMT}}.$$

The coordinates P, E, Φ and the record K are prior outputs of the same enriched state. Their use in the closure of alpha is therefore a subsequent internal composition, not an injection of conventional constants.

For $r \in 6\mathbb{N}$, $r \geq 36$, let $\hat{\mathcal{X}}_r^{\text{enr}, \alpha}$ be the sector of truncated states extending the internal alpha seed fixed at R_{36} . Write

$$x_r^\alpha = (\tilde{x}_r, \chi_\alpha, \varepsilon_r^\alpha, N_r^\alpha),$$

where χ_α is the internal character of the sector, not the expressed scalar value. Each state preserves residue, quotient, carry, sheet, orientation, phase, route, cylinder, boundary, signature, record, memory, survival, provenance, and depth. If

$$d_r^\alpha = \text{Dig}_{729}(\varepsilon_r^\alpha), \quad \varepsilon_{r+6}^\alpha = 729\varepsilon_r^\alpha - d_r^\alpha, \quad N_{r+6}^\alpha = 729N_r^\alpha + d_r^\alpha,$$

the effective transition is

$$u_r^\alpha = \text{Upd}_r^\alpha \circ \text{Tra}_r^\alpha \circ \text{Sel}_r^\alpha, \quad \text{Ext}_{r+6, r}^{\alpha, \rightarrow} = \text{graph}(u_r^\alpha).$$

The functional character of these extensions, with the seed R_{36} fixed, produces a unique compatible section: phase returns while memory and depth advance. The global domain of both determinations is

$$\mathcal{X}_\alpha^{\text{enr}} := \hat{\Omega}_\alpha := \varprojlim_{r \geq 36} (\hat{\mathcal{X}}_r^{\text{enr}, \alpha}, \text{Ext}_{r+6, r}^{\alpha, \rightarrow}). \quad (10.15)$$

10.3.1 Dodecaphase closure as a complete section

For $N \geq 1$, set

$$\mathcal{W}_N := \{0, \dots, 999\}^N, \quad \mathsf{T}_{\alpha, N}(\mathbf{s}) := (A_1, \dots, A_N), \quad \mathbf{s} \in \mathcal{X}_\alpha^{\text{enr}},$$

where the blocks A_j are obtained through (10.1) with the compatible boundary $c_{N+1}^\infty = [R_{N+1}(Z)]$. For $N' \geq N$, let $\rho_{N', N} : \mathcal{W}_{N'} \rightarrow \mathcal{W}_N$ be restriction to the first N blocks.

Theorem 10.8 (Complete dodecaphase output and projective compatibility). *The collection $(\mathsf{T}_{\alpha, N})_{N \geq 1}$ is compatible:*

$$\rho_{N', N} \circ \mathsf{T}_{\alpha, N'} = \mathsf{T}_{\alpha, N} \quad (N' \geq N). \quad (10.16)$$

Its canonical cylinders

$$C_N^A(\mathbf{s}) := \left[\sum_{j=1}^N A_j 1000^{-j}, \sum_{j=1}^N A_j 1000^{-j} + 1000^{-N} \right) \cap I_\alpha \quad (10.17)$$

are nested, have diameter at most 1000^{-N} , and satisfy

$$\bigcap_{N \geq 1} C_N^A(\mathbf{s}) = \{\alpha_A(\mathbf{s})\}, \quad \alpha_A = \pi_{\text{HMT}} + e_{\text{HMT}} - \varphi_{\text{HMT}} - 4 - \kappa_{\text{per}}. \quad (10.18)$$

Consequently, the first route expresses a unique infinite section, denoted by $\alpha_{\text{HMT}} := \alpha_A$.

Proof. Proposition 10.3 provides the unique canonical expansion and the unique bounded carry sequence. The local identity (10.3) and its telescoping form (10.4) preserve the remote boundary; Proposition 10.5 proves that restricting an extended execution recovers precisely the previous prefix. This gives (10.16). Membership of the complete sum in every cylinder follows from the tail identity, and $\text{diam } C_N^A \leq 1000^{-N} \rightarrow 0$; the intersection contains a single point. Finally, (10.7) identifies this point with the expression in (10.18). All its coefficients and signs are fixed before alpha is expressed. $\square$

The window $\alpha_{12}^{(0)}$, closed with $c_{13} = 0$, remains an exact finite calculation; it must not be confused with the restriction of the complete section, whose compatible boundary satisfies $c_{13}^\infty = -1$. In particular,

$$\rho_{36, 12}(\alpha_{A, 36}) = \alpha_{A, 12} \quad (10.19)$$

is an equality of compatible words, not an equality between words of different lengths or an identification of the isolated window with the limit.

10.3.2 Exact analytic characterization of order nine

Let

$$\mathbb{K} := \mathbb{Q}\left(\pi_{\text{HMT}}, \log_{10} \frac{10}{9}\right), \quad E_9(x) := E_{21/22}^{(9)}(x) = B_9(x) - \delta.$$

Equations (10.10) and (10.11) construct $B_9 = P_6 + T_{7;9}$ from the state's signature and the nilpotent resolvent; Proposition 10.7 proves that E_9 has a unique simple root $\xi_9 \in I_\alpha$, isolated by the rational interval (10.14).

Finite bidirectionality can be expressed without introducing an analytic tail. Define

$$B_9(x) := \delta + \frac{7}{4}x^2 - \frac{x^4}{20} + \frac{2x^5}{21} + \frac{x^6}{46} + \frac{x^7}{120} - \frac{x^8}{45} - \frac{2x^9}{495}, \quad (10.20)$$

$$q_x(T) := xT^2 - B_9(x)T + x^3, \quad x \in I_\alpha = (0, 10^{-2}), \quad (10.21)$$

$$J_x : \mathbb{R}^\times \longrightarrow \mathbb{R}^\times, \quad J_x(T) := \frac{x^2}{T}. \quad (10.22)$$

Proposition 10.9 (Finite quadratic equivalence and involution). *For every $x \in I_\alpha$ and every $T \in \mathbb{R}^\times$,*

$$\frac{q_x(2\pi_{\text{HMT}})}{2\pi_{\text{HMT}}} = B_9(x) - \delta = E_9(x), \quad (10.23)$$

$$J_x(J_x(T)) = T, \quad T^2 q_x(J_x(T)) = x^2 q_x(T). \quad (10.24)$$

Let $Z_x = \{T \in \mathbb{R}^\times : q_x(T) = 0\}$. Consequently, $J_x|_{Z_x} : Z_x \rightarrow Z_x$ is involutive. When $E_9(x) = 0$, it exchanges the roots $2\pi_{\text{HMT}}$ and $x^2/(2\pi_{\text{HMT}})$. This is an exact involution on the scalar fibre of the order-nine jet; it does not act on the enriched state.

Proof. Dividing $q_x(T)$ by T and substituting $T = 2\pi_{\text{HMT}}$ reproduces the definition of $B_9(x) - \delta$ term by term. The first identity in (10.24) is immediate. For the second,

$$T^2 q_x(x^2/T) = x^5 - B_9(x)x^2T + x^3T^2 = x^2 q_x(T).$$

Thus J_x preserves the root set and, by Viète's formula, exchanges the two roots whose product is x^2 . $\square$

The finite decimal witness thus retains its exact force:

$$\text{Tr}_9(\xi_9^{-1}) = \text{Tr}_9((\alpha_{12}^{(0)})^{-1}) = 137.035999084. \quad (10.25)$$

Equality (10.25) certifies the finite characterization and the preceding involution. It does not turn the root of a truncated polynomial into the root of every compatible extension.

10.3.3 The jet alone does not determine an analytic extension

Let

$$j_9 : \mathbb{K}[[x]] \longrightarrow \mathbb{K}[x]/(x^{10})$$

be truncation, and for $h \in \mathbb{K}[x]$ set

$$E_h(x) := E_9(x) + x^{10}h(x). \quad (10.26)$$

All these polynomials satisfy $j_9(E_h) = j_9(E_9)$.

Theorem 10.10 (Non-uniqueness of the root from the jet). *There is no map*

$$\mathbf{r}_9 : \mathbb{K}[x]/(x^{10}) \longrightarrow I_\alpha$$

that assigns, from the jet alone, the positive root of every polynomial extension of E_9 having a unique root in I_α . In particular, E_0 and $E_{1/729}$ have the same jet and distinct positive roots.

Proof. The bound established in the proof of 10.7 gives $E'_9(x) > 6.24$ on $[0, 10^{-2}]$. For $t \geq 0$, write $E_t(x) = E_9(x) + tx^{10}$. Then

$$E'_t(x) = E'_9(x) + 10tx^9 > 6.24,$$

whereas $E_t(0) = -\delta < 0$ and $E_t(10^{-2}) > E_9(10^{-2}) > 0$. Each E_t therefore has a unique positive root $\eta(t) \in I_\alpha$. For $t = 1/729$,

$$E_{1/729}(\xi_9) = \frac{\xi_9^{10}}{729} > 0;$$

since $E_{1/729}$ is strictly increasing and changes sign only once, its root satisfies $\eta(1/729) < \xi_9 = \eta(0)$. However, $j_9(E_{1/729}) = j_9(E_0)$. A map that factored the root through j_9 would have to assign the same point to both polynomials, a contradiction. $\square$

The variation is not merely existential. The implicit function theorem applies because $\partial_x E_t(\eta(t)) > 6.24$, and gives

$$\eta'(t) = -\frac{\eta(t)^{10}}{E'_9(\eta(t)) + 10t\eta(t)^9} < 0. \quad (10.27)$$

Thus coefficients beyond degree nine are proved to change the root even while leaving the entire jet constructed through that degree invariant. This conclusion does not deny the tail produced by the enriched state: it proves only that the root does not factor through the bare truncation j_9 . Choosing a tail h from an external conventional root would be circular. The completion below consults neither that root nor a metrological table: it directly evaluates the regional and terminal coordinates of the state. Since that evaluation agrees algebraically with the integer representation, the result is a correlated representation, not an assertion of independence between two selectors.

10.3.4 Analytic representation of the alpha coordinate

The preceding jet control requires the tail to be exhibited: merely naming a complete reader is insufficient. Let $s \in \mathcal{X}_\alpha^{\text{enr}}$. Before decimal normalization, the same state already expresses the precoordinate

$$a(s) := p(s) + e_0(s) - f(s) - \kappa_{\text{per}}(s) \in I_\alpha, \quad (10.28)$$

with $p = \pi_{\text{HMT}} - 3$, $e_0 = e_{\text{HMT}} - 2$ and $f = \varphi_{\text{HMT}} - 1$. Equivalently, $a = \pi_{\text{HMT}} + e_{\text{HMT}} - \varphi_{\text{HMT}} - 4 - \kappa_{\text{per}}$. Conventional α is not introduced here: p, e_0, f are the three regional characters already generated, and κ_{per} is the rational reading of the terminal register. Integer closure applies to (10.28) the normalization of (10.1); the following chart expresses a compatible analytic representation on the same precoordinate. Its dependency graph receives $(p, e_0, f, \kappa_{\text{per}})$ directly: it contains no edge $\alpha_A \rightarrow \lambda$, although the equality proved below subsequently permits the two representation names to be identified.

Write $q = 1/729$, let F_s be the ordered field generated by the coordinates of s , and put $E_{9,s}(X) = E_{21/22}^{(9)}(s; X)$. Define the linking coefficient

$$\lambda(s) := -\frac{E_{9,s}(a(s))(1 - qa(s)^3)}{a(s)^{10}}. \quad (10.29)$$

All its arguments belong to the state before metrological recognition. In particular, λ is not obtained by fitting a target decimal string. Equation (10.29) must be read with its exact type: once the first representation of the state has produced $a(s)$, it fixes the unique prolongation in the geometric class defined below. It is not a second independent selection of $a(s)$, but a correlated analytic representation of the same enriched state.

Indeed, for $\mu \in F_s$, put

$$E_{s,\mu}(X) := E_{9,s}(X) + \mu \frac{X^{10}}{1 - qX^3}.$$

Since $0 < a(s) < 10^{-2}$ and $qa(s)^3 < 1$, we have

$$E_{s,\mu}(a(s)) = 0 \iff \mu = \lambda(s). \quad (10.30)$$

Thus the coefficient seed contains no freedom once the state and the class of q -geometric prolongations have been fixed. This uniqueness is relative to the declared class; it does not assert that the ninth-order jet alone selects a tail among all possible analytic series.

The local action on blocks of three coefficients is

$$\mathcal{C}_\alpha : F_s^3 \longrightarrow F_s^3, \quad \mathcal{C}_\alpha(u, v, w) = q(u, v, w), \quad c^{(0)} = (\lambda(s), 0, 0). \quad (10.31)$$

Consequently, for every $n \geq 0$,

$$b_{10+3n} = q^n \lambda(s), \quad b_{11+3n} = b_{12+3n} = 0. \quad (10.32)$$

This is the rule that computes every coefficient beyond degree nine. No free parameter remains at each prolongation.

For $M \geq 0$, let

$$E_{s,M}(X) := E_{9,s}(X) + \lambda(s) \sum_{n=0}^M q^n X^{10+3n}. \quad (10.33)$$

The complete function is

$$E_{s,\infty}(X) := E_{9,s}(X) + \lambda(s) \frac{X^{10}}{1 - X^3/729}. \quad (10.34)$$

To formulate the projective system without implicit types, fix $d_M = 12 + 3M$ and define the jet fibration

$$\mathfrak{J}_M := \prod_{s \in \mathcal{X}_\alpha^{\text{enr}}} \{s\} \times F_s[X]/(X^{d_M+1}). \quad (10.35)$$

If $M' \geq M$, restriction is

$$\tau_{M',M}(s, [P]_{d_{M'}}) := (s, [P]_{d_M}). \quad (10.36)$$

Proposition 10.11 (Naturality of the analytic recurrence). *Let*

$$\Gamma_M(s) = (s, [E_{s,M}]_{d_M}).$$

If $M' \geq M$, the truncation $\tau_{M',M}$ satisfies

$$\tau_{M',M} \circ \Gamma_{M'} = \Gamma_M, \quad \Gamma_{E21}(s) = \varprojlim_M \Gamma_M(s). \quad (10.37)$$

Projection onto the three new degrees is precisely $\mathcal{C}_\alpha^{M+1}(c^{(0)})$.

Proof. Equation (10.32) fixes the block of degrees $10 + 3n, 11 + 3n, 12 + 3n$. Passing from M to $M + 1$ adds only the next block; truncating it recovers the previous polynomial literally. Compatibility for $M' \geq M$ follows by composition. Thus the inverse limit is not a section chosen from the bare jet, but the unique orbit of $c^{(0)}$ under $\mathcal{C}_\alpha$. $\square$

10.3.5 Uniform convergence, a simple root, and rational intervals

The regional cylinders and periodic register provide bounds through rational interval arithmetic. For each coordinate

$$c \in \{\pi_{\text{HMT}}, e_{\text{HMT}}, \varphi_{\text{HMT}}\},$$

let P_c be its certified thousand-digit nonadic prefix. The datum used is not the rational P_c treated as the complete value, but the half-open cylinder and its outer closure

$$\mathcal{C}_c^{(1000)} = [P_c, P_c + \varepsilon), \quad \overline{\mathcal{C}_c}^{(1000)} = [P_c, P_c + \varepsilon], \quad \varepsilon = 10^{-1000}.$$

Directed arithmetic operates conservatively with these closures. If P_π, P_e, P_φ denote the three prefixes, this gives

$$\begin{aligned} a_- &:= P_\pi + P_e - (P_\varphi + \varepsilon) - 4 - \kappa_{\text{per}}, \\ a_+ &:= (P_\pi + \varepsilon) + (P_e + \varepsilon) - P_\varphi - 4 - \kappa_{\text{per}}, \\ A_{1000} &:= [a_-, a_+], \quad a(s) \in [a_-, a_+] = A_{1000}, \quad \text{diam } A_{1000} = 3\varepsilon. \end{aligned} \quad (10.38)$$

The natural interval extension of (10.29) subsequently transports A_{1000} , the cylinder of π_{HMT} , and the directed rational bound on δ into a closed interval $\Lambda_{1000} \ni \lambda(s)$; none of the three tails is frozen. In particular,

$$\begin{aligned} \frac{7297352569283800997285105472380662}{10^{36}} &< a(s) < \frac{7297352569283800997285105472380663}{10^{36}}, \\ -8.711 \times 10^{-6} &< \lambda(s) < -8.709 \times 10^{-6}. \end{aligned} \quad (10.39)$$

These inequalities use certified prefixes with their rational tail bound; they do not receive an alpha table. The package verifier reproduces them from the three nonadic outputs and the terminal chart.

Let

$$r := \frac{10^{-6}}{729} < 1.372 \times 10^{-9}.$$

For $0 \leq x \leq 10^{-2}$, the tail beyond M satisfies

$$|E_{s,\infty}(x) - E_{s,M}(x)| \leq 8.711 \times 10^{-26} \frac{r^{M+1}}{1-r}. \quad (10.40)$$

Moreover,

$$\left| \frac{d}{dx} \left(\lambda \frac{x^{10}}{1 - x^3/729} \right) \right| \leq 8.711 \times 10^{-6} \left(\frac{10 \cdot 10^{-18}}{1-r} + \frac{3 \cdot 10^{-24}/729}{(1-r)^2} \right) < 8.712 \times 10^{-23}. \quad (10.41)$$

The derivative tail beyond block M also admits the depth-dependent estimate

$$\sup_{0 \leq x \leq 10^{-2}} |E'_{s,\infty}(x) - E'_{s,M}(x)| \leq 8.711 \times 10^{-24} r^{M+1} \left(\frac{13 + 3M}{1-r} + \frac{3r}{(1-r)^2} \right) \xrightarrow{M \rightarrow \infty} 0. \quad (10.42)$$

Theorem 10.12 (Limit function and unique simple root of the correlated chart). *The sequence $E_{s,M}$ converges uniformly, together with its derivatives, to $E_{s,\infty}$ on $\bar{I}_\alpha = [0, 10^{-2}]$. We have*

$$E_{s,\infty}(a(s)) = 0, \quad \inf_{x \in \bar{I}_\alpha} E'_{s,\infty}(x) > 6.239. \quad (10.43)$$

Consequently, $a(s)$ is the unique simple root of the complete function in I_α .

Proof. The ratio of two consecutive terms of the tail is bounded by $r < 1$, which yields (10.40); termwise differentiation and summation of the two geometric series yield (10.41) and (10.42). The jet proof establishes $E'_{9,s} > 6.24$ on $\bar{I}_\alpha$; hence the complete perturbation leaves the derivative above 6.239. Finally, substituting (10.29) into (10.34) makes the two terms evaluated at $a(s)$ cancel exactly. The function is strictly increasing; its unique root is therefore simple and coincides with this precoordinate. $\square$

To make the identification effective at every depth, directed endpoint evaluations first certify

$$E_{s,\infty}(0) < 0 < E_{s,\infty}(10^{-2}), \quad E'_{s,\infty} \geq \mu := 6239/1000 > c := 6 \quad \text{on } [0, 10^{-2}].$$

Continuity ensures the existence of a root in the interval, and the second inequality ensures uniqueness. Only after establishing this existence, monotonicity, and the invariant that the bracket contains the root is the mean-value-theorem fallback applied. Start with $D_0 = [0, 10^{-2}]$. If $D_j = [r_j, s_j]$, let $m_j = (r_j + s_j)/2$, and let $[F_j^-, F_j^+]$ be a directed rational enclosure of $E_{s,\infty}(m_j)$, obtained by propagating the cylinders of $\pi_{\text{HMT}}, e_{\text{HMT}}, \varphi_{\text{HMT}}, \delta$, and λ , with

$$0 \leq F_j^+ - F_j^- \leq \frac{c}{4} (s_j - r_j). \quad (10.44)$$

When $F_j^+ < 0$, retain the right half; when $F_j^- > 0$, retain the left half. If $F_j^- \leq 0 \leq F_j^+$, no attempt is made to decide whether the midpoint is exactly the root. Let $w_j = s_j - r_j$. The uniform bound $E'_{s,\infty} \geq c = 6$ and the mean value theorem imply that any root contained in D_j belongs to

$$D_{j+1} = D_j \cap \left[m_j - \frac{w_j}{4}, m_j + \frac{w_j}{4} \right]. \quad (10.45)$$

Indeed, let ξ_j be the root and let $y_j = E_{s,\infty}(m_j) \in [F_j^-, F_j^+]$. Since the enclosure contains zero and satisfies (10.44), we have $|y_j| \leq cw_j/4$. The mean value theorem gives $|m_j - \xi_j| \leq |y_j|/c \leq w_j/4$, proving (10.45). The argument includes the case $y_j = 0$: the root then lies at the midpoint, but the algorithm does not need to recognize this equality in order to retain it.

If the available enclosure does not satisfy (10.44), the source cylinders are first refined and evaluation is repeated; no sign is declared and no insufficient contraction is accepted. In each of the three branches, the new interval retains the root and has diameter at most $w_j/2$. For every finite target depth, directed refinement permits the width bound to be met without an uncertified comparison of a real number with zero. Monotonicity proves inductively

$$a(s) \in D_{j+1} \subseteq D_j, \quad \text{diam } D_{j+1} \leq \frac{1}{2} \text{diam } D_j, \quad \text{diam } D_j \longrightarrow 0, \quad (10.46)$$

where the inequality is a contraction bound, not an imposed equality. The executable certificate applies this rule at twenty-four base-thousand levels and verifies that the complete outer closure A_{1000} , not merely its lower endpoint, lies within every expressed cylinder. Its negative tests reject an excessively wide enclosure, a disordered enclosure, a point other than the midpoint, a domain without a sign change, and a degenerate bracket. Coinductive prolongation permits a thousand to be replaced by any required depth.

Specifically, for $S_N = 1000^N$, the verifier computes

$$j_N^- := \lfloor S_N a_- \rfloor, \quad j_N^+ := \lfloor S_N a_+ \rfloor \quad (10.47)$$

and expresses the level only if $j_N^- = j_N^+$. This equality, which includes the upper endpoint of the outer closure, proves $A_{1000} \subset [j_N^-/S_N, (j_N^- + 1)/S_N]$ and prevents an endpoint lying exactly on the right boundary from being assigned to the left cell.

Let $C_N^A(s)$ be the integer cylinder of (10.17). Recursively choose $m(N) > m(N-1)$ such that $\text{diam } D_{m(N)} < 1000^{-N}/4$, and define

$$I_{B,N} := D_{m(N)} \cap C_N^A(s). \quad (10.48)$$

Define $\ell_N := \inf I_{B,N}$ and $u_N := \sup I_{B,N}$. Both are rational; the interval is nonempty because both factors contain $a(s)$. Since the cylinder is half-open, u_N may not belong to $I_{B,N}$. Evaluation at this endpoint is understood through the continuous extension of $E_{s,\infty}$ to the outer closure. Thus,

$$I_{B,N+1} \subseteq I_{B,N} \subseteq C_N^A(s), \quad \text{diam } I_{B,N} \longrightarrow 0, \quad E_{s,\infty}(\ell_N) \leq 0 \leq E_{s,\infty}(u_N). \quad (10.49)$$

Bound (10.40) permits a truncation to be chosen that preserves the sign at every endpoint where the complete function does not vanish. If an endpoint coincides with the root, the identity of the complete function and the membership already proved are used; the same sign is not attributed to its truncations. Indeed, the exact remainder is

$$E_{s,\infty}(x) - E_{s,M}(x) = \lambda(s)x^{10} \frac{(qx^3)^{M+1}}{1 - qx^3}.$$

On the positive domain considered, where $\lambda(s) < 0$, this gives at the root

$$E_{s,M}(a(s)) = -\lambda(s)a(s)^{10} \frac{(qa(s)^3)^{M+1}}{1 - qa(s)^3} > 0 \quad (M < \infty).$$

Consequently, certification of the weak signs in (10.49) refers to $E_{s,\infty}$. Its evaluation uses the complete function or the truncation accompanied by the signed remainder. This covers both the included infimum and the potentially excluded supremum of the cylinder.

Proposition 10.13 (Compatibility squares and common cylinders). *The restrictions of words and analytic representations commute:*

$$\begin{array}{ccc}
 \mathcal{X}_\alpha^{\text{enr}} & \xrightarrow{T_{\alpha, N'}} & \mathcal{W}_{N'} \\
 \parallel & & \downarrow \rho_{N', N} \\
 \mathcal{X}_\alpha^{\text{enr}} & \xrightarrow{T_{\alpha, N}} & \mathcal{W}_N
 \end{array}
 \qquad
 \begin{array}{ccc}
 \mathcal{X}_\alpha^{\text{enr}} & \xrightarrow{\Gamma_{M'}} & \mathfrak{F}_{M'} \\
 \parallel & & \downarrow \tau_{M', M} \\
 \mathcal{X}_\alpha^{\text{enr}} & \xrightarrow{\Gamma_M} & \mathfrak{F}_M.
 \end{array}$$

The collection $I_{B, N}$ provides the effective connection between both squares and concretely satisfies $I_{B, N} \subseteq C_N^A$ for every N .

Proof. The first square is the compatibility of integer normalization; the second is Proposition 10.11. The inclusion follows from definition (10.48), and nonemptiness and signs follow from Theorem 10.12. No step identifies the root of a truncation with that of an arbitrary tail. $\square$

Now define, without leaving the prefix reader implicit,

$$\alpha_B(s) := \arg \text{zero}_{x \in I_\alpha} E_{s, \infty}(x), \quad \text{pref}_N(x) := \frac{\lfloor 1000^N x \rfloor}{1000^N}, \quad \alpha_{B, N} := \text{pref}_N(\alpha_B). \quad (10.50)$$

Theorem 10.14 (Projective equality of the two correlated representations). *For every N , the integer route and the analytic route express the same prefix:*

$$\boxed{\alpha_{A, N} = \text{pref}_N(\alpha_A) = \text{pref}_N(\alpha_B) = \alpha_{B, N}.} \quad (A7_0)$$

In the limit,

$$\boxed{\alpha_A = \arg \text{zero}_{x \in I_\alpha} E_{s, \infty}(x) = \alpha_B = \alpha_{\text{HMT}}.} \quad (A8_0)$$

Proof. Integer normalization expresses exactly $a(s)$. Theorem 10.12 proves that the analytic chart represents the same point as its unique simple root. The half-open cylinders C_N^A fix a canonical prefix, and $I_{B, N} \subseteq C_N^A$ contains $\alpha_B = \alpha_A$. Their prefixes therefore coincide for every N ; nesting and diameters tending to zero yield equality of the limits. Metrological recognition occurs afterwards. $\square$

The analytic function constitutes a correlated representation of the same state, not a second derivation causally independent of integer closure. This precision does not diminish the equality proved: it identifies its exact mechanism, prevents a tail that the jet does not determine from being attributed to it, and avoids promoting a conventional value to a generator.

11 Exceptional prolongation: from dodecaphasic incidence to Moonshine

The numerical expression does not exhaust the state that generated it. In the preceding construction, APP preserves the two sheets and division with remainder and quotient; the TRIT determines the local regime and its orientation; the TPK composes selection, transport, carry, memory, and return. The Archimedean reading contracts compatible cylinders. The reading now under study instead preserves the incidence of positions and their transport between frames. Both start from the same arithmetic state with trajectory memory. An exceptional geometry is not reconstructed from the final decimals of π , φ , e , or α .

This point clarifies the meaning of the prolongation $\alpha/K \rightarrow$ exceptional incidence. The arrow does not assert that an isolated real number determines a code, a lattice, or a group. It transports the dodecaphasic record and the signed data that entered the determination of α , together with the words and frames arising from APP–TRIT–TPK. In particular, the ordered record K can be recovered from its rational constant κ under the previously fixed base, length, and origin conventions. Projection onto a support, by contrast, preserves only the selected incidences. For its part, K_{ph} denotes a different, reduced-memory reading, subsequent to the holonomy Γ_9 .

The exposition separates three mathematical operations: constructing the finite objects and their maps; prolonging them to a lattice and a vertex operator algebra; and subsequently recognizing the resulting objects through the relevant classical theorems. The names Paley, Witt, Mathieu, Golay, Leech, and Monster designate those recognitions and the explicit realizations that make them verifiable. They are not retrospective selectors of the seeds or coefficients of the generation.

11.1 Structural function of K and α in exceptional incidence

The link between numerical determination and exceptional realization admits an operational description. In the arithmetic reading, K acts in the normalization of regional records. In the incidence reading, its components select the support

$$B_K = \{j : K_j \geq 729\} = \{4, 6, 9, 10\}.$$

The same ordered record therefore participates in two specified functions: a positional composition and an incidence selection. Its rational realization preserves the record when base, length, and origin are fixed, while passage to the support retains the information needed for the combinatorial flag.

The signed record preceding normalization of α supplies the hexad H_{α}^- . The intersections of this hexad with the regional supports and APP incidence determine the marked point. That point individualizes the norm-54 vector used in the lattice neighbor. Thus the orientation entering arithmetic normalization acts again in geometric selection. The relation does not require deriving a lattice from a decimal approximation: it uses the signed data preceding its evaluation.

The exceptional realization successively preserves different types of relation. The code preserves linear combinations; the design preserves incidences between supports; gluing preserves discriminant classes; the lattice preserves its bilinear form; the vertex algebra preserves grading and the corresponding products. Their automorphism groups are defined with respect to these structures. The appearance of the Monster at the end of the construction expresses a property of the algebra $V^{\natural}$ obtained through the cited procedure, and Moonshine relates the graded traces of that action to modular functions.

This hierarchy gives a concrete mathematical meaning to the linking function attributed to K and α . The former is a reversible integral record with positional and incidence realizations; the latter is the arithmetic-compatibility coordinate whose oriented record also enters the marked incidence. The definitions acquire content through these operations and the identities presented in this chapter. The incidence development is a substantive part of the article's argument: it shows which relations persist when the construction is no longer evaluated as a real coordinate and is instead realized as an algebraic structure.

Articulation of regional coordinates and closure incidence. The role of this prolongation is understood through a distinction between three operations: preserving a record, evaluating a coordinate, and extracting an incidence relation. All three act on information produced by APP-TRIT-TPK. Their articulation makes explicit the article's structural thesis: a numerical realization has an operator provenance, and certain data from that provenance additionally admit a combinatorial and lattice realization. The equations allow both paths to be followed with their domains and the information remaining in each.

The starting point is the terminal state with its compatible prolongations. It preserves the residues and quotients of the two APP sheets, the TRIT regime and orientation, and TPK transport with its memory. The closure, propagation, and self-scale regions are determined in that domain before their readers establish the coordinates π, e, φ . At the same time, the oriented record of the twelve windows determines the signed vector U . The subsequent appearance of K arises from the integral transformation proved in §9.7:

$$U = \mathcal{H}_{12}K, \quad K = \frac{1}{4}\mathcal{H}_{12}U, \quad \mathcal{H}_{12}^2 = 4I_{12}. \quad (11.1)$$

The congruence defining $\mathcal{L}_H$ ensures integrality of the inverse. Thus K is a reversible coordinatization of the signed record in the relevant sublattice. Its positions preserve phase order; its differences D_3K, D_4K preserve transverse relations; the sum $Q(K)$ determines the uniform component. That structure explains why the record enters both numerical composition and the selection of a boundary configuration.

The periodic evaluation

$$K \mapsto \kappa_{\text{per}} = \frac{\sum_{m=1}^{12} K_m 1000^{12-m}}{1000^{12} - 1}$$

keeps the twelve blocks recoverable when base, length, and phase origin are fixed. Extracting a support performs a different operation: it preserves membership of each position in a class defined by a threshold and groups records with the same support. The distinction between these two maps is decisive. The scalar precision of κ_{per} and the incidence of B_K are two specific mathematical contents, related through the complete record from which they arise.

Alpha normalization and its signed configuration. The coordinate of α is determined in the positional route through composition of the regional blocks and the dodecaphase record. Before positional division, the oriented combination has components $s_j = P_j + E_j - \Phi_j - K_j^{\text{per}}$. Normalization satisfies

$$A_j = s_j + c_{j+1} - 1000c_j, \quad 0 \leq A_j < 1000. \quad (11.2)$$

This identity simultaneously preserves the normalized block and the contribution of the transported quotient. The limit of compatible prefixes determines the scalar coordinate of closure. The signed configuration, in turn, preserves which positions of the combination before normalization have a negative defect. In the initial window, that configuration is the vector s_0 of (11.12), and its negative support is H_{α}^- .

Two contents of the same operation are thereby defined. Positional normalization determines a real coordinate through its compatible remainders and quotients; the negative support determines a configuration of positions in the twelve-component cycle. The sign refers to the balance of records before normalization. Its incidence reading remains available because the state preserves that balance together with the normalized result. This articulation corresponds to the proved positional route; comparison with the analytic vacancy operators preserves the precise scope established in the chapter on α .

From ternary coding to the marked flag. The regions also supply the codes $w_{\pi}, w_e, w_{\varphi}$, and w_{APP} in the inherited frame. The map

$$\gamma_{A_W} : w \mapsto (w, wA_W)$$

preserves the six-component word as its first coordinate and adds its transformation by A_W . The relation $A_W^2 = -I_6$ fixes the quadratic compatibility of this realization. The next step, $c \mapsto \text{supp}(c)$, preserves the nonzero positions. For weight-six words, the support identifies exactly the pair $\{c, -c\}$, as proved by the minimum distance of the code. Thus, the sign quotient produces the hexads, whereas the complete code remains available for operations requiring its symbols.

In this chart, the two constructions of the tetrad coincide:

$$\{i : K_i \geq 729\} = H_e \cap P_\pi = B_K = \{4, 6, 9, 10\}. \quad (11.3)$$

The left-hand side arises from an integer comparison of the components of K ; the right-hand side, from the intersection of the coded supports of propagation and closure. The threshold belongs to the declared block chart and remains explicit in the reader's definition. The complete record preserves the values on which it acts, even when different thresholds produce the same subset. The equality therefore expresses compatibility between specified maps.

The support H_α^- completes that tetrad to a hexad. The condition of membership in support P_π , together with the intersection sizes $(4, 3, 2)$ relative to $H_e, H_\varphi, H_{\text{APP}}$, selects a unique hexad. The following proofs bring together the two determinations: that obtained through the sign of s_0 and that obtained through incidence. The result is organized as

$$\tilde{J}_* = (B_K \subset H_\alpha^-; P_\pi, H_e, H_\varphi, H_{\text{APP}}). \quad (11.4)$$

The inclusion is the flag; the four additional supports constitute its marked frame. Simultaneous permutations transport the frame and preserve its inclusions and intersections. The relevant information resides in this transported configuration and its isomorphism class.

The five regions of π recover their internal differentiation here. They share w_π as a visible ternary code but preserve distinct emissions, signatures, and multiplicities. In the binary realization on a marked dodecad, the transverse region corresponds to the octad whose intersection with the dodecad is the tetrad; the other four regions correspond to the lifts of the four hexads of its star. The negative region is distinguished by H_α^- , and chart order individualizes the three positive regions. This correspondence preserves the regional roles and distribution $432 + 4 \cdot 144 = 1008$, while also realizing the incidence partition $1 + 4$.

From the incidence origin to lattice modification. The marked flag determines a position through an equivariant set-theoretic operation:

$$o_* = (H_e \cap H_\varphi) \setminus (H_\alpha^- \cup H_{\text{APP}}) = \{1\}.$$

Its role is to distinguish one component among the twelve entering the lattice realization. The complete ternary code acts as a glue code in the discriminant module $(A_2^*/A_2)^{12}$. Its preimage defines $N(C_W)$, and the self-duality, weights, and distance of the code translate, respectively, into integrality and determinant properties, parity, and norm bounds for the glue classes. Here the symbolic information of the code performs a role that the isolated support no longer performs.

The A_2 metric and positive radial chamber are fixed in the lattice development. Within that chamber, the congruence needed for an even index-three neighbor has twelve realizations of minimum norm 54, permuted by changing the distinguished component. The origin o_* selects

$$v_\alpha = 4\rho_1 + \rho_2 + \cdots + \rho_{12}, \quad v_\alpha^2 = 2(4^2 + 11) = 54.$$

The coefficient 4 belongs to that solution of the congruence in the declared chamber. The subscript α records the incidence provenance of the selection: its content is the frame that entered closure and now individualizes a lattice direction.

Pairing with v_α determines the index-three sublattice $N_{\alpha,0}$, and adjoining the class $v_\alpha/3$ produces the neighbor N_α of (11.17). This modification preserves rank and recovers unimodularity and evenness; congruence selection eliminates the preceding roots, and local bounds exclude roots in the new classes.

Identification with Leech appears after those verifications. The binary realization of the five regions and this lattice construction are two prolongations of the common frame: the latter directly receives the ternary code as glue data.

Grading, unit, and automorphisms. The Frenkel–Lepowsky–Meurman construction is applied to the lattice N_α once its properties have been verified. It incorporates the oscillator space, central extension of the lattice, vertex product, and sector twisted by the lift of negation. The result is the algebra $V_{\text{HMT}}^{\natural}$ of (11.18), whose automorphism group is isomorphic to the Monster. Its action domain is thus fixed by an algebraic structure with grading and product. Borchers’ Moonshine theorem subsequently determines the graded traces with insertions of those automorphisms [7, 8].

At this level, the unit acquires a precise realization: the weight-zero component is the vacuum line and contributes the coefficient 1 of q^{-1} in the graded character. At the cylinder level, refinement maps preserve the constant function through precomposition. Every conservation statement therefore has its object and its map. The development of the proofs makes this continuity of construction visible, from arithmetic operations and memory to incidence, metric, and vertex products.

This articulation places K and the closure of α in the main argument. K reversibly preserves an integral coordinate of the history; normalization of α composes the regional records with that coordinate; the signed configuration determines a flag; and the marked flag selects the lattice modification on which Moonshine is realized. The following proofs develop these maps, distinguishing reversible equivalences, information quotients, and realizations that add metric or algebraic structure. A provable relation between structure, equation, and value is thus expressed, preserving the attribution of each construction and the scope of its theorems.

11.2 Compatibility of the Archimedean and incidence realizations

Let X_n^{cal} be the space of admissible prefixes of length n of the arithmetic state with trajectory memory, with truncations $p_{n,m} : X_n^{\text{cal}} \rightarrow X_m^{\text{cal}}$ for $m \leq n$. The superscript indicates that the declared initial frame and its transport are preserved, not that a parameter is fitted against an experimental value. We write (w_j, f_j) for the visible six-trit word and the frame at step j . The frames satisfy a causal law

$$f_{j+1} = g_j(x)f_j,$$

where $g_j(x)$ depends on the prefix already constructed. Consequently, a prolongation does not alter earlier frames.

The chain

$$w_6 \longrightarrow w_{12} \longrightarrow w_{18} \longrightarrow w_{24} \longrightarrow w_{30} \longrightarrow R_{36} \longrightarrow G_9$$

is preserved in this reading. The return of the phase resets neither the frame nor the memory of the state. Likewise, the census of 468 decimal emissions, the 243 visible six-trit words, and the ambient space $\mathbb{F}_3^6$, with 729 words, remain distinct. The ambient space will be used to construct a linear code; it does not replace the admissible domain of emissions.

For a six-trit block b , the Archimedean reading takes its integer address $\nu(b) \in \{0, \dots, 728\}$. A prefix determines

$$q_n = m + \sum_{j=0}^{n-1} \frac{\nu(b_j)}{729^{j+1}}, \quad I_n = [q_n, q_n + 729^{-n}].$$

Prefix compatibility gives nested cylinders whose diameter tends to zero. Their Cauchy class belongs first to the internal completion $\mathbb{R}_{\text{HMT}}$; its subsequent identification with the usual real line does not enter the choice of blocks.

For the incidence reading we introduce, on the same prefix, the graph map

$$\gamma_A : \mathbb{F}_3^6 \longrightarrow \mathbb{F}_3^{12}, \quad w \longmapsto (w, wA). \quad (11.5)$$

Words are written as row vectors. If $A^f = fA f^{-1}$, the change of frame satisfies the concrete identity

$$\gamma_{A^f}(w f^{-1}) = \gamma_A(w)(f^{-1} \oplus f^{-1}). \quad (11.6)$$

The calibrated codomain retains f together with the encoded word. We therefore define

$$Q_n(x) = ((f_j, \gamma_{f_j A_W f_j^{-1}}(w_j)))_{0 \leq j < n}. \quad (11.7)$$

Proposition 11.1 (Compatibility and prolongation of the incidence reading). *If $r_{n,m}$ truncates the encoded sequences, then*

$$r_{n,m} Q_n = Q_m p_{n,m}.$$

Thus the Q_n determine a unique map Q_∞ between the respective inverse limits. In each frame, the map preserves the visible word exactly, since $\text{pr}_1 \gamma_A = \text{id}$.

Proof. Two compatible prefixes have the same words and the same transport operators through index $m - 1$. By induction, they have the same frames at those indices. The first m components of (11.7) agree, proving the square. The universal property of the inverse limit gives Q_∞ . The recovery claim is the projection onto the first six coordinates of each graph. $\square$

This local injectivity has a precise domain. It does not prove that Q_∞ recovers all the deep memory of the TPK state if only the sequence of visible words and frames has been retained. Still less does it prove injectivity after passing from words to supports. Moreover, a general change of basis in $\text{GL}_6(\mathbb{F}_3)$ does not by itself preserve ordinary Hamming weight. In a general chart, the incidence and metric of the initial chart are also transported. The support calculations below are performed in the declared Paley chart; monomial relabelings directly preserve their coordinate interpretation.

11.3 The finite realization and the ternary code

The elliptic operator of the TRIT admits the following finite realization in the six-position chart. These positions coordinatize the six units $\{1, 2, 4, 5, 7, 8\}$ modulo nine, preserved by modular transport. On $\mathbb{P}^1(\mathbb{F}_5)$, with the order $\infty, 0, 1, 2, 3, 4$ declared, let C be the matrix with zero diagonal, $C_{\infty,a} = C_{a,\infty} = 1$, and $C_{a,b} = \chi(b - a)$ at the five finite positions. Here $\chi(1) = \chi(4) = 1$, $\chi(2) = \chi(3) = -1$, and $\chi(0) = 0$. This is a coordinate realization of the quadratic structure already transported; identifying the six positions with the projective line is chart data, not a new numerical input into the generator.

The quadratic-sum identities give $C^2 = 5I_6$. Reducing modulo three yields

$$A_W = \begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 \\ 1 & 0 & 1 & 2 & 2 & 1 \\ 1 & 1 & 0 & 1 & 2 & 2 \\ 1 & 2 & 1 & 0 & 1 & 2 \\ 1 & 2 & 2 & 1 & 0 & 1 \\ 1 & 1 & 2 & 2 & 1 & 0 \end{pmatrix}, \quad A_W^2 = 2I_6 = -I_6. \quad (11.8)$$

The quadratic equality realizes the elliptic regime; it does not identify the field $\mathbb{F}_3$ with an Archimedean realization of the imaginary unit. They are distinct realizations of a common operator relation, with their corresponding domains.

Definition 11.2 (Code of the dodecaphasic chart). The ternary code associated with (11.8) is

$$C_W = \{(w, wA_W) : w \in \mathbb{F}_3^6\} \subset \mathbb{F}_3^{12}.$$

Its set of hexads is

$$\mathcal{H} = \{\text{supp}(c) : c \in C_W, \text{wt}(c) = 6\}.$$

Proposition 11.3 (Self-duality, distance, and the Witt design). *The code C_W is self-dual with parameters $[12, 6, 6]_3$. Its weight enumerator is*

$$\sum_{c \in C_W} z^{\text{wt}(c)} = 1 + 264z^6 + 440z^9 + 24z^{12}. \quad (11.9)$$

The set $\mathcal{H}$ has 132 elements and constitutes an $S(5, 6, 12)$ system: every set of five positions is contained in a unique hexad.

Proof. The generator matrix $[I_6 \mid A_W]$ has rank six and, by the symmetry of A_W , satisfies

$$[I_6 \mid A_W][I_6 \mid A_W]^\top = I_6 + A_W^2 = 0.$$

The code is self-orthogonal with dimension half its length and is therefore self-dual. Finite evaluation of the 3^6 words gives (11.9); this is an enumeration over the explicit matrix, without real values or external targets. In particular, the minimum distance is six.

If two weight-six words have the same support, one of the signs 1, -1 occurs at least three times among the six ratios of their nonzero coordinates. Subtracting the corresponding multiple would produce a nonzero word of weight at most three, which is impossible. Hence a support corresponds to exactly the pair $\{c, -c\}$, and there are $264/2 = 132$ supports. If two distinct supports shared five positions, the same argument would cancel at least three coordinates in a union of size seven, producing weight at most four. Thus a five-element set belongs to at most one hexad. Finally,

$$132 \binom{6}{5} = 792 = \binom{12}{5},$$

so it belongs to one and only one. □

The subsequent recognition is the extended ternary Golay code and the small Witt design [9]. The preceding proof explains which object is recognized. It does not replace the APP-TRIT-TPK genealogy with a search among known codes.

11.4 Dodecaphasic flag associated with K and α

In the frame inherited from the dodecaphasic record, the data are

$$\begin{aligned} K &= (234, 543, 140, 729, 659, 824, 621, 058, 914, 794, 146, 601), & (11.10) \\ w_\pi &= 010211, & w_e &= 201101, \\ w_\varphi &= 121200, & w_{\text{APP}} &= 022020. \end{aligned}$$

Applying γ_{A_W} yields the following incidences:

$$\begin{aligned} P_\pi &= \{2, 4, 5, 6, 7, 8, 9, 10, 11\}, \\ H_e &= \{1, 3, 4, 6, 9, 10\}, \\ H_\varphi &= \{1, 2, 3, 4, 7, 9\}, \\ H_{\text{APP}} &= \{2, 3, 5, 10, 11, 12\}. \end{aligned}$$

The first is the support of a weight-nine word; the other three are hexads. Through its high-block reading, the record supplies

$$B_K = \{i : K_i \geq 729\} = \{4, 6, 9, 10\} = H_e \cap P_\pi. \quad (11.11)$$

Here the integer 729 marks a boundary of the declared block format. It is not identified merely by cardinality with the ternary ambient space used in Definition 11.2. In this particular record, all integer thresholds between 660 and 729 produce the same support: the support recovers neither the threshold nor the twelve blocks.

Before carry, the arithmetic route to α preserves the signed vector

$$s_0 = (7, 297, 353, -430, -715, -1199, -3, 286, -894, -528, 380, 663). \quad (11.12)$$

Its negative support is

$$H_{\alpha}^{-} = \{4, 5, 6, 7, 9, 10\}. \quad (11.13)$$

These signs are not extracted from the decimal expansion of the scalar α ; they are retained in the state preceding its numerical expression.

Proposition 11.4 (Agreement of the two flag selectors). *Within $\mathcal{H}$, the support (11.13) is the unique hexad H satisfying*

$$B_K \subset H \subset P_{\pi}, \quad (|H \cap H_e|, |H \cap H_{\varphi}|, |H \cap H_{\text{APP}}|) = (4, 3, 2).$$

Thus the signed reading and the incidence reading select the same flag $B_K \subset H_{\alpha}^{-}$.

Proof. The four hexads containing B_K are obtained by adjoining, respectively, the pairs

$$\{1, 3\}, \quad \{2, 11\}, \quad \{5, 7\}, \quad \{8, 12\}.$$

Only the second and third pairs are contained in P_{π} . For the second, the intersection with H_{φ} has size three, and that with H_{APP} has size three; it does not meet the last condition. For the third, the three intersections have sizes 4, 3, 2. This last pair produces precisely the negative support of (11.12). $\square$

It is useful to retain both the flag and its marked frame:

$$\mathcal{J}_* = (B_K \subset H_{\alpha}^{-}), \quad \tilde{\mathcal{J}}_* = (\mathcal{J}_*; P_{\pi}, H_e, H_{\varphi}, H_{\text{APP}}).$$

Union, intersection, and difference are equivariant under a simultaneous relabeling of the design. The marked class, not the position numbers taken in isolation, is the transported object.

Lemma 11.5 (Origin selected by the marked incidence). *For the preceding frame,*

$$o_* = (H_e \cap H_{\varphi}) \setminus (H_{\alpha}^{-} \cup H_{\text{APP}}) = \{1\}.$$

The rule is equivariant under every automorphism applied simultaneously to the data of $\tilde{\mathcal{J}}_$.*

Proof. The first intersection is $\{1, 3, 4, 9\}$. The subtracted union contains 3, 4, 9 and does not contain 1. Equivariance follows because a bijection commutes with set operations. $\square$

This origin will select a direction of the lattice neighbor once the A_2 metric and the positive chamber are declared. Incidence alone does not contain that metric. The lattice step will explicitly preserve both data.

11.5 Isometric intertwining of the incidence representations

Incidence defines a linear map whose action can be checked position by position:

$$I : \mathbb{R}^{12} \longrightarrow \mathbb{R}^{\mathcal{H}}, \quad (Iv)(H) = \sum_{p \in H} v_p. \quad (11.14)$$

Let $V_{11} = \mathbf{1}^{\perp} \subset \mathbb{R}^{12}$. The two Euclidean realizations are subsequent languages for the same finite design here.

Proposition 11.6 (Isometry of the dodecaphasic screen). *We have*

$$I^{\top} I = 36I_{12} + 30J_{12},$$

where J_{12} is the all-ones matrix. Consequently,

$$U_W = \frac{1}{6} I|_{V_{11}} : V_{11} \xrightarrow{\sim} E_{\text{inc}} := I(V_{11})$$

is an equivariant isometry for the permutation actions of the automorphisms of the design.

Proof. Each point belongs to $r = \binom{11}{4}/\binom{5}{4} = 66$ hexads, and each pair to $\lambda_2 = \binom{10}{3}/\binom{4}{3} = 30$. These are, respectively, the diagonal and off-diagonal entries of $I^\top I$. On $\mathbf{1}^\perp$, the J_{12} term vanishes, leaving $36I$. Finally, if g is an automorphism, $p \in H$ is equivalent to $g(p) \in g(H)$; hence $I\rho_{12}(g) = \rho_{\mathcal{H}}(g)I$. $\square$

The incidence information also has a concrete spectral reading. Define

$$G_{H,H'} = \binom{|H \cap H'|}{4}.$$

Lemma 11.7 (Action of the intersection kernel). *If $J_{132,12}$ is the rectangular all-ones matrix,*

$$GI = 30I + 15J_{132,12}.$$

Thus G acts by the scalar 30 on E_{inc} .

Proof. The (H, p) entry of GI counts pairs (T, H') with $T \subset H \cap H'$, $|T| = 4$, and $p \in H'$. If $p \in H$, there are ten four-element sets T containing p , each contained in four hexads, and five not containing it, each with a unique extension of the five-element set $T \cup \{p\}$. The total is $10 \cdot 4 + 5 = 45$. If $p \notin H$, the fifteen four-element subsets of H each give one extension: the total is 15. This yields the identity. The rectangular term vanishes on V_{11} . $\square$

There is indeed a material intertwiner here: its domain, codomain, entries, and normalization are determined. For example, any orthogonal projector P already constructed on V_{11} is transported to $Q = U_W P U_W^*$ on E_{inc} , with $U_W P = Q U_W$. This statement does not identify all the operators acting on these spaces solely by their trace.

11.6 The Witt star and the generation of M_{12}

Definition 11.8 (Star on a four-element set). For $B \in \binom{[12]}{4}$, the Witt star is the collection of the four hexads containing B . Its petals are the pairs $H \setminus B$. The permutation σ_B fixes B and exchanges the two points of each petal.

The existence of four hexads follows from

$$\lambda_4 = \frac{\binom{8}{1}}{\binom{2}{1}} = 4.$$

Two petals cannot share a point, since two hexads would then contain the same five-element set. They are therefore four disjoint pairs partitioning $[12] \setminus B$. This description is neither a five-pointed star nor a geometric metaphor: it is a four-to-one incidence of the design.

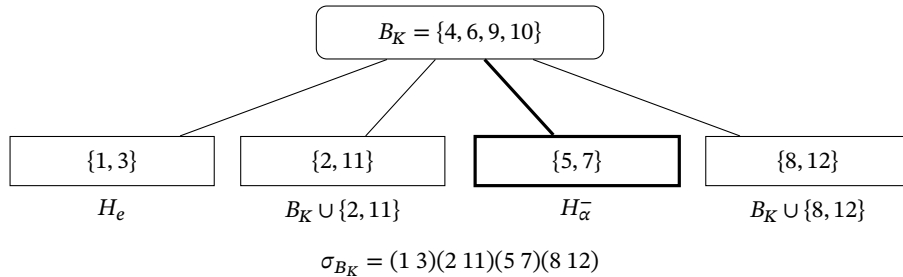


Figure 7: The concrete star of the dodecaphasic flag. Each line indicates the union of the central four-element set with the pair at its endpoint; the thick line marks the hexad also selected by the signs preceding the carry of α . The four pairs exhaust the eight outer positions.

Proposition 11.9 (The collection of stars). *The $\binom{12}{4} = 495$ permutations σ_B preserve $\mathcal{H}$. The group they generate has order 95040 and is the Mathieu group M_{12} in its action on the small Witt design.*

Proof. Preservation reduces to Definition 11.2. The 3^6 words $c(w) = (w, wA_W)$ are enumerated, the 132 supports of weight six are retained, and, for every $B \in \binom{[12]}{4}$, the four pairs $H \setminus B$ with $B \subset H$ are formed. These pairs partition $[12] \setminus B$; replacing each coordinate by its partner maps the 132 supports onto themselves again. This finite verification proves $\{\sigma_B(H) : H \in \mathcal{H}\} = \mathcal{H}$ for the 495 four-element sets.

Six stars sufficient for generation are as follows; the second column displays the four-element set fixed by each permutation:

i	B_i	$g_i = \sigma_{B_i}$	
1	$\{1, 2, 3, 4\}$	$(5\ 12)(6\ 11)(7\ 9)(8\ 10)$	
2	$\{1, 2, 3, 5\}$	$(4\ 12)(6\ 10)(7\ 11)(8\ 9)$	
3	$\{1, 2, 3, 6\}$	$(4\ 11)(5\ 10)(7\ 8)(9\ 12)$	(M12.1)
4	$\{1, 2, 4, 5\}$	$(3\ 12)(6\ 9)(7\ 8)(10\ 11)$	
5	$\{1, 3, 4, 5\}$	$(2\ 12)(6\ 8)(7\ 10)(9\ 11)$	
6	$\{2, 3, 4, 5\}$	$(1\ 12)(6\ 7)(8\ 11)(9\ 10)$	

Let $G_i = \langle g_1, \dots, g_i \rangle$. Its finite closure is determined by

$$G_i^{(0)} = \{1\}, \quad G_i^{(n+1)} = G_i^{(n)} \cup \{g_j h : 1 \leq j \leq i, h \in G_i^{(n)}\}, \quad G_i = \bigcup_{n \geq 0} G_i^{(n)}. \quad (\text{M12.2})$$

Stabilization of this recurrence gives

i	1	2	3	4	5	6	
$ G_i $	2	6	18	360	7920	95040	(M12.3)

A second check successively fixes the points 1, 2, 3, 4, 5. For $G = G_6$ one obtains

fixed points	0	1	2	3	4	5	
$ G_{(1, \dots, k)} $	95040	7920	720	72	8	1	(M12.4)
size of the next orbit	12	11	10	9	8	—	

The entries of (M12.3) and (M12.4) are obtained exclusively by composing the image lists in (M12.1); by orbit–stabilizer, their product is $12 \cdot 11 \cdot 10 \cdot 9 \cdot 8 = 95040$.

Since every g_i preserves $\mathcal{H}$, one has $G \leq \text{Aut}(\mathcal{H})$. Uniqueness of the hexad containing a five-element set and the incidence enumeration in Definition 11.2 make the pointwise stabilizer of a five-element set trivial; consequently, $|\text{Aut}(\mathcal{H})| \leq 12 \cdot 11 \cdot 10 \cdot 9 \cdot 8$. Equality of orders proves $G = \text{Aut}(\mathcal{H})$ and membership of the 495 stars in G . The automorphism group of the system $S(5, 6, 12)$ thus obtained is, by definition and recognition of its natural action, the Mathieu group M_{12} . $\square$

An individual star generates a cyclic group of order two. It does not by itself generate M_{12} , still less the Monster. Nor does the existence of 495 generators of a permutation group prove that each is the transport of a specific TPK loop. That assertion requires the corresponding loop map; the result established here is the finite incidence action.

Remark 11.10 (Equality of traces does not identify operators). On V_{11} , a star has spectrum $+1$ with multiplicity seven and -1 with multiplicity four. Its normalized trace is $3/11$. A rank-three projector on the same space also has normalized trace $3/11$, but spectrum $1^3, 0^8$. They are not conjugate: their minimal polynomials and spectra differ. The intertwiner (11.14) transports defined operators; it does not turn this scalar coincidence into an equivalence of operators.

11.7 The binary residue and the five regions of π

The next recognition uses the extended binary Golay code G_{24} , with parameters $[24, 12, 8]_2$. Its weights are 0, 8, 12, 16, 24, and the weight-eight supports form the design $S(5, 8, 24)$. Fix a dodecad D , the support of a weight-twelve word. This datum and the incidence isomorphism to be declared below belong to the realization chart; they are not used to generate or select the decimals of π .

Proposition 11.11 (Residue of a dodecad and lifting of hexads). *The set*

$$\mathcal{H}(D) = \{O \cap D : O \text{ is an octad, } |O \cap D| = 6\}$$

is an $S(5, 6, 12)$ system on D . Each of its hexads has a unique lift to an octad. For any four-element set $B \subset D$, the five octads containing it split into four residual lifts and a unique octad $O_B^\perp$ with $O_B^\perp \cap D = B$.

Proof. A five-element set $A \subset D$ belongs to a unique octad O . If $j = |O \cap D|$, the binary word given by the sum of O and D has weight $20 - 2j$. Among the permitted weights, with $0 \leq j \leq 8$, only $j = 2, 4, 6$ are possible. Since $A \subset O \cap D$, we have $j \geq 5$ and therefore $j = 6$. This gives existence and uniqueness of the residual hexad, as well as uniqueness of its lift by selecting five of its points. Moreover,

$$\lambda_4(S(5, 8, 24)) = \frac{\binom{20}{1}}{\binom{4}{1}} = 5, \quad \lambda_4(S(5, 6, 12)) = 4.$$

The four residual hexads on B give four distinct octads. The fifth cannot meet D in six points, since it would be another residual hexad; because it contains B , its intersection has size four and equals B . $\square$

A hexad-preserving chart $\ell : [12] \rightarrow D$ identifies $\mathcal{H}$ with $\mathcal{H}(D)$. Its existence is the recognition of the design already constructed as the small Witt design; a concrete choice of ℓ fixes the relabeling. The lift

$$\iota_D : \mathcal{H} \longrightarrow \{\text{octads of } G_{24}\}, \quad H \longmapsto O \text{ such that } O \cap D = \ell(H),$$

is injective because intersection with D recovers $\ell(H)$. The domain of this injectivity is the hexads, not the record K .

The relation to the five regions of π preserves richer data than the number five. The emissions and their signatures are:

Region	Multiplicity	Emission U_6	Signature
$\perp$	432	(501, 614, 498, 169, 272, 272)	555555
$++_1$	144	(810, 923, 870, 169, 272, 272)	339555
$++_2$	144	(810, 983, 810, 169, 272, 272)	393555
$++_3$	144	(870, 923, 810, 169, 272, 272)	933555
$--$	144	(870, 923, 810, 418, 674, 521)	933717

In each three-digit decimal block, the signature uses the modular difference of the first two digits. The five emissions have the same visible tritic word $w_\pi = 010211$, but not the same regional genealogy. The three positive regions preserve the tail 169|272|272; the negative region changes it to 418|674|521; the transverse region has a constant signature and greater multiplicity.

The marked extension of the regional chart sends the transverse region to $O_{\ell(B_K)}^\perp$, the negative region to $\iota_D(H_\alpha^-)$, and the three positive regions to the other three lifts of the star. This realizes the $4 + 1$ partition of the five octads on $\ell(B_K)$. The individual correspondence between the three positive names and the three remaining petals requires a chart order; incidence does not determine it without that datum. The precise claim is therefore a marked correspondence of regions, compatible with their roles and with the flag, not a canonical bijection inferred from the cardinality five.

The identity $432 + 4 \cdot 144 = 1008$ preserves the regional multiplicities. Also, $1008 = 36 \cdot 28$, and an octad has $\binom{8}{2} = 28$ pairs. These equalities are arithmetic catalog checks, not substitutes for the map ι_D or the regional chart.

11.8 From the ternary code to the Niemeier lattice

The passage to dimension twenty-four need not first go through the binary code. It is performed directly from C_W by discriminant gluing. Two compatible prolongations thus remain distinct: one preserves the residue $W_{12} \subset W_{24}$; the other constructs the lattice on which the vertex operator algebra will be realized.

Let $A_2 = \mathbb{Z}a_1 \oplus \mathbb{Z}a_2$ with Gram matrix

$$\begin{pmatrix} 2 & -1 \\ -1 & 2 \end{pmatrix}, \quad \rho = a_1 + a_2, \quad \rho^2 = 2.$$

Represent the three classes of A_2^*/A_2 by

$$\varpi_0 = 0, \quad \varpi_1 = \frac{2a_1 + a_2}{3}, \quad \varpi_2 = \frac{a_1 + 2a_2}{3}.$$

The map $j \mapsto \varpi_j + A_2$ additively identifies $\mathbb{F}_3$ with this discriminant. For $c \in C_W$, write $\phi(c) = (\varpi_{c_1}, \dots, \varpi_{c_{12}})$ and define

$$N(C_W) = \bigcup_{c \in C_W} (A_2^{12} + \phi(c)). \quad (11.15)$$

Theorem 11.12 (Ternary gluing). *The object (11.15) is a positive, even, unimodular lattice of rank 24. Its root system is exactly A_2^{12} ; it is therefore recognized as the Niemeier lattice with that root system.*

Proof. Linearity of the code makes the union of classes stable under addition. Self-orthogonality annihilates the discriminant pairing: the latter is a multiple of $c \cdot d/3$ modulo $\mathbb{Z}$. Thus the lattice is integral. The norm of a glue class is $2 \text{wt}(c)/3$ modulo $2\mathbb{Z}$. The weights in (11.9) are multiples of three, so the lattice is even.

The index of A_2^{12} in $N(C_W)$ is $|C_W| = 3^6$. Consequently,

$$\det N(C_W) = \frac{3^{12}}{(36)^2} = 1.$$

Positivity follows from the orthogonal sum of the twelve A_2 metrics. In a nonzero class of A_2^*/A_2 , the minimum norm is $2/3$; every nonzero word has at least six nonzero positions. Hence a vector in a nontrivial glue class has norm at least $6 \cdot 2/3 = 4$. No norm-two roots occur outside A_2^{12} . Within it, the roots are precisely those of its twelve components. $\square$

11.9 Incidence selection of the rootless unimodular neighbor

The label $o_* = 1$ of the flag is now realized as one of the twelve A_2 components. This operation declares the preceding metric and the positive radial chamber

$$v(a) = \sum_{i=1}^{12} a_i \rho_i, \quad a_i = 1 + 3b_i, \quad b_i \in \mathbb{Z}_{\geq 0}.$$

The evenness requirement for the index-three neighbor is $v(a)^2 \equiv 0 \pmod{18}$. Since

$$v(a)^2 = 2 \sum_i (1 + 3b_i)^2,$$

this congruence is equivalent to $\sum_i b_i \equiv 1 \pmod{3}$. The least norm in the chamber is obtained with a single $b_i = 1$ and all others zero; it is 54. The marked origin selects

$$v_\alpha = 4\rho_1 + \rho_2 + \dots + \rho_{12}, \quad v_\alpha^2 = 54. \quad (11.16)$$

The word “least” refers to this positive chamber. It does not assert minimality throughout the residual class: the vector $(a_1 - 2a_2, \rho, \dots, \rho)$ represents the same class modulo three and has norm $14 + 11 \cdot 2 = 36$. The chamber restriction is part of the construction data, not an omissible conclusion.

Let $N = N(C_W)$. Define

$$N_{\alpha,0} = \{x \in N : \langle x, v_\alpha \rangle \equiv 0 \pmod{3}\}, \quad N_\alpha = N_{\alpha,0} + \mathbb{Z} \frac{v_\alpha}{3}. \quad (11.17)$$

Theorem 11.13 (Leech neighbor determined by the marked flag). *With the declared metric and chamber, N_α is a positive, even, unimodular lattice of rank 24, with no vectors of norm two. By the corresponding classification theorem, it is isometric to the Leech lattice Λ_{24} .*

Proof. The character $x \mapsto \langle x, v_\alpha \rangle \pmod{3}$ is nonzero: a simple root of a component pairs with $a_i \rho_i$ to give $a_i \equiv 1 \pmod{3}$. Hence $[N : N_{\alpha,0}] = 3$ and $\det N_{\alpha,0} = 9$. Moreover, $v_\alpha/3 \in N_{\alpha,0}^*$ and $v_\alpha \in N_{\alpha,0}$. If $v_\alpha/3$ belonged to N , the integrality of N would make all pairings with v_α divisible by three, contradicting the nonzero character. The class of $v_\alpha/3$ therefore has order three. It follows that

$$[N_\alpha : N_{\alpha,0}] = 3, \quad \det N_\alpha = 9/3^2 = 1.$$

The extension is integral, and for $x \in N_{\alpha,0}$,

$$\left(x + m \frac{v_\alpha}{3}\right)^2 = x^2 + 2m \frac{\langle x, v_\alpha \rangle}{3} + 6m^2 \in 2\mathbb{Z}.$$

Hence it is even.

It remains to exclude roots, the decisive step. The roots of N belong to A_2^{12} . Their pairings with v_α are congruent to ± 1 or ± 2 modulo three; none belongs to $N_{\alpha,0}$. For the other two classes of the neighbor, each component belongs to one of the local classes

$$A_2 + \varpi_j \pm \rho/3, \quad j = 0, 1, 2.$$

Since $4\rho/3 \equiv \rho/3 \pmod{A_2}$, this description also holds in the distinguished component. Writing a local vector as $(ua_1 + va_2)/3$, its norm is $2(u^2 - uv + v^2)/9$. The six preceding classes have nonzero residues and minimum $2/9$: representatives minimizing the quadratic form are $(\pm 1, 0)$, $(0, \pm 1)$, $(1, 1)$, $(-1, -1)$ in the appropriate residues. A vector in a new class therefore has norm at least $12 \cdot 2/9 = 8/3 > 2$. There are no roots in any of the three classes. The classification of positive even unimodular lattices of rank twenty-four identifies the unique rootless case with Λ_{24} . $\square$

The argument does not recognize Leech from a coincidence of dimension. It constructs the gluing, proves integrality, evenness, and the determinant, and eliminates roots through a local bound. Only then does it apply the classification theorem. The dependence on α is preserved in the flag selecting o_* and in the vector (11.16); the scalar value of the constant is not converted into a length of the lattice.

11.10 The vertex operator algebra and the Monster

Let $\Lambda = N_\alpha$, already recognized as Leech. The next step uses its integral lattice, not a decimal expansion. The lattice vertex operator algebra construction is written

$$V_\Lambda = M(1) \otimes \mathbb{C}_\varepsilon[\Lambda].$$

Here $M(1)$ is the rank-twenty-four Heisenberg oscillator space; $\mathbb{C}_\varepsilon[\Lambda]$ is the group algebra twisted by the cocycle of the central extension of the lattice. The cocycle and the lift of the isometries are part of the construction: a bare vector-space sum does not determine the vertex products.

Negation $\lambda \mapsto -\lambda$ admits the standard lift θ . With its twisted module and the compatible parity choice, the Frenkel–Lepowsky–Meurman construction produces

$$V_{\text{HMT}}^h = V_\Lambda^+ \oplus (V_\Lambda^T)^+. \quad (11.18)$$

The sign $+$ denotes the even part of the lifted action. The expression includes the orbifold vertex product; it is not merely a sum of graded spaces.

Theorem 11.14 (Prolongation to the Moonshine module). *The algebra (11.18), constructed on the neighbor (11.17), is a realization of the Moonshine algebra $V^{\natural}$. It has central charge 24, zero weight-one component, and automorphism group isomorphic to the Monster $\mathbb{M}$. Its graded character is*

$$\mathrm{Tr}_{V_{\mathrm{HMT}}^{\natural}} q^{L_0-1} = J(\tau) = j(\tau) - 744 = q^{-1} + 196884q + 21493760q^2 + \cdots, \quad q = e^{2\pi i\tau}. \quad (11.19)$$

Proof. Theorem 11.13 verifies the lattice hypotheses: positivity, evenness, unimodularity, rank twenty-four, and absence of roots. The Frenkel–Lepowsky–Meurman construction and theorem for the Leech lattice then apply. They identify the orbifold with $V^{\natural}$ and establish its character and its automorphism group. Composing the flag, the neighbor, and the VOA construction functor gives the realization stated here. The group is not inferred from the coefficient 196884. $\square$

The theorem used is a subsequent classical recognition with verified hypotheses, not a new proof of the FLM construction in this article [7]. Moreover, the notation $q = e^{2\pi i\tau}$ is the conventional Fourier variable used to recognize the character. It does not introduce π or e as inputs into the coinductive generator that has already determined them.

Part of the trace mechanism can be seen before the orbifold. Only the zero lattice vector is fixed by negation; each oscillator contributes an alternating sign. Hence

$$\mathrm{Tr}_{V_{\Lambda}} \theta q^{L_0-1} = t_2(\tau) := q^{-1} \prod_{n \geq 1} (1 + q^n)^{-24}.$$

This trace in the lattice algebra must not be confused with the character of $V^{\natural}$ without insertion. Nor does it by itself identify a Monster conjugacy class: the space, insertion, and lift must be specified before functions are compared.

11.11 Graded Moonshine traces and representation of the Monster

For $g \in \mathrm{Aut}(V_{\mathrm{HMT}}^{\natural})$, the trace with insertion is

$$T_g(\tau) = \mathrm{Tr}_{V_{\mathrm{HMT}}^{\natural}} g q^{L_0-1}.$$

The monstrous Moonshine theorem identifies these series as Hauptmoduln of the corresponding genus-zero groups [8, Theorem 1.1]. The character J , the group $\mathbb{M}$, and the collection $g \mapsto T_g$ are realizations of the algebra already constructed. They do not form a causal sequence in which one first guesses J and then infers the Monster.

The unit has a precise algebraic location here. The weight-zero subspace is the vacuum line, $V_0^{\natural} = \mathbb{C}\mathbf{1}$, and the coefficient of q^{-1} in J is therefore 1. In weight one the dimension is zero, accounting for the absence of the constant term in $J = j - 744$. A morphism of vertex algebras preserves the vacuum vector. This property provides the technical meaning of preservation of the unit at this level; the unit of cylindrical observables is preserved by the precomposition proved earlier. The relation between the two levels must follow the construction maps with their units.

In weight two, the equality

$$196884 = 1 + 196883 = 196560 + 324 = 54(5 \cdot 729 + 1).$$

appears. The first decomposition corresponds to the trivial conformal line and the irreducible component of dimension 196883 in the Monster representation. The other expressions have lattice-theoretic or arithmetic readings. Their equality does not automatically identify their summands with TPK regions, words, or routes. Doing so would require presenting the maps between those spaces and proving that they respect the relevant actions.

In particular, an isometry $N_{\alpha} \simeq \Lambda_{24}$ induces a realization of (11.18), and a choice of isomorphism with $V^{\natural}$ transports its automorphism group. This does not construct an action of the Monster on the

digits of π , on the blocks of K , or on all TPK histories. Such an additional assertion would require a representation ρ_{TPK} and a map F with an explicit square

$$F\rho_{\text{TPK}}(g) = \rho_{V^{\natural}}(g)F,$$

including domain, codomain, and preserved information. The absence of that step from the present claim does not remove the incidence intertwiner U_W , the flag, or the neighbor already constructed; it only delimits a different promotion, namely an action on the source state.

The chain established in this section can be summarized, keeping its transport data visible, as

$$\begin{aligned} &\text{APP-TRIT-TPK and arithmetic state with trajectory memory} \longrightarrow (w_j, f_j; K, s_0) \\ &\longrightarrow (C_W, \mathcal{H}; B_K \subset H_{\alpha}^-, P_{\pi}, H_e, H_{\varphi}, H_{\text{APP}}) \\ &\longrightarrow (N(C_W), o_*, \text{metric and chamber}) \longrightarrow N_{\alpha} \longrightarrow V_{\text{HMT}}^{\natural}. \end{aligned}$$

At the incidence level, the stars generate M_{12} , and the residual chart realizes the regional $4 + 1$ structure in W_{24} . At the lattice and VOA level, the construction gives Leech and the Moonshine module. These are prolongations of the same marked source, with different objects and maps. Causal continuity does not require identifying their codomains or erasing the explicit choices of frame, chamber, and lift.

The result of this section is an assembled derivation: the finite, incidence, lattice, and vertex-operator proofs realize the passage from dodecaphasic incidence to the hypotheses of the recognition theorems. In the domains and with the declared chart choices, the chain constructs the Witt design, generates its Mathieu group, selects the Leech neighbour, and prolongs the structure to the Moonshine module; each stage preserves the object, operation, and transported information that make the subsequent recognition applicable.

12 Planck constants and the determinantal structure of action

12.1 Determinantal action scale and decimal valuation

The absolute scale and the relative corrector originate in distinct objects. The former uses the local block of the step-three dodecaphase record,

$$H_4 = \begin{pmatrix} 1 & 1 & 1 & 1 \\ 1 & 1 & -1 & -1 \\ 1 & -1 & 1 & -1 \\ 1 & -1 & -1 & 1 \end{pmatrix}, \quad G_H = \mathbb{Z}^4 / H_4 \mathbb{Z}^4. \quad (12.1)$$

The operation uses a local quotient. The three blocks of the full record are not replaced by a tensorization that would multiply the number of sheets of this reader once again.

Lemma 12.1 (Number of sheets of the local quotient). *The Smith normal form of H_4 is $\text{diag}(1, 2, 2, 4)$. Consequently,*

$$G_H \simeq \mathbb{Z}/2\mathbb{Z} \oplus \mathbb{Z}/2\mathbb{Z} \oplus \mathbb{Z}/4\mathbb{Z}, \quad |G_H| = 16.$$

Proof. The greatest common divisor of the entries is one. The minors of order two are even, and one has value -2 , so their greatest common divisor is two. Since $H_4 H_4^T = 4I$ and $\det H_4 = -16$, the cofactors are ± 4 ; the third determinantal divisor is four. The fourth is 16. The successive quotients of the divisors 1, 2, 4, 16 are 1, 2, 2, 4. Their product gives the order of the quotient. $\square$

Definition 12.2 (Determinantal action reader). The multiplicative norm of the scalar section α , constant over the sheets of G_H , and one application of the self-scale φ produce

$$\Sigma_H = \varphi N_{G_H}(\alpha), \quad N_{G_H}(\alpha) = \prod_{g \in G_H} \alpha = \alpha^{16}. \quad (12.2)$$

The associated decimal chart determines $\text{ord}_{10}(\Sigma_H) = \lfloor \log_{10} \Sigma_H \rfloor$.

The sixteenth power thus arises from the number of sheets of the quotient, once this local reader has been fixed. The factor φ acts once on the basis of the action line. The Smith computation alone would not select any possible functional on the quotient: the multiplicative norm and the self-scale form part of the declared operation.

Theorem 12.3 (Decade of the action section). *Reader (12.2), applied to the already generated HMT coordinates, satisfies*

$$10^{-34} < \Sigma_H < 10^{-33}, \quad \text{ord}_{10}(\Sigma_H) = -34. \quad (12.3)$$

Proof. The following rational intervals suffice; they are much coarser than the available cylinders:

$$\frac{1618}{1000} < \varphi < \frac{1619}{1000}, \quad \frac{7297}{10^6} < \alpha < \frac{7298}{10^6}.$$

The function $(x, a) \mapsto xa^{16}$ is increasing in both positive arguments. The two integer inequalities

$$1618 \cdot 7297^{16} > 10^{65}, \quad 1619 \cdot 7298^{16} < 10^{66}$$

imply

$$10^{-34} < \frac{1618}{1000} \left(\frac{7297}{10^6} \right)^{16} < \Sigma_H < \frac{1619}{1000} \left(\frac{7298}{10^6} \right)^{16} < 10^{-33}.$$

The definition of decimal order completes the proof. No SI unit or reference mass enters these inequalities. $\square$

The normalized section begins with $\Sigma_H = 1.0462438381 \dots \times 10^{-34}$. Its mantissa is not identified with the action coordinate before or after return. The determinantal reader fixes the decimal order; the two return coordinates are obtained through the following readers.

12.2 Action sections and dimensionless return quotient

Let $\mathcal{L}_S$ be an oriented action line, and let $\mathcal{U}_S$ be its positive basis. The fifth-order character and the regional return term have the expressions

$$H_5 = \frac{1}{2} \sqrt{\frac{1000\alpha}{\varphi}} - \alpha + \frac{9}{16}\alpha^2 - \frac{5}{9}\alpha^3 + \frac{7}{48}\alpha^4 - \frac{1}{54}\alpha^5, \quad (12.4)$$

$$D_A = \exp\left(-\frac{100\pi\alpha}{9}\right), \quad (12.5)$$

$$c_\pi(\alpha) = 169\alpha^6 + \frac{D_A\alpha^7}{1 - D_A\alpha}, \quad (12.6)$$

$$\eta_{\text{ret}} = H_5 - \frac{90}{\pi} c_\pi(\alpha). \quad (12.7)$$

These are subsequent readings of the dodecaphase closure and its regional incidence. The integer 169 is retained as the regional selector already constructed, and the coefficients of H_5 as those of its local character; they are not replaced by an interpolation of the Planck constant. The formulas make explicit exactly which readings are needed for the electronic stage. It is not necessary to introduce other functionals of mass theory here.

For $0 < \alpha < 1$, one has $0 < D_A < 1$, so $D_A\alpha < 1$. The rational term in (12.6) preserves the complete sum of the geometric tail

$$\frac{D_A\alpha^7}{1 - D_A\alpha} = \sum_{n=0}^{\infty} D_A^{n+1} \alpha^{n+7}. \quad (12.8)$$

It is not a truncation that may be silently modified when the precision changes. After the term of index N , the remaining tail is $D_A^{N+2}\alpha^{N+8}/(1 - D_A\alpha)$, allowing evaluation to be controlled without recourse to a target quantity.

Definition 12.4 (Action sections and return reader). The pre-return and post-return sections are

$$\hbar_{\text{pre}} = H_5 10^{-34} \mathcal{U}_S, \quad \hbar_{\text{ret}} = \eta_{\text{ret}} 10^{-34} \mathcal{U}_S. \quad (12.9)$$

Whenever $\eta_{\text{ret}} > 0$, define

$$\delta_{\text{ret}} = H_5 - \eta_{\text{ret}}, \quad R_{\text{act}} = \frac{\hbar_{\text{pre}}}{\hbar_{\text{ret}}} = \frac{H_5}{\eta_{\text{ret}}}. \quad (12.10)$$

Proposition 12.5 (Positivity and invariance of the ratio). *In the intervals used in the proof of (12.3), $0 < \eta_{\text{ret}} < H_5$, and therefore $R_{\text{act}} > 1$. The ratio does not change when the basis $\mathcal{U}_S$ is replaced by any other common positive basis.*

Proof. The terms of c_π are positive. For an elementary bound, it suffices to use $\alpha < 0.008$, $\varphi < 1.619$, $\alpha > 0.007$, $D_A < 1$, and $\pi > 3$. The root in (12.4) is greater than one, whereas the sum of the negative terms is less than 0.008001; in particular, $H_5 > 0.99$. Likewise,

$$0 < \frac{90}{\pi} c_\pi < 30 \left(169(0.008)^6 + \frac{(0.008)^7}{1 - 0.008} \right) < 10^{-6}.$$

Thus $\eta_{\text{ret}} > 0.989999 > 0$ and $\eta_{\text{ret}} < H_5$. Finally, the basis change divides both action coordinates by the same positive factor, which cancels in their quotient. $\square$

Subsequent evaluation gives

$$\begin{aligned} H_5 &= 1.0545718176461544696 \dots, \\ \eta_{\text{ret}} &= 1.0545718169150390611 \dots, \\ R_{\text{act}} &= 1.0000000006932817630 \dots. \end{aligned}$$

The difference and the quotient are two readings of the same sections, one additive and the other multiplicative. They do not represent two independent Planck constants. The passage from angular

action to action per complete turn is expressed, in each section, by $h = 2\pi\hbar$. The subsequent chart $\mathcal{U}_S \mapsto \mathcal{J}$ assigns a unit but does not act retroactively on the decimal order already proved.

Regional return can also be written in an angular chart. This reparametrization is not an additional indispensable dependency: (12.4)–(12.7) directly supply H_5 , η_{ret} , and R_{act} . If angle and counter-angle are introduced, both must be transported in the same chart; degrees and radians are not mixed within a difference.

12.2.1 Action character and provenance of its coefficients

The reduced Planck constant, $\hbar$, is the physical object that motivates realization of the angular-action section. Its position in the electronic dependency must be specified at two levels. The determinantal norm Σ_H determines the decade of the scale. The character H_5 and the regional return then determine the coordinates of the action sections. This separation allows the result needed for the electron to be presented without anticipating the full developments of other sectors of dimensional mechanics.

The provenance of the fifth-order character can be made explicit through invariants already fixed in the support. In the central chart, consider

$$B_c = \begin{pmatrix} 7 & 2 \\ 2 & 7 \end{pmatrix}, \quad \lambda_+ = 9, \quad \lambda_- = 5.$$

The calendar has length 12, the step-3 orbit has length 4, and the relevant census quotient is $104976/1944 = 54$. The character coefficients satisfy

$$\frac{9}{16} = \frac{\lambda_+}{4^2}, \quad \frac{5}{9} = \frac{\lambda_-}{\lambda_+}, \quad \frac{7}{48} = \frac{\text{tr}(B_c)/2}{4 \cdot 12}, \quad \frac{1}{54} = \frac{1944}{104976}. \quad (12.11)$$

The assignment of these invariants to degrees 2, 3, 4, 5, with the alternating signs of (12.4), belongs to the definition of the analytic character used. Making it explicit is essential: the collection of cardinalities, considered without the character and without the ordering of its degrees, would admit other expressions. The constructive content lies in the composition of the support, orientation, and declared analytic rule.

12.2.2 Regional continuation and renewal equation

The selector 169 determines the degree-six impulse. Subsequent prolongation is organized by the transfer D_A and a unit normalization of the determinant line. These rules admit a formulation in formal series before substituting $z = \alpha$, with D_A already constructed. Write

$$\mathcal{C}(z) = 169z^6 + R(z), \quad R(z) \in z^7\mathbb{R}[[z]].$$

Concatenating a prolongation increases the degree by one and multiplies its weight by D_A . The first strict prolongation has weight D_A . Its renewal equation is therefore

$$R(z) = D_A z^7 + D_A z R(z). \quad (12.12)$$

Proposition 12.6 (Formal uniqueness of the normalized continuation). *Equation (12.12) determines a unique series, and*

$$\mathcal{C}(z) = 169z^6 + \frac{D_A z^7}{1 - D_A z} = 169z^6 + \sum_{n=7}^{\infty} D_A^{n-6} z^n.$$

The realization is analytic for $|D_A z| < 1$.

Proof. The element $1 - D_A z$ has constant term one and is invertible in the ring of formal series. Solving the equation produces the displayed series. Equivalently, the degree-seven coefficient is D_A , and each subsequent coefficient is D_A times the preceding one. The geometric series converges absolutely in the indicated disk. $\square$

Normalization of the first prolongation is an indispensable operative datum. If only the degree-six impulse and the concatenation ratio were retained, the series

$$169z^6 + \lambda \frac{D_A z^7}{1 - D_A z}$$

would still share both data for every λ . The unit rule fixes $\lambda = 1$ before evaluation is performed. This locates the contribution of each rule to the uniqueness of the reader, and the rational expression preserves the full continuation.

12.2.3 Reduced Planck constant and return of the action section

In the dimensional chart of (12.9), the pre-return section is the model construction intended for comparison with the reduced Planck constant:

$$\hbar_{\text{pre}} = H_5 10^{-34} \mathcal{U}_S.$$

The post-return section additionally preserves the regional compensation $90\mathcal{C}(\alpha)/\pi$. Both belong to the same action line, and their quotient R_{act} measures the multiplicative effect of return. The passage to h through 2π uses the relation between angular parametrization and a complete turn.

13 Angular coordinates and boundary observables

13.1 Angular coordinates, normalization of the turn, and units of action

The quantity of action and the angular coordinate belong to distinct spaces. The section $\hbar_{\text{ret}}$ occupies the action line $\mathcal{L}_S$; η_{ret} is its coefficient in the declared basis. Angular representation acts on the dimensionless pair $(\alpha, \eta_{\text{ret}})$, after its construction:

$$\mathcal{P}(\alpha, \eta_{\text{ret}}) = ([1000\alpha]_{360}, [2(\eta_{\text{ret}} + \alpha)]_{360}). \quad (13.1)$$

Nonadic completion determines the factor 1000; bidirectional closure determines the factor two. The circular chart of 360 units makes the partitions $12 \cdot 30 = 4 \cdot 90 = 3 \cdot 120$ visible. On the positive branch of the article, the representatives are

$$A_{\text{deg}} = 1000\alpha, \quad C_{\text{deg}}^* = 2(\eta_{\text{ret}} + \alpha).$$

Proposition 13.1 (Shift of the base-thousand representation). *Let $\alpha = \sum_{n \geq 1} b_n 1000^{-n}$, with $b_n \in \{0, \dots, 999\}$, be the compatible expansion already generated. The lifted coordinate $A_{\text{deg}} = 1000\alpha$ satisfies*

$$A_{\text{deg}} = b_1 + \sum_{n \geq 1} b_{n+1} 1000^{-n}.$$

The representation preserves the block sequence from the second block onward; the initial block becomes the integer part.

Proof. The series converges absolutely. Multiplying each term by 1000 and separating the initial term gives the identity. The operation acts on the constructed expansion, with its blocks and memory; the subsequent circular representation takes its class modulo 360. $\square$

Definition 13.2 (Charts of the angular coordinate). The degree chart expresses a complete turn in 360 units. The radian chart expresses it as 2π . The coordinate changes and their inverses are

$$x = \frac{\pi}{180} A_{\text{deg}}, \quad y = \frac{\pi}{180} C_{\text{deg}}^*, \quad A_{\text{deg}} = \frac{180}{\pi} x, \quad C_{\text{deg}}^* = \frac{180}{\pi} y.$$

On a path with preserved winding number, the lifted coordinates are used; the circular quotient expresses its class modulo the corresponding turn.

The definition characterizes degrees and radians through their exact relation to the turn. The record also retains orientation, turns, and transport memory. Representation of its angular class preserves phase, whereas the lift distinguishes paths with the same final phase. The physical reading of action requires specifying which of these coordinates enters the integral.

Proposition 13.3 (Covariance of the action reader). *For $h_a = 2\pi\hbar_a$, the action of a lifted coordinate satisfies*

$$\mathcal{S}_a(\theta) = \hbar_a \theta_{\text{rad}} = h_a \frac{\theta_{\text{deg}}}{360}. \quad (13.2)$$

A positive turn expresses h_a .

Proof. Substituting $\theta_{\text{rad}} = \pi\theta_{\text{deg}}/180$ gives $\hbar_a \pi\theta_{\text{deg}}/180 = (2\pi\hbar_a)\theta_{\text{deg}}/360$. In one turn $\theta_{\text{deg}} = 360$, yielding h_a . $\square$

The periodicity of a representation is declared separately. If the transported representation satisfies $U(2\pi) = -I$, its operator restoration occurs at 4π . This property of the lift coexists with the per-turn normalization of (13.2). Each preserves its own object: integrated quantity, circular phase, and represented state.

13.2 Covariance of the channels and the incidence functional

The channels are written equivalently as

$$q_{\pm} = \exp[-(x \pm y)] = \exp\left[-\frac{\pi}{180}(A_{\text{deg}} \pm C_{\text{deg}}^*)\right]. \quad (13.3)$$

Proposition 13.4 (Recovery of the even and odd coordinates). *In the lifted real chart, the positive channels uniquely determine the two coordinates:*

$$A_{\text{deg}} = -\frac{90}{\pi} \log(q_+ q_-), \quad C_{\text{deg}}^* = \frac{90}{\pi} \log \frac{q_-}{q_+}.$$

The exchange $(q_+, q_-) \mapsto (q_-, q_+)$ preserves A_{deg} and reverses C_{deg}^* .

Proof. The sum and difference of the logarithms of (13.3) are, respectively, $-\pi A_{\text{deg}}/90$ and $-\pi C_{\text{deg}}^*/90$. The real logarithm is single-valued on the positive channels. The product is symmetric, and the quotient is inverted when they are exchanged. $\square$

In the twelve-position record, the antipodal involution $r \mapsto r + 6 \pmod{12}$ forms six pairs. The uniform distribution of the odd coordinate over these pairs is $C_{\text{deg}}^*/6$: summing their six contributions recovers C_{deg}^* . This sectorial coordinate is distinguished from the Planck constant h , which belongs to the action line.

The conversion is applied only once to each argument. The quantity α remains dimensionless, and the quantity $\hbar$ continues to belong to the action line; their preceding coordinates enter the map $\mathcal{P}$.

In addition to the channels, the Barbero–Immirzi functional contains a component bilinear in the angular representatives. Its transport is

$$\frac{\pi}{180} A_{\text{deg}} C_{\text{deg}}^* = \frac{180}{\pi} xy. \quad (13.4)$$

The identity follows by substituting both degree representatives with their expressions in radians. The complete formula in analytic coordinates is therefore

$$\gamma_{\text{HMT}} = \frac{180}{\pi} xy + 12[S_{90}(e^{-x+y}) - S_{90}(e^{-x-y})] - [S_{120}(e^{-x+y}) - S_{120}(e^{-x-y})].$$

Equivalence of the two charts transports the bilinear component and the exponential functions simultaneously. This transformation makes explicit the precise role of the degree in the construction and allows the evaluation to be reproduced without ambiguity in angular units.

14 Hexadic–octadic incidence and the Barbero–Immirzi functional

Let $H \hookrightarrow O$ be a marked inclusion, $|H| = 6$, $|O| = 8$. The incidence objects

$$\mathcal{F}_6 = H \times \binom{H}{2}, \quad \mathcal{F}_8 = O \times \binom{H}{2}$$

have cardinalities 90 and 120. Their normalized difference gives

$$\frac{|\mathcal{F}_6| - |\mathcal{F}_8|}{24\binom{6}{2}} = -\frac{1}{12}.$$

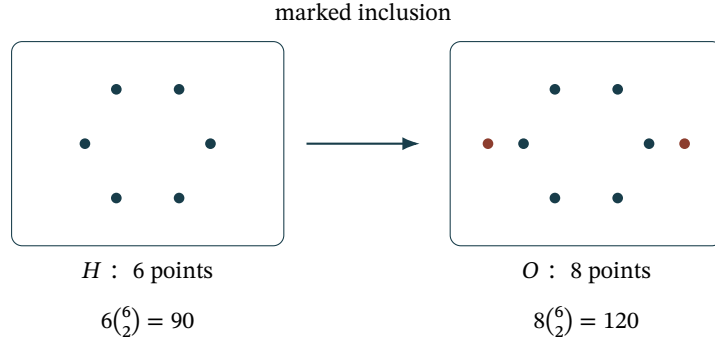


Figure 8: Marked incidence from six to eight points. The drawing represents the sets and the inclusion; it does not purport to turn the faces or vertices of a cube into Steiner blocks without their incidence map.

14.1 The weights precede the pole coefficient

In the free module of oriented events, the source fixes the fundamental chain of the turn

$$c_{12}^+ = \sum_{j \in \mathbb{Z}/12\mathbb{Z}} [j, \mathcal{F}_6] - [\partial_{\text{ret}}, \mathcal{F}_8].$$

Multiplicity twelve corresponds to the sectors, and multiplicity one to the oriented seam. The involution changes the orientation of the complete chain. Let

$$S_m(q) = \frac{q^m}{1 - q^{3m}}, \quad 0 < q < 1.$$

The linear reader assigns S_{90} to each sector and S_{120} to the seam; consequently,

$$\mathcal{R}(c_{12}^+) = 12S_{90} - S_{120}.$$

Theorem 14.1 (Pole coefficient and conjugate evaluation). *The image of the fundamental chain has coefficient $1/24$ with respect to $(1 - q)^{-1}$ and residue $-1/24$ with respect to $(q - 1)^{-1}$. Its evaluation in the previously generated channels determines*

$$\gamma_{\text{HMT}} = \frac{\pi AC^*}{180} + 12[S_{90}(q_-) - S_{90}(q_+)] - [S_{120}(q_-) - S_{120}(q_+)]. \quad (14.1)$$

Proof. The factorization $1 - q^{3m} = (1 - q) \sum_{j=0}^{3m-1} q^j$ implies $\lim_{q \uparrow 1} (1 - q)S_m(q) = 1/(3m)$. By linearity,

$$\frac{12}{270} - \frac{1}{360} = \frac{1}{24}.$$

The coordinate $q - 1$ reverses the sign of the residue. Conjugate evaluation and the angular component of the source produce (14.1). $\square$

The Diophantine certificate $4a - 3b = 45$ admits the pairs $(12 + 3t, 1 + 4t)$, $t \geq 0$; it certifies minimality of $(12, 1)$, but does not replace the origin of the multiplicities in the chain. The evaluation of (14.1) is

$$\gamma_{\text{HMT}} = 0.27400767269445927474725526077398041940 \dots$$

This number is an output of the source functional; it is not an external value used to select A , C^* , or their coefficients.

14.2 Parity and structural sensitivity

Reversing C^* exchanges q_+ and q_- and changes the sign of the angular component. Hence

$$\gamma_{\text{HMT}}(A, -C^*) = -\gamma_{\text{HMT}}(A, C^*).$$

The functional preserves orientation, whereas other readers, such as q_+q_- , are even. This difference precedes their physical interpretation.

15 Simultaneous rigidity of the longitudinal reader and radial coupling

The construction of Triadic Modular Holography transmits to the radial realization an enriched state together with its coordinates, orientation, carry, memory and incidences. The sequence

$$\begin{aligned} & \text{APP} \longrightarrow \text{TRIT} \longrightarrow \text{TPK} \longrightarrow \text{enriched state} \\ & \longrightarrow \text{joint discrete structure of the continuum} \end{aligned}$$

preserves the Archimedean projection and the incidence projection, together with the five constitutive constructions of the continuum. The prolongation

$$w_6 \longrightarrow w_{12} \longrightarrow w_{18} \longrightarrow w_{24} \longrightarrow w_{30} \longrightarrow R_{36} \longrightarrow G_9$$

is coordinated with the operator order $U_t \longrightarrow \Gamma_9 \longrightarrow K_{\text{ph}}$: the phase returns and the memory record advances. The regional coordinates, the selected record K , the coordinate α and the return action are received with this common provenance and their previously constructed readers.

This section classifies realizations that simultaneously preserve the generated data and the longitudinal, metric and temporal rules specified below. Each positive character determines a radial response; the coefficient relating that response to mass is common to all admissible characters. The gravitational interpretation is formulated after both readers have been constructed on the same space.

15.1 Incidence record and constitutive restrictions

Let V_{in} and V_{out} be complex Hilbert spaces of dimension N , equipped with fixed inner products. The marked cardinal specified in R1 is positive; in particular, both fibres are nonzero. Adjoint operators are taken with respect to these products. The symbol U will denote the normalized archive transport, while U_t retains its meaning as the evolution of the enriched state.

R1. Marked record. The incidence set is

$$\Omega = \left\{ (j, k, q, u, v) : \begin{array}{l} j, k \in \mathbb{Z}/12\mathbb{Z}, \quad k - j \in \{0, 3, 6, 9\}, \\ 1 \leq q \leq 8, \quad u < v, \quad u, v \in \{1, 2, 4, 5, 7, 8\} \end{array} \right\}. \quad (15.1)$$

The twelve phases, four relative positions, eight positions of the index q and fifteen pairs from the final set give

$$N = |\Omega| = 12 \cdot 4 \cdot 8 \binom{6}{2} = 12 \cdot 4 \cdot 120 = 5760. \quad (15.2)$$

Each incidence retains its individual mark. In the corresponding basis $(e_i)_{i \in \Omega}$ of V_{out} , let $P_i = e_i e_i^*$. The common mode u is normalized by $|u_i|^2 = 1/N$, and $P = uu^*$. The marked algebra is $\mathcal{A}_\Omega = \text{span}\{P_i : i \in \Omega\}$.

R2. Normalization and complete archive. The generated inverse map $B : V_{\text{in}} \rightarrow V_{\text{out}}$ satisfies

$$B^* B = \frac{1}{4} I_{\text{in}}, \quad B B^* = \frac{1}{4} I_{\text{out}}. \quad (15.3)$$

Both components are preserved:

$$C_0 = (I - P)B, \quad M_0 = PB, \quad C_0 + M_0 = B, \quad U = 2B. \quad (15.4)$$

The first records variation relative to the common mode; the second preserves that mode. Their images are orthogonal, and their sum recovers the complete transport. By (15.3), U is unitary.

R3. Individual inscription. The map $\mathcal{E} : \text{End}(V_{\text{out}}) \rightarrow \mathcal{A}_\Omega$ is a linear conditional expectation that fixes each P_i and satisfies

$$\mathcal{E}(A_1 T A_2) = A_1 \mathcal{E}(T) A_2, \quad A_1, A_2 \in \mathcal{A}_\Omega. \quad (15.5)$$

The inscription of the result retains the complete construction data in a complementary record. The expectation determines the inscribed observable; preservation of the archive determines its subsequent reconstruction.

R4. Longitudinal composition. Fix the generated coordinate $\alpha > 0$ and the longitudinal reference section $L_* > 0$. Define the longitudinal transport by

$$D = L_* \alpha^{16} \mathcal{E}(C_0 C_0^*) U : V_{\text{in}} \longrightarrow V_{\text{out}}. \quad (15.6)$$

This rule applies the inscribed coefficient linearly to the longitudinal section. Its type fixes the position of the power α^{16} and the return coefficient within the composition. Reading the square root of an intensity is a different operation with its own transformation rule.

R5. Positive radial metric. Let $X = X^* > 0$ be the complete character on V_{in} . The positive radial reader R_X on V_{out} is defined by

$$\|R_X v\|^2 = \|XD^* v\|^2 \quad (v \in V_{\text{out}}). \quad (15.7)$$

This identity fixes the metric on every vector of the fibre. In polar notation, R_X is the arrival modulus $|(DX)^*| = \sqrt{DX^2 D^*}$.

R6. Action and clock. The generated action section $\hbar_{\text{ret}} > 0$ and the real phase lift satisfy

$$S(\theta) = \hbar_{\text{ret}} \theta. \quad (15.8)$$

The constitutive velocity $c > 0$ is preserved. Once the intensive length L has been obtained from (15.6), the conjugate clock satisfies $\omega L = c$. For $\theta(t) = \omega t$, the action rate determines the energy scale

$$E_L = \frac{dS}{dt} = \hbar_{\text{ret}} \omega = \frac{\hbar_{\text{ret}} c}{L}. \quad (15.9)$$

The lift and its refinements remain associated with the path memory.

R7. Readers of the same character. On $Y = UXU^* > 0$, define

$$H_X = E_L Y, \quad M_X = \frac{H_X}{c^2}, \quad \Lambda_X = LY^{-1}. \quad (15.10)$$

Energy, mass and conjugate length preserve the same character and the same transport. The inverse acts on the positive sector; the null sector retains its separate treatment.

R8. Reduced radius and dimensional sections. The physical identification of this realization is

$$R_X = \frac{G}{c^2} M_X. \quad (15.11)$$

Thus R_X represents the reduced gravitational radius. The corresponding Schwarzschild radius is $2R_X$. The reference L_* , the action section and the constitutive velocity section are preserved jointly; a change of units transforms their coordinates according to their dimensions.

Provenance of the restrictions. Hadamard transport, archive preservation, inscription, phase action and reciprocity have antecedents in the corpus. The enlarged record (15.1) and the compositions (15.6)–(15.7) are explicitly incorporated into the assembled construction. R4 and R5 are constitutive rules of this realization; the following proof establishes their joint consequence with the remaining antecedents. R8 specifies the physical identification that fixes the meaning of the coefficient.

15.2 Simultaneous determination theorem

Theorem. For the same generated data $B, (P_i), P, \alpha, L_*, \hbar_{\text{ret}}, c$ and the same character X , restrictions R1–R8 uniquely determine the inscription, longitudinal transport and radial response. The coefficient G is common to all admissible positive characters. One has

$$\mathcal{E} = \Delta_\Omega, \quad \Delta_\Omega(T) = \sum_{i \in \Omega} R_i T R_i, \quad (15.12)$$

$$r_N = \frac{N-1}{4N} = \frac{5759}{23040}, \quad L = L_* \alpha^{16} r_N, \quad (15.13)$$

$$D = LU, \quad R_X = LUXU^*, \quad (15.14)$$

$$R_X \Lambda_X = L^2 I, \quad H_X \Lambda_X = \hbar_{\text{ret}} c I. \quad (15.15)$$

The coupling is determined by

$$G = \frac{c^3 L^2}{\hbar_{\text{ret}}} = \frac{c^3 L_*^2}{\hbar_{\text{ret}}} \left(\frac{5759}{23040} \right)^2 \alpha^{32}. \quad (15.16)$$

Proof: uniqueness of inscription. Let $E_{ij} = e_i e_j^*$ be the matrix units. Bimodularity gives

$$\mathcal{E}(E_{ij}) = R_i \mathcal{E}(E_{ij}) R_j.$$

For $i \neq j$, membership of $\mathcal{E}(E_{ij})$ in the diagonal algebra implies $R_i \mathcal{E}(E_{ij}) R_j = 0$. For $i = j$, fixation of R_i gives $\mathcal{E}(E_{ii}) = R_i$. Linearity therefore determines (15.12) on all matrices. The marked inscription is thereby fixed by its algebraic properties.

Proof: return coefficient and archive. The normalization of B , together with $P^2 = P = P^*$, yields

$$C_0 C_0^* = \frac{1}{4}(I - P), \quad M_0 M_0^* = \frac{1}{4}P.$$

Since $R_i P R_i = |u_i|^2 R_i = R_i/N$, it follows that

$$\Delta_\Omega(C_0 C_0^*) = \frac{N-1}{4N} I, \quad \Delta_\Omega(M_0 M_0^*) = \frac{1}{4N} I. \quad (15.17)$$

The two readers sum to $I/4$. The first is scalar, so every normalized distribution (p_i) on the incidences, with $p_i \geq 0$ and $\sum_i p_i = 1$, gives $\sum_i p_i r_N = r_N$. Hence an arbitrary diagonal weighting preserves the same coefficient.

Proof: length and metric. Substitution of (15.17) into R4 gives $D = LU$, with L as in (15.13). Unitarity of U implies $DD^* = L^2 I$. By R5, the quadratic forms of R_X^2 and $DX^2 D^*$ coincide. The polarization identity determines

$$R_X^2 = DX^2 D^* = L^2 UX^2 U^*.$$

The operator $LUXU^*$ is positive and its square is the operator on the right-hand side. Uniqueness of the positive square root proves (15.14). In particular, $R_I = LI$.

Proof: action, reciprocity and coupling. R6 and R7 give

$$H_X = \frac{\hbar_{\text{ret}} c}{L} Y, \quad M_X = \frac{\hbar_{\text{ret}}}{cL} Y, \quad \Lambda_X = LY^{-1}.$$

Multiplication gives both products in (15.15). R8 then becomes

$$LY = \frac{G \hbar_{\text{ret}}}{c^3 L} Y.$$

Invertibility of Y determines (15.16). The same substitution verifies the identity for every positive character. If G_1 and G_2 satisfy the identity with the same data, then $(G_1 - G_2)Y = 0$, hence $G_1 = G_2$. This conclusion distinguishes the state-dependent response R_X from the common coefficient G . $\square$

15.3 Reciprocity and phase rigidity

The power-logarithmic antecedent uses, for $x > 0$,

$$\ell_p(x) = \frac{x^p - 1}{p} \quad (p \neq 0), \quad \ell_0(x) = \log x.$$

Direct substitution proves $\ell_{-p}(x^{-1}) = -\ell_p(x)$. On the positive domain of their inverses, this identity is equivalent to

$$\exp_p(s) \exp_{-p}(-s) = 1.$$

Spectral composition on a positive character transmits this product to the pair LY, LY^{-1} . In the present realization, once R1–R4 and R7 have been fixed, R5 can equivalently be written as $R_X \Lambda_X = L^2 I$: multiplication by Λ_X^{-1} determines $R_X = LUXU^*$, and this expression satisfies R5.

The two products in (15.15) also give a direct form of rigidity:

$$R_X = \frac{L^2}{\hbar_{\text{ret}} c} H_X. \quad (15.18)$$

For a positive rescaling s , the pair $(sR_X, \Lambda_X/s)$ preserves the radial product; the product $H_X(\Lambda_X/s) = \hbar_{\text{ret}} c I/s$, with fixed energy and action, requires $s = 1$. Joint preservation determines the scale. Likewise, every positive radial operator with the same metric R5 coincides with R_X by uniqueness of the positive square root.

The lifted action is also determined exactly. For $\theta \neq 0$, the identity $S/\hbar_{\text{ret}} = \theta$ fixes S . Even when one preserves the rotations of all subdivisions $\theta/9^n$, a fixed real rescaling a satisfying

$$\exp(-ia\theta/9^n) = \exp(-i\theta/9^n) \quad (n \geq 0)$$

must obey $(a - 1)\theta/9^n \in 2\pi\mathbb{Z}$. This sequence tends to zero; for sufficiently large n , its only possible value is zero. Since $\theta \neq 0$, it follows that $a = 1$.

15.4 Planck scales and the unit-character specialization

We write $L_\alpha = L$; in the return section, $E_{P,\text{ret}} = E_L$.

The constructed length $L_\alpha > 0$, the action section $\hbar_a > 0$, and the speed $c > 0$ determine the scales

$$t_P = \frac{L_\alpha}{c}, \quad m_{P,a} = \frac{\hbar_a}{cL_\alpha}, \quad E_{P,a} = \frac{\hbar_a c}{L_\alpha}.$$

Indeed, substituting $G_a = c^3 L_\alpha^2 / \hbar_a$ into $\sqrt{\hbar_a G_a / c^5}$, $\sqrt{\hbar_a c / G_a}$, and $\sqrt{\hbar_a c^5 / G_a}$ yields these three identities, respectively. Positivity fixes each square root. The length precedes the recognition of these scales; their conventional expressions are identities of the constructed realization.

For a mode with character $y > 0$, the mass and longitudinal readings are

$$m(y) = m_{P,a} y, \quad r_g(y) = L_\alpha y, \quad \bar{\lambda}(y) = \frac{L_\alpha}{y}.$$

Thus $r_g(y) \bar{\lambda}(y) = L_\alpha^2$ for every positive mode. Equality of the two lengths characterizes the mode $y = 1$, since $y = y^{-1}$ and $y > 0$ imply $y = 1$. In this mode, the mass is $m_{P,a}$ and both lengths equal L_α . This distinguishes general reciprocity from its unit-character specialization.

The chronological step and the Planck time satisfy $t_0 = (\pi_{\text{HMT}}/54)t_P$. The complete 108-step phase period is $108t_0 = 2\pi_{\text{HMT}}t_P$, with angular frequency $\omega_{108} = 1/t_P = c/L_\alpha$. A horizon of radius $R_h = \zeta_h L_\alpha$, where $\zeta_h > 0$, retains its own geometric variable. The Schwarzschild radius associated with a mass is $2r_g$.

When the two action sections are compared at fixed L_α and c , the ratio $\rho = \hbar_b / \hbar_a$ gives $G_b = G_a / \rho$, $m_{P,b} = \rho m_{P,a}$, and $E_{P,b} = \rho E_{P,a}$; L_α and t_P remain unchanged. This comparison between sections remains distinct from temporal evolution of the constants.

15.5 Archive covariance and change of units

Let $T : V_{\text{in}} \rightarrow V'_{\text{in}}$ and $S : V_{\text{out}} \rightarrow V'_{\text{out}}$ be unitary. Joint transport of the data is given by

$$B' = SBT^*, \quad P' = SPS^*, \quad P'_i = SP_iS^*, \quad X' = TXT^*.$$

The transported expectation $\mathcal{E}'(A') = S\mathcal{E}(S^*A'S)S^*$ fixes P'_i and is bimodular over the transformed marked algebra. Substitution gives

$$\begin{aligned} U' &= SUT^*, & D' &= SDT^*, \\ R'_{X'} &= SR_XS^*, & H'_{X'} &= SH_XS^*, & \Lambda'_{X'} &= S\Lambda_XS^*. \end{aligned}$$

Conjugation preserves the identities R1–R8 and the same coefficient G . Marks, orientations and records are transported together with the bases. Permutations of incidences are a particular case.

Adjoining an auxiliary memory space $\mathcal{F}$ transports P to $P \otimes I_{\mathcal{F}}$, U to $U \otimes I_{\mathcal{F}}$, and Δ_{Ω} to $\Delta_{\Omega} \otimes \text{id}$. The inscription of $C_0C_0^* \otimes I_{\mathcal{F}}$ is $r_N I \otimes I_{\mathcal{F}}$; the coefficient retains the original marked cardinality.

To specify dimensional covariance, let ℓ , τ and m be the positive factors of the new length, time and mass units relative to the previous ones. The numerical coordinates satisfy

$$L' = L/\ell, \quad c' = c\tau/\ell, \quad \hbar'_{\text{ret}} = \hbar_{\text{ret}}\tau/(m\ell^2).$$

Consequently,

$$\frac{(c')^3(L')^2}{\hbar'_{\text{ret}}} = G \frac{m\tau^2}{\ell^3},$$

which is the transformation law for a quantity of dimension $\text{length}^3 \text{mass}^{-1} \text{time}^{-2}$. The section L_* transforms as L , while α and r_N remain dimensionless.

15.6 Exact elimination of coupled memory

Let $\mathcal{K}$ and $\mathcal{H}$ be finite-dimensional Hilbert spaces, let $W = W^* > 0$ be a global weight, and let $\mathcal{C} : \mathcal{K} \rightarrow \mathcal{H}$ be surjective. The action of residuals r , with boundary defect z , is

$$\mathcal{S}(r) = \alpha r^* W r, \quad \mathcal{C}r = z, \quad \alpha > 0.$$

The global weight retains its cross-blocks between incidences. Define $\mathcal{R} = \mathcal{C}W^{-1}\mathcal{C}^*$. For $v \neq 0$, injectivity of $\mathcal{C}^*$ gives

$$v^* \mathcal{R} v = \|W^{-1/2} \mathcal{C}^* v\|^2 > 0.$$

Thus $\mathcal{R}$ is invertible. The minimum-action residual is

$$r_*(z) = W^{-1} \mathcal{C}^* \mathcal{R}^{-1} z.$$

It satisfies $\mathcal{C}r_* = z$. Every admissible residual can be written as $r = r_* + k$, where $\mathcal{C}k = 0$, and $k^* W r_* = k^* \mathcal{C}^* \mathcal{R}^{-1} z = 0$. Therefore,

$$\mathcal{S}(r) = \alpha z^* \mathcal{R}^{-1} z + \alpha k^* W k. \quad (15.19)$$

Positivity of W proves uniqueness of the minimum. The pair (z, k) preserves and reconstructs the complete residual.

For a path with transports U_j , the residuals $r_j = f_j - U_j f_{j-1}$ reconstruct the fields through $f_j = U_j f_{j-1} + r_j$. The map $\mathcal{C}$ collects the subsequent transports of each residual to the endpoint. If its blocks are $\mathcal{C}_j$, then

$$\mathcal{R} = \sum_{j,k} \mathcal{C}_j (W^{-1})_{jk} \mathcal{C}_k^*.$$

This expression preserves correlations between segments of the path. For an invertible longitudinal transport D , the boundary $y = Dz$ has response

$$\mathcal{S}_{\text{ef},L}(y) = \alpha y^* (D \mathcal{R} D^*)^{-1} y = \alpha z^* \mathcal{R}^{-1} z.$$

The same action is obtained by eliminating memory first or transporting the boundary first. The prefactor retains the normalization of the source action; when that action has $\alpha = \hbar_{\text{ret}}$, reduction transmits the same section.

15.7 Joint reduction of radius, energy and conjugate length

Write the same positive character in a boundary–memory decomposition:

$$Y = \begin{pmatrix} A & F \\ F^* & C \end{pmatrix}, \quad C > 0, \quad Y_{\text{ef}} = A - FC^{-1}F^*.$$

The identity

$$\begin{pmatrix} b \\ y \end{pmatrix}^* Y \begin{pmatrix} b \\ y \end{pmatrix} = b^* Y_{\text{ef}} b + \eta^* C \eta, \quad \eta = y + C^{-1}F^*b,$$

proves $Y_{\text{ef}} > 0$, characterizes the minimum-energy extension and preserves the reconstruction $y = \eta - C^{-1}F^*b$. For $s > 0$, substitution into the Schur complement gives $\text{Schur}(sY) = sY_{\text{ef}}$. Consequently,

$$R_{\text{ef}} = LY_{\text{ef}}, \quad H_{\text{ef}} = E_L Y_{\text{ef}}.$$

To compute the compressed conjugate length, solve $Y(x, y) = (f, 0)$. The second equation gives $y = -C^{-1}F^*x$, and the first gives $x = Y_{\text{ef}}^{-1}f$. The boundary block of Y^{-1} is therefore Y_{ef}^{-1} . Thus,

$$\Lambda_b = LY_{\text{ef}}^{-1}, \quad R_{\text{ef}}\Lambda_b = L^2I, \quad R_{\text{ef}} = \frac{L}{E_L}H_{\text{ef}}. \quad (15.20)$$

Reduction acts on the same blocks in both readers and retains the residual η . The effective character may depend on memory through a matrix-valued expression, while L/E_L remains fixed.

15.8 Dynamic memory and induced temporal norm

Dynamic elimination retains an additional spectral dependence. In a stationary representation, let

$$H = \begin{pmatrix} H_{\text{bb}} & V \\ V^* & H_{\text{ii}} \end{pmatrix}, \quad H_{\text{ii}} > 0.$$

For z in the relevant resolvent domain, eliminating the interior component in $(H - zI)(b, y) = (f, 0)$ yields

$$\mathcal{K}_H(z) = H_{\text{bb}} - zI - V(H_{\text{ii}} - zI)^{-1}V^*. \quad (15.21)$$

When both inverses exist, the boundary block of $(H - zI)^{-1}$ is $\mathcal{K}_H(z)^{-1}$. For $|z| < \lambda_{\min}(H_{\text{ii}})$, the convergent resolvent series gives

$$\mathcal{K}_H(z) = H_{\text{est}} - zZ - z^2VH_{\text{ii}}^{-3}V^* - \dots, \quad (15.22)$$

$$H_{\text{est}} = H_{\text{bb}} - VH_{\text{ii}}^{-1}V^*, \quad Z = I + VH_{\text{ii}}^{-2}V^* \geq I. \quad (15.23)$$

Positivity follows from $Z - I = (H_{\text{ii}}^{-1}V^*)^*(H_{\text{ii}}^{-1}V^*)$. The static extension $(b, -H_{\text{ii}}^{-1}V^*b)$ has squared norm b^*Zb ; energy and temporal norm arise from the same extension. The physical parameter is $z = \hbar_{\text{ret}}\omega$.

At first temporal order, $T = Z^{1/2}$ and $\psi = Tb$ transform the pair (H_{est}, Z) into $(T^{-*}H_{\text{est}}T^{-1}, I)$. The action section $\hbar_{\text{ret}}$ remains fixed. If T depends on time, the transformation also retains the term $\dot{T}b$ in $\dot{\psi} = \dot{T}b + T\dot{b}$.

Radial proportionality is preserved in the complete resolvent. Writing $\chi_{\text{rad}} = L/E_L$ and $R = \chi_{\text{rad}}H$, substitution in each block proves

$$\mathcal{K}_R(\chi_{\text{rad}}z) = \chi_{\text{rad}}\mathcal{K}_H(z). \quad (15.24)$$

This identity transports the operator and the spectral parameter together, preserving every order of dynamic memory. For the effective forms in (15.20), canonical normalization requires the dual transport of inverse length:

$$H_{\text{can}} = T^{-*}H_{\text{ef}}T^{-1}, \quad R_{\text{can}} = T^{-*}R_{\text{ef}}T^{-1}, \quad \Lambda_{\text{can}} = T\Lambda_bT^*.$$

Multiplication gives

$$R_{\text{can}} = \frac{L}{E_L}H_{\text{can}}, \quad R_{\text{can}}\Lambda_{\text{can}} = L^2I.$$

The proof allows noncommuting T , R_{ef} and Λ_b . The induced temporal norm and the dual length jointly preserve the same G .

15.9 Refinement, spectral domains and scope

Let J and K be isometric inscriptions between two levels satisfying $X'J = JX$ and $U'J = KU$, with the same intensive section L . Then

$$R'K = LU'X'U'^*K = KLUXU^* = KR,$$

and the same calculation proves $H'K = KH$. The intertwining of inverses is formulated on the corresponding domains. When the compatible system has a positive injective self-adjoint limit X , spectral calculus defines $Y = UXU^*$, $R = LY$, $H = E_L Y$ and $\Lambda = LY^{-1}$. The proportionality $R = (L/E_L)H$ holds on $\text{Dom}(Y)$, while $R\Lambda = L^2I$ and $H\Lambda = \hbar_{\text{ret}}cI$ hold on $\text{Dom}(Y^{-1})$, with bounded extensions of the products. The spectral cutoffs $\mathbf{1}_{[\varepsilon, M]}(Y)$ verify the identities before passing to the limit on these domains. A refined edge length $L/9$ carries its own scale factor; the intensive section is preserved according to the declared intertwining.

The classification distinguishes transformations preserving the realization from those modifying a constitutive rule. Diagonal weightings, unitary equivalences, the auxiliary archive and joint reduction preserve G . Changing the marked resolution modifies R1; replacing r_N by $\sqrt{r_N}$, α^{16} by its square, or the inscribed component by the total norm modifies R4. An isolated radial rescaling changes R5, and an isolated action rescaling changes R6. R8 maintains the distinction between reduced radius, Schwarzschild radius and a change of units.

This establishes the mathematical uniqueness of realization R1–R8: the generated data and the character fix the response, and their common composition fixes G . The longitudinal and metric rules appear explicitly in this conclusion as incorporated compositions; the gravitational identification retains its physical status. The proof covers the declared class with its constitutive definitions and compatible transformations. The rational checks of the development verify finite instances of the displayed identities; the general proof is contained in the preceding equations and arguments.

16 Action, gravity, and thermal information

16.1 Dodecaphasic period and circular realization of action

The length in Section 15 fixes, within the same realization, the compatible clock

$$t_0 = \frac{\pi_{\text{HMT}} L}{54 c_{\text{int}}}.$$

Its circular representation then satisfies $R_{108} = L$. This substitution expresses the constructed length chronologically and retains the longitudinal reference, action and velocity of that realization. The calendar already constructed determines $T_{108} = 108 t_0$. In its circular realization,

$$\omega_{108} = \frac{2\pi}{108 t_0}, \quad R_{108} = \frac{c_{\text{int}}}{\omega_{108}} = \frac{54}{\pi} \ell_0, \quad \ell_0 = c_{\text{int}} t_0.$$

The unit ℓ_0 and the time and velocity charts remain explicit; the SI chart does not select the cycle. For each oriented action section $a \in \{\text{pre}, \text{ret}\}$,

$$E_{108,a} = \hbar_a \omega_{108}, \quad m_{108,a} = \frac{E_{108,a}}{c_{\text{int}}^2}.$$

The reduced and full lengths are, respectively,

$$\frac{\hbar_a}{m_{108,a} c_{\text{int}}} = R_{108}, \quad \frac{h_a}{m_{108,a} c_{\text{int}}} = 2\pi R_{108}.$$

The same action and duration enter both readings .

Proposition 16.1 (Radius-coincidence selector). *In the dimensional collection*

$$G_a(\vartheta) = \vartheta \frac{c_{\text{int}}^5 t_0^2}{\hbar_a}, \quad \vartheta > 0,$$

the circular-realization condition $G_a(\vartheta) m_{108,a} / c_{\text{int}}^2 = R_{108}$ has the unique solution

$$\vartheta_{108}^\circ = (54/\pi)^2.$$

Proof. The first radius is $r_g(\vartheta) = \vartheta(\pi/54)\ell_0$. Comparing it with $(54/\pi)\ell_0$, with $\ell_0 > 0$, reduces the equality to a strictly increasing linear equation. $\square$

The coincidence condition is an explicit part of this realization; the theorem solves for its coefficient without using a metrological value of G . The radius used here is Gm/c^2 , so the factor two from the convention $2Gm/c^2$ is not inadvertently transferred.

Proposition 16.2 (Reciprocal action and gravity, preserved area). *In the two preceding charts,*

$$\frac{G_{\text{pre}}}{G_{\text{ret}}} = R_{\text{act}}^{-1}, \quad \hbar_a G_a = \vartheta_{108}^\circ c_{\text{int}}^5 t_0^2, \quad \ell_P^2 = \frac{\hbar_a G_a}{c_{\text{int}}^3} = \vartheta_{108}^\circ \ell_0^2.$$

Proof. The definition of G_a contains $\hbar_a^{-1}$, and the remaining factors are common to both charts. Taking the quotient and product gives the three equalities. $\square$

This reciprocity explains why orientation information may vary while area remains unchanged. A precise relation is thus available between the reciprocal action sections, the Planck scale, and the areal operator of section 17.

16.2 Constitutive composition and evaluation of the gravitational section

The evaluation reuses the generated regional coordinates and record K selected in the preceding kernel. Coordinate α determines the action sections in (12.9); in the return section write $\eta = \eta_{\text{ret}}$ and $S_* = 10^{-34} \text{ J s}$, so that $\hbar_{\text{ret}} = S_* \eta$. The previously constructed angle and counterangle give, in the degree representation,

$$A_{\text{deg}} = 1000\alpha, \quad C_{\text{deg}}^* = 2(\eta + \alpha), \quad x = \frac{\pi_{\text{HMT}} A_{\text{deg}}}{180}, \quad y = \frac{\pi_{\text{HMT}} C_{\text{deg}}^*}{180}.$$

Thus $\eta = (C_{\text{deg}}^* - 2\alpha)/2$. The difference is taken between coordinates in the same angular representation.

On the generated domain $0 < y < x$, the labelled channels are

$$q_{\pm} = e^{-(x \pm y)}, \quad r_{\pm} = \frac{1 - q_{\pm}^{90}}{1 - q_{\pm}^{120}}, \quad 0 < q_{\pm} < 1.$$

The quotient comes from the two extension sectors: for $S_n(q) = \sum_{j=0}^{n-1} q^j$, the identity $(1 - q)S_n(q) = 1 - q^n$ proves $r_{\pm} = S_{90}(q_{\pm})/S_{120}(q_{\pm})$. The squared channels determine the constitutive readings

$$\hat{\varepsilon} = r_+^2, \quad \hat{\mu} = r_-^2, \quad \hat{c} = \frac{1}{r_+ r_-}, \quad \hat{Z} = \frac{r_-}{r_+}.$$

For positive velocity, action, and charge bases c_* , u_S , and u_Q , the dimensional sections are

$$\mu = \hat{\mu} \frac{u_S}{c_* u_Q^2}, \quad \varepsilon = \hat{\varepsilon} \frac{u_Q^2}{u_S c_*}, \quad c = c_* \hat{c} = \frac{1}{\sqrt{\mu \varepsilon}}.$$

Indeed, the product of the first two equations cancels the action and charge bases, leaving $\mu \varepsilon = (r_+ r_-)^2 / c_*^2$; its positive square root proves the last equality. The factor $\hat{c}$ transforms the base c_* once. The SI evaluation below represents the same c by $299792458 \text{ m s}^{-1}$ and restores $c_* = c / \hat{c}$. This operation verifies equivalence of the dimensional and constitutive representations.

With $r_N = 5759/23040$ and the stated reference $L_* = 1 \text{ m}$, substitution of these equations into the radial coefficient gives

$$G = \frac{c^3 L_*^2}{S_*} \frac{r_N^2 \alpha^{32}}{\eta} = \frac{2c^3 L_*^2 r_N^2 \alpha^{32}}{S_* (C_{\text{deg}}^* - 2\alpha)}, \quad (16.1)$$

$$= \frac{L_*^2 r_N^2 \alpha^{32}}{S_* \eta (\mu \varepsilon)^{3/2}}. \quad (16.2)$$

These express the same coefficient in the radial realization. The longitudinal reference L_* , constructed length L_{α} , chronological edge ct_0 , and horizon radius retain their distinct types.

Evaluation with the archived regional expansions yields

$$\begin{aligned} L_{\alpha} &= 1.6162549826251263301682625013 \dots 10^{-35} \text{ m}, \\ \hbar_{\text{ret}} &= 1.0545718169150390611336905847 \dots 10^{-34} \text{ J s}, \\ t_0 &= 3.1364999625365490734 \dots 10^{-45} \text{ s}, \\ G &= 6.6742996594140227063677761891 \dots 10^{-11} \text{ m}^3 \text{ kg}^{-1} \text{ s}^{-2}. \end{aligned}$$

Calculations at 100 and 140 working digits preserve 71 significant digits of the evaluation. This numerical stability is distinguished from experimental uncertainty. Against the CODATA 2022 reference $G_{\text{ref}} = 6.67430(15) 10^{-11} \text{ m}^3 \text{ kg}^{-1} \text{ s}^{-2}$, the difference is $G - G_{\text{ref}} = -3.4058597729 \dots 10^{-18} \text{ m}^3 \text{ kg}^{-1} \text{ s}^{-2}$, approximately -0.00227057 times its standard uncertainty. The reference is compared after evaluation; rules R4 and R5 and identification R8 retain the scope proved in the preceding section.

The reproduction sources include the regional expansions, record K , action and counterangle equations, exact verifiers, and decimal evaluation. They run from the local directory `reproduccion_gravitatoria_20260926` through `verificar_reproduccion.py`. Each test states its domain; reuse of archived coordinates is distinguished from a new execution of the entire generator.

17 Area operator and reconstruction of sectorial information

Regrouping six trits constructs a bijection between $(\mathbb{Z}/9\mathbb{Z})^3$ and $(\mathbb{Z}/27\mathbb{Z})^2$, both of cardinality 729. The bijection is not presented as an additive isomorphism. On a 9×9 boundary block, the source decomposes 36 links into two sets of 18, labeled by channels 90 and 120.

17.1 Tritic regrouping and boundary incidence

The correspondence preserves an ordered word, in addition to its cardinality. Write each coordinate $x_j \in \mathbb{Z}/9\mathbb{Z}$, for $j = 1, 2, 3$, through its unique positional representatives $x_j = r_{2j-1} + 3r_{2j}$, with $r_k \in \{0, 1, 2\}$. The map is

$$\beta_{729}(x_1, x_2, x_3) = (r_1 + 3r_2 + 9r_3, r_4 + 3r_5 + 9r_6) \in (\mathbb{Z}/27\mathbb{Z})^2. \quad (17.1)$$

Proposition 17.1 (Positional correspondence between interior and boundary). *The map β_{729} is bijective and preserves the order of the six trits. Its inverse extracts the two triples of digits and regroups them into three pairs.*

Proof. The positional expansion of the representatives $0, \dots, 8$ and $0, \dots, 26$ is unique. Extracting the six digits, regrouping them, and extracting them again returns the same word. The inverse operation thus restores each of the three initial coordinates. $\square$

Carry preserves its position in the enriched word. The additive operations of the two packagings have different boundaries: for example, $\beta_{729}(0, 2, 0) = (18, 0)$ and $\beta_{729}(0, 1, 0) = (9, 0)$, whose sum in the order-twenty-seven torus is $(0, 0)$, whereas $\beta_{729}(0, 3, 0) = (0, 1)$. This distinction allows the positional correspondence to be reused while keeping the operations typed.

In the torus $(\mathbb{Z}/27\mathbb{Z})^2$, the block $B_{\partial} = \{0, \dots, 8\}^2$ has the oriented link sets

$$\partial B_{90} = \{((-1, j), (0, j)), ((8, j), (9, j)) : 0 \leq j \leq 8\}, \quad (17.2)$$

$$\partial B_{120} = \{((i, -1), (i, 0)), ((i, 8), (i, 9)) : 0 \leq i \leq 8\}. \quad (17.3)$$

All coordinates are read modulo twenty-seven. Each collection contains eighteen links, and the two are disjoint. Their labels preserve the incidence channels; the number of links is $18 + 18$.

Theorem 17.2 (Minimum capacity of the cubic cut). *In the cubic graph with vertices $\{0, \dots, N-1\}^3$, with $N \geq 2$, links between neighbors, and unit capacity per link, the minimum cut between two opposite faces has capacity N^2 . For $N = 9$, a maximally mixed product state on the trits of a minimum layer has information*

$$\text{cap}_{\min} = 81, \quad s_{\text{corte}} = 81 \log 3. \quad (17.4)$$

Proof. The N^2 lines parallel to the normal of the faces form edge-disjoint paths. Every separating cut intercepts each path, so it contains at least N^2 links. The links crossing a transverse layer constitute a cut of that cardinality. Each uniform trit contributes $\log 3$ nats; entropy is additive for the product state of the eighty-one trits of the layer. $\square$

The capacity of the cubic cut and the state of the thirty-six links in (17.3) belong to observables defined on different cuts. The genealogy preserves both domains and their measures.

17.2 Transport of the collective incidence modes

The spaces $\mathcal{F}_6$ and $\mathcal{F}_8$ of Section 14 provide the uniform vectors

$$u_{90} = \frac{1}{\sqrt{90}} \sum_{f \in \mathcal{F}_6} (\delta_f, 0), \quad u_{120} = \frac{1}{\sqrt{120}} \sum_{f \in \mathcal{F}_8} (0, \delta_f)$$

in $\ell^2(\mathcal{F}_6) \oplus \ell^2(\mathcal{F}_8)$. In $\ell^2(\partial B_{90}) \oplus \ell^2(\partial B_{120})$, construct

$$v_{90} = \frac{1}{\sqrt{18}} \sum_{e \in \partial B_{90}} (\delta_e, 0), \quad v_{120} = \frac{1}{\sqrt{18}} \sum_{e \in \partial B_{120}} (0, \delta_e).$$

The two pairs are orthonormal. Denote their respective spanned planes by $\mathcal{H}_{\text{inc}}^{\text{coll}}$ and $\mathcal{H}_{\partial}^{\text{coll}}$.

Theorem 17.3 (Collective intertwining of the sectorial labels). *There is a unique surjective linear isometry between these planes such that*

$$\mathcal{J}_{6 \rightarrow 8} u_m = v_m \quad (m = 90, 120). \quad (17.5)$$

For the operators

$$\begin{aligned} D_{\text{inc}} &= 90|u_{90}\rangle\langle u_{90}| + 120|u_{120}\rangle\langle u_{120}|, \\ D_{\partial} &= 90|v_{90}\rangle\langle v_{90}| + 120|v_{120}\rangle\langle v_{120}|, \end{aligned}$$

the following identity holds:

$$\mathcal{J}_{6 \rightarrow 8} D_{\text{inc}} = D_{\partial} \mathcal{J}_{6 \rightarrow 8}. \quad (17.6)$$

The sectorial projectors and their oriented combination with coefficients 12 : -1 are also transported.

Proof. The image of a basis fixes the unique linear map. Norms and inner products are preserved because both bases are orthonormal. Identity (17.6) holds on each vector u_m , since its two sides are mu_m . The projectors are expressed as $(120I - D)/30$ and $(D - 90I)/30$; their linear functional calculus and any combination of the two preserve intertwining. $\square$

This transport acts on the two uniform modes and their spectral labels. Individual incidences and fluctuations within each sector remain in their complete spaces. The link preserves exactly the collective datum that the areal weighting will use.

17.3 Angular character and area operator

Let $N_e = \text{diag}(0, 1, 2)$ on each link, and let N_{90}, N_{120} be the sector sums. With the scale

$$\theta_{\text{clk}} = \frac{C^*}{6} \frac{\pi}{180}, \quad a_0 = \frac{1 + 2 \cos(10^\circ)}{3} \theta_{\text{clk}},$$

fixed, the angular factor is the normalized real character of the triplet

$$\chi_{10} = \frac{1}{3} \text{Tr} \text{diag}(1, e^{i\pi/18}, e^{-i\pi/18}) = \frac{1 + 2 \cos(10^\circ)}{3}, \quad a_0 = \chi_{10} \theta_{\text{clk}}. \quad (17.7)$$

The sum of the conjugate phases proves the equality. On the positive branch considered, $\theta_{\text{clk}} > 0$, $\chi_{10} > 0$, and $\gamma_{\text{HMT}} > 0$; hence a_0 and the areal scale are positive. The angular triplet and its normalization precede the entropic distribution on the links. With these quantities, the normalized area operator is

$$\hat{\mathcal{A}}/\ell_P^2 = \gamma_{\text{HMT}} a_0 (N_{90} + N_{120}).$$

The radial composition fixes $\ell_P^2 = L_\alpha^2$ here, so that $\hat{\mathcal{A}} = \gamma_{\text{HMT}} a_0 L_\alpha^2 (N_{90} + N_{120})$. The same constructed length is transmitted to the area quantum, preserving the occupation operators and their sectorial recovery. The spectrum of the normalized operator $\hat{\mathcal{A}}/L_\alpha^2$ has values $n\gamma_{\text{HMT}} a_0$, $0 \leq n \leq 72$, with multiplicities given by $(1 + z + z^2)^{36}$. The proof consists in summing the eigenvalues 0, 1, 2 of the 36 tensor factors.

Theorem 17.4 (Recovery of the boundary sectors). *Let*

$$N = N_{90} + N_{120}, \quad D = 90N_{90} + 120N_{120}.$$

Then

$$N_{90} = 4N - \frac{D}{30}, \quad N_{120} = \frac{D}{30} - 3N. \quad (17.8)$$

Thus the total and the weighted reader jointly recover the two occupations.

Proof. The matrix $\begin{pmatrix} 1 & 1 \\ 90 & 120 \end{pmatrix}$ has determinant 30. Its inversion gives (17.8). On the domain of integer occupations, the image satisfies $D \in 30\mathbb{Z}$ and $90N \leq D \leq 120N$. $\square$

The modular Hamiltonian in the source is $K_A = -\log \rho_A = cI + \Delta_{\text{mod}} D$. For $\Delta_{\text{mod}} \neq 0$ and known normalization, $D = (K_A - cI)/\Delta_{\text{mod}}$ expresses the second observable. This operator describes the spectral structure of the reduced state; it is not identified by notation with the generator of temporal evolution. Area preserves total occupation, and the pair adds provenance. For example, $(N_{90}, N_{120}) = (1, 0)$ and $(0, 1)$ share $N = 1$, but have $D = 90$ and $D = 120$.

17.4 State normalization and the meaning of the modular Hamiltonian

The second observable is made explicit through the state that defines it. The preceding tensor decomposition is retained: eighteen qutrits per sector, each with number operator $\text{diag}(0, 1, 2)$; N_m is the sum of these operators, and the sectors act on distinct factors. Let

$$z_m = 1 + e^{-m\Delta_{\text{mod}}} + e^{-2m\Delta_{\text{mod}}}, \quad m \in \{90, 120\}.$$

The state collection in the source has the product realization

$$\rho_A = Z^{-1} e^{-\Delta_{\text{mod}}(90N_{90} + 120N_{120})}, \quad Z = z_{90}^{18} z_{120}^{18}.$$

Commutativity of the number operators and tensor factorization prove the expression for Z . The state has full rank for real, finite Δ_{mod} , and its spectral logarithm gives

$$K_A = -\log \rho_A = (\log Z)I + \Delta_{\text{mod}} D.$$

For $\Delta_{\text{mod}} \neq 0$, K_A weights the excitations of the two sectors differently. The joint reading with N recovers the two occupations through equation (17.8).

The need for the joint reading is seen in two complementary examples. The occupation states $(1, 0)$ and $(0, 1)$ have the same total $N = 1$ and different modular weights. Conversely, $(4, 0)$ and $(0, 3)$ have the same $D = 360$ and different totals. The pair (N, D) separates both cases. This recovery concerns sectorial occupations, not each microscopic vector within a degeneracy of those occupations.

When $\Delta_{\text{mod}} = 0$, the state is proportional to the identity and K_A is scalar: that reading no longer distinguishes the sectors. For $\Delta_{\text{mod}} \neq 0$, the known normalization makes it possible to recover D and compose it with area. This defines the role of the modular Hamiltonian: it characterizes the reduced state and its information distribution. Its inclusion in the article is motivated by that mathematical role, whereas the evolution Hamiltonian belongs to the dynamical development of action.

17.5 Reduced state, Schmidt decomposition, and entanglement entropy

If $\rho_A = \sum_{\nu} p_{\nu} |\nu\rangle\langle\nu|$, the purification

$$|\Psi\rangle = \sum_{\nu} \sqrt{p_{\nu}} |\nu\rangle_A |\nu\rangle_B$$

satisfies $\text{Tr}_B |\Psi\rangle\langle\Psi| = \rho_A$. Thus $-\text{Tr}(\rho_A \log \rho_A)$ is the entanglement entropy of this particular pure bipartition. The enlarged state declares which system carries that interpretation.

For a two-sheet involution with one eigenvector of each sign,

$$\rho_y = \frac{e^{-yR}}{2 \cosh y}, \quad R^2 = I,$$

diagonalization gives

$$S(\rho_y) = \log(2 \cosh y) - y \tanh y, \quad D(\rho_y \| \rho_{-y}) = 2y \tanh y.$$

Entropy is even in y ; the distance explicitly compares the two orientations. This two-level entropy is not identified with the capacity $81 \log 3$ of the triadic cut.

18 Genealogical memory, information reduction, and thermal transduction

The thermal construction uses two representations from the same APP–TRIT–TPK development. The first preserves states, admissible routes, and record information; the second provides oriented action and the duration of return. Their composition determines an energy per tritic decision. Temperature and energy coordinates are introduced afterward through explicit dimensional maps. This sequence makes it possible to identify what information is preserved, what information a reader expresses, and what quantity converts that information into dimensional entropy.

18.1 Reader, memory, and state recovery

Fix a finite level of the development, and denote by Ω the set of admissible enriched states. The record preserves sheet, orientation, residue, quotient, carry, route, phase, and memory in the domains where each coordinate enters. Let

$$q : \Omega \longrightarrow Y, \quad M : \Omega \longrightarrow \mathcal{M}$$

be the visible representation and retained memory, respectively. For a distribution p on Ω , write $H(p) = -\sum_x p_x \log p_x$, with $0 \log 0 = 0$. The logarithm is applied to dimensionless probabilities.

Definition 18.1 (Genealogical recovery of a representation). The pair $\hat{q} = (q, M)$ is a recoverable reader on support Ω_p when there exists

$$d : \hat{q}(\Omega_p) \longrightarrow \Omega_p, \quad d \circ \hat{q} = \text{id}_{\Omega_p}.$$

The fiber information of q is $H_{\text{fib}}(q; p) = H(X \mid q(X))$, where X has law p .

Proposition 18.2 (Recovery identity and cost of reduction). *For a recoverable reader,*

$$H(X \mid q(X), M(X)) = 0, \tag{18.1}$$

$$H(X) = H(q(X)) + H(X \mid q(X)), \tag{18.2}$$

$$H(X \mid q(X)) = H(M(X) \mid q(X)). \tag{18.3}$$

Proof. The map d recovers X from the pair $(q(X), M(X))$, so the first conditional entropy is zero. Since q is a function of X , the chain rule gives the second equality. For the third, expand $H(X, M(X) \mid q(X))$ in two ways. One gives $H(X \mid q(X))$, because M is a function of X ; the other gives $H(M(X) \mid q(X)) + H(X \mid q(X), M(X))$. $\square$

Euclidean division of the APP sheets displays the elementary arithmetic mechanism. With $i, j \in \{1, \dots, 9\}$, one preserves

$$i + j = 9q_{ij}^+ + R_{ij}^+, \quad ij = 9q_{ij}^\times + R_{ij}^\times, \quad 1 \leq R_{ij}^\sigma \leq 9 \quad (\sigma \in \{+, \times\}).$$

The residue–quotient pair recovers the integer value of the operation. Summing the differences over the eighty-one positions gives

$$2025 - 810 = 54 + 9 \cdot 129.$$

Quotient memory restores the contribution not expressed by residual reduction. Recoverability of the complete state additionally requires preserving the relevant position and route coordinates; the residue–quotient pair here identifies the result of the operation.

18.2 Reversible update, tritic decision, and erasure

Let $U : \Omega \rightarrow \Omega$ be the TPK update on a domain where enriched transport has an inverse. The transported law is $p'_y = p_{U^{-1}y}$. The bijection preserves the multiset of probabilities, and hence

$$H(U(X)) = H(X), \quad H(X | U(X)) = 0. \quad (18.4)$$

If the observed output is $q(U(X))$, the unpublished information is quantified by $H(X | q(U(X)))$. The complete state, its representation, and the reset of a record are thus three distinct maps.

For the three oriented TRIT states, let $p = (p_{-1}, p_0, p_{+1})$ be a normalized distribution. Its information is

$$I_{\text{TRIT}}(p) = - \sum_{a=-1}^1 p_a \log p_a, \quad 0 \leq I_{\text{TRIT}}(p) \leq \log 3. \quad (18.5)$$

The upper bound corresponds to $p_a = 1/3$. Indeed,

$$D(p \| (1/3, 1/3, 1/3)) = \sum_a p_a \log(3p_a) = \log 3 - I_{\text{TRIT}}(p) \geq 0;$$

the inequality follows from convexity of $t \log t$, and equality characterizes the uniform distribution. The lower bound corresponds to a distribution concentrated on one state.

The ideal Landauer realization uses a memory with energetically degenerate logical states, an isothermal environment at temperature Θ , and a balance that preserves the logical map's output and accounts for auxiliary records. In that model, the decrease in logical entropy is $k_B[H(X) - H(F(X))] = k_B H(X | F(X))$ for a deterministic map F . Entropy balance requires transferring at least that amount to the environment; its energy expression is $Q \geq k_B \Theta H(X | F(X))$. The distinction between reversible logical evolution and physical erasure corresponds to the Landauer–Bennett realization [6]. The invertible update (18.4) has zero logical contribution; resetting a uniform triple requires transferring $\log 3$ nats to the environment and satisfies $Q \geq k_B \Theta \log 3$. Control energy and dissipation in a particular physical realization belong to its additional thermodynamic balance.

18.3 Action, frequency, and decision energy

Let $\mathcal{L}_{\text{act}}$ and $\mathcal{L}_T$ be the dimensional lines of action and duration; the energy line is $\mathcal{L}_E = \mathcal{L}_{\text{act}} \otimes \mathcal{L}_T^*$. The preceding development supplies the positive sections $\hbar_a \in \mathcal{L}_{\text{act}}$, with $a \in \{\text{pre}, \text{ret}\}$, and the elementary duration $t_0 \in \mathcal{L}_T$. The complete calendar provides

$$T_{108} = 108t_0, \quad \omega_{108} = \frac{2\pi}{T_{108}}, \quad h_a = 2\pi\hbar_a.$$

The dimensional contraction between action and frequency defines

$$\epsilon_a := \hbar_a \omega_{108} = \frac{h_a}{108t_0} = \frac{\hbar_a}{t_P}, \quad t_P = \frac{54}{\pi} t_0. \quad (18.6)$$

The last member uses the same reduced circular period as the gravitational section. The three expressions correspond to one energy section, with an orientation a maintained in all factors. The normalized information of a uniform decision produces the energy-per-nat section

$$\epsilon_a := \frac{\epsilon_a}{\log 3} \in \mathcal{L}_E^+. \quad (18.7)$$

The previously constructed action determines the energy scale once duration is fixed; the cardinality of TRIT alternatives determines its information normalization. The value $\log 3$ enters through the count and measure declared in (18.5).

18.4 Dimensional definition of Boltzmann and coordinate selection

Let $\mathcal{L}_\Theta$ be the oriented temperature line. The dimensional entropy space is $\mathcal{L}_{\text{ent}} = \mathcal{L}_E \otimes \mathcal{L}_\Theta^*$. A positive linear map $k_B : \mathcal{L}_\Theta \rightarrow \mathcal{L}_E$ is, equivalently, a positive element of $\mathcal{L}_{\text{ent}}$. The same object can therefore act on a temperature or multiply dimensionless information.

Definition 18.3 (Thermal normalization of the tritic decision). A thermal pair compatible with oriented section a consists of a positive transducer k_B and a positive section $\Theta_{\text{clk},a} \in \mathcal{L}_\Theta$ such that

$$k_B(\Theta_{\text{clk},a}) = \epsilon_a, \quad k_B \Theta_{\text{clk},a} \log 3 = \epsilon_a. \quad (18.8)$$

The Boltzmann constant serves as the energy–temperature transducer of tritic information in this realization.

Theorem 18.4 (Uniqueness relative to the thermal section and dimensional covariance). *With a positive section $\Theta_{\text{clk},a}$ fixed, there is a unique positive linear map satisfying (18.8). For positive bases u_E, u_Θ , if*

$$\epsilon_a = \epsilon_a^{\text{num}} u_E, \quad \Theta_{\text{clk},a} = \theta_a u_\Theta, \quad k_B(u_\Theta) = b u_E,$$

its coordinate is

$$b = \frac{\epsilon_a^{\text{num}}}{\theta_a \log 3}. \quad (18.9)$$

Under $u'_E = s_E u_E$, $u'_\Theta = s_\Theta u_\Theta$, with $s_E, s_\Theta > 0$, one has

$$b' = \frac{s_\Theta}{s_E} b, \quad \theta'_a = \frac{\theta_a}{s_\Theta}, \quad (\epsilon_a^{\text{num}})' = \frac{\epsilon_a^{\text{num}}}{s_E}.$$

Proof. A nonzero section spans a line. Prescribing its image determines the unique linear map; positivity of both sections proves its positivity. Substituting the coordinates into (18.8) gives $b \theta_a \log 3 = \epsilon_a^{\text{num}}$. The three transformation laws follow by expressing the same sections and the same map in the new bases. $\square$

The same Boltzmann constant must act in both orientations. Since t_P is common, simultaneous compatibility is equivalent to

$$\frac{\Theta_{\text{clk,pre}}}{\Theta_{\text{clk,ret}}} = \frac{\epsilon_{\text{pre}}}{\epsilon_{\text{ret}}} = \frac{\hbar_{\text{pre}}}{\hbar_{\text{ret}}}. \quad (18.10)$$

Indeed, an isomorphism of lines preserves quotients of sections. Conversely, if the preceding equality holds, the isomorphism determined in one orientation also satisfies the normalization of the other. Thus return modifies the thermal section of the cycle in the same proportion as energy, while the transducer remains common to both readings.

The preceding equation makes explicit the choice required to express an individual value. If the thermal section has not yet been selected by an independent reader, then for every $\lambda > 0$ the pair

$$(k_B, \Theta_{\text{clk},a}) \longmapsto (\lambda k_B, \lambda^{-1} \Theta_{\text{clk},a})$$

preserves ϵ_a . Consequently, presenting an internally selected individual coordinate requires providing the reader of $\Theta_{\text{clk},a}$, the bases, and their dimensional transport. The action and information equations remain determined before that specification. Subsequent identification of the bases with joules and kelvins belongs to the metrological representation of the result.

18.5 Dimensional entropy and information of the reduced state

For a distribution p of admitted routes, or a reduced state with density matrix ρ , let

$$s(p) = - \sum_x p_x \log p_x, \quad s(\rho) = - \text{Tr}(\rho \log \rho).$$

Their dimensional representations are

$$S(p) = k_B s(p), \quad S(\rho) = k_B s(\rho) \quad \text{in } \mathcal{L}_{\text{ent}}. \quad (18.11)$$

Evaluation at Θ produces an energy $\Theta S = k_B \Theta s$. For the uniform triple at the cycle temperature, (18.8) gives exactly

$$S_{\text{TRIT}} = k_B \log 3, \quad \Theta_{\text{clk},a} S_{\text{TRIT}} = \epsilon_a.$$

If ρ_A is the reduction of the bipartite purification constructed in the areal section, $s(\rho_A)$ measures the entanglement of that bipartition. Unit transport preserves its eigenvalues, orientation, and sectorial occupations. The areal scale $\gamma a_0 \ell_P^2$ converts occupations into area; the transducer k_B converts information into dimensional entropy. Both readings use the same record through different operators.

18.6 Spectral normalization of thermal histories

Let $\mathcal{G}$ be an already constructed finite graph of admissible extensions, with adjacency matrix A . Each edge e preserves its route coordinates and receives a dimensionless action increment $a_*(e) > 0$ from the corresponding reader. With energy section ϵ_* chosen, put $t = \beta \epsilon_*$, where $\beta = (k_B \Theta)^{-1}$. The thermal operator is

$$(L_t f)(y) = \sum_{e: x \rightarrow y} \exp[-t a_*(e)] f(x). \quad (18.12)$$

All its exponents are dimensionless. The condition $\rho(L_t) = 1$ normalizes the weighted growth of histories.

Proposition 18.5 (Existence and uniqueness of normalized pressure). *Suppose that A is irreducible, $\rho(A) > 1$, and $0 < m \leq a_*(e) \leq M$ for all edges. There is a unique $t_* > 0$ such that $\rho(L_{t_*}) = 1$, and*

$$\frac{\log \rho(A)}{M} \leq t_* \leq \frac{\log \rho(A)}{m}. \quad (18.13)$$

Proof. Entrywise comparison gives $e^{-tM} A \leq L_t \leq e^{-tm} A$ for $t \geq 0$. Monotonicity of the spectral radius of nonnegative matrices implies the corresponding bounds for $\rho(L_t)$. The spectral radius is continuous, equals $\rho(A) > 1$ at zero, and tends to zero at infinity. For $s > 0$, moreover, $L_{t+s} \leq e^{-sm} L_t$, so $\rho(L_{t+s}) \leq e^{-sm} \rho(L_t) < \rho(L_t)$. Continuity and strict decrease prove existence and uniqueness. Substituting the value one into the initial bounds proves (18.13). $\square$

The local realization of three equally weighted alternatives, represented by a vertex with three loops and $a_*(e) = 1$, satisfies $\rho(L_t) = 3e^{-t}$. Its normalization gives $t_* = \log 3$. With $\epsilon_* = \epsilon_a$, one obtains $\beta_{\text{clk}} = \log 3 / \epsilon_a$, exactly relation (18.8). This computation identifies the local tritic confluence. For a history graph with additional constraints, confluence with the same clock requires checking

$$\rho(L_{(\log 3 / \epsilon_a) \epsilon_*}) = 1$$

with its effective increments. Local normalization and the pressure of the complete system thus preserve their respective domains.

The relation obtained brings together structure, value, and meaning. The record determines recoverable information; probabilities on continuations determine its entropic reading; action and period determine ϵ_a ; the energy–temperature transducer expresses dimensional entropy. Area and entanglement are related through the reduced state and its boundary operators, keeping visible which datum produces each quantity.

19 Oriented incidence and gravitational tensor representation

The exceptional representation of the state preserves the incidence accompanying its arithmetic reading. The ternary matrix A_W in (11.8) provides a precise material antecedent here. Its balanced lift $C = \text{bal}(A_W)$, with representatives $0, 1, -1$, satisfies

$$C = C^T, \quad C^2 = 5I_6, \quad \text{Tr } C = 0.$$

The six positions are ordered as $(\infty, 0, 1, 2, 3, 4)$. This incidence first produces a three-dimensional space; its quadratic reading then produces the five-component space on which the gravitational coupling acts.

19.1 Signed covering and calendar transport

We define

$$E_C = \frac{1}{2}(I_6 + C/\sqrt{5}), \quad \mathcal{H}_C = \text{im } E_C, \quad v_{u,\sigma} = \sigma\sqrt{2}E_C e_u, \quad \sigma \in \{1, -1\}.$$

Proposition 19.1 (Twelve oriented positions). *The vectors $v_{u,\sigma}$ are twelve distinct unit vectors in a three-dimensional space. Their incidence is determined by $\langle v_{u,\sigma}, v_{v,\tau} \rangle = 1/\sqrt{5}$. Reversal of orientation acts by antipodality.*

Proof. The quadratic identity for C implies $E_C^2 = E_C = E_C^*$; its trace is three. Since the diagonal of C is zero,

$$\langle v_{u,\sigma}, v_{v,\tau} \rangle = \begin{cases} \sigma\tau, & u = v, \\ \sigma\tau C_{uv}/\sqrt{5}, & u \neq v. \end{cases}$$

This proves normalization, distinctness of the twelve vectors, and incidence. Changing σ to $-\sigma$ reverses the sign of the vector. $\square$

An explicit Euclidean expression is obtained by setting $v = \sqrt{1 + \varphi^2}$ and

$$B = \frac{1}{v} \begin{pmatrix} 0 & -1 & -\varphi & 0 & \varphi & 1 \\ -1 & -\varphi & 0 & 1 & 0 & -\varphi \\ -\varphi & 0 & -1 & -\varphi & -1 & 0 \end{pmatrix}.$$

The relation $\varphi^2 = \varphi + 1$, for the already generated coordinate, gives $B^*B = 2E_C$. Consequently, $B/\sqrt{2}|_{\mathcal{H}_C}$ is an isometry and sends $v_{u,\sigma}$ to $\sigma B e_u$. The image is

$$\mathcal{V} = \frac{1}{v}(\{0\} \times \{\pm 1\} \times \{\pm \varphi\} \cup \{\pm 1\} \times \{\pm \varphi\} \times \{0\} \cup \{\pm \varphi\} \times \{0\} \times \{\pm 1\}).$$

The internal coordinate, the incidence, and its twelve geometric representatives are thus related.

The calendar coordinate is written $(b, r) \in \mathbb{Z}/12\mathbb{Z} \times \{0, \dots, 8\}$. A route of length n induces

$$k(r, n) = \left\lfloor \frac{r+n}{9} \right\rfloor, \quad (b, r) \mapsto (b + k(r, n) \bmod 12, (r+n) \bmod 9).$$

The explicit alignment of its positions $p = b + 1$ with the signed chart is

p	1	2	3	4	5	6	7	8	9	10	11	12
u	∞	0	0	∞	1	3	3	1	2	4	4	2
σ	+	-	+	-	-	+	-	+	-	+	-	+

This table fixes a trivialization of the carrier. Each change of position transports the vectors and their incidence matrix jointly.

Proposition 19.2 (Covariance of the frame under return). *If $Se_p = e_{p+1 \bmod 12}$, the matrix representation of an operator P on the carrier is transported as $R_k = S^k P S^{-k}$. This transport preserves inner products, incidence, and route composition. After twelve returns the chart returns, while the memory accumulated in the source state is preserved.*

Proof. S is orthogonal and $S^{12} = I$; the metric properties and periodicity of the chart follow. For composition, with $r' = (r + n) \bmod 9$, Euclidean division gives $k(r, n + m) = k(r, n) + k(r', m)$. The exponents of S add exactly with the carry. The operation acts on the calendar coordinate; the route's memory updates remain present in its enriched domain. $\square$

19.2 Quadratic reading and five-component space

Let $\mathbb{R}^{\mathcal{V}}$ be the space of the twelve coordinates, $Ae_v = e_{-v}$ its antipodal involution, and $J = \mathbf{11}^T$. Define

$$P_5 = \frac{1}{2}(I + A) - \frac{1}{12}J, \quad Q_v = vv^T - \frac{1}{3}I_3, \quad \Gamma_0 x = \sum_{v \in \mathcal{V}} x_v Q_v.$$

Theorem 19.3 (Five-component projector and tensor isometry). P_5 is the rank-five orthogonal projector onto the space of coordinates that are even under antipodality and have zero sum. Moreover,

$$\Gamma_0^* \Gamma_0 = \frac{8}{5}P_5, \quad \Gamma_0 \Gamma_0^* = \frac{8}{5}I_{\text{Sym}_0^2(\mathbb{R}^3)}.$$

Therefore

$$\Gamma_5 = \sqrt{\frac{5}{8}} \Gamma_0|_{\text{im } P_5}$$

is an isometry onto the space of traceless symmetric tensors.

Proof. The even subspace has one coordinate for each antipodal pair, so its dimension is six. The constant vector belongs to it; subtracting its rank-one projector gives P_5 , of rank five.

Since $\langle Q_v, Q_w \rangle_F = (v^T w)^2 - 1/3$, the entries of $\Gamma_0^* \Gamma_0$ are $2/3$ for $w = \pm v$ and $-2/15$ at the other ten positions in each row. These are exactly the entries of $(8/5)P_5$.

The identity can also be checked in the tensor space without appealing to a classification of representations. The vertex list gives

$$\sum_v v_i^4 = \frac{12}{5}, \quad \sum_v v_i^2 v_j^2 = \frac{4}{5} \quad (i \neq j),$$

and moments with an odd power vanish by the sign changes. Indeed, $(1 + \varphi^2)^2 = 5\varphi^2$ and $1 + \varphi^4 = 3\varphi^2$. For a symmetric tensor Q , these sums give

$$\sum_v (v^T Q v) v v^T = \frac{8}{5}Q + \frac{4}{5}(\text{tr } Q)I.$$

With $\text{tr } Q = 0$, and using $\sum_v v^T Q v = 4 \text{tr } Q = 0$, it follows that $\Gamma_0 \Gamma_0^* Q = (8/5)Q$. The two identities prove the normalized isometry and its surjectivity. $\square$

The gravitational section in 16, with the normalization proved in 15, determines the operator $G_a P_5$. In the return section, $G_a = c^3 L^2 / \hbar_{\text{ret}}$. The tensor isometry transports this same operator through

$$\Gamma_5(G_a P_5)|_{\text{im } P_5} \Gamma_5^* = G_a I_{\text{Sym}_0^2(\mathbb{R}^3)}.$$

The equality follows from $P_5|_{\text{im } P_5} = I$ and $\Gamma_5 \Gamma_5^* = I$. Thus $\sqrt{5/8}$ fixes tensor normalization and the radial composition fixes the scalar coefficient, with their respective proofs. The spectrum of the operator $G_a P_5$ on $\mathbb{R}^{12}$ has G_a with multiplicity five and zero with multiplicity seven. Orientation is preserved in the carrier and its transport; G_a fixes the common coupling.

19.3 Transverse projection and helicities

For $n \in S^2$, set $\Pi = I - nn^T$ and

$$\Lambda_n Q = \Pi Q \Pi - \frac{1}{2} \text{tr}(\Pi Q \Pi) \Pi.$$

Proposition 19.4 (Rank-two transverse reduction). Λ_n is the orthogonal projector onto $\{Q \in \text{Sym}_0^2(\mathbb{R}^3) : Qn = 0\}$, of dimension two.

Proof. $\Pi n = 0$ gives transversality and $\text{tr} \Pi = 2$ gives zero trace. A transverse traceless tensor is fixed. The formula is self-adjoint for the Frobenius inner product. In a frame (e_1, e_2, n) , the image has orthonormal basis

$$E_+ = \frac{e_1 e_1^T - e_2 e_2^T}{\sqrt{2}}, \quad E_\times = \frac{e_1 e_2^T + e_2 e_1^T}{\sqrt{2}}.$$

□

The other three vectors of an orthonormal basis of the carrier are

$$E_1 = \frac{e_1 n^T + n e_1^T}{\sqrt{2}}, \quad E_2 = \frac{e_2 n^T + n e_2^T}{\sqrt{2}}, \quad E_0 = \frac{2nn^T - e_1 e_1^T - e_2 e_2^T}{\sqrt{6}}.$$

When the transverse frame is rotated through an angle θ , the pairs $(E_+, E_\times)$ and (E_1, E_2) rotate through 2θ and θ , respectively, while E_0 remains fixed. The complexification thus has weights $2, -2, 1, -1, 0$. Λ_n preserves the first two and annihilates the others. This establishes the chain

$$\mathbb{R}^{12} \xrightarrow{P_5} \text{im } P_5 \xrightarrow{\Gamma_5} \text{Sym}_0^2(\mathbb{R}^3) \xrightarrow{\Lambda_n} \mathcal{H}_{\text{TT}}(n), \quad 12 \longrightarrow 5 \longrightarrow 5 \longrightarrow 2.$$

The five-component representation and its reduction are algebraic results preceding the choice of field equations. In the massless gravitational realization, the gauge constraints select the transverse image. This distinction preserves both the five structural coordinates and the two radiative helicities.

20 Phase return and representation of accumulated memory

The chronology of enriched routes distinguishes positional return, orientational return, and phase return. The observable representation identifies certain arrival states, while the cumulative record preserves the sequence traversed. The geometric representation must transport both data jointly.

Preservation of the complement in (15.4) and reconstruction of residuals in (15.19) use this complete temporal record. Two paths with the same observable phase may have different memory cocycles. Transport of the radial reader preserves both coordinates: periodic return acts on phase, while concatenation increments the record appearing in the following formulas.

20.1 Inversion cocycle and change of sheet

Let F_{obs} be the chronological representation of TPK transport. Its canonical blocks have the hierarchy

27	positional return with reversal of orientation,
54	positional and orientational return,
108	complete return of the observable phase.

The record counts reversals of orientation $g(a)$ and effective changes of sheet $s(a)$. For a route,

$$c(\gamma) = \sum_{a \in \gamma} (g(a), s(a)), \quad c(\gamma_2 \gamma_1) = c(\gamma_2) + c(\gamma_1). \quad (20.1)$$

The canonical computation of the block of length 27 expresses $c_{27} = (1, 18)$. Composition preserves those increments in the four blocks completing the observable period. On forward-oriented routes the record is cumulative; its extension to the route groupoid uses $\mathbb{Z}^2$ and $c(\bar{\gamma}) = -c(\gamma)$.

Proposition 20.1 (Memory of the complete observable circuit). *Writing $\mathcal{K}_{27} = F_{\text{obs}}^{27}$, the lift to the record, for $x \in \mathcal{O}_{108}$ and $z \in \mathbb{Z}^2$, satisfies*

$$\tilde{\mathcal{K}}_{27}(x, z) = (\mathcal{K}_{27}x, z + (1, 18)), \quad \tilde{\mathcal{K}}_{27}^4(x, z) = (x, z + (4, 72)).$$

After N complete circuits, the increment is $N(4, 72)$.

Proof. The observable projection of $\mathcal{K}_{27}$ has order four. The preceding canonical computation fixes $c_{27} = (1, 18)$ constantly on that observable orbit. Additivity of (20.1) sums the four increments and gives $c_{108} = 4(1, 18) = (4, 72)$. Repetition applies the same concatenation law and produces Nc_{108} . Return of the first coordinate preserves the increment of the second. $\square$

20.2 Affine representation of the cocycle

The discrete realization uses an element R_4 of order four in $\text{Spin}^+(1, 3)$, a unit timelike vector u fixed by its vector action, and a length unit $\ell_0 > 0$. The scale ℓ_0 preserves its provenance in the dimensional section. We define

$$\rho_C^{\text{disc}}(\gamma) = (R_4^{c_g(\gamma)}, \ell_0 c_s(\gamma)u) \in \text{Spin}^+(1, 3) \ltimes \mathbb{R}^{1,3}. \quad (20.2)$$

The power acts on the vector through the associated vector representation of the spin group.

Theorem 20.2 (Composition and translational amplitude of return). *The map ρ_C^{disc} preserves the identity, concatenation, and inversion. At complete returns,*

$$\rho_C^{\text{disc}}(\gamma_{108}) = (I, 72\ell_0 u), \quad \rho_C^{\text{disc}}(\gamma_{108}^N) = (I, 72N\ell_0 u). \quad (20.3)$$

Inversion reverses the sign of the translation.

Proof. The semidirect product is $(R, a)(S, b) = (RS, a + Rb)$. Since $R_4 u = u$,

$$\begin{aligned}\rho_C^{\text{disc}}(\gamma_2)\rho_C^{\text{disc}}(\gamma_1) &= (R_4^{c_g(\gamma_2)+c_g(\gamma_1)}, \ell_0(c_s(\gamma_2) + c_s(\gamma_1))u) \\ &= \rho_C^{\text{disc}}(\gamma_2\gamma_1).\end{aligned}$$

The cocycle of the identity is zero and that of the inverse is its negative. For the complete circuit, substitute $c_{108} = (4, 72)$ and $R_4^4 = I$. Concatenation of N circuits keeps the rotational part equal to the identity and sums their translations. $\square$

The amplitude $72\ell_0$ is a length associated with a finite route. A realization through a cotetrad and connection transports it to its affine holonomy. The local contribution to the gravitational equations is then obtained through the curvature of that connection, variation of the material action, and normalization of its density. The representation thus preserves the exact cumulative content that must enter that realization.

21 Boundary soldering and energetic response

The cubic realization of transport preserves six oriented incidences. Its geometric quotient, Lorentzian form, and energetic response are obtained from the same incidence matrix. Transport originates in the enriched APP-TRIT-TPK route: the face, sign, carry, and memory accompany each edge. This section develops the linear operation acting on that boundary representation.

21.1 Cubic incidence and the rank-four quotient

Let $H_F = \mathbb{R}^F$, with $F = \{(i, \epsilon) : 1 \leq i \leq 3, \epsilon \in \{-1, 1\}\}$, its Euclidean inner product, and its basis $e_{i,\epsilon}$. Face opposition is the orthogonal involution $Re_{i,\epsilon} = e_{i,-\epsilon}$. Define

$$u = \sum_{i,\epsilon} e_{i,\epsilon}, \quad P_u = \frac{uu^T}{6}, \quad P_- = \frac{I - R}{2}, \quad P_0 = \frac{I + R}{2} - P_u, \quad P_{\square} = P_u + P_-.$$

The vertices are $V = \{-1, 1\}^3$. The incidence matrix $M : \mathbb{R}^V \rightarrow H_F$ is determined by

$$M_{(i,\epsilon),v} = \mathbf{1}_{\{v_i=\epsilon\}}.$$

Proposition 21.1 (Incidence decomposition). *One has*

$$MM^T = 12P_u + 4P_-, \quad \ker M^T = P_0 H_F.$$

Consequently, the quotient $Q = H_F / P_0 H_F$, orthogonally identified with $P_{\square} H_F$, has dimension four.

Proof. A face contains four vertices. Opposite faces have no common vertices, and two faces on different axes have two common vertices. Writing $J = uu^T$ gives

$$MM^T = 4I + 2(J - I - R) = 2(I - R) + 2J = 4P_- + 12P_u.$$

The three projectors P_u, P_-, P_0 are orthogonal and sum to I ; their ranks are one, three, and two, respectively. The identity $\|M^T x\|^2 = \langle x, MM^T x \rangle$ identifies the stated kernel. The complementary rank is four. $\square$

On Q , the form

$$B = P_- - P_u$$

has signature $(3, 1)$. Time orientation is fixed by u . With

$$t = \frac{u}{\sqrt{6}}, \quad d_i = \frac{e_{i,+} - e_{i,-}}{\sqrt{2}}, \quad W = \sqrt{6}P_u + \sqrt{2}P_-,$$

one has $We_{i,\epsilon} = t + \epsilon d_i$. These six vectors are null for B , whereas

$$3t + \nu_1 d_1 + \nu_2 d_2 + \nu_3 d_3$$

has squared norm -6 . The same incidence satisfies $MM^T = 2W^*W$ on Q . The proper rotations of the cube commute with the projectors, preserve both forms, and fix t . These identities determine the metric of the cubic representation.

21.2 Cellular soldering and memory transport

For each oriented edge a , the route representation provides the face $\lambda(a)$, the sign $\sigma(a)$, the source and target fibers, and the transport $U(a) : Q_{s(a)} \rightarrow Q_{t(a)}$. Reversal satisfies $U(\bar{a}) = U(a)^{-1}$. Boundary half-edges preserve their formal reversal and their incidence on the boundary. With $\ell_m = \ell_0 9^{-m} > 0$, soldering is

$$\beta_m(a) = \ell_m \sigma_m(a) n_{\lambda(a)}^{s(a)}, \quad n_{i,\epsilon} = t + \epsilon d_i. \quad (21.1)$$

The face and sign are transported together with the route memory. The reversal convention requires

$$\beta_m(\bar{a}) = -U_m(a)\beta_m(a). \quad (21.2)$$

Applying this relation twice recovers $\beta_m(a)$; the carry record separately preserves the ordered composition.

Proposition 21.2 (Naturality of soldering under refinement). *Let $a = a_1 \cdots a_9$ be a compatible refinement of an edge. Suppose that the nine successor incidences, transported to the predecessor fiber, preserve the face and sign, and that $\ell_{m+1} = \ell_m/9$. Let $\mathcal{U}_j : Q_{m,s(a)} \longrightarrow Q_{m+1,s(a_j)}$ be the accumulated transport, including the refinement identification and the path to the source of successor j . Then*

$$\beta_m(a) = \sum_{j=1}^9 \mathcal{U}_j^{-1} \beta_{m+1}(a_j).$$

The equality is compatible with reversal and with concatenation of the memory record.

Proof. Each summand, expressed in the source fiber of the predecessor, is $(\ell_m/9)\sigma_m(a)n_{\lambda(a)}$. The sum of nine terms reproduces the predecessor soldering. Reversal reverses the order of the transports and replaces each by its inverse; relation (21.2) supplies the overall sign. Memory is concatenated in the same order, so refinement preserves both the vector representation and its enriched record. $\square$

21.3 Positive response, uniqueness, and minimal extension

The linear screen soldering is $\bar{\beta}_m = \ell_m W|_Q$. It is invertible, with

$$\bar{\beta}_m^{-1} = \ell_m^{-1} \left(\frac{1}{\sqrt{6}} P_u + \frac{1}{\sqrt{2}} P_- \right).$$

The eigenvalue $5/9$ of the antisymmetric sector of the normalized block $B_c/9$, constructed in Section 5.4, determines the compression coefficient $q_{\text{vis}} = 5/9$. We reserve this notation to distinguish it from the Euclidean quotient $q_9(n)$ of the arithmetic kernel. Its realization on the screen uses an isometry $V_9 : Q \longrightarrow \mathcal{K}$. The visible contribution $C_9 = q_{\text{vis}} V_9$ and its complement $D_9 = \sqrt{1 - q_{\text{vis}}^2} V_9$ satisfy

$$C_9^* C_9 + D_9^* D_9 = I_Q, \quad D_9^* D_9 = \frac{56}{81} I_Q.$$

The coefficient 56 is $9^2 - 5^2$; it belongs to the quadratic defect of that compression reader.

Theorem 21.3 (Energetic response determined by soldering). *There exists a unique operator $\Lambda_m : Q \longrightarrow Q$ such that $\bar{\beta}_m^* \Lambda_m \bar{\beta}_m = D_9^* D_9$. It is positive definite and*

$$\Lambda_m = \frac{28}{243 \ell_m^2} (P_u + 3P_-) = \frac{112}{81 \ell_m^2} (MM^T|_Q)^{-1}. \quad (21.3)$$

On H_F , the corresponding identity has right-hand side $(56/81)P_{\square}$.

Proof. Congruence by the invertible soldering necessarily determines

$$\Lambda_m = \bar{\beta}_m^* \frac{56}{81} I_Q \bar{\beta}_m^{-1}.$$

Substituting the inverse, the coefficient of P_u is $56/(486 \ell_m^2) = 28/(243 \ell_m^2)$; that of P_- is $28/(81 \ell_m^2)$. Both are positive. By Proposition 21.1, $(MM^T|_Q)^{-1} = P_u/12 + P_-/4$, which gives the second expression. The extension to six faces annihilates $P_0 H_F$; the identity therefore extends through $P_{\square}$. $\square$

Proposition 21.4 (Characterization by minimal extension). *For every $b \in Q$,*

$$\langle b, \Lambda_m b \rangle = \frac{112}{81\ell_m^2} \min_{Mx=b} \|x\|^2.$$

The minimizing extension is unique: $x_{\min} = M^\top(MM^\top|_Q)^{-1}b$.

Proof. The stated vector satisfies $Mx_{\min} = b$ and belongs to $\text{im } M^\top = (\ker M)^\perp$. Every other solution is $x_{\min} + z$, with $z \in \ker M$; by orthogonality, its squared norm is $\|x_{\min}\|^2 + \|z\|^2$. Moreover,

$$\|x_{\min}\|^2 = \langle b, (MM^\top|_Q)^{-1}b \rangle.$$

The formula for Λ_m completes the proof. $\square$

The operator $MM^\top|_Q$ realizes the Dirichlet-to-Neumann response of this incidence formulation; its inverse is the Neumann-to-Dirichlet response. The preceding energy corresponds to the latter, with the normalization of the soldering and the compression defect. The Schur complement of $\begin{pmatrix} I & M^\top \\ M & 0 \end{pmatrix}$ is $-MM^\top$, giving the same variational elimination.

21.4 Preservation of density under refinement

Consider the sector of cubic transports, orthogonal for the energetic inner product and compatible with the form B . Fix a common label space Q^{lab} , isometrically identified with the predecessor screen. The solderings have types $\tilde{\beta}_m : Q^{\text{lab}} \rightarrow Q_m$ and $\tilde{\beta}_{m+1,j} : Q^{\text{lab}} \rightarrow Q_{m+1,j}$; the operators Λ act on their respective target fibers. If $U_j : Q_j \rightarrow Q$ transports the successor fiber to the predecessor fiber, then

$$\tilde{\beta}_{m+1,j} = \frac{1}{9}U_j^{-1}\tilde{\beta}_m, \quad \Lambda_{m+1,j} = 81U_j^{-1}\Lambda_m U_j.$$

The factor 81 comes from $\ell_{m+1}^{-2} = 81\ell_m^{-2}$.

Theorem 21.5 (Invariance of the normalized density). *Each successor preserves the quadratic response of the predecessor:*

$$\tilde{\beta}_{m+1,j}^* \Lambda_{m+1,j} \tilde{\beta}_{m+1,j} = \frac{56}{81} I_{Q^{\text{lab}}}.$$

For r refinements, the joint response, expressed in the predecessor fibers, is

$$E_{m+r} = \frac{56}{81} I_{Q^{\text{lab}}} \otimes I_{9^r}.$$

The expectation with normalized trace $\tau_{9^r}(A) = 9^{-r} \text{Tr}(A)$ satisfies

$$(\text{id} \otimes \tau_{9^r})(E_{m+r}) = \frac{56}{81} I_{Q^{\text{lab}}}$$

at every finite depth.

Proof. The congruence for a successor contains the factors $1/9, 81, 1/9$; their product is one. Orthogonality cancels the intermediate transports. The direct sum of nine equal responses identifies the first refinement with $(56/81)I_Q \otimes I_9$. Iteration produces the tensor with I_{9^r} . Finally, $\tau_{9^r}(I_{9^r}) = 1$. The expectations compose by the identity $\tau_{9^{r+s}} = \tau_{9^r} \otimes \tau_{9^s}$. $\square$

The unital inclusions $A \mapsto A \otimes I_9$ are compatible with that collection of traces. In the inductive limit of the matrix algebras, the response is the central element $(56/81)I_Q \otimes \mathbf{1}$. The unnormalized extensive sum instead produces $9^r(56/81)I_Q$. This distinction specifies the quantity preserved: the quadratic screen density.

21.5 Variational realization of the response

The material realization uses an isometry $J_{\text{scr}} : Q \longrightarrow \mathcal{H}_{\partial}$ compatible with the spin action and refinement. The boundary co-tetrad is $e = J_{\text{scr}} \tilde{\beta}_m$. Let Λ_{CE} be the boundary response obtained by the second variation of the material and geometric action, with its boundary conditions and the elimination of the interior variables. The compatibility defect is

$$\mathcal{R}_{\text{grav}} = \tilde{\beta}_m^* (J_{\text{scr}}^* \Lambda_{\text{CE}} J_{\text{scr}} - \Lambda_m) \tilde{\beta}_m = e^* \Lambda_{\text{CE}} e - D_9^* D_9. \quad (21.4)$$

By the invertibility of $\tilde{\beta}_m$,

$$\mathcal{R}_{\text{grav}} = 0 \iff J_{\text{scr}}^* \Lambda_{\text{CE}} J_{\text{scr}} = \Lambda_m.$$

The isometry J_{scr} transports the boundary states; the variational spin current is a degree-three form with values in bivectors. Each enters the realization of the response with its own domain.

The construction determines a positive energetic form, its minimizing interior extension, and the density preserved under refinement. The amplitude of a specific material contribution is obtained by evaluating the current of that realization in its variational quadratic form. The operator fixing the response, the current on which it acts, and the gravitational observable publishing the evaluation are thus distinguished.

22 Geometric realization of transport and soldering

The discrete representation of return and screen soldering determine two aspects of transport: accumulation of memory along routes and comparison of their fibres. Their geometric realization preserves both operations. This section fixes the target space, the dimensional normalization of the connection, and the identities that its variation will use.

22.1 Screen, cotetrad, and connection

The screen quotient $Q_{\square}$, its Lorentzian form, and the soldering β_m come from 21. In each cell, a screen realization $J_{\text{scr},m}$ transports the form and soldering to the corresponding geometric fibre. Its cellular image is

$$e_m = J_{\text{scr},m} \beta_m.$$

The scale ℓ_m remains within the soldering. Frame changes act jointly on $J_{\text{scr},m}$, the connection, and cell transport.

A smooth realization of these data on an oriented manifold $\mathcal{U}$ of dimension four consists of a nondegenerate cotetrad $e = (e^I)$, a Lorentz connection $\omega = (\omega^I_J)$, and a route realization $\mathfrak{R}_C$ satisfying

$$\text{Hol}_{\mathcal{A}_{\ell_0}}(\mathfrak{R}_C(\gamma)) = \rho_C(\text{Hol}_{\text{TPK}}(\gamma)). \quad (22.1)$$

In this matrix equality, the representation is normalized by

$$N_{\ell_0}(R, a) = (\Lambda(R), a/\ell_0), \quad \rho_C = N_{\ell_0} \circ \rho_C^{\text{disc}},$$

where $\Lambda : \text{Spin}^+(3, 1) \rightarrow \text{SO}^+(3, 1)$ is the vector representation. The map N_{ℓ_0} preserves the semidirect product: dividing the translation $a + \Lambda(R)b$ by ℓ_0 is equivalent to composing the two normalized translations. The complete circuit therefore expresses $72u$ in the normalized connection and $72\ell_0 u$ in the dimensional chart. The spin lift preserves, separately, $R_4^{c_g(\gamma)}$; the vector representation has central kernel $\{1, -1\}$, so that lift is retained when spinorial fields are used. Identity, concatenation, inversion, and subdivision are maintained in the corresponding realization. The screen pairing and holonomy condition specify the data of the realization used in the field equations.

With $\eta = \text{diag}(-1, 1, 1, 1)$, define $g = \eta_{IJ} e^I \otimes e^J$. For a positive elementary scale ℓ_0 , constant in the chart, the affine connection is

$$\mathcal{A}_{\ell_0} = \begin{pmatrix} \omega & \ell_0^{-1} e \\ 0 & 0 \end{pmatrix}. \quad (22.2)$$

The cotetrad has dimensions of length; the factor ℓ_0^{-1} makes its translational component dimensionally homogeneous.

Proposition 22.1 (Curvature and torsion of the realized connection). *The curvature of (22.2) is*

$$\mathcal{F}_{\ell_0} = d\mathcal{A}_{\ell_0} + \mathcal{A}_{\ell_0} \wedge \mathcal{A}_{\ell_0} = \begin{pmatrix} F_{\omega} & \ell_0^{-1} T \\ 0 & 0 \end{pmatrix}, \quad F_{\omega} = d\omega + \omega \wedge \omega, \quad T = de + \omega \wedge e. \quad (22.3)$$

Proof. The product of matrices of forms produces $\omega \wedge \omega$ in the diagonal block and $\ell_0^{-1} \omega \wedge e$ in the translational block. The exterior derivative contributes $d\omega$ and $\ell_0^{-1} de$, since ℓ_0 is constant. Their sum proves the equality. $\square$

Curvature describes the transport defect of frames and torsion that of the cotetrad. Both belong to the same realized connection; their components preserve distinct tensor domains.

Proposition 22.2 (Covariance and Bianchi identities). *Under a frame change $g_L : \mathcal{U} \rightarrow \text{SO}^+(3, 1)$,*

$$e' = g_L e, \quad \omega' = g_L \omega g_L^{-1} - dg_L g_L^{-1},$$

one has $T' = g_L T$ and $F_{\omega'} = g_L F_{\omega} g_L^{-1}$. Moreover,

$$D_{\omega} T = F_{\omega} \wedge e, \quad D_{\omega} F_{\omega} = 0. \quad (22.4)$$

Proof. In $d(g_L e) + \omega' \wedge g_L e$, the terms containing $dg_L \wedge e$ cancel. The same substitution into the curvature gives its transformation by conjugation. For the first Bianchi identity, expand

$$D_\omega(de + \omega \wedge e) = (d\omega + \omega \wedge \omega) \wedge e.$$

In $dF_\omega + \omega \wedge F_\omega - F_\omega \wedge \omega$, the terms containing $d\omega$ and the cubic products cancel in pairs. This proves the second identity. $\square$

22.2 The three quadratic realizations of curvature

The TRIT sheet $\tau \in \{+1, 0, -1\}$ is represented in the Cartan algebra by generators $J_{ab} = -J_{ba}$ and $P_a^{(\tau)}$, with relations

$$\begin{aligned} [J_{ab}, J_{cd}] &= \eta_{bc} J_{ad} - \eta_{ac} J_{bd} - \eta_{bd} J_{ac} + \eta_{ad} J_{bc}, \\ [J_{ab}, P_c^{(\tau)}] &= \eta_{bc} P_a^{(\tau)} - \eta_{ac} P_b^{(\tau)}, \\ [P_a^{(\tau)}, P_b^{(\tau)}] &= -\tau J_{ab}. \end{aligned}$$

For constant $\ell > 0$, write

$$\mathcal{A}_\tau^{\text{Cart}} = \frac{1}{2} \omega^{ab} J_{ab} + \ell^{-1} e^a P_a^{(\tau)}.$$

Proposition 22.3 (Curvature of the trit-typed realization). *With the preceding relations,*

$$\mathcal{F}_\tau^{\text{Cart}} = \frac{1}{2} (F_\omega^{ab} - \tau \ell^{-2} e^a \wedge e^b) J_{ab} + \ell^{-1} T^a P_a^{(\tau)}. \quad (22.5)$$

Proof. Expand $d\mathcal{A} + \frac{1}{2}[\mathcal{A}, \mathcal{A}]$. The bracket of the Lorentz sector produces F_ω , the mixed bracket produces $D_\omega e$, and the translational bracket contributes $-\frac{1}{2} \tau \ell^{-2} e^a \wedge e^b J_{ab}$. $\square$

The central sheet recovers the affine connection; the outer sheets preserve the two signs of the constant-curvature term. The representation thus maintains the route's tritic regime before applying the variational equations. The holonomy equality (22.1) corresponds to the central sheet. Each outer sheet uses the representation $\rho_{C,\tau}$ in its Cartan group and its respective holonomy condition. The current and metric tensor use the fields and representation of the same realization.

II. Amplitude, evolution, and persistence

23 Variational response of memory energy

Transport along a route identifies the fibres at its endpoints and preserves the orientation and memory entering that identification. The associated energy compares the final field with the transported initial field. Its differential quantifies how that comparison responds to a variation of transport. This section obtains the response directly from the discrete functional and develops its composition, covariance, and inversion.

23.1 Discrete functional and space of variations

At a finite level m , let $\mathcal{V}_m$ be the vertices of the route representation and $\mathcal{E}_m^+$ one orientation of each edge. Each vertex carries a finite-dimensional Hermitian fibre H_v . Define

$$\mathcal{H}_m = \bigoplus_{v \in \mathcal{V}_m} H_v, \quad \mathcal{K}_m = \bigoplus_{a \in \mathcal{E}_m^+} H_{t(a)}.$$

The Hermitian inner product is antilinear in its first variable. For unitary $U_a : H_{s(a)} \rightarrow H_{t(a)}$, the difference operator is

$$(D_m f)_a = f_{t(a)} - U_a f_{s(a)}, \quad E_m(f, U) = \langle D_m f, W_m D_m f \rangle, \quad W_m = W_m^* > 0. \quad (23.1)$$

This is the energy of the term $D_m^* W_m D_m$ in the discrete generator. The weight W_m may couple different edges. The soldering operator W of Section 21 and this energy weight have different domains and roles.

This possible coupling is retained in the radial reduction of 15: for a surjective incidence $\mathcal{C}$, the response uses $\mathcal{R} = \mathcal{C} W_m^{-1} \mathcal{C}^*$, including the cross-blocks of W_m^{-1} . Identity (15.19) also preserves the memory complement. The following differential formulas act on this global weight; independent edge weights are specified later as a specialization.

Transport U is a representation of the enriched state. The field f and weight W_m remain explicit arguments of the functional; each material application preserves the rules determining them. We consider differentiable collections of unitary realizations of transport,

$$\dot{U}_a = A_a U_a, \quad A_a^* = -A_a.$$

The physically used variations form the subspace preserving the conditions of the corresponding realization. The differential is calculated first in the unitary tangent space and then restricted to that subspace.

Proposition 23.1 (Transport-response covector). *Write*

$$v_a = U_a f_{s(a)}, \quad r = D_m f, \quad q = W_m r.$$

At fixed field and weight, the energy differential is

$$\dot{E}_m(A) = -2 \operatorname{Re} \sum_{a \in \mathcal{E}_m^+} \langle q_a, A_a v_a \rangle. \quad (23.2)$$

Its representative for the real Hilbert–Schmidt inner product is

$$\mathcal{J}_a = v_a q_a^* - q_a v_a^*, \quad \mathcal{J}_a^* = -\mathcal{J}_a, \quad \dot{E}_m(A) = \sum_a \operatorname{Re} \operatorname{Tr}(\mathcal{J}_a^* A_a). \quad (23.3)$$

For simultaneous variations of field, transport, and weight, one has

$$\dot{E}_m = 2 \operatorname{Re} \sum_a \langle q_a, \dot{f}_{t(a)} - U_a \dot{f}_{s(a)} - A_a v_a \rangle + \langle r, \dot{W}_m r \rangle. \quad (23.4)$$

Proof. The first-order update, written in the algebra of dual numbers with $\epsilon^2 = 0$, gives

$$r + \epsilon \dot{r}, \quad \dot{r}_a = \dot{f}_{t(a)} - U_a \dot{f}_{s(a)} - A_a v_a.$$

The coefficient of ϵ in the energy is

$$\langle \dot{r}, W_m r \rangle + \langle r, W_m \dot{r} \rangle + \langle r, \dot{W}_m r \rangle.$$

Self-adjointness of the weight yields (23.4). Fixing field and weight leaves (23.2). Moreover,

$$\text{Tr}(\mathcal{J}_a^* A_a) = v_a^* A_a q_a - q_a^* A_a v_a.$$

Anti-Hermiticity of A_a implies $v_a^* A_a q_a = -\overline{q_a^* A_a v_a}$, giving the stated representative. $\square$

The response is thus evaluated: on each edge, the field is transported, its difference from the final field is calculated, and the energy weight is applied. The antisymmetrized product of those two vectors determines $\mathcal{J}_a \in \mathfrak{u}(H_{t(a)})$. Through the real Hilbert–Schmidt inner product, this operator represents the response covector on unitary variations of the target fibre.

23.2 Composition and transport of the response

Proposition 23.2 (Composition rule along a route). *For a route formed by N transports, let $U_\gamma = U_N \cdots U_1$ and $B_k = U_N \cdots U_{k+1}$, with $B_N = I$. If $\dot{U}_k = A_k U_k$, then*

$$\dot{U}_\gamma U_\gamma^{-1} = \sum_{k=1}^N B_k A_k B_k^{-1}. \quad (23.5)$$

Therefore, a representative $\mathcal{J}_\gamma$ in the final fibre induces the contribution $B_k^ \mathcal{J}_\gamma B_k$ in factor k . When a factor belongs to several routes of the functional, its contributions add with the incidences used by that functional.*

Proof. The product rule gives

$$\dot{U}_\gamma = \sum_k U_N \cdots U_{k+1} A_k U_k \cdots U_1.$$

Multiplication by U_γ^{-1} eliminates the factors to the right of A_k and yields (23.5). By unitarity and cyclicity of the trace,

$$\text{Re Tr}(\mathcal{J}_\gamma^* B_k A_k B_k^{-1}) = \text{Re Tr}((B_k^* \mathcal{J}_\gamma B_k)^* A_k).$$

Linearity of the differential proves the sum of contributions. $\square$

This formula preserves the position of each operation within the route. Transport subsequent to an incidence determines how its contribution is represented in the final fibre; the local response is recovered by transporting it to the fibre of that incidence.

Proposition 23.3 (Covariance under unitary frame changes). *Let g_v be unitary frame changes independent of the variation parameter, and $G_\mathcal{E} = \bigoplus_a g_{t(a)}$. Under the transformations*

$$f'_v = g_v f_v, \quad U'_a = g_{t(a)} U_a g_{s(a)}^{-1}, \quad W'_m = G_\mathcal{E} W_m G_\mathcal{E}^{-1}, \quad A'_a = g_{t(a)} A_a g_{t(a)}^{-1},$$

one has

$$\mathcal{J}'_a = g_{t(a)} \mathcal{J}_a g_{t(a)}^{-1}, \quad \dot{J}'_m(A') = \dot{J}_m(A). \quad (23.6)$$

Proof. One obtains $r' = G_\mathcal{E} r$, $q' = G_\mathcal{E} q$, and $v'_a = g_{t(a)} v_a$. Substitution into (23.3) proves the first identity. Invariance of the Hermitian inner product, or cyclicity of the trace, proves the second. $\square$

When frame changes depend on the parameter, their derivatives also act on the field and weight. The corresponding covariant expression is then the complete differential (23.4).

23.3 Oriented inversion and transport of the weight

Inversion can be seen explicitly in the case of an energy local to each edge, $W_m = \bigoplus_a W_a$. For $a : s \rightarrow t$, its inverse carries

$$U_a = U_a^{-1}, \quad r_a = -U_a^{-1}r_a, \quad W_a = U_a^{-1}W_aU_a.$$

These rules preserve the edge energy exactly.

Proposition 23.4 (Differential compatibility with inversion). *If W_a and the fields at both endpoints remain fixed, then*

$$A_a = -U_a^{-1}A_aU_a, \quad \dot{W}_a = U_a^{-1}[W_a, A_a]U_a.$$

The complete variation of the inverted edge energy agrees with the original one. In particular, if $[W_a, A_a] = 0$, the partial transport covector satisfies

$$j_a(U_a^{-1}A_aU_a) = -j_a(A_a). \quad (23.7)$$

Proof. The derivative of U_a^{-1} is $-U_a^{-1}A_a$, yielding both formulas. Writing $v_a = U_a f_s$ and $f_t = r_a + v_a$, the part due exclusively to inverted transport is

$$j_{\bar{a}}(A_{\bar{a}}) = -2 \operatorname{Re}\langle r_a, W_a A_a f_t \rangle = j_a(A_a) - 2 \operatorname{Re}\langle r_a, W_a A_a r_a \rangle.$$

Variation of the weight contributes

$$\langle r_a, \dot{W}_a r_a \rangle = \langle r_a, [W_a, A_a] r_a \rangle = 2 \operatorname{Re}\langle r_a, W_a A_a r_a \rangle.$$

The two terms cancel. If the commutator vanishes, that contribution disappears; linearity in A_a gives (23.7). $\square$

Energy is summed over one orientation per edge. A sum over both orientations carries the factor 1/2. This convention preserves the same functional and hence the same response.

23.4 Compatibility with refinement

Theorem 23.5 (Transport of the differential by refinement). *Let $J_m : \mathcal{H}_m \rightarrow \mathcal{H}_{m+1}$ and $K_m : \mathcal{K}_m \rightarrow \mathcal{K}_{m+1}$ be fixed maps. For a C^1 collection, assume the refinement identities*

$$D_{m+1}(t)J_m = K_m D_m(t), \quad K_m^* W_{m+1}(t) K_m = W_m(t). \quad (23.8)$$

Then $E_{m+1}(J_m f) = E_m(f)$. The transport covectors agree on variations linked by the first identity. When the weights vary, their contributions to the differential also agree.

Proof. By substitution,

$$\langle D_{m+1}J_m f, W_{m+1}D_{m+1}J_m f \rangle = \langle D_m f, K_m^* W_{m+1} K_m D_m f \rangle = E_m(f).$$

Differentiating the first identity in (23.8) gives $\dot{D}_{m+1}J_m = K_m \dot{D}_m$. Consequently,

$$2 \operatorname{Re}\langle D_{m+1}J_m f, W_{m+1}\dot{D}_{m+1}J_m f \rangle = 2 \operatorname{Re}\langle D_m f, W_m \dot{D}_m f \rangle.$$

The derivative of the second identity is $K_m^* \dot{W}_{m+1} K_m = \dot{W}_m$, proving agreement of the weight-variation terms. $\square$

The statement uses compatibility of the transport collection. If J_m depends on the parameter, the chain rule also retains the term

$$2 \operatorname{Re}\langle D_{m+1}J_m f, W_{m+1}D_{m+1}\dot{J}_m f \rangle.$$

The naturality obtained transports the functional, its incidences, and its differential jointly.

23.5 Material realization of the response

The result of this section is the explicit covector j_m , with its representative $\mathcal{J}_a$. The material realization composes that covector with variation of transport induced by the connection and with normalization of the action. If $\mathcal{L}_m$ denotes that differential map, the covector on connection variations is $\mathcal{L}_m^* j_m$. The composition rule (23.5) evaluates its part corresponding to finite products of transports.

In Section 26, the current σ_{IJ} is a three-form defined by variation of the material action. Identification between the two realizations preserves the field, representation, weight, dimensional normalization, and passage to the limit used by that action. These data make it possible to evaluate the contraction entering torsional density. The present development supplies the discrete differential composed in that calculation.

24 Realization of the current in connection coordinates

Memory energy assigns each route a response preserving its incidences and the order of transports. To insert it into a geometric action, its components with respect to connection variations are required. This step follows by differentiating the transport already realizing the route: the connection current is the composition of that differential with the energetic covector of the preceding section.

The elliptic, parabolic, and hyperbolic TRIT regimes require preserving the complete differential before restricting it to a particular representation. Unitary rotations constitute a domain of that construction. The following extension also preserves the dilation and transport directions admitted by the geometric realization used.

24.1 Differential on invertible transports

The fibers, fields, and weight of (23.1) are retained. Consider now invertible transports and differentiable collections with $\dot{U}_a = A_a U_a$, where A_a belongs to the tangent space of the chosen representation. Write

$$v_a = U_a f_{s(a)}, \quad r_a = f_{t(a)} - v_a, \quad q = W_m r, \quad E_m = \langle r, W_m r \rangle.$$

Proposition 24.1 (Complete representative of the differential). *With field and weight fixed, one has*

$$d_U E_m(A) = -2 \operatorname{Re} \sum_a \langle q_a, A_a v_a \rangle = \sum_a \operatorname{Re} \operatorname{Tr}(\mathcal{G}_a^* A_a), \quad \mathcal{G}_a = -2 q_a v_a^*. \quad (24.1)$$

The anti-Hermitian and Hermitian projections of this representative are

$$\frac{\mathcal{G}_a - \mathcal{G}_a^*}{2} = v_a q_a^* - q_a v_a^* = \mathcal{J}_a, \quad \frac{\mathcal{G}_a + \mathcal{G}_a^*}{2} = -(q_a v_a^* + v_a q_a^*). \quad (24.2)$$

Restriction to anti-Hermitian generators recovers exactly (23.3).

Proof. The identity $\dot{r}_a = -A_a v_a$ and self-adjointness of the weight give $\dot{E}_m = 2 \operatorname{Re} \langle q, \dot{r} \rangle$. Moreover,

$$\operatorname{Tr}(\mathcal{G}_a^* A_a) = -2 \operatorname{Tr}(v_a q_a^* A_a) = -2 q_a^* A_a v_a.$$

The two projections follow by taking the adjoint. For the real Hilbert–Schmidt product, the Hermitian and anti-Hermitian subspaces are orthogonal; the Hermitian part therefore has zero contraction with an anti-Hermitian generator. $\square$

The formula allows weights coupling edges: their action is evaluated in $q = W_m r$ before separating the contributions of each incidence. If the field or weight varies, the complete expression (23.4) is retained; it is also valid for the invertible transports considered here.

24.2 Quadratic representations of TRIT

In a two-dimensional matrix realization of $\mathbb{A}_\tau$, one may take

$$J_+ = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}, \quad J_- = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad J_0 = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}.$$

Direct multiplication gives

$$J_+^2 = -I, \quad J_-^2 = I, \quad J_0^2 = 0.$$

Their exponentials realize the elliptic, hyperbolic, and parabolic regimes, respectively. The first is orthogonal for the fixed positive inner product. The second preserves the form $B = \operatorname{diag}(-1, 1)$, since $J_-^\top B + B J_- = 0$. The third represents the algebra of dual numbers. These representations fix examples of the quadratic types; using them on a screen of another dimension retains the corresponding representation map.

A calculation distinguishes the components of the differential. On one edge, take $U = I$, $f_s = (1, 1)^T$, $f_t = 2f_s$, and $W = I$. Then $q = v = f_s$, $\mathcal{J} = 0$, whereas

$$dE(J_-) = -4, \quad dE(J_0) = -2. \quad (24.3)$$

The complete representative preserves both responses. This distinction identifies the space of variations before the material contraction is performed.

24.3 Route composition and oriented reversal

Proposition 24.2 (Contragredient transport of the differential). *For $U_\gamma = U_N \cdots U_1$, let $B_k = U_N \cdots U_{k+1}$ and $B_N = I$. One has*

$$\dot{U}_\gamma U_\gamma^{-1} = \sum_k B_k A_k B_k^{-1}.$$

The representative $\mathcal{G}_\gamma$ in the final fiber induces at incidence k

$$\mathcal{G}_k = B_k^* \mathcal{G}_\gamma (B_k^{-1})^*. \quad (24.4)$$

Proof. The product rule, followed by multiplication by U_γ^{-1} , proves the first equality. Cyclicity of the trace gives

$$\operatorname{Re} \operatorname{Tr}(\mathcal{G}_\gamma^* B_k A_k B_k^{-1}) = \operatorname{Re} \operatorname{Tr}((B_k^* \mathcal{G}_\gamma (B_k^{-1})^*)^* A_k).$$

For unitary B_k , the conjugation of 23.2 is recovered. $\square$

Proposition 24.3 (Reversal by energetic congruence). *For a local energy per edge, reversal is represented by*

$$U_{\bar{a}} = U_a^{-1}, \quad r_{\bar{a}} = -U_a^{-1} r_a, \quad W_{\bar{a}} = U_a^* W_a U_a. \quad (24.5)$$

It preserves energy. If W_a remains fixed,

$$A_{\bar{a}} = -U_a^{-1} A_a U_a, \quad \dot{W}_{\bar{a}} = U_a^* (A_a^* W_a + W_a A_a) U_a.$$

The complete variation of the reversed energy agrees with the original.

Proof. Substitution into $r_{\bar{a}}^* W_{\bar{a}} r_{\bar{a}}$ gives $r_a^* W_a r_a$. The derivative of the inverse is $\dot{U}_a^{-1} = -U_a^{-1} \dot{A}_a$; that of the weight follows from the product rule. Equality of energies holds throughout the collection, so their derivatives agree, including the term $\langle r_{\bar{a}}, \dot{W}_{\bar{a}} r_{\bar{a}} \rangle$. $\square$

Congruence preserves positivity for every invertible transport. In the unitary representation it reduces to the reversal rule of 23.4. The sum uses one orientation per edge, or half the sum over both orientations.

24.4 Evaluation of current components

Let $\theta = (\theta_b)$ be a real variation of the connection coordinates of the realization used. The effective differential of transport defines the operators

$$A_a(\theta) = \dot{U}_a U_a^{-1} = \sum_b B_{ab} \theta_b. \quad (24.6)$$

For a finite composition, their contributions follow from (24.4). The operators B_{ab} are the derivatives of the connection representation realizing the route.

Theorem 24.4 (Connection current of the memory action). *For a material term $S_{\text{mem}} = \mathfrak{a}_m E_m$, with $\mathfrak{a}_m$ its normalization on the action line, at fixed field, weight, and normalization one has*

$$\delta S_{\text{mem}} = -\frac{1}{2} \sum_b \theta_b s_b, \quad s_b = 4\mathfrak{a}_m \operatorname{Re} \sum_a \langle q_a, B_{ab} v_a \rangle. \quad (24.7)$$

If the cellular pairing between the connection and current bases is an invertible matrix M , defined by $\delta S_{\text{mem}} = -\frac{1}{2}\theta^\top M\sigma$, the current coordinates are

$$\sigma = M^{-1}s. \quad (24.8)$$

Proof. Substitute (24.6) into (24.1), multiply by $\mathbf{a}_m$, and collect the coefficients of θ_b . The variational convention $-1/2$ fixes the factor 4. Comparing both expressions for every variation θ gives $M\sigma = s$; invertibility of the pairing proves existence and uniqueness of these coordinates. $\square$

The formula explicitly realizes the composition described in 23.5. To evaluate each component, the field is transported, the residual is computed, the weight is applied, and contraction is performed with the connection variation of the same realization. The pairing M preserves cell orientation and volume. A diagonal pairing gives the corresponding division by its volumes; other bases retain the complete matrix.

If the connection also enters f , W_m , or $\mathbf{a}_m$, apply (23.4) and add $E_m\delta\mathbf{a}_m$. The contributions of all material terms are added before composing the Cartan response.

Corollary 24.5 (Compatibility of the current with refinement). *For the compatible collections of 23.5, let P_m be the fixed map taking connection variations from level m to level $m+1$. If the memory actions agree on those refined fields and transports, their covectors satisfy*

$$s_m = P_m^\top s_{m+1}, \quad M_m \sigma_m = P_m^\top M_{m+1} \sigma_{m+1}.$$

Proof. Differentiating equality of actions in the direction $P_m\theta$ gives $-\frac{1}{2}\theta^\top s_m = -\frac{1}{2}\theta^\top P_m^\top s_{m+1}$. The arbitrariness of θ and the cellular representations of both covectors prove the two identities. Composing these maps iterates the equality through every finite number of refinements. $\square$

24.5 Insertion into the Cartan equation

Section 26 normalizes the material three-form by $\delta_\omega S_m = -\frac{1}{2} \int \delta\omega^{IJ} \wedge \sigma_{IJ}$. The preceding theorem provides its cellular components for the stated memory action and pairing. The Cartan response uses this current together with the co-tetrad, the Barbero–Immirzi parameter, and the gravitational coupling of the same realization.

The material dependence on contorsion determines how that equation is solved. For affine matter, the current is independent of contorsion, and the linear inversion and quadratic elimination proved in Section 26 apply. For a holonomy action, the differential is evaluated at the current connection, and this dependence is retained when solving the equation. For example, an edge with $f_s = f_t = 1$, unit weight, and $U(t) = e^{it}$ has

$$E(t) = 2 - 2 \cos t, \quad E'(t) = 2 \sin t.$$

Here the current depends on t . The complete formula retains this dependence instead of replacing it by its value at a particular connection.

Finally, the gravitational density is obtained from the metric variation of the effective action, with its field and normalization. The current representation, the contorsion it produces, and the resulting density are the successive operations of this composition. The subsequent dilution law transports the amplitude thus evaluated in the corresponding material section.

25 Differential of the minimal boundary extension

Cubic incidence determines the minimal-extension field and the weight of its response simultaneously. For a boundary datum, both are calculated through the inverse of the incidence Gramian. We now develop the differential of that same operation, preserving variations of the boundary, transport, and normalization scale. In this way, the field entering the response is selected by the screen variational problem.

25.1 Minimizing field and response weight

Let H and Q be finite-dimensional Hermitian spaces, with fixed inner products antilinear in the first variable. The real case is obtained by replacing adjoints with transposes. On an open set of real parameters Ω , consider C^1 collections

$$C(\theta) : H \longrightarrow Q, \quad b(\theta) \in Q, \quad \chi(\theta) > 0, \quad \text{Ran } C(\theta) = Q.$$

Surjectivity is equivalent to full row rank in orthonormal bases. Define the Gramian, conjugate potential, and extension

$$L = CC^*, \quad y = L^{-1}b, \quad f_{\min} = C^*y, \quad \Lambda = \chi L^{-1}. \quad (25.1)$$

The incidence C and the cellular pairing M in (24.8) are different operators.

Proposition 25.1 (Variational selection of the extension). *The extension $f_{\min}$ is the unique norm minimizer among fields $f \in H$ satisfying $Cf = b$. Its energy is*

$$E(\theta) = \chi \min_{Cf=b} \|f\|^2 = \chi \langle b, L^{-1}b \rangle = \langle b, \Lambda b \rangle. \quad (25.2)$$

The field $f_{\min}$, potential y , and energy E depend C^1 -smoothly on the parameters.

Proof. Surjectivity of C makes C^* injective, so $\langle z, Lz \rangle = \|C^*z\|^2 > 0$ for $z \neq 0$. Thus L is invertible and $Cf_{\min} = Ly = b$. Every solution is $f_{\min} + z$, with $z \in \ker C$. Since $f_{\min} \in \text{Ran } C^* = (\ker C)^\perp$,

$$\|f_{\min} + z\|^2 = \|f_{\min}\|^2 + \|z\|^2, \quad \|f_{\min}\|^2 = \langle y, Ly \rangle = \langle b, L^{-1}b \rangle.$$

These identities prove the minimum, its uniqueness, and its value. Inversion is differentiable on the open set of invertible matrices; the expressions in (25.1) provide the stated regularity. $\square$

The cubic realization of 21.1 corresponds to $H = \mathbb{R}^{\{-1,1\}^3}$, the screen quotient $Q = P_\square \mathbb{R}^6$, and incidence $C = M : H \rightarrow Q$. Its Gramian and normalization are

$$L_\square = 12P_u + 4P_-, \quad \chi_m = \frac{112}{81\ell_m^2}, \quad \Lambda_m = \frac{28}{243\ell_m^2}(P_u + 3P_-). \quad (25.3)$$

The factor 112/81 comes from soldering and the quadratic compression defect in (21.3). If b has dimensions of length, so do y and $f_{\min}$; incidence is dimensionless, and E is dimensionless. Conversion to action uses the normalization on the action line specified below.

The weighted formulation in 15 preserves the same mechanism with a global energetic inner product $W > 0$:

$$\mathcal{R} = CW^{-1}C^*, \quad f_{\min} = W^{-1}C^*\mathcal{R}^{-1}b, \quad f = f_{\min} + k, \quad Ck = 0.$$

Weighted orthogonality proves $\langle f, Wf \rangle = \langle b, \mathcal{R}^{-1}b \rangle + \langle k, Wk \rangle$. The pair (b, k) reconstructs the complete field. For $W = I$, these are the expressions of the preceding theorem. Longitudinal transport sends $\mathcal{R}$ to $D\mathcal{R}D^*$, preserving the minimum action with the transported boundary. This static reduction is distinguished from dynamic elimination in (15.21), where the spectral parameter and the temporal norm of (15.23) enter.

25.2 Complete differential of the minimum

Theorem 25.2 (Response to variation of incidence and boundary). *For every real tangent direction $\delta\theta$, one has*

$$\delta E = \delta\chi \langle b, y \rangle + 2\chi \operatorname{Re} \langle y, \delta b - (\delta C)f_{\min} \rangle. \quad (25.4)$$

If $\chi = 112/(81\ell_m^2)$, with variable $\ell_m > 0$,

$$\delta E = -2E \frac{\delta\ell_m}{\ell_m} + 2\chi_m \operatorname{Re} \langle y, \delta b - (\delta C)f_{\min} \rangle. \quad (25.5)$$

Proof. Differentiating $LL^{-1} = I_Q$ gives

$$\delta L^{-1} = -L^{-1}(\delta L)L^{-1}, \quad \delta L = (\delta C)C^* + C(\delta C)^*.$$

Applying the product rule to (25.2) and using self-adjointness of L^{-1} yields

$$\delta E = \delta\chi \langle b, y \rangle + 2\chi \operatorname{Re} \langle y, \delta b \rangle - \chi \langle y, (\delta L)y \rangle.$$

The two terms in the final product are complex conjugates, and $C^*y = f_{\min}$; hence

$$\langle y, (\delta L)y \rangle = 2 \operatorname{Re} \langle y, (\delta C)f_{\min} \rangle.$$

This gives (25.4). Finally, $\delta\chi_m = -2\chi_m \delta\ell_m/\ell_m$ produces (25.5). $\square$

The formula incorporates variation of the minimizing field through the equation determining it. It can also be checked directly: $C\delta f_{\min} = \delta b - (\delta C)f_{\min}$ and $f_{\min} = C^*y$ imply

$$\delta \|f_{\min}\|^2 = 2 \operatorname{Re} \langle f_{\min}, \delta f_{\min} \rangle = 2 \operatorname{Re} \langle y, \delta b - (\delta C)f_{\min} \rangle.$$

For example, if C is fixed and $b = \ell_m b_0$, with fixed b_0 , the two terms in (25.5) cancel and the energy remains constant. This check jointly preserves the boundary dimension and its normalization.

25.3 Fixed quotient and frame changes

The preceding derivatives are taken in fixed spaces H, Q with fixed Hermitian inner products. In the six-face presentation, the inverse always acts on $Q = P_{\square}H_F$. For a collection of quotients $Q(\theta)$, first fix an isometric trivialization $V_Q(\theta) : Q \rightarrow Q(\theta)$, and analogously for the domain. Derivatives of these maps are included when differentiating expressions transported to the fixed spaces. The formulas apply on each open set where rank is preserved. Changes to the energy inner product also require differentiating that inner product; a Lorentzian metric and the positive inner product used for minimization retain their different types here.

Proposition 25.3 (Covariance of the minimum). *Let g_H, g_Q be unitary. Under*

$$C' = g_Q C g_H^*, \quad b' = g_Q b, \quad \chi' = \chi,$$

one has $y' = g_Q y$, $f'_{\min} = g_H f_{\min}$, and $E' = E$. When frame changes depend on the parameter, the complete differential preserves that equality. In particular, a pure frame variation has $\delta E = 0$.

Proof. The transformed Gramian is $L' = g_Q L g_Q^*$, yielding the transformations of $y, f_{\min}$ and equality of energies. For a pure frame variation, at a point with identity frames, let $A_Q^* = -A_Q$ and $A_H^* = -A_H$ be its generators. Then

$$\delta C = A_Q C - C A_H, \quad \delta b = A_Q b, \quad \delta b - (\delta C)f_{\min} = C A_H f_{\min}.$$

Its contribution to (25.4) is $2\chi \operatorname{Re} \langle f_{\min}, A_H f_{\min} \rangle = 0$. The proof thus includes terms due to moving frames. $\square$

25.4 Transported incidence and connection components

Connection dependence is specified before performing the contraction. An explicit block realization of incidence is obtained as follows. Let $\mathcal{K}$ be a common Hermitian fibre, $F = \{(i, \epsilon) : 1 \leq i \leq 3, \epsilon = \pm 1\}$, and $V = \{-1, 1\}^3$. For each incidence $\nu_i = \epsilon$, the realization used declares a route $\gamma_{i,\epsilon,\nu}$ from the vertex fibre to the face fibre, with transport $U_{i,\epsilon,\nu}$. Trivializations identify those fibres with $\mathcal{K}$. Define

$$\begin{aligned} (B_U f)_{i,\epsilon} &= \sum_{\nu_i=\epsilon} U_{i,\epsilon,\nu} f_\nu, \\ C_U &= (q_\square \otimes I_{\mathcal{K}}) B_U : \bigoplus_{\nu \in V} \mathcal{K} \longrightarrow Q \otimes \mathcal{K}, \quad q_\square = P_\square : H_F \rightarrow Q. \end{aligned} \quad (25.6)$$

This is an explicit variational continuation of cubic incidence in the declared route representation. For identity transports, $C_U = M \otimes I_{\mathcal{K}}$, and (25.3) is recovered, tensored with the fibre. The Gramian remains positive in a neighbourhood of that realization, since positive definiteness is an open condition. On each full-rank domain, the weight and field are fixed by (25.1). The routes realizing the incidences, their orientations, and their memories are data of that realization of APP-TRIT-TPK transport and accompany the construction.

If $\omega = (\omega_j)$ are real connection coordinates, the derivative of C_U is calculated by substituting in each block

$$\partial_j U_\gamma = \sum_{k=1}^N U_N \cdots U_{k+1} (\partial_j U_k) U_{k-1} \cdots U_1, \quad U_\gamma = U_N \cdots U_1. \quad (25.7)$$

Empty products are identities. The formula follows from the product rule and preserves the place of each operation within the route. Thus $C_j := \partial_j C_U$ is obtained from a finite sum of known matrices in the chosen realization, followed by the projection $q_\square \otimes I$. If a moving quotient is used, its derivative is incorporated according to 25.3. The fixed integer cubic matrix corresponds to the case $C_j = 0$; the connection response comes from the transport dependence actually declared in (25.6).

Theorem 25.4 (Variational components of the minimal extension). *For the screen action $S_{\min} = \mathbf{a}_m E$, where $\mathbf{a}_m$ is its normalization on the action line, write*

$$b_j = \partial_j b, \quad \chi_j = \partial_j \chi, \quad a_j = \partial_j \mathbf{a}_m.$$

With the convention $\delta S_{\min} = -\frac{1}{2} \sum_j s_j \delta \omega_j$, the components are

$$\begin{aligned} s_j &= 4\mathbf{a}_m \chi \operatorname{Re} \langle y, C_j f_{\min} - b_j \rangle \\ &\quad - 2\mathbf{a}_m \chi_j \langle b, y \rangle - 2a_j E. \end{aligned} \quad (25.8)$$

For cubic normalization, the second term equals $4\mathbf{a}_m E \partial_j \ell_m / \ell_m$. If the cellular pairing of this same action has invertible matrix M_{cel} , its current coordinates are $\sigma = M_{\text{cel}}^{-1} s$.

Proof. The product rule gives $\partial_j S_{\min} = a_j E + \mathbf{a}_m \partial_j E$. Substituting (25.4) and multiplying by -2 proves (25.8). The derivative of χ_m produces the scale expression. Finally, comparing $\delta S_{\min} = -\frac{1}{2} \delta \omega^\top s$ with $-\frac{1}{2} \delta \omega^\top M_{\text{cel}} \sigma$ for every variation gives $M_{\text{cel}} \sigma = s$, and its inverse determines the coordinates. $\square$

The construction recovers a field and weight selected jointly by incidence, and evaluates their response to variations of that same realization. The boundary datum b , route configuration, scale ℓ_m , normalization $\mathbf{a}_m$, and cellular pairing preserve their representation rules. Interpretation as a material current uses these objects and the complete action. The Cartan response and its subsequent metric density are composed with it in the corresponding matter domain, preserving any dependence of the current on the connection.

26 Spin current and algebraic Cartan–Holst response

The realization of enriched transport in Section 22 provides the co-tetrad and connection; the dimensional sections and incidence parameter in Sections 14 and 16 fix their couplings. Their composition determines a first-order variational problem. The material current has a precise place in it: it is the derivative of the action with respect to the connection and determines the torsional response with the same normalization.

26.1 Geometric domain and normalization of the current

We work on a smooth oriented four-dimensional manifold M , with internal Lorentzian bundle E , metric $\eta = \text{diag}(-1, 1, 1, 1)$, and nondegenerate co-tetrad $e \in \Omega^1(M; E)$. In a local frame we write e^I , with $I = 0, 1, 2, 3$. The metric connection $\omega^{IJ} = -\omega^{JI}$ acts through D_ω . When the fields Ψ are spinorial, a spin structure and the corresponding material representation are retained. Define

$$\Sigma^{IJ} = e^I \wedge e^J, \quad F^{IJ} = d\omega^{IJ} + \omega^I_K \wedge \omega^{KJ}, \quad T^I = D_\omega e^I. \quad (26.1)$$

The operator $\star$ acts on the internal bivector indices,

$$(\star X)^{IJ} = \frac{1}{2} \epsilon^{IJ}_{KL} X^{KL}, \quad \star^2 = -I, \quad P_\gamma = \star + \gamma^{-1}I. \quad (26.2)$$

This distinguishes this duality from the exterior duality of forms on M . Internal indices are raised and contracted with η .

Fix a chart with $G_{\text{int}} > 0$, $c_{\text{int}} > 0$, $\gamma = \gamma_{\text{HMT}} \in \mathbb{R} \setminus \{0\}$, and Λ_{int} constant on it. Abbreviate $\kappa = 8\pi G_{\text{int}}/c_{\text{int}}^4$ and $c = c_{\text{int}}$. These sections remain fixed during variation. In the radial realization of 15, substitution of $G_{\text{int}} = c^3 L^2 / \hbar_{\text{ret}}$ explicitly determines

$$\kappa = \frac{8\pi L^2}{\hbar_{\text{ret}} c}, \quad \frac{1}{2\kappa c} = \frac{\hbar_{\text{ret}}}{16\pi L^2}, \quad \frac{\Lambda_{\text{int}}}{\kappa c} = \frac{\Lambda_{\text{int}} \hbar_{\text{ret}}}{8\pi L^2}.$$

These identities transmit the previously constructed length and action section to the Cartan–Holst action. The geometric realization, volume convention, and material fields are those specified in this section. The action is

$$S[e, \omega, \Psi] = \frac{1}{2\kappa c} \int_M \Sigma_{IJ} \wedge (P_\gamma F)^{IJ} - \frac{\Lambda_{\text{int}}}{\kappa c} \int_M \text{vol}_e + S_{\text{m}}[e, \omega, \Psi]. \quad (26.3)$$

The volume and orientation convention is the one giving $\Sigma_{IJ} \wedge (\star F)^{IJ} = R \text{vol}_e$ in the Levi–Civita sector. The preceding dimensional type makes every summand an action section.

For connection variation, variations compactly supported in the interior are used, or one fixes $\delta\omega|_{\partial M} = 0$. One may also use a boundary completion whose variational problem annihilates the same boundary term. With co-tetrad and material fields fixed, the spin current is the degree-three form, valued in dual internal bivectors, determined by

$$\delta_\omega S_{\text{m}} = -\frac{1}{2} \int_M \delta\omega^{IJ} \wedge \sigma_{IJ}. \quad (26.4)$$

Pairing with an arbitrary compactly supported variation fixes σ uniquely. This definition preserves the action unit of the current and the normalization factor entering its geometric response.

26.2 Connection variation and Holst inversion

Theorem 26.1 (Variational Cartan–Holst equation). *In the preceding domain, stationarity with respect to the connection is equivalent to*

$$D_\omega(P_\gamma \Sigma)^{IJ} = \kappa c \sigma^{IJ}. \quad (26.5)$$

Proof. The curvature variation is $\delta F = D_\omega \delta \omega$. Internal duality is parallel and self-adjoint for bivector contraction; since γ is constant, $D_\omega P_\gamma = P_\gamma D_\omega$. Setting $B_{IJ} = (P_\gamma \Sigma)_{IJ}$ yields

$$B_{IJ} \wedge D_\omega \delta \omega^{IJ} = d(B_{IJ} \wedge \delta \omega^{IJ}) - D_\omega B_{IJ} \wedge \delta \omega^{IJ}.$$

The last form changes sign when its degree-three and degree-one factors are interchanged. The boundary conditions eliminate the integral of the exact term. The cosmological term is independent of ω . Therefore,

$$\delta_\omega S = \frac{1}{2\kappa c} \int_M \delta \omega^{IJ} \wedge [D_\omega B_{IJ} - \kappa c \sigma_{IJ}].$$

The arbitrariness of the variation proves the equation, including its sign and coefficient. $\square$

Lemma 26.2 (Real inverse of the Holst operator). *For $\gamma \in \mathbb{R} \setminus \{0\}$,*

$$P_\gamma^{-1} = \frac{\gamma^2}{1 + \gamma^2} (-\star + \gamma^{-1} I). \quad (26.6)$$

Proof. Both factors are polynomials in $\star$ and commute. Their product before normalization is

$$(\star + \gamma^{-1} I)(-\star + \gamma^{-1} I) = -\star^2 + \gamma^{-2} I = (1 + \gamma^{-2}) I.$$

The denominator $1 + \gamma^2$ is positive in the fixed real domain. Multiplication by $\gamma^2/(1 + \gamma^2)$ gives the identity. $\square$

The current thus determines the degree-three bivector form

$$Y^{IJ} := \kappa c \frac{\gamma^2}{1 + \gamma^2} (-\star + \gamma^{-1} I) \sigma^{IJ}, \quad D_\omega \Sigma^{IJ} = Y^{IJ}. \quad (26.7)$$

26.3 Pointwise reconstruction of torsion and contorsion

The Leibniz rule gives

$$D_\omega \Sigma^{IJ} = T^I \wedge e^J - e^I \wedge T^J =: \mathbb{L}_e(T)^{IJ}. \quad (26.8)$$

At each point, $\mathbb{L}_e$ acts from $\Lambda^2 T^* M \otimes E$ to $\Lambda^3 T^* M \otimes \Lambda^2 E$. Both spaces have dimension twenty-four. Let E_I be the dual frame of the co-tetrad, so that $\iota_{E_I} e^J = \delta_I^J$.

Theorem 26.3 (Algebraic inversion of the torsional equation). *For a nondegenerate co-tetrad, $\mathbb{L}_e$ is an isomorphism. Its inverse on the form Y is written*

$$B^I = \sum_J \iota_{E_J} Y^{IJ}, \quad b = \frac{1}{4} \sum_I \iota_{E_I} B^I, \quad T^I = B^I - e^I \wedge b. \quad (26.9)$$

Equation (26.5) therefore determines torsion pointwise from e and the current appearing in it.

Proof. First let $Y = \mathbb{L}_e(T)$ and set $t = \sum_J \iota_{E_J} T^J$. Contracting (26.8), using the fact that T^I has degree two and the co-tetrad has four elements, gives

$$\sum_J \iota_{E_J} (T^I \wedge e^J) = 2T^I, \quad \sum_J \iota_{E_J} (e^I \wedge T^J) = T^I - e^I \wedge t.$$

Hence $B^I = T^I + e^I \wedge t$. A second contraction produces

$$\sum_I \iota_{E_I} B^I = t + 3t = 4t.$$

This recovers $t = b$, and then $T^I = B^I - e^I \wedge b$. In particular, the kernel of $\mathbb{L}_e$ is zero. Equality of dimensions proves surjectivity and makes the formula valid for every form Y in the codomain. $\square$

The connection decomposes uniquely as

$$\omega^{IJ} = \hat{\omega}^{IJ}(e) + K^{IJ}, \quad K^{IJ} = -K^{JI}, \quad T^I = K^I_J \wedge e^J, \quad (26.10)$$

where $\hat{\omega}(e)$ is the Levi-Civita connection and K the contorsion. To make the inverse explicit, write $K_{IJ} = K_{IJK}e^K$ and $T_I = \frac{1}{2}T_{IJK}e^J \wedge e^K$. Then

$$T_{IJK} = K_{IKJ} - K_{IJK}, \quad K_{IJK} = \frac{1}{2}(T_{KIJ} - T_{IJK} - T_{JKI}). \quad (26.11)$$

The second equality follows by cyclically combining the first and using $K_{IJK} = -K_{JIK}$; substitution recovers the components of T . Torsion and contorsion are therefore two equivalent presentations of the algebraic response at fixed co-tetrad.

Corollary 26.4 (Current-free sector and minimal linear response). *If $\sigma = 0$, one obtains $T = 0$ and $\omega = \hat{\omega}(e)$. For a material coupling whose current $\sigma[e, \Psi]$ is independent of K , equations (26.7), (26.9), and (26.11) provide a unique response $K_*(e, \Psi)$, linear in σ .*

Proof. The three reconstruction maps are linear with e , γ , and κc fixed. The zero current produces $Y = 0$, then $T = 0$, and finally $K = 0$. $\square$

In the minimal action, the connection equation contains this pointwise response. Its evolution follows by evolving e and Ψ : $K_* = K_*(e, \Psi)$ changes with them. If a material action retains additional algebraic dependence on K , the Cartan equation keeps its variational form and must be solved with that current; the preceding linear response belongs to the specified minimal domain.

Geometric propagation in the current-free sector. The preceding corollary identifies a precise regime: when the spin current vanishes, the algebraic torsional response disappears and the connection is determined by the co-tetrad. Curvature need not vanish. The identity $F(\hat{\omega} + K) = F(\hat{\omega}) + D_{\hat{\omega}}K + K \wedge K$ reduces to $F(\omega) = F(\hat{\omega})$, retaining propagating geometric degrees of freedom. The expression *clean wave* denotes propagation without a local torsional source, not flat geometry or the erasure of the construction's memory. The classical formulation uses co-tetrad, connection and curvature, without adding an elementary graviton as a primitive constituent. A collective mode obtained by quantizing the geometric sector is a further description of those degrees of freedom.

26.4 Elimination of contorsion and the quadratic contribution

Now consider the minimal coupling affine in K , expressed by

$$S_m[e, \hat{\omega} + K, \Psi] = S_m[e, \hat{\omega}, \Psi] - \frac{1}{2} \int_M K^{IJ} \wedge \sigma_{IJ}[e, \Psi]. \quad (26.12)$$

This condition identifies the material action to which quadratic elimination applies; the current remains the one defined by (26.4).

The curvature expansion is

$$F(\hat{\omega} + K) = F(\hat{\omega}) + D_{\hat{\omega}}K + K \wedge K, \quad (K \wedge K)^{IJ} = K^I_L \wedge K^{LJ}. \quad (26.13)$$

Since $D_{\hat{\omega}}\Sigma = 0$, the linear term is the exterior derivative of $\Sigma_{IJ} \wedge (P_\gamma K)^{IJ}$. The corresponding boundary contribution remains explicit:

$$S_\partial[e, K] = \frac{1}{2\kappa c} \int_{\partial M} \Sigma_{IJ} \wedge (P_\gamma K)^{IJ}. \quad (26.14)$$

In an interior domain, or with the compatible boundary completion, the action density depending on contorsion is

$$\mathcal{Q}_{e,\gamma}(K) - \frac{1}{2} K^{IJ} \wedge \sigma_{IJ}, \quad \mathcal{Q}_{e,\gamma}(K) = \frac{1}{2\kappa c} \Sigma_{IJ} \wedge P_\gamma(K \wedge K)^{IJ}. \quad (26.15)$$

The density $\mathcal{Q}_{e,\gamma}$ is a degree-four differential form, homogeneous of degree two in K .

Theorem 26.5 (Effective response quadratic in the current). *Under (26.12), substituting the algebraic solution K_* produces the effective spin density*

$$\mathcal{L}_{\text{spin}}^{\text{eff}} = -\frac{1}{4} K_*^{IJ} \wedge \sigma_{IJ} = -\mathcal{Q}_{e,\gamma}(K_*). \quad (26.16)$$

With co-tetrad and couplings fixed, it is homogeneous of degree two in the current. The effective action also retains $S_\partial[e, K_]$, or its stated boundary completion.*

Proof. By Corollary 26.4, K_* is linear in σ . The connection equation is precisely stationarity of (26.15). Since it is algebraic, the radial variation $\delta K = K_*$ applies pointwise; it may also be multiplied by a compactly supported function. Euler's identity for a quadratic form gives

$$2\mathcal{Q}_{e,\gamma}(K_*) - \frac{1}{2} K_*^{IJ} \wedge \sigma_{IJ} = 0.$$

Substitution into the action density proves (26.16). If $\sigma \mapsto \lambda\sigma$, then $K_* \mapsto \lambda K_*$, and the effective density is multiplied by λ^2 . Expansion (26.13) separately identifies the boundary term accompanying this elimination. $\square$

The specific current, its Lorentzian contraction, duality, and the conventions of the material action determine the sign and coefficient of this contribution. The quadratic identity provides the composition rule allowing them to be evaluated once these data are specified.

26.5 Effective metric source and covariant balance

The normalization of the metric source is fixed on the same action line. After eliminating K , define

$$\delta_g S_m[e, \hat{\omega}, \Psi] = -\frac{1}{2} \int_M \text{vol}_e \mathcal{T}_{\mu\nu}^{S,\text{LC}} \delta g^{\mu\nu}, \quad (26.17)$$

$$\delta_g S_{\text{spin}}^{\text{eff}} = -\frac{1}{2} \int_M \text{vol}_e \mathcal{U}_{\mu\nu}^{S,\text{spin}} \delta g^{\mu\nu}. \quad (26.18)$$

Here interior variations are considered, with boundary terms treated according to the fixed variational problem. For the affine coupling, the second source is the metric variation of (26.16). The variation includes the dependence of the material current on e and Ψ .

Proposition 26.6 (Metric equation and conservation of the total source). *With the preceding conventions, the metric equations are*

$$G_{\mu\nu}[g] + \Lambda_{\text{int}} g_{\mu\nu} = \kappa c (\mathcal{T}_{\mu\nu}^{S,\text{LC}} + \mathcal{U}_{\mu\nu}^{S,\text{spin}}) = \kappa (T_{\mu\nu}^{\text{phys,LC}} + U_{\mu\nu}^{\text{phys,spin}}), \quad (26.19)$$

where $T^{\text{phys,LC}} = c\mathcal{T}^{S,\text{LC}}$ and $U^{\text{phys,spin}} = c\mathcal{U}^{S,\text{spin}}$ when the co-tetrad takes values on the length line. Every solution satisfies

$$\hat{\nabla}^\mu (T_{\mu\nu}^{\text{phys,LC}} + U_{\mu\nu}^{\text{phys,spin}}) = 0. \quad (26.20)$$

Proof. Eliminating K preserves the equations of the remaining variables: in the chain rule, the term multiplying δK_* vanishes by the connection equation. In the Levi-Civita sector, the first Bianchi identity annihilates the term $\Sigma_{IJ} \wedge F^{IJ}$, and the gravitational part is written $(2\kappa c)^{-1} \int_M (R - 2\Lambda_{\text{int}}) \text{vol}_e$. Its interior variation is

$$\frac{1}{2\kappa c} \int_M \text{vol}_e (G_{\mu\nu} + \Lambda_{\text{int}} g_{\mu\nu}) \delta g^{\mu\nu}.$$

Adding (26.18) gives (26.19). The contracted Bianchi identity for $\hat{\nabla}$, metric compatibility, and constancy of κ and Λ_{int} in the chart produce (26.20). $\square$

Before eliminating the connection, the geometric identities retain the form

$$D_\omega T^I = F^I{}_J \wedge e^J, \quad D_\omega F^{IJ} = 0. \quad (26.21)$$

The first is $D_\omega^2 e = F \wedge e$; the second follows by expanding $D_\omega(d\omega + \omega \wedge \omega)$ and canceling the terms. For a covariant material action, the total metric balance agrees with the Noether identity on the material equations. Geometry and matter thus preserve a joint source; the division between its Levi-Civita contribution and its spin contribution is determined by the specialized action.

27 Torsional elimination and metric response of the minimal extension

The minimal extension provides a field, a response weight, and a current through a single variational operation. Its transported incidence depends on the connection through the route products in (25.7). That dependence is preserved when the current is composed with the Cartan–Holst equation. The connection is therefore eliminated from the complete action of that extension. The affine case of 26.5 is contained as an exact specialization of the following construction.

27.1 Variational composition at a cellular level

Fix a finite level, its routes, the transport representation, and the cellular pairing of 25.4. Let p be real coordinates of the cotetrad and material fields in a declared local trivialization, and let $k \in \mathbb{R}^d$ be the coordinates of contorsion K . The collection $p \mapsto e(p)$ preserves nondegeneracy of the cotetrad. The connection acting on each route is

$$\omega(p, k) = \hat{\omega}(e(p)) + K(k), \quad F(p, k) = S_{\min}[e(p), \omega(p, k), \Psi(p)] = \mathbf{a}(p, k)E(p, k). \quad (27.1)$$

The spaces and bases in this expression are transported to fixed spaces before differentiation. We work on an open set where the incidence $C(p, k)$ is surjective and the collections are C^2 . This additional regularity permits examination of the action Hessian. The couplings $G_{\text{int}}, c_{\text{int}}, \gamma_{\text{HMT}}$ preserve the chart and variation convention of 26.1.

Let $Q(p, k)$ be the integral, or the declared cellular realization, of the quadratic form (26.15). In real bases,

$$Q(p, k) = \frac{1}{2}k^\top A(p)k, \quad A(p)^\top = A(p). \quad (27.2)$$

The operator A comes from the gravitational action and its pairing, whereas the Gramian CC^* comes from the extension problem. Positivity of the latter guarantees existence of the minimizing field; the gravitational form preserves its own signature. Boundary terms are maintained according to (26.14); the following interior formulas apply with admissible variations annihilating their contribution, or with the corresponding boundary completion.

Define the response covector by

$$d_k F(p, k)[h] = -\frac{1}{2}h^\top s(p, k), \quad s(p, k) = M_{\text{cel}}(p, k)\sigma(p, k). \quad (27.3)$$

Here s and σ preserve the distinct types in (25.8): the former contains the coefficients of the differential, and the latter the coordinates of the current in the material pairing. When the basis in k changes, that pairing is transported as well. Differentiation of $s = M_{\text{cel}}\sigma$ includes derivatives of both factors when the pairing depends on the parameter.

Proposition 27.1 (Composed equation of extension and contorsion). *Interior stationarity of the action with respect to contorsion is*

$$\mathcal{R}(p, k) := A(p)k - \frac{1}{2}s(p, k) = 0. \quad (27.4)$$

Each component of s is calculated from the same extension:

$$\begin{aligned} s_j = & 4\mathbf{a}\chi \operatorname{Re}\langle y, (\partial_{k_j} C)f_{\min} - \partial_{k_j} b \rangle \\ & - 2\mathbf{a}(\partial_{k_j} \chi)\langle b, y \rangle - 2(\partial_{k_j} \mathbf{a})E. \end{aligned} \quad (27.5)$$

When the cellular pairing realizes the connection equation of 26.1, this equation is equivalent to composing $\sigma(p, k)$ with the Holst inverse, the inverse of L_e , and reconstruction of contorsion:

$$K = C_e^{\text{tor}} L_e^{-1}(\kappa c P_\gamma^{-1} \sigma[e, \hat{\omega}(e) + K, \Psi]). \quad (27.6)$$

The symbol C_e^{tor} denotes exclusively the map $T \mapsto K$ of (26.11), with fixed cotetrad.

Proof. The differential of $Q + F$ in the arbitrary direction h is

$$d_k(Q + F)[h] = h^\top (Ak - \frac{1}{2}s).$$

Its vanishing in every direction proves (27.4). Theorem 25.4, applied to ∂_{k_j} , gives (27.5). With p fixed, $\delta\omega = \delta K$. The Cartan equation is transformed successively through (26.6), (26.9), and (26.11); each of these operations is invertible on its domain. This proves equivalence with (27.6) when both expressions use the same material and cellular realization. $\square$

The composition preserves the connection on both sides. Routes traversing several cells may couple the current components between them. Pointwise reconstruction of T from a given current and joint solution of (27.4) are, in that case, successive operations. This finite formulation keeps explicit the route representation entering the functional.

27.2 Local existence of a stationary branch

Theorem 27.2 (Local continuation of the torsional solution). *Suppose (p_0, k_0) belongs to the preceding open set, $\mathcal{R}(p_0, k_0) = 0$, and the total Hessian*

$$H_0 = A(p_0) + \partial_k^2 F(p_0, k_0) = A(p_0) - \frac{1}{2}\partial_k s(p_0, k_0) \quad (27.7)$$

is invertible. There exist neighbourhoods of p_0 and k_0 in which a unique C^1 branch, $k = k_(p)$, satisfies (27.4) and passes through k_0 . On that branch,*

$$\partial_{p_i} k_* = -[A + \partial_k^2 F]^{-1} [(\partial_{p_i} A)k_* + \partial_{p_i} \partial_k F]. \quad (27.8)$$

Proof. Gramian inversion is C^2 on the full-rank open set. Thus F is C^2 , and $\mathcal{R}$ is C^1 . Its derivative with respect to k , at (p_0, k_0) , is H_0 . The implicit function theorem provides the branch, its regularity, and uniqueness in a neighbourhood of the point. Existence can also be seen through the map

$$\mathcal{T}_p(k) = k - H_0^{-1} \mathcal{R}(p, k).$$

Its derivative with respect to k vanishes at (p_0, k_0) . In a sufficiently small ball its norm is at most $q < 1$. Shrinking the neighbourhood of p_0 gives $\|\mathcal{T}_p(k_0) - k_0\| < (1 - q)r$, where r is the ball's radius. Thus $\mathcal{T}_p$ maps the closed ball into itself and is contractive. Its fixed point is the local solution. Differentiating $\mathcal{R}(p, k_*(p)) = 0$ and solving for $\partial_{p_i} k_*$ gives (27.8). $\square$

The hypothesis is checked on the composed action: Gramian rank ensures the screen minimum, and the total Hessian controls the torsional branch. The asserted uniqueness is local around the specified solution; other branches may coexist in the same full-rank domain.

27.3 Effective action and non-affine contribution

Theorem 27.3 (Exact elimination identity). *Let $k_*(p)$ be a stationary branch. Suppose also that the segment $t \mapsto (p, tk_*)$, $0 \leq t \leq 1$, remains in the open set where F is defined. The action correction relative to the Levi–Civita realization is*

$$\begin{aligned} \Delta S(p) &:= Q(p, k_*) + F(p, k_*) - F(p, 0) \\ &= \frac{1}{4}k_*^\top s(p, k_*) - \frac{1}{2} \int_0^1 k_*^\top s(p, tk_*) dt. \end{aligned} \quad (27.9)$$

Equivalently,

$$\Delta S = -\frac{1}{4}k_*^\top s(p, k_*) + \frac{1}{2} \int_0^1 k_*^\top [s(p, k_*) - s(p, tk_*)] dt. \quad (27.10)$$

These identities concern the integrated or cellular action. In a realization with local density, they apply to the density with its corresponding pairing of forms. The effective action preserves the declared boundary term separately.

Proof. By the chain rule and (27.3),

$$\frac{d}{dt}F(p, tk_*) = -\frac{1}{2}k_*^\top s(p, tk_*).$$

Integration from 0 to 1 gives the difference of material actions. The stationarity equation, evaluated in the direction k_* , and quadratic homogeneity of Q give

$$2Q(p, k_*) = \frac{1}{2}k_*^\top s(p, k_*).$$

Adding both identities proves (27.9); adding and subtracting $\frac{1}{2}k_*^\top s(p, k_*)$ gives (27.10). $\square$

Corollary 27.4 (Affine specialization). *If $s(p, k) = s(p)$, then*

$$\Delta S = -\frac{1}{4}k_*^\top s(p) = -Q(p, k_*).$$

This is the integrated realization of (26.16).

Proof. The difference inside the integral in (27.10) vanishes. The second equality follows from the radial stationarity identity. $\square$

27.4 Metric differential of the eliminated action

Theorem 27.5 (Response of the geometric variables). *On the branch of Theorem 27.2,*

$$\partial_{p_i}\Delta S = \frac{1}{2}k_*^\top (\partial_{p_i}A)k_* + (\partial_{p_i}F)(p, k_*) - (\partial_{p_i}F)(p, 0), \quad (27.11)$$

where the partial derivatives hold the coordinates k fixed. For fixed Hermitian inner products, each material derivative is calculated by

$$\begin{aligned} \partial_{p_i}F &= (\partial_{p_i}a)E + a(\partial_{p_i}\chi)\langle b, y \rangle \\ &\quad + 2a\chi \operatorname{Re}\langle y, \partial_{p_i}b - (\partial_{p_i}C)f_{\min} \rangle. \end{aligned} \quad (27.12)$$

If p_i varies the cotetrad, $\partial_{p_i}C$ includes variation of $\hat{\omega}(e(p)) + K(k)$, the realized routes, and the trivializations used, according to their declared dependencies.

Proof. The total derivative of $Q(p, k_*) + F(p, k_*)$ contains, in addition to its partial derivatives, the term

$$[\partial_k Q(p, k_*) + \partial_k F(p, k_*)]\partial_{p_i}k_*.$$

The bracket is zero by the connection equation. Subtracting the derivative of $F(p, 0)$ proves (27.11). The product rule and (25.4) prove (27.12). The composition (27.1) requires differentiating the Levi-Civita connection when e varies with k fixed. $\square$

When the positive extension inner product also varies, its contribution is preserved explicitly. In coordinates, let $h(p, k) > 0$ be its Hermitian matrix and consider the minimum of f^*hf subject to $Cf = b$. Then

$$L = Ch^{-1}C^*, \quad y = L^{-1}b, \quad f_{\min} = h^{-1}C^*y, \quad E = \chi b^*y. \quad (27.13)$$

Here y represents the constraint covector in fixed bases. The positive matrix h preserves its energetic role, distinct from that of the Lorentzian metric.

Lemma 27.6 (Variation of the extension inner product). *With the preceding conventions,*

$$\delta E = (\delta\chi)b^*y + 2\chi \operatorname{Re}[y^*(\delta b - \delta C f_{\min})] + \chi f_{\min}^*(\delta h)f_{\min}. \quad (27.14)$$

Proof. The formula $\delta h^{-1} = -h^{-1}(\delta h)h^{-1}$ gives

$$\delta L = (\delta C)h^{-1}C^* + Ch^{-1}(\delta C)^* - Ch^{-1}(\delta h)h^{-1}C^*.$$

When $E = \chi b^* L^{-1} b$ is differentiated, the term $-\chi y^*(\delta L)y$ produces the two terms with δC and the positive term with δh . This proves the complete identity. $\square$

In a covariant material realization whose variation admits the metric pairing of (26.18), the correction determines

$$\delta_g \Delta S = -\frac{1}{2} \int_M \text{vol}_e \mathcal{U}_{\mu\nu}^{S,\min} \delta g^{\mu\nu}, \quad \mathcal{U}_{\mu\nu}^{\text{phys},\min} = c \mathcal{U}_{\mu\nu}^{S,\min}. \quad (27.15)$$

Equations (27.11) and (27.12), supplemented by (27.14) where appropriate, evaluate its coefficients in the declared metric variations. The screen minimum, torsional solution, and metric derivative are three consecutive operations on the same action. The reduced action preserves any couplings between cells present in the original routes.

The homogeneous specialization of 29 additionally uses the realization of the tensor obtained: its spatial symmetry, the sign of the contraction with the observer, and the scaling law. When that evaluation gives $\rho_{\text{corr}}(a) = -s_0 a^{-6}$, the amplitude is read in the reference state as $s_0 = -a_{\text{ref}}^6 \rho_{\text{corr}}(a_{\text{ref}})$. This reading preserves the material state and its normalization; the elimination theorem supplies the action to be contracted.

27.5 Exact non-affine verification model

The following finite model checks the coefficients and non-affine dependence of the identities. Its parameters belong to the algebraic example and keep the problem of material realization of routes separate.

Proposition 27.7 (Exact elimination of a rank-one extension). *Let $H = \mathbb{R}^2$, $Q = \mathbb{R}$, $C(k) = (1 \ k)$, $b = 1$, $\chi = 1$, $\mathfrak{a} = \lambda > 0$, and $Q_{\text{ex}}(k) = k^2/2$. Then*

$$f_{\min} = \frac{(1, k)^T}{1 + k^2}, \quad F(\lambda, k) = \frac{\lambda}{1 + k^2}, \quad s(\lambda, k) = \frac{4\lambda k}{(1 + k^2)^2}, \quad \mathcal{R}(\lambda, k) = k \left(1 - \frac{2\lambda}{(1 + k^2)^2} \right). \quad (27.16)$$

For $\lambda = 25/2$, the stationary solutions are 0, 2, -2. At $k_* = 2$,

$$H_* = \frac{16}{5}, \quad s_* = 4, \quad Q_{\text{ex}}(k_*) = 2, \quad F(\lambda, k_*) = \frac{5}{2}, \quad \Delta S = -8. \quad (27.17)$$

The affine expression $-k_* s_*/4 = -2$ differs from the exact correction. On the positive branch,

$$\partial_\lambda \Delta S = -\frac{4}{5}, \quad \partial_\lambda k_* = \frac{1}{20} \quad \text{at } \lambda = 25/2. \quad (27.18)$$

Proof. The Gramian is $1 + k^2 > 0$, so the minimum has the stated expressions. Differentiating F gives $s = -2\partial_k F$, and then $\mathcal{R} = k - s/2$. For $\lambda = 25/2$, the equation is $k[(1 + k^2)^2 - 25] = 0$; since $1 + k^2 > 0$, its roots are precisely 0, ± 2 . The total Hessian is

$$1 - \frac{2\lambda(1 - 3k^2)}{(1 + k^2)^3},$$

equal to 16/5 at both nonzero roots. The correction is $2 + 5/2 - 25/2 = -8$. In the radial identity,

$$\int_0^1 k_* s(\lambda, tk_*) dt = \int_0^1 \frac{200t}{(1 + 4t^2)^2} dt = 20,$$

since an antiderivative is $-25/(1 + 4t^2)$. Thus $\Delta S = 8/4 - 20/2 = -8$, including the non-affine term. For $\lambda > 1/2$, the positive branch satisfies $k_*^2 = \sqrt{2\lambda} - 1$ and $\Delta S = \sqrt{2\lambda} - 1/2 - \lambda$. Their derivatives give (27.18). The envelope formula gives the same result: $(\partial_\lambda F)(\lambda, k_*) - (\partial_\lambda F)(\lambda, 0) = 1/5 - 1 = -4/5$. $\square$

28 Spinorial current and torsional amplitude of the material state

The preceding development and the spinorial realization below articulate two variational problems with common geometric antecedents. The minimal-extension action $S_{\min} = \alpha_m E$, constructed in Section 25, incorporates the dependence of transported incidence and its minimum on the connection; its torsional elimination preserves that complete dependence in Section 27. The spinorial action S_D in (28.3) couples the same connection to the fields of its representation and is affine in contorsion when the co-tetrad and fields are fixed. Each current is obtained by varying its own action. A minimizing preparation may determine the initial material state of the spinorial realization; substituting connection-dependent fields also retains the terms of the composed differential. This distinction places the following amplitude calculation within its specific action and variational domain.

The spinorial representation of transport and the algebraic Cartan–Holst response allow torsional intensity to be calculated as a functional of the material state. Its evaluation preserves three data: the field representation, the second moment of its current, and the volume with respect to which density is defined. The coupling coefficient follows from the action and its inversion; the initial state determines the material amplitude.

28.1 Spinorial representation and affine material action

We retain the co-tetrad, orientation, and couplings of Section 26. The spinorial realization uses the four-dimensional complex module obtained by complexifying the real matrices

$$J = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}, \quad R = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad Z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}, \quad (28.1)$$

$$g^0 = J \otimes I_2, \quad g^1 = R \otimes I_2, \quad g^2 = Z \otimes R, \quad g^3 = Z \otimes Z.$$

Multiplication gives $\{g^I, g^J\} = 2\eta^{IJ}I_4$, with $\eta = \text{diag}(-1, 1, 1, 1)$. The matrices are distinguished from the scalar parameter γ of the Holst operator. Define

$$g_I = \eta_{IJ}g^J, \quad \mathbb{B}_{IJ} = \frac{1}{4}[g_I, g_J], \quad A_D = -ig^0, \quad \bar{\psi} = \psi^\dagger A_D, \quad \Omega = \frac{1}{2}\omega^{IJ}\mathbb{B}_{IJ}. \quad (28.2)$$

The Clifford product satisfies

$$[\mathbb{B}_{IJ}, g^K] = \delta_J^K g_I - \delta_I^K g_J.$$

Thus the same metric connection acts on the fields through $D\psi = d\psi + \Omega\psi$ and $D\bar{\psi} = d\bar{\psi} - \bar{\psi}\Omega$. Transport along a route is obtained by integrating this connection in its spinorial representation; it preserves the connection and route of the preceding geometric realization.

Let $\eta_I = \iota_{E_I} \text{vol}_e$, so that $e^J \wedge \eta_I = \delta_I^J \text{vol}_e$. For a field of dimensional type $[\psi] = L^{-3/2}$, take the minimal material action

$$S_D = \int_M \left\{ \frac{\hbar}{2} [\bar{\psi} g^I D\psi - (D\bar{\psi}) g^I \psi] \wedge \eta_I - mc \bar{\psi} \psi \text{vol}_e \right\}. \quad (28.3)$$

Here $\hbar = \hbar_{\text{int}}$, $c = c_{\text{int}}$, and the mass m is the reading of the material species in the chosen chart. These values are outputs preceding this realization. The matrix A_D is Hermitian and $A_D g^I$ is anti-Hermitian. The symmetric kinetic term therefore has the real convention corresponding to (28.1). In this signature, the torsion-free equation has principal term $\hbar g^I D_I$; the equivalent notation $i\hbar \Gamma^I D_I$, with $\Gamma^I = -ig^I$, uses the Clifford relation of opposite signature. This precision fixes the adjoint, signature, and imaginary factor of the action jointly.

The material fields and co-tetrad remain fixed during connection variation. A preparation from the minimal extension in Section 25 may determine the initial field or state datum. Each action retains its own variation: if a connection-dependent minimizing field is substituted into an action, its composed differential also applies, or the envelope theorem under its stationarity hypotheses. Action (28.3) realizes here a coupling affine in contorsion.

Proposition 28.1 (Spinorial current with variational normalization). *Let $B^{IJK} = \bar{\psi} g^{[I} g^J g^{K]} \psi$, where brackets denote normalized antisymmetrization. The current defined in (26.4) by action (28.3) is*

$$\sigma_{JK} = -\frac{\hbar}{2} \bar{\psi} \{g^I, \mathbb{B}_{JK}\} \psi \eta_I = -\frac{\hbar}{2} B^I_{JK} \eta_I. \quad (28.4)$$

It is independent of contorsion when e and ψ are fixed. The inversion in Section 26.3 therefore applies to this current.

Proof. The variation of Ω is $\delta\Omega = \frac{1}{2} \delta\omega^{JK} \mathbb{B}_{JK}$. The two kinetic summands give, respectively, the product with g^I on the left and on the right. Consequently,

$$\delta_\omega S_D = \frac{\hbar}{4} \int_M \delta\omega^{JK} \wedge \eta_I \bar{\psi} \{g^I, \mathbb{B}_{JK}\} \psi.$$

Comparison with (26.4) fixes the factor and sign in (28.4). Expanding the anticommutator and using the Clifford relation cancels the degree-one contractions, leaving $\{g^I, \mathbb{B}_{JK}\} = g^I g_J g_K$. The action is linear in Ω , which proves the final statement. $\square$

28.2 Effective contraction and second-moment coefficient

The Lorentzian quadratic form of the trivector is written

$$q(B) = \frac{1}{3!} B^{IJK} B_{IJK} = -(B^{012})^2 - (B^{013})^2 - (B^{023})^2 + (B^{123})^2. \quad (28.5)$$

This contraction preserves the signature of the spinorial module. Its second moment depends on the state and on the realization of field products.

Theorem 28.2 (Effective interaction of the spinorial current). *With material action (28.3), algebraic elimination of contorsion gives*

$$\mathcal{L}_{\text{spin}}^{\text{eff}} = C_\gamma q(B) \text{vol}_e, \quad C_\gamma = \frac{3\kappa\hbar^2}{16} \frac{\gamma^2}{1+\gamma^2}. \quad (28.6)$$

The coefficient is obtained from the current and the Holst inverse with the normalizations of the preceding action.

Proof. By (26.16), it suffices to evaluate $-\frac{1}{4} K_*^{IJ} \wedge \sigma_{IJ}$, with

$$Y^{IJ} = \kappa c \frac{\gamma^2}{1+\gamma^2} (-\star + \gamma^{-1} I) \sigma^{IJ}, \quad K_* = C_e(Y).$$

Here C_e is the explicit composition of (26.9) with (26.11). Linearity of these two inverses reduces the calculation to the $-\star$ and I parts. To give the complete contraction, order the trivector components as (012, 013, 023, 123), factor out $\kappa\hbar^2\gamma^2/(1+\gamma^2)$, and substitute $\sigma_{JK} = -\frac{1}{2} B^I_{JK} \eta_I$. The two matrices of the remaining quadratic form are

$$B_{-\star} = \frac{3}{16} \text{diag}(-1, -1, -1, 1), \quad B_I = 0_{4 \times 4}. \quad (28.7)$$

These matrices are obtained using only $e^I \wedge \eta_J = \delta_J^I \text{vol}_e$: for each component B^{abc} , form σ , compute Y , then $T^I = \sum_J \iota_{E_J} Y^{IJ} - \frac{1}{4} e^I \wedge \sum_{L,J} \iota_{E_L} \iota_{E_J} Y^{LJ}$, and finally $K_{IJK} = (T_{KIJ} - T_{IJK} - T_{JKI})/2$. Contraction gives the four displayed diagonal entries; the six polarizations between distinct components are zero for both operators. This calculates every coefficient of the two quadratic forms. The $\gamma^{-1} I$ part vanishes, and the $-\star$ part produces exactly (28.5), with the coefficient in (28.6). $\square$

28.3 Finite fermionic realization and the second moment of the state

Fix a periodic comoving cell, an orthonormal frame adapted to its observer, and the homogeneous spatial mode of the four spinorial components. This finite realization has state space $\mathcal{F} = \bigoplus_{n=0}^4 \Lambda^n \mathbb{C}^4$. Let $a_r, a_r^\dagger, r = 1, \dots, 4$, be its canonical operators, with $\{a_r, a_s^\dagger\} = \delta_{rs}$ and $\{a_r, a_s\} = 0$. The occupation vacuum fixes normal ordering. For a matrix M , write

$$d\Gamma(M) = \sum_{r,s} a_r^\dagger M_{rs} a_s, \quad \hat{N} = d\Gamma(I_4).$$

The letter Γ in this notation denotes second quantization; nonadic holonomy retains its own notation.

The four matrices of the bilinears are calculated directly:

$$\begin{array}{c|cccc} abc & 012 & 013 & 023 & 123 \\ \hline M_{abc} = A_D g^a g^b g^c & iJ \otimes R & iJ \otimes Z & iI_2 \otimes J & iZ \otimes J \end{array} \quad (28.8)$$

All are Hermitian, have zero trace, and satisfy $M_{abc}^2 = I_4$. The quartic observable corresponding to (28.5), with this stated normal ordering, is

$$\begin{aligned} \hat{Q}_{\text{sp}} &= \sum_{a < b < c} \eta_{aa} \eta_{bb} \eta_{cc} : d\Gamma(M_{abc})^2 : \\ &= \sum_{a < b < c} \eta_{aa} \eta_{bb} \eta_{cc} d\Gamma(M_{abc})^2 + 2\hat{N}. \end{aligned} \quad (28.9)$$

Normal ordering is part of the specified finite realization; its definition makes explicit the subtraction of single-occupation contractions.

Proposition 28.3 (Evaluation by the canonical relations). *The operator $\hat{Q}_{\text{sp}}$ is self-adjoint, preserves each occupation sector, and has restrictions*

$$\hat{Q}_{\text{sp}}|_{\Lambda^0 \oplus \Lambda^1} = 0, \quad \hat{Q}_{\text{sp}}|_{\Lambda^3} = 4I_4, \quad \hat{Q}_{\text{sp}}|_{\Lambda^4} = 8I_1. \quad (28.10)$$

In the basis of $\Lambda^2 \mathbb{C}^4$ whose occupation subsets are (12, 13, 23, 14, 24, 34), its matrix is

$$\begin{pmatrix} 0 & 0 & 0 & 0 & 0 & -4 \\ 0 & 2 & 0 & 0 & 6 & 0 \\ 0 & 0 & 2 & -6 & 0 & 0 \\ 0 & 0 & -6 & 2 & 0 & 0 \\ 0 & 6 & 0 & 0 & 2 & 0 \\ -4 & 0 & 0 & 0 & 0 & 0 \end{pmatrix}. \quad (28.11)$$

Proof. The relation $a_s a_t^\dagger = \delta_{st} - a_t^\dagger a_s$ applied to $d\Gamma(M)^2$ separates its single-occupation contraction and gives

$$: d\Gamma(M)^2 := d\Gamma(M)^2 - d\Gamma(M^2).$$

Since the first three signs in (28.8) are negative and the last positive, the contractions sum to $-2\hat{N}$. This proves (28.9). Self-adjointness and preservation of occupation follow from the same expressions.

On Λ^0 , all summands vanish. On Λ^1 , $d\Gamma(M) = M$, and the squares sum to $-2I_4$, which cancels with $2\hat{N}$. On Λ^4 , $d\Gamma(M) = \text{tr } M = 0$, leaving eight. The identification by complements between $\Lambda^3 \mathbb{C}^4$ and the dual of $\mathbb{C}^4$ takes $d\Gamma(M)$ to $(\text{tr } M)I - M^\top = -M^\top$. Their squares again sum to $-2I_4$, whereas $2\hat{N} = 6I_4$; the restriction is therefore $4I_4$. Finally, applying the four explicit matrices to the six exterior products of the indicated basis produces (28.11). The operator is thus evaluated on all sixteen dimensions of $\mathcal{F}$. $\square$

For a state operator $\varrho \geq 0$, $\text{Tr } \varrho = 1$, define $q_\varrho = \text{Tr}(\varrho \hat{Q}_{\text{sp}})$. Its dependence on second moments is explicit:

$$q_\varrho = \sum_{a < b < c} \eta_{aa} \eta_{bb} \eta_{cc} \{ \langle \hat{B}_{abc} \rangle_\varrho^2 + \text{Var}_\varrho(\hat{B}_{abc}) - \langle \hat{N} \rangle_\varrho \}, \quad \hat{B}_{abc} = d\Gamma(M_{abc}). \quad (28.12)$$

The covariance and normal-ordering subtraction are preserved even when the mean currents vanish. In particular, if P_3 is the projection onto $\Lambda^3 \mathbb{C}^4$,

$$\varrho_3 = \frac{1}{4}P_3, \quad \langle \hat{B}_{abc} \rangle_{\varrho_3} = 0, \quad q_{\varrho_3} = 4 > 0. \quad (28.13)$$

This is a positive trace-one state and provides an explicit witness for the positive domain of the functional. The two-occupation sector also contains negative values; for example, the vector $(|12\rangle + |34\rangle)/\sqrt{2}$ has value -4 . The sign of the intensity thus originates in the contraction and the state, and is kept separate from the selection of cosmological conditions.

28.4 Density, pressure, and preservation of comoving amplitude

In the spatially homogeneous realization, fix

$$e^0 = N(t)c \, dt, \quad e^j = a(t) \, dx^j, \quad \mathcal{V}(t) = a(t)^3 V_c,$$

with lapse $N(t) > 0$, dimensionless scale factor $a(t) > 0$, and comoving volume $V_c > 0$. The frame and periodic mode of Subsection 28.3 remain specified. The normalization of the homogeneous field is $\hat{\psi} = \mathcal{V}^{-1/2}(a_1, a_2, a_3, a_4)^\top$. Each bilinear carries a factor $\mathcal{V}^{-1}$, and its normally ordered product carries $\mathcal{V}^{-2}$. The symmetric temporal kinetic form preserves this canonical normalization: the terms arising from differentiating $\mathcal{V}^{-1/2}$ cancel between its two summands.

Theorem 28.4 (Torsional amplitude as a functional of the state). *Holding the canonical state and the product realization fixed during metric variations, the effective spin term has reduced action*

$$S_{\text{spin}}^{\text{hom}} = \int dt \, N(t) \frac{cC_\gamma}{a(t)^3 V_c} q_{\varrho}. \quad (28.14)$$

Its energy density and isotropic pressure are

$$\rho_{\text{spin}} = p_{\text{spin}} = -\frac{cC_\gamma}{V_c^2} q_{\varrho} a^{-6}. \quad (28.15)$$

In any evolution preserving q_{ϱ} , the corresponding constant amplitude is

$$s_0[\varrho, V_c] = \frac{3\kappa c^2 \hbar^2}{16} \frac{\gamma^2}{1 + \gamma^2} \frac{\text{Tr}(\varrho \hat{Q}_{\text{sp}})}{V_c^2}. \quad (28.16)$$

Proof. Spatial integration of (28.6) multiplies the factor $\mathcal{V}^{-2}$ by $Nc\mathcal{V} \, dt$, giving (28.14). In canonical variables, lapse variation defines energy through $\delta_N S_m = -\int E \, \delta N \, dt$. Therefore,

$$E_{\text{spin}} = -\frac{cC_\gamma q_{\varrho}}{a^3 V_c}, \quad \rho_{\text{spin}} = \frac{E_{\text{spin}}}{a^3 V_c}.$$

The isotropic spatial variation satisfies $\delta_a S_m = \int 3Na^2 V_c p \, \delta a \, dt$. Applied to (28.14), it gives

$$3Na^2 V_c p_{\text{spin}} = -\frac{3NcC_\gamma q_{\varrho}}{a^4 V_c}.$$

This proves simultaneously the sign, pressure, and density in (28.15). When q_{ϱ} is preserved, the coefficient of $-a^{-6}$ is constant, yielding (28.16). The dimensions are consistent: $[cC_\gamma] = \text{energy } L^3$, so s_0 has the dimension of energy density. $\square$

Corollary 28.5 (Preserved domain of positive amplitude). *Let a unitary evolution of $\mathcal{F}$ preserve $\Lambda^3\mathbb{C}^4$. For every trace-one initial state supported on that sector, one has $q_{\varrho(t)} = 4$, and*

$$s_0 = \frac{3\kappa c^2 \hbar^2}{4V_c^2} \frac{\gamma^2}{1 + \gamma^2} > 0. \quad (28.17)$$

Proof. Sector invariance preserves the support of the state; on that sector, (28.10) gives $\hat{Q}_{\text{sp}} = 4I$. Trace preservation then fixes its expectation at four, independently of the internal evolution of its coherences. Positivity of the couplings, $V_c > 0$, and $\gamma \neq 0$ proves the formula and its sign. $\square$

For general states, preservation of $\langle \hat{N} \rangle$ determines dilution of the occupation density, $n(t) = \langle \hat{N} \rangle / (a^3 V_c)$. Preservation of the quartic moment is checked separately. If $i\hbar \dot{\varrho} = [H(t), \varrho]$, then

$$\frac{dq_{\varrho}}{dt} = \frac{i}{\hbar} \text{Tr}(\varrho[H(t), \hat{Q}_{\text{sp}}]). \quad (28.18)$$

Commutation, invariance of a sector on which the observable is scalar, or a conservation law proved for the specific state provides the relevant conditions. If the moment evolves, (28.15) retains its instantaneous evaluation; its balance includes

$$\dot{\rho}_{\text{spin}} + 6H\rho_{\text{spin}} = -\frac{cC_{\gamma}}{V_c^2} a^{-6} \dot{q}_{\varrho}, \quad H = \dot{a}/a$$

in proper time. This term records exchange with the remaining material degrees of freedom.

The composition is determined: spinorial representation of transport, material action, variational current, Cartan–Holst inversion, second moment, and normalized density. The domain $s_0 > 0$ in Section 29 thus has an evaluable functional and explicit witness states. Its use in cosmological dynamics also preserves the material, geometric, and conservation conditions stated in that section.

29 Memory dilution and persistence of the bounce

Memory evolution and spatial expansion are two readings of one and the same sequence of states. The refinement index can be eliminated between them; that elimination determines the dilution exponent transmitted by the cosmological realization to its torsional contribution. The initial amplitude depends on evaluation of the concrete material current in its variational response. The scaling law and bounce theorem below preserve that dependency.

Theorem 28.4 determines that amplitude as a functional of the material state and comoving volume. Corollary 28.5 provides an explicit domain of states with positive conserved amplitude. In that domain, evaluation of s_0 precedes the cosmological integration developed below. Ordinary density $\rho_0 > 0$ and its barotropic law are additionally specified in Section 29.2: the amplitude's positivity witness does not by itself evaluate that material density.

29.1 The a^{-6} dilution law

Proposition 29.1 (Elimination of the memory index). *Let $a_{\text{ref}} > 0$, $\varphi > 1$, and let the route readers be*

$$\frac{a_n}{a_{\text{ref}}} = \varphi^{n/3}, \quad M_n = \varphi^{-2n}.$$

Then

$$M_n = \left(\frac{a_n}{a_{\text{ref}}} \right)^{-6}$$

for every index of the sequence. If the realization of the quadratic density is $s_n^2 = s_^2 M_n$, with $s_*^2 > 0$, one obtains*

$$s_n^2 = s_*^2 \left(\frac{a_n}{a_{\text{ref}}} \right)^{-6}.$$

Proof. Raising the first equality to the power -6 gives φ^{-2n} , exactly the second reader. Multiplication by the initial amplitude reproduces the second assertion. The exponent is determined by the ratio between the exponents of both readers; the amplitude is the evaluation of the reference state. $\square$

This constitutive law is preserved in the continuous homogeneous realization: torsional density is written $s_0 a^{-6}$, where normalization of the factor a , the material coefficient, and the reference amplitude have been absorbed into s_0 . The scaling law and evaluation of s_0 are successive operations of the construction.

29.2 Homogeneous threshold and transversality

Consider the homogeneous, spatially flat chart in units $c = 1$, with constant $G > 0$, positive densities ρ_0, s_0 , and constant barotropic parameter $w < 1$. The specified realization satisfies $\Lambda_{\text{eff}} = 0$ and the equations

$$\rho_m(a) = \rho_0 a^{-3(1+w)}, \quad \rho_s(a) = s_0 a^{-6}, \quad (29.1)$$

$$H^2 = \frac{8\pi G}{3}(\rho_m - \rho_s) =: F_0(a), \quad \dot{H} = -4\pi G((1+w)\rho_m - 2\rho_s). \quad (29.2)$$

The couplings belong to the dimensional section already constructed. The expression for ρ_s incorporates the sign of its contribution in this gravitational realization.

Theorem 29.2 (Existence and uniqueness of the transverse threshold). *There exists a unique $a_b > 0$ with $F_0(a_b) = 0$:*

$$a_b^{3(1-w)} = \frac{s_0}{\rho_0}.$$

The region allowed by $H^2 \geq 0$ is $a \geq a_b$. At the threshold, with $\rho_b = \rho_m(a_b) = \rho_s(a_b) > 0$,

$$\dot{H}_b = 4\pi G(1-w)\rho_b > 0.$$

The solution crossing $H = 0$ has a strict minimum of a , with local expansion

$$a(t) = a_b \left[1 + 2\pi G(1-w)\rho_b(t-t_b)^2 + O((t-t_b)^3) \right].$$

Proof. The factorization

$$F_0(a) = \frac{8\pi G}{3} a^{-6} (\rho_0 a^{3(1-w)} - s_0)$$

reduces its zeros to that of a strictly increasing function from $(0, \infty)$ onto $(-s_0, \infty)$. This proves existence, uniqueness, and the sign of F_0 on either side of the threshold. The second equation in (29.2), evaluated at equality of densities, gives the formula for $\dot{H}_b$. Since $\dot{a} = aH$, at the threshold $\ddot{a}_b = a_b \dot{H}_b > 0$. The equations are smooth for $a > 0$, so their Taylor expansion gives the stated expansion. The sign change of H is transverse: it passes from contraction to expansion. $\square$

For $w = 1$, both terms have the same exponent, and equality of amplitudes determines a degenerate case. For $w > 1$, the sign of $\dot{H}$ at a coincidence of densities is negative. The theorem's domain $w < 1$ expresses exactly the order between the two dilution exponents.

29.3 Local persistence under controlled perturbations

Persistence of the threshold uses two independent estimates: control of the sign at the endpoints and of the derivative on the interval. Let $I = [a_-, a_+] \subset (0, \infty)$, with $a_- < a_b < a_+$, be chosen so that

$$F_0(a_-) < 0 < F_0(a_+), \quad m_I := \inf_{a \in I} F'_0(a) > 0.$$

Such an interval exists because

$$F'_0(a_b) = \frac{8\pi G}{a_b} (1-w)\rho_b > 0.$$

Define $\delta_I = \min\{-F_0(a_-), F_0(a_+)\} > 0$.

Theorem 29.3 (Local persistence of the transverse bounce). *Let $R \in C^1(I)$ be a perturbation satisfying*

$$\|R\|_{\infty, I} < \delta_I, \quad \|R'\|_{\infty, I} < m_I.$$

Then $F_R = F_0 + R$ has a unique zero $a_R \in (a_-, a_+)$, and $F'_R(a_R) > 0$. The realization $H^2 = F_R(a)$ admits a local transverse bounce, with

$$\dot{H}_R = \frac{a_R}{2} F'_R(a_R) > 0$$

at the threshold. If $R(a, \eta)$ is jointly C^1 , the persistent zeros depend locally C^1 -smoothly on the parameter η .

Proof. The first bound preserves $F_R(a_-) < 0 < F_R(a_+)$; the intermediate value theorem provides an interior zero. The second gives $F'_R(a) \geq m_I - \|R'\|_{\infty, I} > 0$ on I , so the zero is unique and simple.

For $a > a_R$ near the zero, $F_R(a) = F'_R(a_R)(a - a_R) + o(a - a_R)$. The integral

$$t - t_b = \int_{a_R}^a \frac{d\xi}{\xi \sqrt{F_R(\xi)}}$$

is finite and strictly increasing. Its inverse defines the expanding branch; time reflection defines the contracting branch. Both join with $\dot{a}(t_b) = 0$ and

$$a(t) - a_R = \frac{a_R^2 F'_R(a_R)}{4} (t - t_b)^2 + o((t - t_b)^2).$$

Away from the threshold, differentiating $H^2 = F_R(a)$ gives $\dot{H} = aF'_R(a)/2$; its continuous limit gives the positive expression at a_R . The C^1 regularity of F_R guarantees continuity of acceleration at this join. Finally, $\partial_a F_R(a_R) > 0$ permits application of the implicit function theorem when the perturbation depends jointly on (a, η) . $\square$

The first bound controls existence; the second ensures local uniqueness and transversality. The proved stability concerns that threshold on the selected interval. The torsional amplitude enters its location, whereas the relation between exponents determines the temporal orientation of the crossing.

30 Explicit homogeneous solution and causal completeness

Determination of the amplitude permits development of a complete realization of the bounce, beyond the local argument. The spatially flat chart of (29.2) is preserved, in units $c = 1$, with $\rho_0, s_0, G > 0$, and the material sector is specialized to dust, $w = 0$, with $\Lambda_{\text{eff}} = 0$. The coefficient s_0 is that obtained from the material state in the corresponding realization; the following integration uses that coefficient and preserves its domain.

Theorem 30.1 (Exact integration of the dust bounce). *Set*

$$a_b^3 = \frac{s_0}{\rho_0}, \quad \tau^2 = \frac{s_0}{6\pi G \rho_0^2}, \quad u = t - t_b.$$

The solution of the homogeneous system with $a(t_b) = a_b$ and $H(t_b) = 0$ is, for every $t \in \mathbb{R}$,

$$a(t) = a_b \left(1 + \frac{u^2}{\tau^2}\right)^{1/3}, \quad H(t) = \frac{2u}{3(\tau^2 + u^2)}, \quad \dot{H}(t) = \frac{2(\tau^2 - u^2)}{3(\tau^2 + u^2)^2}. \quad (30.1)$$

It has a unique minimum, at t_b ; it contracts before and expands afterward. Evolution is smooth and satisfies $a(t) \geq a_b > 0$.

Proof. Let $y = a^3$. The Friedmann equation gives

$$\dot{y}^2 = 24\pi G(\rho_0 y - s_0).$$

The function $y = s_0/\rho_0 + 6\pi G \rho_0 u^2$ satisfies that identity and the initial conditions. Its derivatives give the three expressions in (30.1). To check the evolution equation as well, write

$$\rho_b = \frac{\rho_0^2}{s_0}, \quad x = \frac{u}{\tau}.$$

Then

$$\rho_m = \frac{\rho_b}{1 + x^2}, \quad \rho_s = \frac{\rho_b}{(1 + x^2)^2}, \quad 4\pi G \rho_b = \frac{2}{3\tau^2}.$$

Substitution into $\dot{H} = -4\pi G(\rho_m - 2\rho_s)$ gives exactly the third formula. The complete system $(\dot{a}, \dot{H})$, whose vector field is smooth for $a > 0$, is therefore satisfied. Its local uniqueness, prolonged along the explicit solution, fixes evolution by those initial conditions. The denominator is positive for every t , and the assertions about the minimum and sign follow from H . $\square$

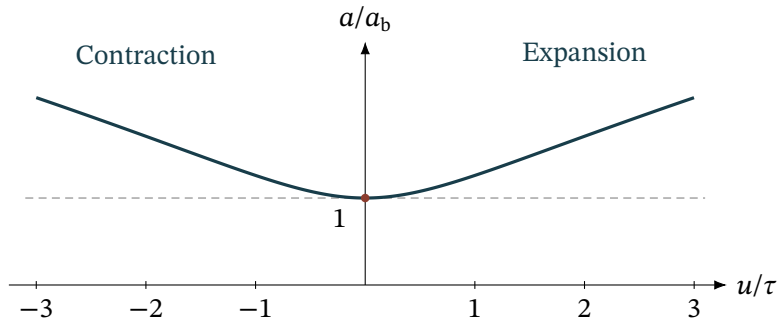


Figure 9: Dimensionless profile of the proved solution: $a/a_b = (1 + (u/\tau)^2)^{1/3}$. The parameters a_b and τ are calculated from ρ_0, s_0, G ; the diagram represents the entire rescaled collection.

30.1 Finite curvature and geodesic extension

The metric realized in this chart is $g = -dt^2 + a(t)^2 dx^2$ on $\mathbb{R} \times \mathbb{R}^3$. With the convention in which the flat curvature scalar is $R = 6(\dot{H} + 2H^2)$, the solution gives

$$R(t) = \frac{4\tau^2 + \frac{4}{3}u^2}{(\tau^2 + u^2)^2}.$$

In particular, $R(t_b) = 4/\tau^2$ is finite. In an orthonormal frame, the Riemann components depend on H^2 and $\dot{H} + H^2$, which are bounded functions on $\mathbb{R}$. The same holds for polynomial contractions of those components; for example,

$$R_{\mu\nu\rho\sigma}R^{\mu\nu\rho\sigma} = 12[(\dot{H} + H^2)^2 + H^4].$$

Proposition 30.2 (Completeness of causal geodesics). *The preceding metric realization is geodesically complete for timelike and null geodesics of its Levi-Civita connection, in both directions.*

Proof. Spatial translations provide the constants $p_i = a^2 dx^i/d\lambda$. Causal normalization gives

$$\left| \frac{dt}{d\lambda} \right| = \sqrt{\epsilon + \frac{|\mathbf{p}|^2}{a(t)^2}}, \quad \epsilon = 1 \text{ in the timelike case, } \epsilon = 0 \text{ in the null case.}$$

A nonconstant null geodesic has $|\mathbf{p}| > 0$. Since $a \geq a_b$, in the timelike case

$$\left| \frac{d\lambda}{dt} \right| \geq \left(1 + \frac{|\mathbf{p}|^2}{a_b^2} \right)^{-1/2} > 0,$$

and in the null case $|d\lambda/dt| = a/|\mathbf{p}| \geq a_b/|\mathbf{p}| > 0$. Thus reaching $t = \pm\infty$ requires infinite affine length. On a compact time interval, a and its derivatives are smooth, and a is separated from zero. The preceding explicit equations and $dx^i/d\lambda = p_i/a^2$ allow continuation of the geodesic through its finite endpoints. Both possibilities for prolongation are covered. $\square$

The completeness result belongs to this homogeneous dust realization, with its conditions on space, matter, and amplitude. Its local persistence under threshold perturbations preserves the bounds of Theorem 29.3; that local property and global completeness of the explicit solution are distinct conclusions. The minimum scale comes from the relation between material density and torsional response. Memory thus enters a verifiable geometric evolution, with a regular realization of contraction and expansion.

31 Residual tensor, trace decomposition, and covariant balance

The geometric realization of the enriched state provides a metric, while the material transducer expresses the source to which the reading refers. The residual tensor preserves the difference between both representations in a single tensor space. Its decomposition distinguishes the sector proportional to the metric from the trace-free sector and determines the exchange current corresponding to that division.

31.1 Chart data and construction of the residual tensor

Fix an HMT state x and a connected chart $\mathcal{U}$ of its geometric realization. On $\mathcal{U}$, the nondegenerate co-tetrad of Section 22 determines a smooth metric $g = g_x$ of signature $(-+++)$. Denote its Levi-Civita connection by $\overset{\circ}{\nabla}$ and write

$$G_{\mu\nu}[g] := \text{Ric}_{\mu\nu}[g] - \frac{1}{2}R[g]g_{\mu\nu}.$$

This distinguishes the Einstein tensor $G[g]$ from the gravitational coupling G_{int} . The derivative used in the balances of this section is $\overset{\circ}{\nabla}$, consistently with the effective metric equation of Section 26.5.

We work in the dimensional chart $c_{\text{int}} = 1$. Assume $G_{\text{int}}(x) > 0$ and $\Lambda_{\text{ref}}(x)$ constant on $\mathcal{U}$, and set $\kappa_x = 8\pi G_{\text{int}}(x)$. The symbol x identifies the state supplying these readers; covariant differentiation acts on the points of $\mathcal{U}$. The genealogy of the couplings remains in Section 16.

Let T_b be the smooth symmetric tensor expressed by the material transducer of the chart; in the baryonic reading it is its baryonic source. Define

$$\mathcal{E}_{\mu\nu}^{\text{ref}} := \kappa_x^{-1}(G_{\mu\nu}[g] + \Lambda_{\text{ref}}g_{\mu\nu}), \quad T_{\mu\nu}^{\text{res}} := \mathcal{E}_{\mu\nu}^{\text{ref}} - T_{b,\mu\nu}. \quad (31.1)$$

The tensor T^{res} is the effective residual tensor, also denoted T_{osc} when interpreting the geometric sector remaining to be expressed relative to T_b . The construction retains g_x , G_{int} , Λ_{ref} , and T_b as effective arguments.

Theorem 31.1 (Uniqueness of the residual and balance with the material source). *For the preceding data there exists a unique symmetric tensor T^{res} satisfying $\mathcal{E}^{\text{ref}} = T_b + T^{\text{res}}$. Its divergence is*

$$\overset{\circ}{\nabla}^{\mu} T_{\mu\nu}^{\text{res}} = -\overset{\circ}{\nabla}^{\mu} T_{b,\mu\nu}. \quad (31.2)$$

In particular, if the material source is separately conserved, $\overset{\circ}{\nabla}^{\mu} T_{b,\mu\nu} = 0$, then T^{res} is also conserved.

Proof. The difference of two tensors satisfying the stated equation is zero; expression (31.1) provides existence. The contracted Bianchi identity and Levi-Civita metric compatibility give

$$\overset{\circ}{\nabla}^{\mu} G_{\mu\nu} = 0, \quad \overset{\circ}{\nabla} g = 0.$$

Since κ_x and Λ_{ref} are constant on the chart, $\overset{\circ}{\nabla}^{\mu} \mathcal{E}_{\mu\nu}^{\text{ref}} = 0$. Differentiating the difference defining T^{res} proves (31.2), including the assertion under separate material conservation. $\square$

Spatial and temporal constancy of the couplings in the chart fixes the domain of this balance. A collection of states may express different couplings in different charts; each preceding identity applies to the chart and state specifying it.

31.2 Canonical decomposition and exchange current

Define the trace function and the two tensors

$$\theta := g^{\mu\nu} T_{\mu\nu}^{\text{res}}, \quad T_{\mu\nu}^{\Lambda} := \frac{1}{4}\theta g_{\mu\nu}, \quad T_{\mu\nu}^0 := T_{\mu\nu}^{\text{res}} - T_{\mu\nu}^{\Lambda}. \quad (31.3)$$

The superscript Λ identifies the component proportional to the metric; Λ_{ref} is the reference parameter in (31.1). Their roles are distinct.

Proposition 31.2 (Trace projectors and uniqueness of the decomposition). *On symmetric covariant tensors of order two, the maps*

$$P_{\text{tr}}(S) = \frac{1}{4} \text{tr}_g(S)g, \quad P_0(S) = S - P_{\text{tr}}(S)$$

satisfy

$$P_{\text{tr}}^2 = P_{\text{tr}}, \quad P_0^2 = P_0, \quad P_{\text{tr}}P_0 = P_0P_{\text{tr}} = 0, \quad P_{\text{tr}} + P_0 = I. \quad (31.4)$$

In particular, $T^{\text{res}} = T^\Lambda + T^0$, $\text{tr}_g T^0 = 0$, and this decomposition is unique.

Proof. In dimension four, $g^{\mu\nu}g_{\mu\nu} = 4$. Therefore, $\text{tr}_g(P_{\text{tr}}S) = \text{tr}_g S$, which proves idempotence of P_{tr} , vanishing of the trace of P_0S , and, by expansion, the other identities. If $S = fg + Q$ with $\text{tr}_g Q = 0$, taking the trace gives $f = \text{tr}_g S/4$. This also determines $Q = S - fg$, hence uniqueness. $\square$

Theorem 31.3 (Covariant exchange between the two sectors). *In the domain of (31.1), let $b_\nu := \mathring{\nabla}^\mu T_{b,\mu\nu}$. The trace exchange current is the one-form*

$$\mathcal{J}_\nu^{\text{tr}} := \frac{1}{4} \partial_\nu \theta = \mathring{\nabla}^\mu T_{\mu\nu}^\Lambda. \quad (31.5)$$

The complete balances are

$$\mathring{\nabla}^\mu T_{\mu\nu}^0 = -b_\nu - \mathcal{J}_\nu^{\text{tr}}, \quad \mathring{\nabla}^\mu (T^\Lambda + T^0)_{\mu\nu} = -b_\nu. \quad (31.6)$$

If $b = 0$, the two sectors exchange exactly opposite currents. In that case they are separately conserved if and only if θ is constant on the connected chart.

Proof. Metric compatibility allows the calculation

$$\mathring{\nabla}^\mu \left(\frac{1}{4} \theta g_{\mu\nu} \right) = \frac{1}{4} g^{\mu\alpha} (\partial_\alpha \theta) g_{\mu\nu} = \frac{1}{4} \partial_\nu \theta.$$

Subtracting this equality from (31.2) gives (31.6). With $b = 0$, separate conservation is equivalent to $d\theta = 0$; a smooth function with zero differential is constant on each connected component. $\square$

The one-form $\mathcal{J}^{\text{tr}}$ measures exchange within the tensor decomposition. It has a type distinct from the transport covector of Section 23 and the spin current $\sigma_{IJ} \in \Omega^3(\mathcal{U}; \Lambda^2 E^*)$ of (26.4).

31.3 Link to the effective Cartan–Holst source

On a solution of (26.19), in the same units $c = 1$, the residual has expression

$$T^{\text{res}} = T^{\text{phys,L C}} + U^{\text{phys,spin}} - T_b + \frac{\Lambda_{\text{ref}} - \Lambda_{\text{int}}}{\kappa_x} g. \quad (31.7)$$

Indeed, substituting $G + \Lambda_{\text{int}}g = \kappa_x(T^{\text{phys,L C}} + U^{\text{phys,spin}})$ into (31.1) gives the equality term by term. When the material transducer expresses precisely $T_b = T^{\text{phys,L C}}$ and $\Lambda_{\text{ref}} = \Lambda_{\text{int}}$, one obtains $T^{\text{res}} = U^{\text{phys,spin}}$. The separate conservation condition on T_b in Theorem 31.1 retains its own role: (26.20) establishes the balance of the total source, and the division between its components depends on the material action used.

31.4 Trace sector and acceleration of the selected charts

The TRIT reader of state x expresses $s = \tau(x) \in \{+1, 0, -1\}$, and its positive clock expresses

$$H_* = \frac{2\pi}{108t_0(x)} > 0. \quad (31.8)$$

Let h_s be a three-dimensional metric of constant sectional curvature s . The three charts of the selected realization are written

$$g_s = -dt^2 + a_s(t)^2 h_s, \quad a_+(t) = H_*^{-1} \cosh(H_* t), \quad a_0(t) = \exp(H_* t), \quad a_-(t) = H_*^{-1} \sinh(H_* t). \quad (31.9)$$

The closed and flat charts are considered for $t \in \mathbb{R}$; the open chart for $t > 0$. The sign s and clock H_* retain their antecedents in the state selecting the realization.

Lemma 31.4 (Curvature of the three charts of the same clock). *For the functions in (31.9),*

$$\frac{\ddot{a}_s}{a_s} = H_*^2, \quad \left(\frac{\dot{a}_s}{a_s} \right)^2 + \frac{s}{a_s^2} = H_*^2, \quad \text{Ric}[g_s] = 3H_*^2 g_s. \quad (31.10)$$

Consequently, $R[g_s] = 12H_*^2$ and $G[g_s] = -3H_*^2 g_s$.

Proof. The first two equalities follow by differentiating the three functions: the closed and open charts use $\cosh^2 z - \sinh^2 z = 1$, and the flat chart uses $\dot{a}_0 = H_* a_0$. To verify the tensor identification, the connection coefficients involving the temporal coordinate are

$$\Gamma^0_{ij} = a_s \dot{a}_s (h_s)_{ij}, \quad \Gamma^i_{0j} = (\dot{a}_s / a_s) \delta^i_j,$$

and the purely spatial coefficients agree with those of h_s . Contracting the curvature gives

$$\text{Ric}_{00} = -3\ddot{a}_s / a_s, \quad \text{Ric}_{0i} = 0, \quad \text{Ric}_{ij} = (a_s \ddot{a}_s + 2\dot{a}_s^2 + 2s)(h_s)_{ij}.$$

Substituting the first two identities of (31.10) produces $\text{Ric}_{00} = -3H_*^2$ and $\text{Ric}_{ij} = 3H_*^2 a_s^2 (h_s)_{ij}$. Taking the trace and forming the Einstein tensor proves the other statements, with the curvature convention of (26.19). $\square$

Theorem 31.5 (Density, pressure, and sign of the selected acceleration). *In the preceding charts, for $\Lambda_{\text{ref}} = 0$ and $T_b = 0$, the residual tensor is*

$$T^{\text{res}} = -\rho_* g_s, \quad \rho_* := \frac{3H_*^2}{8\pi G_{\text{int}}} > 0, \quad T^\Lambda = T^{\text{res}}, \quad T^0 = 0. \quad (31.11)$$

For a unit comoving observer $u = \partial_t$, the density and isotropic pressure are ρ_* and $p_* = -\rho_*$. The acceleration equation evaluates to

$$\frac{\ddot{a}_s}{a_s} = -\frac{4\pi G_{\text{int}}}{3}(\rho_* + 3p_*) = H_*^2 > 0. \quad (31.12)$$

Proof. The preceding lemma and (31.1) give $T^{\text{res}} = -(3H_*^2 / \kappa_x) g_s$. Since $g_s(u, u) = -1$, $T^{\text{res}}_{\mu\nu} u^\mu u^\nu = \rho_*$. The spatial projector $q^{\mu\nu} = g_s^{\mu\nu} + u^\mu u^\nu$ satisfies $q^{\mu\nu} (g_s)_{\mu\nu} = 3$, hence

$$p_* := \frac{1}{3} q^{\mu\nu} T^{\text{res}}_{\mu\nu} = -\rho_*.$$

The trace is $-4\rho_*$, so (31.3) produces the two sectors of (31.11). Finally, $\rho_* + 3p_* = -2\rho_*$, and substitution into the right-hand side of (31.12) gives $\kappa_x \rho_* / 3 = H_*^2$, equal to the geometric value already calculated. Positivity uses exactly $G_{\text{int}} > 0$ and $H_* > 0$. $\square$

In this realization, the trace sector expresses a constant density and an opposite pressure, and the internal clock fixes positive acceleration. In a general chart, T^Λ describes the pointwise isotropic component, while T^0 preserves the trace-free residual and both obey the exchange in (31.6). Identifying these sectors with observational dark sources requires a single realization to reproduce expansion, lensing, rotation, structure growth, and the cosmic background jointly. The present construction fixes the tensor, its balances, and the acceleration of the selected charts entering that comparison.

III. Horizons and relational response

32 Horizons, purification, and information balances

The enriched state preserves routes and their records during evolution; a subsystem reading determines which part of that information remains accessible. The construction in this section uses the angular channels of Section 13.1, the incidence and area operator of Section 17, and the action, area, and temperature units of Sections 16 and 18. These are earlier representations of the same APP–TRIT–TPK construction. Each transmits its domain, orientation, normalization, and realization conditions here.

Three spaces are distinguished: the binary sheet record, the tritic factors of sectorial incidence, and the geometric realization of the exterior regions of a horizon. Purification, transport of observables, and dimensional conversion specify their relationships. This distinction preserves the structure accompanying each entropic value.

32.1 Sheet record and bipartite purification

Write $a = A_{\text{rad}}$ and $y = C_{\text{rad}}^*$ for the angular coordinates already constructed. The channels and their probabilities are

$$q_{\pm} = e^{-a\mp y}, \quad p_{\pm} = \frac{q_{\pm}}{q_{+} + q_{-}} = \frac{e^{\mp y}}{2 \cosh y}. \quad (32.1)$$

The signs index these two channels. On $\mathcal{H}_h = \text{span}\{|+\rangle, |-\rangle\}$, define

$$\rho_h(y) = p_{+}|+\rangle\langle+| + p_{-}|-\rangle\langle-|.$$

The auxiliary space $\mathcal{H}_{\text{ar}} = \text{span}\{|\text{adv}\rangle, |\text{ret}\rangle\}$ carries two orthonormal labels. With a phase χ_0 fixed, the joint state is

$$|\Psi_{\text{ar}}\rangle = \sqrt{p_{+}}|+\rangle \otimes |\text{adv}\rangle + e^{i\chi_0}\sqrt{p_{-}}|-\rangle \otimes |\text{ret}\rangle. \quad (32.2)$$

Proposition 32.1 (Accessible information in the sheet record). *The vector (32.2) is normalized, and its partial trace over $\mathcal{H}_{\text{ar}}$ is $\rho_h(y)$. Its entanglement entropy, expressed in nats, is*

$$s_h(y) = \log(2 \cosh y) - y \tanh y. \quad (32.3)$$

Exchange of the labels + and – conjugates $\rho_h(y)$ to $\rho_h(-y)$ and preserves s_h .

Proof. The squared norm is $p_{+} + p_{-} = 1$. Orthogonality of the auxiliary labels eliminates the cross terms in the partial trace. The squared Schmidt coefficients are precisely $p_{\pm}$. By (32.1), $p_{+} - p_{-} = -\tanh y$; substituting $\log p_{\pm} = \mp y - \log(2 \cosh y)$ into $-\sum_{\pm} p_{\pm} \log p_{\pm}$ proves the formula. Exchange of signs permutes the two eigenvalues and preserves entropy. $\square$

The common factor e^{-a} cancels upon normalization of the binary record. The relative phase χ_0 remains in the joint state even though this diagonal reduction does not express it. Logical reversibility concerns transport of the complete record, with its labels and correlations; phase return preserves the memory advance described in Section 20. A thermal implementation additionally has its own energy and dissipation law. In particular, s_h measures this binary subsystem and its concrete bipartition.

32.2 Purification of the incidence sectors

The boundary incidence constructed earlier distinguishes two sets of eighteen links, with sectorial labels $m = 90, 120$. Each link carries $\mathbb{C}^3$, with basis $|0\rangle, |1\rangle, |2\rangle$ and occupation operator $\text{diag}(0, 1, 2)$. Let N_m be their sectorial sums, and

$$N = N_{90} + N_{120}, \quad D_{\partial} = 90N_{90} + 120N_{120}. \quad (32.4)$$

On the positive branch, γ_{BI} denotes the Barbero–Immirzi parameter already obtained, and $a_0 > 0$ its angular areal factor. This notation distinguishes the parameter from the connection and its holonomies. Area is expressed as

$$\hat{\mathcal{A}}_m = \ell_P^2 \gamma_{\text{BI}} a_0 N_m, \quad \hat{\mathcal{A}} = \hat{\mathcal{A}}_{90} + \hat{\mathcal{A}}_{120}. \quad (32.5)$$

For the finite parameter $\Delta = \Delta_{\text{mod}} \in \mathbb{R}$ of the reduced state, set

$$x_m = e^{-m\Delta}, \quad z_m = 1 + x_m + x_m^2, \quad Z(\Delta) = z_{90}^{18} z_{120}^{18}, \quad (32.6)$$

$$\rho_{\partial}(\Delta) = Z(\Delta)^{-1} e^{-\Delta D_{\partial}}. \quad (32.7)$$

The trace in this section is the ordinary matrix trace, and the states have trace one. The parameter Δ preserves the selection and normalization of the antecedent state.

Proposition 32.2 (Tensor purification of the boundary). *For a finite real weight w , let*

$$|\text{TFD}(w)\rangle = \frac{|0\rangle|0\rangle + e^{-w/2}|1\rangle|1\rangle + e^{-w}|2\rangle|2\rangle}{\sqrt{1 + e^{-w} + e^{-2w}}}. \quad (32.8)$$

The product

$$|\Psi_{\Delta}\rangle = |\text{TFD}(90\Delta)\rangle^{\otimes 18} \otimes |\text{TFD}(120\Delta)\rangle^{\otimes 18} \quad (32.9)$$

is a normalized purification of $\rho_{\partial}(\Delta)$.

Proof. At each link, the sum of the squares of the three coefficients is one. The partial trace eliminates terms with different indices and returns $\text{diag}(1, e^{-w}, e^{-2w})/(1 + e^{-w} + e^{-2w})$. Partial trace commutes with the tensor product. Multiplying the thirty-six reductions gives the exponential and partition function of (32.7). $\square$

32.3 Informational equation and areal fluctuations

The occupation means and variances per link are

$$f_m = \frac{x_m + 2x_m^2}{z_m}, \quad v_m = \frac{x_m + 4x_m^2}{z_m} - f_m^2. \quad (32.10)$$

Write $s(\rho) = -\text{Tr}(\rho \log \rho)$, with $0 \log 0 = 0$. In the preceding purification, $s(\rho_{\partial})$ is its entanglement entropy.

Proposition 32.3 (Mean area, entropy, and sectorial response). *For real finite Δ , one has*

$$\langle N \rangle_{\Delta} = 18(f_{90} + f_{120}), \quad (32.11)$$

$$\langle D_{\partial} \rangle_{\Delta} = 18(90f_{90} + 120f_{120}), \quad (32.12)$$

$$\langle \hat{\mathcal{A}} \rangle_{\Delta} = 18\ell_P^2 \gamma_{\text{BI}} a_0 (f_{90} + f_{120}), \quad (32.13)$$

$$s(\Delta) = \log Z(\Delta) + \Delta \langle D_{\partial} \rangle_{\Delta}, \quad (32.14)$$

$$\frac{d\langle D_{\partial} \rangle_{\Delta}}{d\Delta} = -\text{Var}_{\Delta}(D_{\partial}) = -18(90^2 v_{90} + 120^2 v_{120}), \quad (32.15)$$

$$\frac{ds}{d\Delta} = -\Delta \text{Var}_{\Delta}(D_{\partial}), \quad (32.16)$$

$$\frac{d\langle \hat{\mathcal{A}} \rangle_{\Delta}}{d\Delta} = -18\ell_P^2 \gamma_{\text{BI}} a_0 (90v_{90} + 120v_{120}). \quad (32.17)$$

Proof. The local weights are $(1, x_m, x_m^2)/z_m$, whose first two moments give f_m and v_m . Additivity of occupation operators and factorization of the state give the first three formulas. Taking the trace of $-\rho_{\partial} \log \rho_{\partial} = \rho_{\partial}(\log Z + \Delta D_{\partial})$ gives the entropy. Differentiation of the finite sums yields $\partial_{\Delta} \log Z = -\langle D_{\partial} \rangle_{\Delta}$ and $\partial_{\Delta}^2 \log Z = \text{Var}_{\Delta}(D_{\partial})$. Independent factors add their variances. In the derivative of (32.14), the two first-moment terms cancel. Finally, $\partial_{\Delta} f_m = -mv_m$ gives the area response. $\square$

Loss of uniformity is quantified in dimension $d_\partial = 3^{36}$:

$$I_\partial(\Delta) = 36 \log 3 - s(\Delta) = \mathcal{D}(\rho_\partial(\Delta) \| I/d_\partial) \geq 0. \quad (32.18)$$

Here $\mathcal{D}$ denotes relative entropy, defined and justified below. Substitution of $\log(I/d_\partial) = -36 \log 3$ I proves the identity. Convexity of $t \log t$ gives positivity; equality corresponds to the uniform distribution, which in this collection occurs exactly at $\Delta = 0$. The deficit measures dimensionless information; the scale $\gamma_{\text{BI}} a_0 \ell_P^2$ measures area per occupation.

Theorem 32.4 (Occupation inversion and complementary area). *Let $\mathcal{R}_\partial$ be the involution exchanging $|0\rangle$ and $|2\rangle$ and fixing $|1\rangle$ in each factor. Then*

$$\mathcal{R}_\partial N \mathcal{R}_\partial^* = 72I - N, \quad (32.19)$$

$$\mathcal{R}_\partial D_\partial \mathcal{R}_\partial^* = 7560I - D_\partial, \quad (32.20)$$

$$\rho_\partial(-\Delta) = \mathcal{R}_\partial \rho_\partial(\Delta) \mathcal{R}_\partial^*, \quad (32.21)$$

$$s(-\Delta) = s(\Delta), \quad (32.22)$$

$$\langle \hat{\mathcal{A}} \rangle_{-\Delta} + \langle \hat{\mathcal{A}} \rangle_\Delta = 72 \gamma_{\text{BI}} a_0 \ell_P^2, \quad (32.23)$$

$$\mathcal{D}(\rho_\partial(\Delta) \| \rho_\partial(-\Delta)) = \Delta(7560 - 2\langle D_\partial \rangle_\Delta). \quad (32.24)$$

Proof. Each occupation transforms by $n \mapsto 2 - n$. In each sector, $N_m \mapsto 36I - N_m$, so the constant term of the weighted reader is $36(90 + 120) = 7560$. Moreover, $z_m(-\Delta) = e^{2m\Delta} z_m(\Delta)$ and $Z(-\Delta) = e^{7560\Delta} Z(\Delta)$. Substitution into the normalized state proves conjugation. Its spectrum preserves entropy; the transformation of N gives complementary area. Subtracting the logarithms of the two states and taking the expectation in $\rho_\partial(\Delta)$ gives the final equality. $\square$

At $\Delta = 0$, $s = 36 \log 3$. As $\Delta \rightarrow +\infty$, zero occupation concentrates the state and entropy tends to zero; inversion carries that limit to maximal occupation. For $\Delta > 0$, entropy decreases strictly because the weighted reader has more than one eigenvalue and the state has full rank. The complete configuration, area, and entropy thus preserve differentiated roles.

32.4 Modular law and finite area variations

Fix $\rho_0 = \rho_\partial(\Delta_0)$, with real finite Δ_0 , and hold $\Delta_0, \gamma_{\text{BI}}, a_0, \ell_P$ constant. The normalized areas $\mathcal{A}_m^n = \gamma_{\text{BI}} a_0 N_m$ express the modular operator as

$$K_0 = -\log \rho_0 = c_0 I + \sum_{m=90,120} \beta_m \mathcal{A}_m^n, \quad \beta_m = \frac{m\Delta_0}{\gamma_{\text{BI}} a_0}, \quad c_0 = \log Z(\Delta_0). \quad (32.25)$$

One has $\beta_{120} = \frac{4}{3}\beta_{90}$, including when both are zero. The ratio is 4/3 when $\Delta_0 \neq 0$. K_0 is the logarithm of this reference state; its type distinguishes it from temporal transport and the gravitational connection.

Theorem 32.5 (First modular law and finite correction). *For any normalized state σ on the same space, let $\Delta_\sigma \langle B \rangle = \text{Tr}[(\sigma - \rho_0)B]$. Then*

$$s(\sigma) - s(\rho_0) = \sum_{m=90,120} \beta_m \Delta_\sigma \langle \mathcal{A}_m^n \rangle - \mathcal{D}(\sigma \| \rho_0), \quad (32.26)$$

where $\mathcal{D}(\sigma \| \rho_0) = \text{Tr}[(\log \sigma - \log \rho_0)]$ is finite and nonnegative. If $\sigma(\varepsilon)$ is differentiable, normalized, and $\sigma(0) = \rho_0$, one obtains

$$\left. \frac{ds(\sigma(\varepsilon))}{d\varepsilon} \right|_0 = \sum_{m=90,120} \beta_m \left. \frac{d\langle \mathcal{A}_m^n \rangle_{\sigma(\varepsilon)}}{d\varepsilon} \right|_0. \quad (32.27)$$

Proof. By definition, $\mathcal{D}(\sigma \| \rho_0) = -s(\sigma) + \text{Tr}(\sigma K_0)$, whereas $s(\rho_0) = \text{Tr}(\rho_0 K_0)$. Subtraction and (32.25) prove the finite identity, since normalization cancels the term $c_0 I$.

For positivity, diagonalize σ with eigenvalues p_i and vectors u_i , and ρ_0 with eigenvalues $q_j > 0$ and vectors v_j . If $r_i = \langle u_i, \rho_0 u_i \rangle > 0$, concavity of the logarithm gives

$$\sum_j |\langle u_i, v_j \rangle|^2 \log q_j \leq \log r_i.$$

Consequently, $\mathcal{D}(\sigma \| \rho_0) \geq \sum_i p_i \log(p_i/r_i) \geq 0$. The final inequality is convexity of $t \log t$ with weights r_i , whose sum is one. Faithfulness of ρ_0 keeps its logarithms finite; zero eigenvalues of σ are interpreted through $0 \log 0 = 0$.

For the infinitesimal formula, in a neighbourhood of the faithful state ρ_0 , the derivative of the trace of $-\sigma \log \sigma$ is $-\text{Tr}[\dot{\sigma}(\log \rho_0 + I)]$. This identity follows from the spectral derivative of the trace. The condition $\text{Tr} \dot{\sigma} = 0$ leaves $\text{Tr}(\dot{\sigma} K_0)$, and substitution of (32.25) completes the proof. $\square$

For physical areas, the coefficients are β_m/ℓ_P^2 . Applying the common transducer k_B gives $S = k_B S$ and preserves the term $k_B \mathcal{D}$ in full. The derivative in (32.17) varies Δ ; law (32.27) holds Δ_0 fixed and varies the state around the reference. These are different operations.

32.5 Action, decision energy, and areal unit

Fix one and the same oriented action section and its gravitational and thermal sections, on the positive branch. The composition in 15 identifies $\ell_P = L_\alpha = L$ in that realization. In the return section, the characteristic energy is $E_L = \hbar_{\text{ret}} c/L$. The antecedents give

$$\begin{aligned} \ell_P^2 &= \frac{\hbar G}{c^3}, & t_P &= \frac{\ell_P}{c}, \\ T_{108} &= 2\pi t_P, & h &= 2\pi \hbar, \\ E_P &= \frac{\hbar}{t_P} = k_B \Theta_{\text{clk}} \log 3. \end{aligned} \tag{32.28}$$

The thermal equality preserves the previously constructed pair and normalization; changing its dimensional basis transports k_B and Θ_{clk} jointly. The elementary area increment is $\delta \mathcal{A} = \gamma_{\text{BI}} a_0 \ell_P^2$.

Proposition 32.6 (Conversion coefficient between decision and area). *The ratio of decision energy to the areal increment satisfies*

$$\begin{aligned} \mathcal{J}_{I/A} &:= \frac{E_P}{\delta \mathcal{A}} = \frac{k_B \Theta_{\text{clk}} \log 3}{\gamma_{\text{BI}} a_0 \ell_P^2} \\ &= \frac{c^3}{\gamma_{\text{BI}} a_0 G t_P} = \frac{c^4}{\gamma_{\text{BI}} a_0 G \ell_P}. \end{aligned} \tag{32.29}$$

Its dimension is energy per area.

Proof. Substitution of $E_P = \hbar/t_P$ and $\ell_P^2 = \hbar G/c^3$ cancels the common action section. The identity $\ell_P = c t_P$ gives the final expression. Positivity of the factors ensures that all quotients are defined. $\square$

The capacity 81 of the cubic cut and the bipartition of the thirty-six sectorial links correspond to different domains. In the cubic graph $\{0, \dots, N-1\}^3$, the N^2 parallel lines between two opposite faces are edge-disjoint: every cut intercepts all N^2 , and a transverse layer attains that bound. For $N = 9$, the uniform product state of the trits in that layer has $s_{\text{corte}} = 81 \log 3$. Assigning one decision of the same cycle to each link expresses

$$S_{\text{corte}}^{\text{fis}} = 81 k_B \log 3, \quad E_{\text{corte}}^{(1)} = 81 E_P = \Theta_{\text{clk}} S_{\text{corte}}^{\text{fis}}. \tag{32.30}$$

The proof combines the minimum cut, additivity of product-state entropy, and (32.28). In the sectorial state, the relevant finite equation remains (32.26), with its relative entropy.

32.6 Cosmological normalization of the horizon

For an areal radius $R > 0$, the cosmological realization considered uses the triad

$$E_\Lambda(R) = \frac{c^4 R}{2G}, \quad S_H(R) = \frac{k_B \pi R^2}{\ell_P^2}, \quad T_{\cos}(R) = \frac{\hbar c}{2\pi k_B R}. \quad (32.31)$$

The factor 2π belongs to this temperature normalization. The area $\mathcal{A}_H = 4\pi R^2$ expresses the same entropy as $S_H/k_B = \mathcal{A}_H/(4\ell_P^2)$.

To retain the distinction between the constructed scale and the variable radius, write $R = R_h = \zeta_h L_\alpha$, with $\zeta_h > 0$. In the return section, substituting the same G and reference area gives

$$E_\Lambda(R_h) = \frac{\zeta_h}{2} E_L, \quad \frac{S_H(R_h)}{k_B} = \pi \zeta_h^2, \quad k_B T_{\cos}(R_h) = \frac{E_L}{2\pi \zeta_h}.$$

Each identity follows directly from (32.31). The factor $1/2$ belongs to this horizon energy. The reduced reading of a mass $m_{P,\text{ref}}$ has radius $r_g = L_\alpha y$; its Schwarzschild radius is $2L_\alpha y$. The geometric condition for a cosmological horizon retains its own domain.

Theorem 32.7 (Gravitational half-action cell). *In (32.31),*

$$T_{\cos}(R)S_H(R) = E_\Lambda(R), \quad (32.32)$$

$$T_{\cos}(\ell_P)S_H(\ell_P) = \frac{E_P}{2} = \frac{k_B \Theta_{\text{clk}} \log 3}{2}, \quad (32.33)$$

$$E_\Lambda(\ell_P)t_P = \frac{\hbar}{2}, \quad E_\Lambda(\ell_P)T_{108} = \frac{h}{2}. \quad (32.34)$$

Holding the chart constants fixed and varying R ,

$$T_{\cos} dS_H = 2 dE_\Lambda, \quad S_H dT_{\cos} = -dE_\Lambda. \quad (32.35)$$

Proof. Multiplying temperature and entropy gives $\hbar c R/(2\ell_P^2) = c^4 R/(2G)$. At $R = \ell_P$, the result is $\hbar/(2t_P) = E_P/2$. The two durations in (32.28) give the action relations. For the differentials, $dE_\Lambda = c^4 dR/(2G)$, $dS_H = 2\pi k_B R dR/\ell_P^2$, and $dT_{\cos} = -T_{\cos} dR/R$. Substitution proves both identities. $\square$

The product $E_\Lambda = T_{\cos} S_H$ expresses a state identity. Its radial variations preserve the two terms of (32.35); a complete thermodynamic interpretation additionally specifies the corresponding temporal generator, work, and material realization. In this chart, action and gravitation fix the unit ℓ_P^2 , while Boltzmann converts information into dimensional entropy. Transport of sections with fixed $G\hbar$ preserves that unit when the section c is held fixed, and preserves S_H/k_B at fixed radius.

In the cell $R = \ell_P$, $S_H/k_B = \pi$, so the effective multiplicity is $\Omega_{\text{eff}} = e^\pi$. For the tritic hyperbolic involution H_{trit} , with spectrum $\{1, -1\}$, diagonalization gives $\text{spec}(e^{\pi H_{\text{trit}}}) = \{e^\pi, e^{-\pi}\}$. This spectral relation specifies the expansive value; a microscopic identification of both spaces additionally preserves their realization map and the state.

32.7 Thermal conversion between the calendar and the horizon

Action, the area increment, and angular memory admit an explicit energetic composition. We retain one action orientation, the positive Barbero–Immirzi branch, and the realization of (32.31). Write

$$L = \ell_P = L_\alpha, \quad E_P = \frac{h}{108 t_0} = \frac{\hbar c}{L}, \quad q_{\mathcal{A}} = \delta \mathcal{A} = \gamma_{\text{BI}} a_0 L^2.$$

The symbol $q_{\mathcal{A}}$ denotes an area increment here. The angular factor a_0 , the functional γ_{BI} , and action are the quantities obtained in the preceding sections. Their domain preserves the hexadic–octadic incidence, orientation, and calendar of the same enriched APP–TRIT–TPK state.

Proposition 32.8 (Composition of the areal and thermal scales). *For the tension in (32.29),*

$$\frac{\mathcal{F}_{I/A} q_A}{\log 3} = \frac{E_P}{\log 3} = \frac{h}{108 t_0 \log 3} = k_B \Theta_{\text{clk}}. \quad (32.36)$$

For $R = \zeta L$, with $\zeta > 0$, let $\tau_R = k_B T_{\text{cos}}(R)$. Then

$$\tau_R = \frac{E_P}{2\pi\zeta}, \quad \frac{T_{\text{cos}}(R)}{\Theta_{\text{clk}}} = \frac{\log 3}{2\pi\zeta}. \quad (32.37)$$

In particular, the ratio at $R = L$ is $\log 3/(2\pi)$.

Proof. The definition $\mathcal{F}_{I/A} = E_P/q_A$ and the tritic normalization in (32.28) give (32.36). The gravitational expression of the same tension gives the same result:

$$\frac{c^4}{\gamma_{\text{Bl}} a_0 G L} \gamma_{\text{Bl}} a_0 L^2 = \frac{c^4 L}{G} = \frac{\hbar c}{L} = E_P,$$

using $G\hbar = c^3 L^2$. By (32.31), $\tau_R = \hbar c/(2\pi R) = E_P/(2\pi\zeta)$. Dividing this identity by $k_B \Theta_{\text{clk}} = E_P/\log 3$ proves the temperature ratio. $\square$

In the same action section, the gravitational composition reads

$$G k_B \Theta_{\text{clk}} \log 3 = G E_P = c^4 L, \quad G \tau_R = \frac{c^4 L}{2\pi\zeta}. \quad (32.38)$$

Both identities follow from $G\hbar = c^3 L^2$ and the preceding scales.

The tension converts the area increment into decision energy. Tritic normalization and cosmological periodicity determine two thermal sections of that same energy, related by (32.37). The conversion preserves both normalizations and the role of Barbero–Immirzi in geometric resolution.

32.8 Energetic response of sectorial memory

Fix $R > 0$ and the faithful state $\rho_0 = \rho_{\partial}(\Delta_0)$, with $\Delta_0 > 0$, in (32.7). Its modular operator is $K_0 = (\log Z_0)I + \Delta_0 D_{\partial}$. The horizon realization assigns to each density state σ the response quantities

$$\begin{aligned} \mathcal{E}_R(\sigma) &= E_{\Lambda}(R) + \tau_R \text{Tr}[(\sigma - \rho_0)K_0], \\ \mathcal{S}_R(\sigma) &= S_{\text{H}}(R) + k_B[s(\sigma) - s(\rho_0)]. \end{aligned} \quad (32.39)$$

These describe the response of the modular state in the geometric realization already specified. Radius and action fix its energetic scale τ_R ; occupations preserve the provenance of each channel.

Theorem 32.9 (Response Hamiltonian and informational balance). *The operator of energy differences corresponding to (32.39) may be chosen as*

$$H_R = \tau_R \Delta_0 D_{\partial}. \quad (32.40)$$

Its Gibbs state at energy scale τ_R is ρ_0 . For every density state σ on the sectorial space,

$$\mathcal{E}_R(\sigma) - T_{\cos}(R) \mathcal{S}_R(\sigma) = \tau_R \mathcal{D}(\sigma \| \rho_0) \geq 0. \quad (32.41)$$

For a differentiable state variation about ρ_0 , with R and H_R fixed, one obtains

$$\delta \mathcal{E}_R = \tau_R \delta s = T_{\cos}(R) \delta \mathcal{S}_R. \quad (32.42)$$

Proof. The traces of σ and ρ_0 are one, so their difference eliminates the scalar term $\log Z_0$. Hence $\mathcal{E}_R(\sigma) - E_{\Lambda}(R) = \text{Tr}[(\sigma - \rho_0)H_R]$. Moreover,

$$\frac{e^{-H_R/\tau_R}}{\text{Tr } e^{-H_R/\tau_R}} = \frac{e^{-\Delta_0 D_{\partial}}}{Z_0} = \rho_0.$$

The identity $E_{\Lambda} = T_{\cos} S_H$ cancels the geometric reference in the balance. The finite identity (32.26), written as $s(\sigma) - s(\rho_0) = \text{Tr}[(\sigma - \rho_0)K_0] - \mathcal{D}(\sigma \| \rho_0)$, proves (32.41). Its tangent term is the first modular law already established and gives (32.42). $\square$

For a unit occupation increment, the energy gaps are $\epsilon_{90} = 90\tau_R \Delta_0$ and $\epsilon_{120} = 120\tau_R \Delta_0$. In both sectors,

$$\frac{\epsilon_m}{m\Delta_0} = \tau_R, \quad m \in \{90, 120\}. \quad (32.43)$$

Thus the ratio 4/3 between areal weights records different gaps of the same Hamiltonian. The ratio of an energy gap to the corresponding variation of $-\log p$ preserves a common scale. State normalization eliminates only a scalar constant.

Corollary 32.10 (Energy susceptibility at fixed Hamiltonian). *For $\rho_{\tau} = e^{-H_R/\tau} / \text{Tr } e^{-H_R/\tau}$, with $\tau > 0$,*

$$\frac{d\langle H_R \rangle_{\tau}}{d\tau} = \frac{\text{Var}_{\rho_{\tau}}(H_R)}{\tau^2} > 0. \quad (32.44)$$

At $\tau = \tau_R$, the result is $\Delta_0^2 \text{Var}_{\rho_0}(D_{\partial})$.

Proof. Differentiating the finite sum of exponentials gives the variance divided by τ^2 . The state is faithful and H_R has more than one eigenvalue, so the variance is positive. Substitution of (32.40) gives the stated evaluation. $\square$

This derivative varies the Gibbs scale while holding the Hamiltonian fixed. The radial variation in (32.35) also changes H_R and retains its own geometric terms.

32.9 Thermometric section and preservation of the record

The state ρ_0 and response Hamiltonian determine a thermometric coordinate before any candidate numerical reader is considered. For the fixed radius, let $\mathcal{L}_{\Theta} = \mathbb{R}u_R$ be an oriented line whose positive section u_R corresponds to the reference state. On faithful Gibbs states of the same Hamiltonian,

$$\rho_{\beta} = \frac{e^{-\beta H_R}}{\text{Tr } e^{-\beta H_R}}, \quad \Theta_R(\rho_{\beta}) = \frac{1}{\beta \tau_R} u_R, \quad \beta > 0. \quad (32.45)$$

The operator identity $-\log \rho_{\beta} = \beta H_R + (\log Z_{\beta})I$ determines β : subtracting two eigenvalues with distinct energies gives its unique ratio. In particular, $\Theta_R(\rho_0) = u_R$. The line $\mathcal{L}_E$ is the one-dimensional space of energetic quantities.

Theorem 32.11 (Positive transducer and identification criterion). *There is a unique positive linear map $B_R : \mathcal{L}_\Theta \rightarrow \mathcal{L}_E$ satisfying $B_R(u_R) = \tau_R$. It obeys*

$$B_R(\Theta_R(\rho_\beta)) = \beta^{-1}. \quad (32.46)$$

The compatible chronological section is

$$\Theta_{\text{clk},R} = \frac{E_P}{\tau_R \log 3} u_R = \frac{2\pi R}{L \log 3} u_R, \quad B_R(\Theta_{\text{clk},R}) = \frac{\mathcal{I}_{I/A} \mathcal{Q}_A}{\log 3}. \quad (32.47)$$

For any other linear map B_ between the same lines,*

$$B_* = B_R \iff B_*(\Theta_{\text{clk},R}) = \frac{h}{108t_0 \log 3}. \quad (32.48)$$

Proof. The section u_R is a basis of a line, so prescribing its image defines a unique linear map; positivity of τ_R ensures positivity of the map. Substituting (32.45) proves (32.46). The expression for τ_R and (32.36) give (32.47). Finally, $\Theta_{\text{clk},R}$ is nonzero. Two linear maps from a line coincide if and only if they coincide on that section. $\square$

The criterion compares readers on a section previously obtained from the state, energetic response, and tritic normalization. Application to a concrete numerical coordinate requires evaluating that reader on the specified section. Dimensional bases are transported jointly with the maps and the thermal coordinates they express.

Proposition 32.12 (Transport of the response with memory). *Let J be an isometry preserving the complete record labels, and take its image as the receiving space. For $H'_R = JH_R J^*$, $\rho'_0 = J\rho_0 J^*$, and $\sigma' = J\sigma J^*$, energy expectations, entropies, and balance (32.41) are preserved. Moreover, $JH_R = H'_R J$.*

Proof. On its image, J is unitary and transports functional calculus. Cyclicity of the trace preserves expectations; the same nonzero eigenvalues preserve entropy and relative entropy. The intertwining identity uses $J^* J = I$. Substitution into the balance proves its preservation. $\square$

The same equivalence admits another scale $\tau' = a\tau_R$, with $a > 0$, and another energy origin e' . If $H' = e'I + aJH_R J^*$, then

$$\frac{H' - e'I}{\tau'} = J \frac{H_R}{\tau_R} J^*, \quad \frac{e^{-H'/\tau'}}{\text{Tr}_{\text{im } J} e^{-H'/\tau'}} = J\rho_0 J^*.$$

The scalar factor $e^{-e'/\tau'}$ cancels in the state; energy differences are multiplied by a , while relative entropy remains invariant. The corresponding balance therefore has coefficient τ' . The full energy expectation preserves the origin: $\langle H' \rangle = e' + a\langle H_R \rangle$. Realizations are thus compared through their centered exponents and their own scales. If the expectation supplies the readings $m = E/c^2$ and $r_g = GE/c^4$, keeping c, G fixed gives

$$m' = \frac{e'}{c^2} + am, \quad r'_g = \frac{Ge'}{c^4} + ar_g.$$

Statistical normalization and the transport of quantities therefore retain distinct roles for the same recorded origin.

The composition connects the Barbero–Immirzi area increment, calendar energy, and angular boundary distribution. Its transport retains sectorial provenance and the record: phase return remains accompanied by the memory advance of the enriched state.

32.10 Kruskal chart and exchange of the exterior regions

Fix a expressed state x , its radius $R = R(x) > 0$, and the positive sections c, G . In the following chart, x, R, c, G are fixed parameters; t, r are spacetime coordinates. The exterior realization of the two sheets is specified by

$$f(r) = 1 - \frac{R}{r}, \quad ds^2 = -f(r)c^2 dt^2 + f(r)^{-1} dr^2 + r^2 d\Omega_2^2, \quad r > R. \quad (32.49)$$

For the sheet reader $\sigma \in \{+1, -1\}$, define

$$r_* = r + R \log\left(\frac{r}{R} - 1\right), \quad (32.50)$$

$$\mathcal{U} = -\sigma \exp\left(-\frac{ct - r_*}{2R}\right), \quad \mathcal{V} = \sigma \exp\left(\frac{ct + r_*}{2R}\right). \quad (32.51)$$

The symbols $\mathcal{U}, \mathcal{V}$ are coordinates and are distinguished from route-transport operators.

Theorem 32.13 (Geometric realization and sheet involution). *Chart (32.51) realizes the two sheets as the two exterior regions $\mathcal{U}\mathcal{V} < 0$ of the Kruskal extension. One has*

$$-\mathcal{U}\mathcal{V} = \left(\frac{r}{R} - 1\right) e^{r/R}, \quad ds^2 = -\frac{4R^3}{r} e^{-r/R} d\mathcal{U} d\mathcal{V} + r^2 d\Omega_2^2. \quad (32.52)$$

The involution

$$\mathcal{C}_H : (\sigma, \mathcal{U}, \mathcal{V}) \mapsto (-\sigma, -\mathcal{U}, -\mathcal{V}) \quad (32.53)$$

is isometric, fixes the areal radius, exchanges the exterior regions, and preserves the horizon set $\mathcal{U}\mathcal{V} = 0$. The metric is regular at $r = R$ and satisfies the vacuum equations on the regular domain $r > 0$.

Proof. Multiplying the coordinates gives the first identity. Since $dr_*/dr = f^{-1}$,

$$d\mathcal{U} = \frac{\mathcal{U}}{2R}(-c dt + f^{-1} dr), \quad d\mathcal{V} = \frac{\mathcal{V}}{2R}(c dt + f^{-1} dr).$$

Their product reproduces the radial and temporal part of the metric. The function $z \mapsto (z - 1)e^z$ has derivative $ze^z > 0$ for $z > 0$, determining r from the product. In the exterior region one recovers $\sigma = \text{sign } \mathcal{V}$ and $t = (R/c) \log(-\mathcal{V}/\mathcal{U})$. At $r = R$, the radial coefficient of the extended metric is $-4R^2/e$, finite and nonzero. Inversion preserves the product, radius, and $d\mathcal{U}d\mathcal{V}$, and exchanges the two sign choices.

To check the vacuum equations in the exterior chart, the independent mixed components of the Einstein tensor reduce to

$$G^t_t = G^r_r = \frac{rf' + f - 1}{r^2}, \quad G^\theta_\theta = G^\phi_\phi = \frac{f'}{r} + \frac{f''}{2}.$$

With $f' = R/r^2$ and $f'' = -2R/r^3$, all vanish. The regular expression (32.52) prolongs the result through the horizons on the domain $r > 0$. $\square$

This realization preserves $E_H = c^4 R/(2G)$ and $S_H = k_B \pi R^2/\ell_P^2$, which depend on areal radius. The temperature T_{cos} in (32.31) belongs to the cosmological realization declared there. Exterior chart (32.49) preserves its own temporal normalization. The geometric map establishes exchange of the exterior regions; comparison with formation by collapse, radiation spectra, or quasinormal modes additionally uses their respective physical conditions.

32.11 Transport of states, observables, and refinements

Denote by F_x the chart of the two exterior regions just constructed. In each regular angular chart, its temporal and radial Jacobian is

$$\left| \frac{\partial(\mathcal{U}, \mathcal{V})}{\partial(t, r)} \right| = \frac{cr}{2R^3} e^{r/R} > 0. \quad (32.54)$$

The identity follows from the two differentials in the preceding proof. The remaining state records are transported as fibre coordinates; the chart preserves their identity and memory.

Proposition 32.14 (Unitary equivalence of sheet and exterior readings). *Let μ be a sheet measure and $\nu = (F_x)_*\mu$ its transported measure. Change of variables defines a unitary map*

$$W_H : L^2(\mu) \longrightarrow L^2(\nu), \quad (W_H \psi)(z) = \psi(F_x^{-1} z). \quad (32.55)$$

With densities relative to fixed coordinate measures, the same map incorporates the corresponding Jacobian factor. Conjugation

$$\mathfrak{S}_H(B) = W_H B W_H^* \quad (32.56)$$

is a unital completely positive $$ -isomorphism. It transports the state, its expectations, and its modular functional calculus. For the symmetric measure of the two sheets, it also intertwines their involutions.*

Proof. The definition of pushforward measure gives $\int |\psi \circ F_x^{-1}|^2 d\nu = \int |\psi|^2 d\mu$. The chart is invertible in the exterior regions, so the isometry is surjective. Unitary conjugation preserves product, adjoint, and identity; at each matrix amplification it conjugates by $I_n \otimes W_H$, proving complete positivity. For a state ρ , cyclicity of the trace gives $\text{Tr}(W_H \rho W_H^* \mathfrak{S}_H(B)) = \text{Tr}(\rho B)$. Functional calculus implies $g(W_H \rho W_H^*) = W_H g(\rho) W_H^*$ on its domain, in particular for the logarithm and modular flow of a state faithful on its support. The equality $F_x \circ \mathcal{C}_{\text{hoja}} = \mathcal{C}_H \circ F_x$ proves intertwining of involutions; the symmetric measure realizes them unitarily. $\square$

If refinements preserve (σ, t, r, R) and use the same isometry on fibre records, the inclusions j_m^{hoja} and j_m^{ext} satisfy

$$W_{H, m+1} j_m^{\text{hoja}} = j_m^{\text{ext}} W_{H, m}. \quad (32.57)$$

Indeed, change of variables acts on the preserved geometric coordinates and inclusion on the same fibre coordinates; evaluating both compositions gives the same function and the same transported density. The equality extends the map to the refined system without resetting the records.

For a bipartition $\mathcal{H}_A \otimes \mathcal{H}_B$, a factorized transport $U_A \otimes U_B$ satisfies

$$\rho'_A = U_A \rho_A U_A^*, \quad s(\rho'_A) = s(\rho_A).$$

The first identity follows from partial trace and the second from the spectrum. For a general unitary, the equivalent formulation also transports the subsystem algebra: $\mathcal{B}(\mathcal{H}_A) \otimes I \mapsto U(\mathcal{B}(\mathcal{H}_A) \otimes I)U^*$. This preserves the same operational question concerning accessible information. Entropy equalities apply to states with the relevant finite entropies.

The relationship between information and gravitation is expressed through distinct composable operations: the state and bipartition fix accessible information; incidence and Barbero–Immirzi fix areal resolution; action and gravitation construct the unit ℓ_P^2 ; Boltzmann realizes entropy dimensionally; the horizon chart transports geometry and its observables. Finite laws preserve relative entropy, and geometric realizations preserve their measures, temporal normalization, and domains.

33 Observer, memory, and relational response

The reading operation acts on the transport publishing the enriched APP–TRIT–TPK state. Its domain preserves route, sheet, orientation, carry, and memory; the result accessible to the observer is a map from that domain. This distinction allows three operations to be described jointly: choosing an oriented representative of transport, inscribing its record in an apparatus, and the local response induced by the global boundary. Each operation transmits different information. The following construction specifies their maps and the conditions allowing them to be composed.

33.1 Boundary transport and intrinsic reading

The finite representation of transport provides an oriented cycle C_n , with $n \geq 1$, vertices $0, \dots, n-1$, and edges $e_j : j \rightarrow j+1 \pmod{n}$. Fix the lattice realization L of the record, its vector space $W = L \otimes_{\mathbb{Z}} \mathbb{R}$, a positive inner product, and the nonzero primitive representative $v \in L$ of the distinguished residual class in the stated chamber. The APP sum and product sheets supply the polarizations of transport; TRIT orientation and TPK updating precede this linear representation. The cycle, lattice realization, and residual class are the data this section receives from the transport constructed previously.

A 0-cochain is a function on the vertices with values in W ; a 1-cochain assigns a vector to each oriented edge. The coboundary acts by

$$(d\Phi)(e_j) = \Phi(j+1) - \Phi(j).$$

Denote by $\mathbf{1}_E$ the scalar cochain constant on the edges. For a transport $\omega \in C^1(C_n; W)$, its accumulated representation Γ_ω is obtained by partial sums and affine interpolation; its closure defect is

$$B(\omega) = \Gamma_\omega(1) - \Gamma_\omega(0) = \sum_{j=0}^{n-1} \omega(e_j). \quad (33.1)$$

Theorem 33.1 (Exact decomposition and harmonic mode of the cycle). *Every transport has a unique decomposition*

$$\omega = d\Phi + u\mathbf{1}_E, \quad \Phi(0) = 0, \quad u = \frac{1}{n} \sum_j \omega(e_j).$$

In particular, $B(d\Phi) = 0$ and $B(\omega) = nu$.

Proof. Let $y_j = \omega(e_j) - u$. The sum of the y_j is zero. Define $\Phi(k) = \sum_{j=0}^{k-1} y_j$; the value at $k = n$ agrees with $\Phi(0) = 0$, so the function is well defined on the cycle, including its final edge. One has $d\Phi(e_j) = y_j$. If $d\Phi + u\mathbf{1}_E = d\Psi + w\mathbf{1}_E$, summing the edges gives $nu = nw$. Then $d(\Phi - \Psi) = 0$; connectivity of the cycle and normalization at the initial vertex imply $\Phi = \Psi$. The identity $\sum_j (\Phi(j+1) - \Phi(j)) = 0$ directly proves the telescoping cancellation. $\square$

Definition 33.2 (Admissible transport and normalized reading). The transport is admissible when its harmonic mode belongs to $\mathbb{R}v$. On the space $\mathcal{T}_{\text{adm}}$ of these transports, define the global reading and its density per phase:

$$\mathbf{a}_v(\omega) = \frac{\langle B(\omega), v \rangle}{\langle v, v \rangle}, \quad \mu_v(\omega) = \frac{\mathbf{a}_v(\omega)}{n}. \quad (33.2)$$

One has $B(\omega) = \mathbf{a}_v(\omega)v$.

Theorem 33.3 (Determination of the intrinsic reading). *The quotient $\mathcal{T}_{\text{adm}}/dC^0(C_n; W)$ is one-dimensional, generated by the class of $v\mathbf{1}_E$. The map μ_v is the unique linear functional on $\mathcal{T}_{\text{adm}}$ satisfying*

$$F(\omega + d\Phi) = F(\omega), \quad F(v\mathbf{1}_E) = 1.$$

Proof. The preceding decomposition writes each admissible transport as $d\Phi + \lambda v\mathbf{1}_E$. Its class modulo exact cochains is determined by λ . The class of $v\mathbf{1}_E$ is nonzero because its sum is $nv \neq 0$. By linearity and invariance, every functional with the stated properties satisfies $F(\omega) = \lambda F(v\mathbf{1}_E) = \lambda$. Formula (33.2) gives precisely $\mu_v = \lambda$. The global reading is $\mathbf{a}_v = n\mu_v$. $\square$

Definition 33.4 (Boundary observer). With the polarization of the expressed transport fixed, a boundary observer is a pair $\mathcal{O} = (\varepsilon, \Psi)$, where $\varepsilon \in \{+1, -1\}$ and $\Psi \in C^0(C_n; W)$. Its reading action is

$$\omega^{\mathcal{O}} = \varepsilon\omega + d\Psi. \quad (33.3)$$

The sign fixes orientation and the cochain fixes trivialization. Two choices of polarization are compared through their expressed transport; relation (33.3) describes exactly the representatives differing by orientation and coboundary.

Proposition 33.5 (Covariance of the reading). *The action preserves admissibility and satisfies*

$$B(\omega^{\mathcal{O}}) = \varepsilon B(\omega), \quad \mathfrak{a}_v(\omega^{\mathcal{O}}) = \varepsilon \mathfrak{a}_v(\omega), \quad \mu_v(\omega^{\mathcal{O}}) = \varepsilon \mu_v(\omega).$$

Trivializations with the same orientation express the same closure.

Proof. The sum of the coboundary $d\Psi$ is zero. Therefore, the harmonic mode transforms to εu , which still belongs to $\mathbb{R}v$, and the defect transforms to εB . The two scalar equalities follow from the normalized projection. $\square$

The unit of this reading is the residual representative v , with its internal normalization. Trivialization freedom acts on the cochain representative; closure information resides in its class. The description allows observers to be compared while keeping genealogical transport, its linear representation, and the scalar evaluating the residual class separate.

33.2 Boundary identity and cellular interior

Let K be an oriented finite two-dimensional cell complex, equipped with a fundamental 2-chain c_K such that $\partial c_K = [C_n]$. This condition expresses cancellation of interior incidences with their multiplicities and orientations. Let $\Omega \in C^1(K; W)$ extend ω , and let $F_\Omega = d\Omega$ be its additive discrete curvature.

Theorem 33.6 (Cellular holographic identity of the reading). *With the preceding data,*

$$B(\omega) = \langle \Omega, \partial c_K \rangle = \langle d\Omega, c_K \rangle, \quad \mathfrak{a}_v(\omega) = \frac{\langle \langle d\Omega, c_K \rangle, v \rangle}{\langle v, v \rangle}. \quad (33.4)$$

In particular, $F_\Omega = 0$ implies $\mathfrak{a}_v(\omega) = 0$. If closure is nonzero, every extension of the transport to that complex has nonzero discrete curvature.

Proof. The pairing between chains and cochains satisfies $\langle d\Omega, c_K \rangle = \langle \Omega, \partial c_K \rangle$. On expansion, each interior edge appears with the sum of the incidence numbers of its adjacent cells; the condition on ∂c_K leaves exactly the oriented boundary cycle. The restriction of Ω to that cycle is ω , whose sum is $B(\omega)$. Projection onto v gives the second formula and the flatness implication. $\square$

The equality compares two evaluations of the same extended cochain: the accumulated boundary defect and the coboundary integrated over the interior. Its curvature takes values in W and uses the additive cellular coboundary. Realization by a geometric connection additionally preserves its own coefficient space and transport law.

33.3 Isometric inscription of the record in the apparatus

Let $\mathfrak{M}_n$ be a nonempty finite set of records expressed at level n , with result, route, carry, sheet, and memory. Its record space is $\mathcal{H}_n = \ell^2(\mathfrak{M}_n)$, with basis $|m\rangle$. The visible update F_n has a record lift

$$\hat{F}_n(m) = (F_n(m), m).$$

Take as the next-level records a set containing these pairs. The second coordinate preserves the antecedent and makes the lift injective. This construction specifies the memory retained by inscription, even when the visible update merges values.

In a finite apparatus space $\mathcal{K}_{A,n}$, fix an orthonormal collection $(|a_m\rangle)_{m \in \mathfrak{M}_n}$ and define

$$\iota_n : \mathcal{H}_n \longrightarrow \mathcal{K}_{A,n}, \quad \iota_n |m\rangle = |a_m\rangle, \quad V_n : \mathcal{H}_n \longrightarrow \mathcal{H}_{n+1}, \quad V_n |m\rangle = |\widehat{F}_n(m)\rangle.$$

Let $\mathcal{P}_n = \iota_n \mathcal{H}_n$ be the pointer subspace.

Theorem 33.7 (Compatibility between update and inscription). *The maps ι_n and V_n are isometries. There exists a unique isometry $W_n^0 : \mathcal{P}_n \longrightarrow \mathcal{K}_{A,n+1}$ satisfying*

$$W_n^0 \iota_n = \iota_{n+1} V_n. \quad (33.5)$$

If $\dim \mathcal{K}_{A,n+1} \geq \dim \mathcal{K}_{A,n}$, it extends to an isometry on all of $\mathcal{K}_{A,n}$. Using auxiliary complements to equalize dimensions gives a unitary extension between the enlarged spaces.

Proof. The basis images under ι_n are orthonormal; the images under V_n are also orthonormal by injectivity of $\widehat{F}_n$. For $z = \iota_n \psi$, define $W_n^0 z = \iota_{n+1} V_n \psi$. The representation of z is unique, and the three maps preserve its norm. The formula determines the map on a basis of $\mathcal{P}_n$ and proves the square. If the target dimension is sufficient, the orthogonal complement of $W_n^0 \mathcal{P}_n$ contains an orthonormal collection of the size of $\mathcal{P}_n^\perp$; sending a basis of the latter to that collection constructs the extension. Finally, enlarge both spaces to the same dimension and complete the two prescribed collections to orthonormal bases. The map between these bases is unitary. $\square$

When the record includes labels of a finite instrument, this same inscription realizes its environment: if the operators $K_{y,a} : \mathcal{H}_S \rightarrow \mathcal{H}'_S$ satisfy $\sum_{y,a} K_{y,a}^* K_{y,a} = I$, the map

$$V_{\text{inst}} \psi = \sum_{y,a} K_{y,a} \psi \otimes |y, a\rangle$$

is isometric, since its squared norm is $\sum_{y,a} \langle \psi, K_{y,a}^* K_{y,a} \psi \rangle = \|\psi\|^2$. The inscription ι transports each label to its corresponding pointer. The model thus fixes the premeasurement record; a material device additionally requires its interaction and calibration.

33.4 Accessible algebra and pointer complement

Define $P_m = |a_m\rangle\langle a_m|$, $P = \sum_m P_m$, and $P_\perp = I - P$. The algebra of the pointer subspace is $P\mathcal{B}(\mathcal{K}_{A,n})P$, whose unit is P . On it, the diagonal reader is

$$\mathcal{D}_P(A) = \sum_m P_m A P_m.$$

Proposition 33.8 (Record expectation and unital extension). *The map $\mathcal{D}_P$ is a completely positive, trace-preserving, idempotent conditional expectation on the pointer algebra, with $\mathcal{D}_P(P) = P$. Its image is $\bigoplus_m \mathbb{C}P_m$. On the algebra of the complete apparatus, the map*

$$\mathcal{E}_P(A) = \sum_m P_m A P_m + P_\perp A P_\perp \quad (33.6)$$

is a unital, completely positive, trace-preserving conditional expectation. Its image is

$$\left(\bigoplus_m \mathbb{C}P_m \right) \oplus \mathcal{B}(\mathcal{P}_n^\perp).$$

Both maps are orthogonal projections for the Hilbert–Schmidt product. Their spectra are contained in $\{0, 1\}$. When eliminated components exist, the separation between the fixed subspace and the kernel is one.

Proof. The projectors P_m are mutually orthogonal. On the complete apparatus, the collection including $P_\perp$ sums to I . Each summand $A \mapsto P_m A P_m$ is completely positive: under any matrix amplification it acts by congruence with $I \otimes P_m$. The same holds for $P_\perp$. Their sum retains this property. Cyclicity of the trace

and the sum of projectors prove trace preservation on each domain. Orthogonality annihilates cross terms when iterating the map, so it is idempotent. Its fixed elements are precisely the stated blocks, and the map is bimodular over that subalgebra. Moreover, $\text{Tr}(A^*P_mBP_m) = \text{Tr}((P_mAP_m)^*B)$, which gives Hilbert–Schmidt self-adjointness. A self-adjoint projection has only spectral values zero and one when both subspaces are nonzero. If only the fixed subspace remains, the map is the identity. $\square$

The sum $\sum_m P_mAP_m$, applied without the complement to the entire apparatus, sends I to P . Its natural unit belongs to the pointer algebra. Formula (33.6) also preserves the auxiliary block; an orthonormal resolution of that block provides, when desired, diagonalization of the entire apparatus.

If U is a unitary transformation of the apparatus and $P'_m = UP_mU^*$, then

$$\mathcal{E}_{P'}(UAU^*) = U\mathcal{E}_P(A)U^*.$$

The equality follows by substituting each projector into the sum. Likewise, for $A \in \mathcal{B}(\mathcal{P}_n)$, the square (33.5) implies

$$\mathcal{D}_{P_{n+1}}(W_n^0 A (W_n^0)^*) = W_n^0 \mathcal{D}_{P_n}(A) (W_n^0)^*.$$

Indeed, each matrix unit $|a_m\rangle\langle a_{m'}|$ is transported to the unit corresponding to the distinct records $\hat{F}_n(m), \hat{F}_n(m')$; the diagonal reader preserves exactly the terms with $m = m'$.

33.5 Factorization of the response through the effective boundary

Let $\mathcal{X}_\infty^{\text{enr}}$ be the domain of compatible enriched states, and let $b : \mathcal{X}_\infty^{\text{enr}} \rightarrow B_\infty$ be its global boundary trace. Denote by $B_{\text{ef}} = \text{im } b$ the effectively expressed boundary. For a response space $\mathcal{V}_v$, consider $I_v : \mathcal{X}_\infty^{\text{enr}} \rightarrow \mathcal{V}_v$.

Theorem 33.9 (Relational-response criterion). *There exists a unique map $\bar{I}_v : B_{\text{ef}} \rightarrow \mathcal{V}_v$ such that $I_v = \bar{I}_v \circ b$ if and only if*

$$b(x) = b(x') \implies I_v(x) = I_v(x'). \quad (33.7)$$

Uniqueness extends to all of B_∞ when b is surjective.

Proof. A composition through b is constant on each fiber. Conversely, for $\beta \in B_{\text{ef}}$, define $\bar{I}_v(\beta) = I_v(x)$ with $b(x) = \beta$. Condition (33.7) guarantees independence of the representative; every value of $\bar{I}_v$ is determined by some state in that fiber. If b is surjective, $B_{\text{ef}} = B_\infty$. Otherwise, the composition equation determines only values on B_{ef} . $\square$

This criterion is set-theoretic. For a continuous factorization, one also retains the topologies and the quotient-map condition on $b : \mathcal{X}_\infty^{\text{enr}} \rightarrow B_{\text{ef}}$; under this condition, continuity of I_v is equivalent to continuity of its factor. This precision separates determination of a reading from its regularity properties.

Let $\ell_v : \mathcal{X}_\infty^{\text{enr}} \rightarrow \mathcal{L}_v$ be the restriction accessible to a local neighborhood. Effective global dependence is verified through two states satisfying

$$\ell_v(x) = \ell_v(x'), \quad I_v(x) \neq I_v(x'). \quad (33.8)$$

If I_v factors through b , these two states have different boundaries. Moreover, the response cannot be written as a function of ℓ_v , since that function would assign the same value to both states. The boundary difference thereby acquires a verifiable operator effect.

33.6 Ambivalence weights and self-inertial response

For the general formula, consider positive readers A_∂, V_∂ and an energetic reader Q defined on the stated effective boundary domain. A realization on $B_{av} \subseteq B_{ef}$ is made explicit below. Memory remains as a coordinate $M_\partial(\beta)$ of the boundary. The ambivalence hinge provides the two orientations

$$r_{av} = \frac{\pi_{HMT}}{\varphi_{HMT}^2}, \quad r_{av}^J = \frac{\varphi_{HMT}}{\pi_{HMT}^2}.$$

The superscript J expresses interchange of the generating pair.

Normalized realization of the boundary readers. The APP-TRIT-TPK modal incidence, developed in the regional construction, provides the path envelope with matrix

$$F_{av} = \begin{pmatrix} 0 & 1 \\ 1 & 1 \end{pmatrix}.$$

The positive vector $v = (1, \varphi_{HMT})^T$ satisfies $F_{av}v = \varphi_{HMT}v$. Its normalized transition and stationary weights are

$$P_{ij} = \frac{(F_{av})_{ij}v_j}{\varphi_{HMT}v_i}, \quad P = \begin{pmatrix} 0 & 1 \\ \varphi_{HMT}^{-2} & \varphi_{HMT}^{-1} \end{pmatrix}, \quad (\mu_{ar}, \mu_{vol}) = \frac{(1, \varphi_{HMT}^2)}{1 + \varphi_{HMT}^2}. \quad (33.9)$$

The rows of P sum to one by $\varphi^{-2} + \varphi^{-1} = 1$; the stationary equation $\mu_{ar} = \mu_{vol}\varphi^{-2}$, together with normalization, determines the displayed weights. This Parry measure weights paths of the envelope; the chronological frequency (1/3, 2/3) counts instants of the modal calendar and retains its distinct role.

The common-memory-amplitude realization is defined on the sector $\mathcal{X}_{av} \subset \mathcal{X}_\infty^{enr}$ in which the readers of the two sheets have factorization

$$\mathcal{L}_{vol}(x) = \mu_{vol}\mathcal{M}(x), \quad \mathcal{L}_{ar}(x) = \pi_{HMT}\mu_{ar}\mathcal{M}(x), \quad \mathcal{M}(x) > 0. \quad (33.10)$$

Both values belong to the same positive memory-amplitude line. The areal representation carries the regional mark π_{HMT} of return. The dimensionless reading is obtained by dividing both sections by $\mathcal{L}_{vol}(x)$. The sums of weights are thus taken between quantities of the same type; any realization as physical areas and volumes is composed subsequently with their respective dimensional transducers.

Proposition 33.10 (Descent of normalized readers and the energetic clock). *On $B_{av} = b(\mathcal{X}_{av})$, the rules*

$$A_\partial(b(x)) = \frac{\mathcal{L}_{ar}(x)}{\mathcal{L}_{vol}(x)} = r_{av}, \quad V_\partial(b(x)) = 1 \quad (33.11)$$

define two positive readers independent of the representative. Fixing an orientation a of the action section, the decision energy of (18.6) determines on states

$$q_a(x) = \frac{h_a(x)}{108t_0(x)} = \frac{\hbar_a(x)}{t_P(x)} \in \mathcal{L}_E^+, \quad t_P = \frac{54}{\pi_{HMT}}t_0. \quad (33.12)$$

Write

$$b_{av} := b|_{\mathcal{X}_{av}} : \mathcal{X}_{av} \longrightarrow B_{av}.$$

There exists a unique reader $Q : B_{av} \rightarrow \mathcal{L}_E^+$ such that $q_a|_{\mathcal{X}_{av}} = Q \circ b_{av}$ if and only if, for all $x, x' \in \mathcal{X}_{av}$,

$$b(x) = b(x') \implies \frac{h_a(x)}{t_0(x)} = \frac{h_a(x')}{t_0(x')}. \quad (33.13)$$

In particular, a chart of fixed action and duration satisfies this condition and expresses $Q = h_a/(108t_0)$.

Proof. The quotient in (33.10) cancels the common amplitude and equals $\pi_{HMT}\mu_{ar}/\mu_{vol} = \pi_{HMT}/\varphi_{HMT}^2$. The result is the same for all representatives of a fiber of b_{av} within $\mathcal{X}_{av}$, and is positive. Dividing both readers by the same positive factor preserves the weighted quotients in which they enter. For the clock, the factorization criterion of Theorem 33.9, applied to q_a , is exactly equivalent to (33.13). Fixed sections satisfy it; variable sections whose quotient descends through b_{av} also satisfy it. Uniqueness follows from the definition of B_{av} as the effective image. $\square$

The value of reader Q belongs to the energy line $\mathcal{L}_E = \mathcal{L}_{\text{act}} \otimes \mathcal{L}_T^*$. With E_1, E_2 on that same line, $E_1 E_2 / Q^2$ is dimensionless and independent of the positive energy basis. The symbol Q in this section denotes exclusively that clock energy; the quantity of heat in the Landauer realization belongs to the corresponding thermal balance. The common-amplitude factorization specifies a particular realization of the general readers. Every other domain retains its boundary maps and its own verification of descent. Complete memory remains in the enriched history and its boundary representation, even though it cancels in the normalized quotient.

Define

$$w_A(\beta) = \frac{A_\partial(\beta)r_{\text{av}}}{A_\partial(\beta)r_{\text{av}} + V_\partial(\beta)r_{\text{av}}^J}, \quad w_V(\beta) = 1 - w_A(\beta).$$

In realization (33.11), the weights admit the exact evaluation

$$w_A = \frac{r_{\text{av}}^2}{r_{\text{av}}^2 + r_{\text{av}}^J} = \frac{\pi_{\text{HMT}}^4}{\pi_{\text{HMT}}^4 + \varphi_{\text{HMT}}^5}, \quad w_V = \frac{\varphi_{\text{HMT}}^5}{\pi_{\text{HMT}}^4 + \varphi_{\text{HMT}}^5}. \quad (33.14)$$

Substitution preserves the two factors of the definition: the normalized areal reader and its subsequent weighting by r_{av} . Generation and regional marking of return are performed once. Multiplying numerator and denominator by $\varphi_{\text{HMT}}^4 \pi_{\text{HMT}}^2$ gives the final expression. These weights remain constant in the common-amplitude sector. With projectors and energies fixed, a variation of the response may originate in reader Q ; the witness of global dependence additionally requires the states with equal local restriction and different response in (33.8).

Let P_A, P_V be complementary orthogonal projections on a response space $\mathcal{H}_v$, and let $E_1, E_2 > 0$ be two internal energies fixed for comparison. The response is

$$\bar{I}_v(\beta) = \frac{E_1 E_2}{Q(\beta)^2} (w_A(\beta)P_A + w_V(\beta)P_V), \quad I_v = \bar{I}_v \circ b. \quad (33.15)$$

Energies that are explicit readers of β are also allowed; in that case they are retained as such in the formula.

Proposition 33.11 (Positivity and covariance of the response). *Response (33.15) is self-adjoint, positive definite, and constant on the fibers of b . Its eigenvalues on the nonzero sectors are*

$$\lambda_A = E_1 E_2 Q^{-2} w_A, \quad \lambda_V = E_1 E_2 Q^{-2} w_V.$$

The complete interchange $(A_\partial, r_{\text{av}}, P_A) \leftrightarrow (V_\partial, r_{\text{av}}^J, P_V)$ permutes the two summands and preserves the operator. Unitary transport of the response space conjugates that operator and preserves its spectrum with multiplicities.

Proof. Positivity of the readers gives $0 < w_A, w_V < 1$. For $z = z_A + z_V$, with $z_A = P_A z$, $z_V = P_V z$,

$$\langle z, \bar{I}_v z \rangle = \frac{E_1 E_2}{Q^2} (w_A \|z_A\|^2 + w_V \|z_V\|^2) > 0$$

if $z \neq 0$. The orthogonal decomposition proves the spectral expressions. Exclusive dependence on β gives constancy on the fibers and factorization. Interchanging the triples interchanges numerators, weights, and projectors simultaneously, thus preserving the sum. For unitary transport U , substitution $P_A \mapsto U P_A U^*$, $P_V \mapsto U P_V U^*$ produces $U \bar{I}_v U^*$, with the same spectrum. $\square$

An interchange implemented within one space by a unitary map sending P_A to P_V requires both sectors to have equal dimension. Interchanging the two triples as labels of the formula is valid with the respective dimensions of each sector.

Condition (33.8) requires comparing the responses. For example, multiplying A_∂, V_∂ simultaneously by the same positive factor preserves both weights; if E_1, E_2, Q are also preserved, the response remains identical. By contrast, with equal weights and energies, replacing Q by qQ , $q > 0$, multiplies the response by q^{-2} . If two states with equal local restriction realize that change with $q \neq 1$, they provide the global witness. These are operator separation conditions; existence of the states is checked in the boundary domain being realized.

33.7 Connection of the joint state and expressed acceleration

In a differentiable realization of the joint state $\Gamma = (x, \mathcal{O}, m)$, let N be its realization space, ∇^{tot} its connection, and $\pi_S : N \rightarrow N_S$ a differentiable projection onto the subsystem, equipped with connection ∇^S . The connection and projection preserve the provenance of the transports and readers they realize. The observed frame is obtained through

$$\mathfrak{F}_{\mathcal{O}}(\Gamma) = \text{Res}_{\mathcal{O}}(\Gamma, \nabla^{\text{tot}}).$$

For a twice-differentiable trajectory $\Gamma(t)$, write $\gamma_S = \pi_S \circ \Gamma$.

Definition 33.12 (Self-inertia and projection defect). A trajectory is self-inertial with respect to the stated connection when $\nabla_{\dot{\Gamma}}^{\text{tot}} \dot{\Gamma} = 0$. The defect of its projection is

$$\mathcal{K}_{\pi_S}(\dot{\Gamma}, \dot{\Gamma}) = \nabla_{\dot{\gamma}_S}^S \dot{\gamma}_S - d\pi_S(\nabla_{\dot{\Gamma}}^{\text{tot}} \dot{\Gamma}). \quad (33.16)$$

Proposition 33.13 (Subsystem acceleration). *For a self-inertial trajectory,*

$$a_S = \nabla_{\dot{\gamma}_S}^S \dot{\gamma}_S = \mathcal{K}_{\pi_S}(\dot{\Gamma}, \dot{\Gamma}).$$

In coordinates x^B on N and y^a on N_S , the defect is quadratic in velocity, with components

$$(\mathcal{K}_{\pi_S})^a_{BC} = \partial_B \partial_C \pi_S^a + (\Gamma^S)^a_{bc} \partial_B \pi_S^b \partial_C \pi_S^c - \partial_D \pi_S^a (\Gamma^{\text{tot}})^D_{BC}. \quad (33.17)$$

Proof. The chain rule gives $\dot{\gamma}_S^a = \partial_B \pi_S^a \dot{\Gamma}^B + \partial_B \partial_C \pi_S^a \dot{\Gamma}^B \dot{\Gamma}^C$. Adding the contribution of ∇^S and subtracting the image of total acceleration cancels the terms containing $\ddot{\Gamma}$. The remaining terms are (33.17) contracted with $\dot{\Gamma}^B \dot{\Gamma}^C$. When total acceleration vanishes, the stated equality follows. $\square$

The geometric response factors through b when the connection, projection, and kinematic datum used in the evaluation are determined by $\beta = b(x)$, or when those kinematic data are held fixed in the comparison. Under these conditions, (33.17) gives the same result for any representatives of a fiber; Theorem 33.9 constructs its unique factor on B_{ef} . If the kinematic datum varies, it forms part of the domain of the response being compared.

The self-inertial formulation describes a geodesic trajectory of the joint state and an effective acceleration of its subsystem due to the projection defect. The gravitational realization subsequently identifies that connection, frame, and response with their physical observables. Observer covariance, memory inscription, and factorization through the boundary thus retain complementary roles: the first determines the reading class, the second retains genealogy, and the third specifies the relational dependence of the response.

34 Bidirectional propagation and the absorption condition at the boundary

APP-TRIT-TPK transport preserves the orientation of the route and the memory that distinguishes its outward passage from its return. On this antecedent, three successive operations are constructed: the action of an enriched history, the propagation of a source through the realized transports, and the imposition of a common condition at the boundary. The inscription of the record in an apparatus space, established in Section 33.3, allows these operations to be expressed through linear maps without conflating the observable representation with the state that preserves its antecedents.

Temporal direction, sheet conjugation, and spatial reflection have different domains. Their relations are first specified on the record; the intertwining used by a dynamical realization are then specified. This order determines what information remains in the propagators and what condition is introduced by the choice of an absorbing boundary.

34.1 Route action and involutions of the record

A finite route contains its positions, cardinal directions $(d_1, \dots, d_N)$, sheets $(\mu_1, \dots, \mu_N)$, with $\mu_t \in \{\Sigma, \Pi\}$, and the increments of its enriched record. For $\lambda \geq 0$, we define

$$\mathcal{A}_\lambda(\gamma) = \sum_{t=2}^N (1 - \delta_{d_t, d_{t-1}} + \lambda(1 - \delta_{\mu_t, \mu_{t-1}})). \quad (34.1)$$

The first summand counts changes of direction; the second counts changes of sheet with the stated weight. For $N \leq 1$, the sum is zero. The general action of the sector also retains its boundary and memory terms:

$$\mathcal{S}[\gamma] = \sum_{a \in \gamma} \mathcal{L}_a + \mathcal{B}_{\text{in}}(\partial_- \gamma) + \mathcal{B}_{\text{out}}(\partial_+ \gamma) + \mathcal{M}(\gamma). \quad (34.2)$$

The terms in this expression specify a variational class with typed endpoints and preserved memory. The dynamics of the sector determines $\mathcal{L}_a$, the admissible conditions, and the term $\mathcal{M}$.

On the records we define $\mathcal{C}_r$ as the global exchange of sheets; $\mathcal{P}_r$ as a reflection of the spatial chart and its directions; and $\mathcal{T}_r$ as the reversal of temporal order, the replacement of each direction by its opposite, and the exchange of endpoints. Cardinal reflection is an involutive permutation. Reversal of the signed record is $C_M(a_1, \dots, a_N) = (-a_N, \dots, -a_1)$. The sheets, quotients, and memory are transported with the complete route; the scalar reading of a word is a subsequent operation.

Proposition 34.1 (Invariance of the local route action). *On the preceding domain of records,*

$$\mathcal{A}_\lambda(\gamma) = \mathcal{A}_\lambda(\mathcal{C}_r \gamma) = \mathcal{A}_\lambda(\mathcal{P}_r \gamma) = \mathcal{A}_\lambda(\mathcal{T}_r \gamma) = \mathcal{A}_\lambda(\mathcal{C}_r \mathcal{P}_r \mathcal{T}_r \gamma).$$

Proof. A global permutation of the alphabet of directions preserves equality between consecutive directions. Exchanging the two sheets preserves equality between consecutive sheets. Reversing the order establishes a bijection between adjacent pairs, and replacing directions by their opposites also preserves their equalities. Each operation preserves the two counts in (34.1); their composition preserves them. $\square$

The proposition determines the symmetry of the local functional. For a transformation also to act on a dynamical problem with fixed boundaries, it must transport its class of admissible prolongations and the other three terms in (34.2). In particular, two routes with the same final representation may retain different variational classes because of their memory.

An interaction between two sections realized at a boundary B uses a coupling map

$$\mathcal{J}_B : E_B^{(1)} \otimes E_B^{(2)} \longrightarrow E_B^{(12)}$$

and a term $\mathcal{S}_{\text{int}}$ in the same action. Its covariance is expressed by

$$\rho_{12}(g) \mathcal{J}_B = \mathcal{J}_B(\rho_1(g) \otimes \rho_2(g)).$$

When the histories are composed, the record of the shared boundary is counted once; the memories of both sides are transported to the composite section. The coupling and this covariance condition are data of the specific interaction, distinct from the involutions of the record.

34.2 Reversal of the realized evolution

The realization of the record provides a real Hilbert space $\mathcal{H}_{\mathbb{R}}$, an orthogonal complex structure J_E , with $J_E^2 = -I$, and a self-adjoint Hamiltonian H . Its domain $\mathcal{D}(H)$ is invariant under J_E ; we assume $HJ_E = J_EH$ on that domain. Let $\mathcal{C}$ be an orthogonal involution that preserves $\mathcal{D}(H)$ and satisfies

$$\mathcal{C}J_E\mathcal{C}^{-1} = -J_E, \quad \mathcal{C}H\mathcal{C}^{-1} = H. \quad (34.3)$$

The internal action constant $\hbar_{\text{int}} > 0$ retains its previous generation; here it normalizes the evolution group.

Theorem 34.2 (Temporal conjugation of the group). *The generator $A = -J_EH/\hbar_{\text{int}}$, with domain $\mathcal{D}(H)$, is skew-adjoint. Its orthogonal group, unitary with respect to J_E , satisfies*

$$U(t) = \exp(-J_EHt/\hbar_{\text{int}}), \quad \mathcal{C}U(t)\mathcal{C}^{-1} = U(-t). \quad (34.4)$$

Proof. Orthogonality and $J_E^2 = -I$ imply $J_E^* = -J_E$. Domain invariance and commutation with H allow H to be treated as a complex self-adjoint operator in the space defined by J_E . By functional calculus, $-J_EH/\hbar_{\text{int}}$ is skew-adjoint and generates the stated group. In finite dimension, the same assertions follow from the exponential series and $A^* = -A$. The hypotheses on $\mathcal{C}$ give $\mathcal{C}A\mathcal{C}^{-1} = -A$. Conjugating the group, or its functional calculus, yields $\exp(-tA) = U(-t)$. $\square$

The map realizing the signed involution C_M as $\mathcal{C}$ preserves temporal orientation and the preceding domain. Charge conjugation in a field representation also preserves its own characters and operators. Equality (34.4) specifies the temporal reversal of the group under the stated conditions.

34.3 Propagators of the chronological channel of a route

The isometric inscription of the record admits the unitary prolongations of Theorem 33.7. Fix a bilateral realization of a chronological channel with fibres $\mathcal{H}_n$, $n \in \mathbb{Z}$, and unitary transports $U_n : \mathcal{H}_n \rightarrow \mathcal{H}_{n+1}$. The choice includes auxiliary spaces when required; it preserves the route and its preceding records. We denote by $U(n, k)$ the transport from k to n , with

$$U(k, k) = I, \quad U(n, k)U(k, l) = U(n, l), \quad U(n, k)^* = U(k, n).$$

For $n > k$, it is the product $U_{n-1} \cdots U_k$; for $n < k$, it is the inverse of the transport from n to k .

The source space $\mathcal{J}_c$ contains the finitely supported sections. The response space $\mathcal{F}$ contains all sections of these fibres. The quadratic route action, in the internal units of this realization, uses

$$(D\psi)_n = \psi_{n+1} - U_n\psi_n, \quad E[\psi] = \sum_n \|(D\psi)_n\|^2$$

for finitely supported sections. The kinetic operator with the sign convention $L = -D^*D$ is

$$(L\psi)_n = U_n^*\psi_{n+1} - 2\psi_n + U_{n-1}\psi_{n-1}. \quad (34.5)$$

Indeed, $(D^*z)_n = z_{n-1} - U_n^*z_n$, and substitution yields this formula. The variation of E is $-2 \operatorname{Re}\langle \delta\psi, L\psi \rangle$.

Theorem 34.3 (Exact Green operators of the covariant difference). *Writing $x_+ = \max(x, 0)$, the maps*

$$(G_{\text{ret}}j)_n = \sum_k (n-k)_+ U(n, k)j_k, \quad (34.6)$$

$$(G_{\text{adv}}j)_n = \sum_k (k-n)_+ U(n, k)j_k \quad (34.7)$$

from $\mathcal{J}_c$ to $\mathcal{F}$ satisfy

$$LG_{\text{ret}}j = LG_{\text{adv}}j = j.$$

The retarded response vanishes before and at the first source position; the advanced response vanishes after and at the last. They are the unique solutions that vanish sufficiently far in the respective direction. For finitely supported ψ , one also has $G_{\text{ret}}L\psi = G_{\text{adv}}L\psi = \psi$. With respect to pairing with finitely supported sources, $G_{\text{adv}} = G_{\text{ret}}^\dagger$.

Proof. Each sum has as many terms as the source has positions. Composition of the transports reduces application of L to the scalar identity

$$(n+1-k)_+ - 2(n-k)_+ + (n-1-k)_+ = \delta_{nk}.$$

The function $(k-n)_+$ satisfies the same identity. This proves both equations and the support statements. For uniqueness, a homogeneous solution that vanishes at two consecutive positions is determined to be zero at the next by (34.5); the recurrence, using invertible transports, extends in both directions. The difference of two solutions with the same causal support vanishes sufficiently far away and is therefore zero. Applying this argument to ψ and $G_{\text{ret}}L\psi$, or to the advanced case, gives the right-hand identities on finite support. Finally, for two compactly supported sources, the double sum in the pairing is finite and

$$((k-n)_+U(k,n))^* = (k-n)_+U(n,k).$$

Interchanging the indices proves the stated adjoint relation. $\square$

The responses belong to $\mathcal{F}$; in general they grow linearly outside the source and do not belong to ℓ^2 . The finite-support domain and its pairing specify the Green adjoint relation used here. The result constructs propagation in the chronological channel from its action; a spatial-field realization additionally retains its own operator, geometry, and propagation domain.

Let C_0 be an isometric involution, linear or antilinear, of $\mathcal{H}_0$. Transport from the reference fibre constructs

$$C_n = U(-n, 0)C_0U(0, n) : \mathcal{H}_n \longrightarrow \mathcal{H}_{-n}, \quad (\mathcal{C}\psi)_n = C_{-n}\psi_{-n}.$$

By composition, $C_{-n}C_n = I$ and $C_{n+1}U_n = U_{-n-1}^*C_n$. In particular, $\mathcal{C}^2 = I$, and the substitution $(n, k) \mapsto (-n, -k)$ in the preceding sums proves

$$\mathcal{C}L = L\mathcal{C}, \quad \mathcal{C}G_{\text{ret}}\mathcal{C}^{-1} = G_{\text{adv}}. \quad (34.8)$$

When C_0 is the stated realization of the record involution, these formulas transport that involution to the entire channel, without identifying phase return with a memory reset.

34.4 Causal interchange and decomposition of the response

In a field realization, let $L : \mathcal{D}(L) \subset \mathcal{H}_\Phi \rightarrow \mathcal{H}_J$ be an operator with unique retarded and advanced Green operators on the admissible source domains. We denote by $J^+(S)$ and $J^-(S)$ the future and past causal domains of a support S . The conditions are

$$\begin{aligned} LG_{\text{ret}} &= LG_{\text{adv}} = I, \\ \text{supp}(G_{\text{ret}}j) &\subset J^+(\text{supp } j), \\ \text{supp}(G_{\text{adv}}j) &\subset J^-(\text{supp } j). \end{aligned}$$

Let C_Φ, C_J be isometric involutions, both linear or both antilinear, that transport these domains, reverse their causal orientations, and satisfy $LC_\Phi = C_JL$. A Green pairing is retained for which $G_{\text{adv}} = G_{\text{ret}}^\dagger$.

Proposition 34.4 (Conjugation of the Green operators and parities). *Under the preceding conditions,*

$$C_\Phi G_{\text{ret}} C_J^{-1} = G_{\text{adv}}.$$

On the real space of response operators, define

$$\Theta(K) = C_\Phi K C_J^{-1}, \quad P_\pm(K) = \frac{1}{2}(K \pm \Theta(K)).$$

These are complementary projections for the involution Θ , and

$$G_{\text{sym}} = \frac{1}{2}(G_{\text{ret}} + G_{\text{adv}}), \quad G_{\text{rad}} = \frac{1}{2}(G_{\text{ret}} - G_{\text{adv}}) \quad (34.9)$$

satisfy $\Theta(G_{\text{sym}}) = G_{\text{sym}}$, $\Theta(G_{\text{rad}}) = -G_{\text{rad}}$, $G_{\text{sym}}^\dagger = G_{\text{sym}}$, and $LG_{\text{rad}} = 0$. The support of the symmetric response is contained in the union of the two causal domains.

Proof. The conjugated operator solves the source equation, since

$$LC_\Phi G_{\text{ret}} C_J^{-1} = C_J L G_{\text{ret}} C_J^{-1} = I.$$

Reversal of orientation carries its support to the advanced domain; uniqueness identifies it with G_{adv} . The involutions give $\Theta^2 = I$; expanding $(I \pm \Theta)^2$ and $(I + \Theta)(I - \Theta)$ proves the projection assertions. The Green adjoint relation proves symmetry, and subtracting the two source equations gives $LG_{\text{rad}} = 0$. The sum of two responses has support contained in the union of their supports. $\square$

If the involutions are antilinear, Θ is linear over real scalars; this is the operator space used in the proposition. Conjugation of the temporal group, the support condition, and the Green adjoint relation are three distinct properties, transmitted by the corresponding realization maps.

34.5 Projection of absorbed currents and variation of the action

Let $\gamma_\partial : \mathcal{H}_\Phi \rightarrow \mathcal{H}_\partial$ be the boundary reading, distinct from the route γ . Define

$$\mathcal{A}_\partial = \gamma_\partial G_{\text{rad}}.$$

The finite window fixes the source space and boundary reading; the Green operators retain their response domain. At a finite level with fixed inner products, its Moore–Penrose inverse determines

$$P_{\text{abs}} = I - \mathcal{A}_\partial^\dagger \mathcal{A}_\partial. \quad (34.10)$$

Theorem 34.5 (Absorption condition and boundary response). P_{abs} is the orthogonal projection onto $\ker \mathcal{A}_\partial$. For any source j , the source $j_{\text{abs}} = P_{\text{abs}} j$ satisfies

$$\gamma_\partial G_{\text{ret}} j_{\text{abs}} = \gamma_\partial G_{\text{adv}} j_{\text{abs}}. \quad (34.11)$$

It is the absorbed source closest to j for the stated inner product. Nonzero absorbed sources exist exactly when $\text{rank } \mathcal{A}_\partial < \dim \mathcal{H}_J$.

Proof. The orthogonal decomposition $\mathcal{H}_J = \ker \mathcal{A}_\partial \oplus \text{im } \mathcal{A}_\partial^*$ shows that $\mathcal{A}_\partial^\dagger \mathcal{A}_\partial$ is the identity on the second summand and zero on the first. Its complement is the stated projection. By definition, $2\mathcal{A}_\partial j_{\text{abs}} = \gamma_\partial(G_{\text{ret}} - G_{\text{adv}})j_{\text{abs}} = 0$. For $k \in \ker \mathcal{A}_\partial$, the orthogonal decomposition gives

$$\|j - k\|^2 = \|(I - P_{\text{abs}})j\|^2 + \|P_{\text{abs}}j - k\|^2.$$

The unique minimum corresponds to $k = P_{\text{abs}}j$. The final equivalence is the rank–nullity theorem. $\square$

In the source–field pairing, the symmetric action is $\mathcal{S}[j] = \frac{1}{2}\langle j, G_{\text{sym}} j \rangle$. The stated adjoint relation makes this quantity real. For real variations,

$$d\mathcal{S}[j](h) = \text{Re}\langle h, G_{\text{sym}} j \rangle.$$

The action restricted to absorbed sources is written on the ambient space as

$$\mathcal{S}_{\text{abs}}[j] = \frac{1}{2}\langle P_{\text{abs}}j, G_{\text{sym}} P_{\text{abs}}j \rangle, \quad d\mathcal{S}_{\text{abs}}[j](h) = \text{Re}\langle P_{\text{abs}}h, G_{\text{sym}} P_{\text{abs}}j \rangle. \quad (34.12)$$

When the pairing identifies field and source through Riesz representation, the ambient gradient is $P_{\text{abs}} G_{\text{sym}} P_{\text{abs}} j$. The field response remains $G_{\text{sym}} j_{\text{abs}}$; the gradient projection expresses the restriction of variations. The formulas are proved by expanding the quadratic term of $j + th$ and using self-adjointness. A matter term $\mathcal{S}_{\text{matter}}$ adds its own differential.

34.6 Explicit absorbing condition in a chronological window

The channel in Subsection 34.3 allows this condition to be evaluated without introducing a target response. Let a source be supported on $0, \dots, N$, with $N \geq 1$. Transport its components to the reference fibre:

$$\hat{j}_k = U(0, k)j_k, \quad M_0(j) = \sum_{k=0}^N \hat{j}_k, \quad M_1(j) = \sum_{k=0}^N k \hat{j}_k.$$

The identity $(n - k)_+ - (k - n)_+ = n - k$ gives

$$(G_{\text{ret}} - G_{\text{adv}})j|_n = U(n, 0)(nM_0(j) - M_1(j)). \quad (34.13)$$

Proposition 34.6 (Absorption and two transported moments). *If the trace evaluates the field at two distinct positions a, b , equality of the advanced and retarded traces is equivalent to $M_0(j) = M_1(j) = 0$. In a fibre of dimension d , the space of such sources has dimension $(N - 1)d$. For $N \geq 2$, it contains the nonzero sources with transported components $(z, -2z, z, 0, \dots, 0)$, $z \neq 0$.*

Proof. Invertibility of $U(a, 0)$ and $U(b, 0)$ reduces the boundary condition to $aM_0 - M_1 = 0$ and $bM_0 - M_1 = 0$. Subtracting the equations gives $(a - b)M_0 = 0$, and then $M_1 = 0$. The two-moment map is surjective: for prescribed values (z_0, z_1) , it suffices to take $\hat{j}_0 = z_0 - z_1$, $\hat{j}_1 = z_1$, and all other components zero. Its rank is $2d$; its nullity is $(N + 1)d - 2d$. The indicated source has both moments zero. $\square$

In the transported frame, write

$$B_N = \begin{pmatrix} 1 & 1 & \cdots & 1 \\ 0 & 1 & \cdots & N \end{pmatrix}.$$

The absorbing projection is evaluated by

$$P_N = [I - B_N^\top (B_N B_N^\top)^{-1} B_N] \otimes I_{\mathcal{H}_0}. \quad (34.14)$$

The two rows of B_N are independent for $N \geq 1$, so the inverse exists. Unitary transport between frames carries this projection to the original fibres. For $N = 2$,

$$P_2 = \frac{1}{6} \begin{pmatrix} 1 & -2 & 1 \\ -2 & 4 & -2 \\ 1 & -2 & 1 \end{pmatrix} \otimes I_{\mathcal{H}_0}.$$

This construction produces a nonzero absorbed class and its projection operation in the chronological channel. Its amplitude and its identification with a matter current retain the rules of the corresponding physical sector.

34.7 Refinement and continuity of the boundary condition

Let $R_m^\Phi, R_m^J, R_m^\partial$ be the refinement maps for fields, sources, and boundaries. Suppose that they intertwine both Green operators and the trace. Composition gives

$$R_m^\Phi G_{\text{sym}, m} = G_{\text{sym}, m+1} R_m^J, \quad R_m^\partial \mathcal{A}_{\partial, m} = \mathcal{A}_{\partial, m+1} R_m^J. \quad (34.15)$$

The second equality carries an absorbed source to another absorbed source. Naturality of the projections additionally preserves the orthogonal structure of the realizations.

Proposition 34.7 (Naturality of the absorbing projection). *For a source isometry $R = R_m^J$, one has $P_{\text{abs}, m+1} R = R P_{\text{abs}, m}$ exactly when*

$$R(\ker \mathcal{A}_{\partial, m}) \subset \ker \mathcal{A}_{\partial, m+1}, \quad R((\ker \mathcal{A}_{\partial, m})^\perp) \subset (\ker \mathcal{A}_{\partial, m+1})^\perp.$$

Under these conditions at every level, the projections define an orthogonal projection on the completion of the isometric inductive limit of the source spaces.

Proof. Each source is decomposed into its components in the kernel and its orthogonal complement. The two inclusions imply that applying the projection before or after refinement respectively preserves the first component and annihilates the second. Conversely, the operator equality applied to each component gives the two inclusions. In the inductive limit, the equality allows an operator to be defined on the union of the levels independently of the representative. All projections have norm at most one, so the operator extends to the completion. Its self-adjointness and idempotence identities, valid on the dense union, pass by continuity to the completion. $\square$

To prolong a Green response to a completion as well, its domains and the corresponding continuity estimates are retained. Chronological propagation with finite support and factorization of the response through the boundary image, proved in 33.9, thus retain their respective domains throughout refinement.

The physical realization of emission and absorption composes these operators with the matter currents, action, and causal conditions of the sector. The intertwining must preserve the route involution, source–field pairing, transported action, and boundary conditions at each level. The preceding construction provides group reversal, the Green operators of the chronological channel, temporal decomposition, and an explicit absorbing projection; spatial realizations use their own maps and causality conditions.

35 Conclusions

Torsional amplitude is determined by the spinorial action and material state, with an explicit formula and a sector in which its positivity is preserved. This result yields an energetic contribution proportional to $-a^{-6}$, a unique homogeneous bounce under the proved conditions, and, for flat dust with $\Lambda_{\text{eff}} = 0$, a regular evolution for all time with complete Levi–Civita causal geodesics. The sequence connects a quartic state observable to a minimum cosmological scale. Its origin remains in the APP–TRIT–TPK construction: orientation, remainder, carry, and depth are transported through to the actions, currents, and observables of the geometric realization.

35.1 Determination of the amplitude and cosmological evolution

The central result of the spinorial realization is the determination of torsional amplitude as a functional of the material state:

$$s_0[\varrho, V_c] = \frac{3\kappa c^2 \hbar^2}{16} \frac{\gamma^2}{1 + \gamma^2} \frac{\text{Tr}(\varrho \hat{Q}_{\text{sp}})}{V_c^2}.$$

The formula follows successively from the internal Clifford representation, the spin current, elimination of contorsion, the normally ordered quartic observable, and lapse and volume variations. The quadratic volume factor expresses the dependence of density on the second moment of the current. A zero mean spin retains a second moment that can contribute to geometry.

In the occupation-three sector one obtains $\hat{Q}_{\text{sp}} = 4I$. For trace-one states supported in that sector, every unitary evolution preserving the sector preserves its expectation and determines a positive amplitude. The material state and comoving cell individualize the realization; G , $\hbar$, and γ retain their structural antecedents. The reduced Planck constant enters quadratically into the interaction, Barbero–Immirzi determines the coefficient of the Holst operator, and gravitation converts that interaction into a geometric response. This composition jointly explains the functional value of the amplitude and the role of each factor.

The memory law $M_n = \varphi^{-2n}$ and spatial reading $a_n/a_{\text{ref}} = \varphi^{n/3}$ produce dilution a^{-6} . In the spinorial realization with preserved second moment, the same power appears through material variation. With $\rho_0, s_0 > 0$, $w < 1$, zero spatial curvature, and $\Lambda_{\text{eff}} = 0$, the bounce threshold is unique and transverse. Two distinct bounds locally preserve its existence, uniqueness, and time orientation under perturbations.

The dust sector also admits an exact solution for all time. The scale factor remains bounded away from zero, curvature is finite, and the causal Levi–Civita geodesics are complete in both directions. The global result belongs to that explicit realization; persistence of the bounce is established under the local conditions of its theorem. Both proofs show how the amplitude obtained from the state translates into a minimum scale and an evolution through contraction and expansion.

35.2 From transported memory to gravitational response

Radial normalization follows the longitudinal composition $L_\alpha = (5759/23040)\alpha^{16}L_*$. Normalized archive transport and the positive metric yield $R_X = L_\alpha \cup X U^*$; energy preserves the same character and its conjugate length satisfies $H_X \Lambda_X = \hbar_{\text{ret}} c I$. Under the reduced-radius reading, these identities determine $G = c^3 L_\alpha^2 / \hbar_{\text{ret}}$, common to the positive modes of the realization. The proof retains its longitudinal and metric rules and the stated physical identification. Joint boundary and memory reduction preserves the coefficient; dynamic reduction also transports the spectral parameter and the temporal norm. The circular expression is recovered with $t_0 = \pi_{\text{HMT}} L_\alpha / (54c)$.

The five-component space is obtained through an explicit projection of the twelve vertices and is transported isometrically to symmetric trace-free tensors. Transverse reduction preserves two components. The sequence $12 \rightarrow 5 \rightarrow 5 \rightarrow 2$ distinguishes the initial incidence, the gravitational carrier, its tensor representation, and the transverse reading. The coefficient G_a fixes the common coupling; orientations belong to the operator and its components.

In the same return section, the spinorial-amplitude coefficient has the equivalent expression

$$s_0[\varrho, V_c] = \frac{3\pi c \hbar_{\text{ret}} L_\alpha^2}{2} \frac{\gamma^2}{1 + \gamma^2} \frac{\text{Tr}(\varrho \hat{Q}_{\text{sp}})}{V_c^2}.$$

It follows by substituting $\kappa = 8\pi L_\alpha^2/(\hbar_{\text{ret}} c)$ into the proved functional. Length and action therefore fix its coefficient, while the state and volume retain their material roles in the cosmological solution.

The memory screen provides soldering and an energetic response compatible with refinement. Its minimizing extension, constructed from transported incidence, determines an energy and a differential with respect to the connection. The calculation includes variations of incidence, boundary, weight, and normalization. Upon passage to the Cartan realization, the co-tetrad and the material pairing convert this differential into a current with an explicit geometric domain.

The connection equations admit algebraic inversion of the Holst and torsion operators. In the material sector affine in contorsion, this inversion produces the effective correction to the action. For the minimal extension, whose current generally depends on the connection itself, the complete stationarity equation, the local continuation criterion through its Hessian, and the integral formula for the effective action are proved. The spinorial and minimizing-extension actions therefore retain their own arguments and variation laws. The former supplies the amplitude functional evaluated in the cosmological solution; the latter describes the full variational response of the transported screen.

35.3 Information, horizons, and relational response

The residual balances bring together the trace, the trace-free part, and their divergences in one formulation. With constant couplings in the chart and a conserved material source, the residual tensor covariantly preserves its total balance; variation of its trace determines exchange between the two sectors. A constant trace characterizes separate conservation.

Horizon information uses the areal unit, action, and thermal conversion previously constructed. The reduced state and its purification distinguish accessible information from preserved correlations. In the spherical balance developed here, $TS = E$, $TdS = 2dE$, and $SdT = -dE$ are jointly verified; the dependence of temperature on radius is part of these identities. The Boltzmann constant converts informational content within the thermal section, while the action-gravitation relation determines the area unit of its geometric reading.

The areal and chronological composition determines $\mathcal{J}_{I/A} q_A / \log 3 = h / (108 t_0 \log 3)$. At $R = L_\alpha$, cosmological and chronological temperatures are related by the factor $\log 3 / (2\pi)$. For the response Hamiltonian $H_R = \tau_R \Delta_0 D_\partial$, the weights 90 and 120 correspond to different energy gaps with a common scale. The finite balance is exactly $\mathcal{E}_R - T_{\cos} \mathcal{S}_R = \tau_R \mathcal{D}(\sigma \| \rho_0)$, and its tangent term is the first law at fixed radius. The relative section constructed from that state determines a unique positive transducer; identification of an additional reader reduces to checking its value on the chronological section in (32.48). Record-preserving isometries transport this balance and its observables.

The observer is defined through operations of orientation, trivialization, and restriction of observables. The self-inertial response preserves its factorization through global information and makes the defect produced by projection explicit. Finally, in the realized chronological channel, the covariant difference constructed with route transport admits retarded and advanced Green operators for finitely supported sources. Compatibility of both readings at the boundary specifies a source space and a projection entering the variational response. Bidirectionality is expressed by operators, moments, and boundary conditions that can be checked within the article.

The decisive relationship is that information preserved by the state enters the geometric response through a current and its second moment. Elimination of contorsion and metric variation convert that observable into a density and pressure; the dust solution shows how both determine a minimum scale and a causally complete evolution. Balances and the observer's reading preserve, in turn, the distinction between accessible information and global correlation. Within the constructed domains, memory, material source, and geometry thus form an explicit chain of determinations, from the arithmetic kernel to gravitational evolution and its informational content.

A Notation for the gravitational realizations

The notation distinguishes the objects of the construction, the response operators, and their physical realizations. The positive inner products of the extensions and the Lorentzian metric have different roles.

Symbol	Object and role
$G_a, \hbar_a$	Reciprocal sections of gravitation and action. Their product determines the invariant area $\ell_p^2 = \hbar_a G_a / c^3 = L_\alpha^2$ in the radial realization.
$L_*, L_\alpha = L$	Longitudinal reference section and constructed length $L_\alpha = (5759/23040)\alpha^{16}L_*$.
E_L, t_P, t_0	Return energy $E_L = \hbar_{\text{ret}}c/L_\alpha$, time $t_P = L_\alpha/c$, and step $t_0 = \pi_{\text{HMT}}t_P/54$.
R_X, Λ_X	Positive radial reader and conjugate length of the same character; $R_X\Lambda_X = L_\alpha^2I$.
R_h, ζ_h	Areal horizon radius and coordinate $R_h/L_\alpha = \zeta_h > 0$. They retain the horizon's geometric condition.
$\chi_{\text{rad}} = L_\alpha/E_L$	Dimensional ratio between radius and energy. Its units and role differ from those of record K and of $\kappa = 8\pi G/c^4$.
γ_{HMT}	Evaluation of the hexad–octad functional, reused as a coefficient of the Holst operator.
k_B, Θ	Information-conversion coefficient and temperature in the stated thermal section.
P_5, Γ_5, Λ_n	Projection onto the five coordinates, tensor isometry, and rank-two transverse reduction.
c_s, c_g	Translational-memory cocycle and return component of oriented transport.
$\beta_m^\square, \Lambda_m^\square$	Screen soldering and energetic response, with their quadratic identity and scale.
$C, L, f_{\min}$	Transported incidence, Gramian, and minimum-energy extension for the boundary datum.
$s, \sigma, M_{\text{cel}}$	Covector of the differential, material current, and pairing that converts their coordinates.
$e, \omega, \hat{\omega}, K, T$	Co-tetrad, connection, Levi–Civita connection, contorsion, and torsion.
P_γ, L_e	Internal Holst operator and linear map from torsion to the connection equation.
$Q, F, \Delta S$	Quadratic gravitational term, material action, and effective correction after eliminating contorsion.
H	Hessian of the composed action controlling the local continuation of a torsional branch.
M_n, a, a_{ref}	Memory, scale factor, and reference normalization of its homogeneous realization.
s_0, ρ_0, w	Torsional amplitude, reference material density, and barotropic parameter of the bounce domain.
$\varrho, V_c, \hat{Q}_{\text{sp}}$	Normalized material state, comoving volume, and normally ordered quartic observable; they determine the functional $s_0[\varrho, V_c]$.
C_γ	Coefficient of the effective spinorial interaction obtained by eliminating contorsion.
τ, t_b	Characteristic duration and time of the minimum of the homogeneous dust solution.
F_0, R, a_b	Friedmann function, perturbation, and transverse threshold of homogeneous evolution.
T_Λ, T_0	Trace and traceless parts of the residual tensor; their divergences record exchange between sectors.
$G_{\text{ret}}, G_{\text{adv}}$	Retarded and advanced propagators. Here the letter G denotes operators, not the gravitational constant.
P_{abs}	Projection onto the sources satisfying equality of the two causal readings at the boundary.

References

- [1] *Simultaneous rigidity of the radial reader; metric composition, memory, and coupling*. HMT–MD developments of 25–26 September 2026. Proofs incorporated in 15; complete texts and verifiers in `reproduccion_gravitatoria_20260926/demostraciones` and `reproduccion_gravitatoria_20260926/pruebas`.
- [2] *Coupled action, phase, and boundary memory; rigidity and structural explanation*. HMT–MD developments of 25–26 September 2026. The proofs preserve lifted phase, cross-blocks of the weight, and the conditions of the radial realization. Local reproduction distinguishes exact checks, decimal evaluation, and realization assumptions.
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